



PORTFOLIO

2026

By Precious ELaish Nathan

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Student, B.Arch Year 4

About Me

Aspiring architect specializing in **flexible, organic forms and deconstructive design**. My work blends immersive gaming aesthetics with minimalist architecture across diverse typologies. Proficient in **ArchiCAD, SketchUp, and D5 Render**, I am seeking an internship to contribute to innovative, flexible, recreative projects.

Education

SSC (2021)– N.H.S.S., Airoli

HSC (2023)– N.H.S.S. Junior College, Airoli

B.Arch (Present)– V.E.S. College of Architecture

Languages

English

Hindi

Marathi (passive fluency)

Workshops

EDGE Design

Resin Art

Pottery

Photography

Wall Painting

Arch Construction

Water Colour

Origami

QGIS Mapping

Charcoal

Communication

Archicad Workshop

Mandala Art

Tensegrity

Parametric Modelling

Electives

Design Thinking

Disaster risk reduction

Research Writing

Digital Sketching

Model Making

Competitions

Solvathon (2025)

Solar Decathlon India (2026)

Soft Skills

Adept computer user

Conceptual sketching and design

Vivid learner

Proficient researcher

Good observation & detailing

Collaborative teamwork

Hobbies

Playing Video Games

Reading through ArchDaily

Listening to Music



Contacts

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preciouselaish@gmail.com

6689.precious

D.O.B.

12th August, 2005

Design Projects

SEM 1: Memory Lane, Museum (hypothetical site)

SEM 2: Playcare Centre, Karade Khurd, Raigad District

SEM 3: Skill Development Centre, Gagode, Pen

SEM 4: Community Housing, Bhuj, Gujarat

SEM 5: Science Centre, Nashik

SEM 6: Commercial Office (SDI), Ghansoli, Navi Mumbai

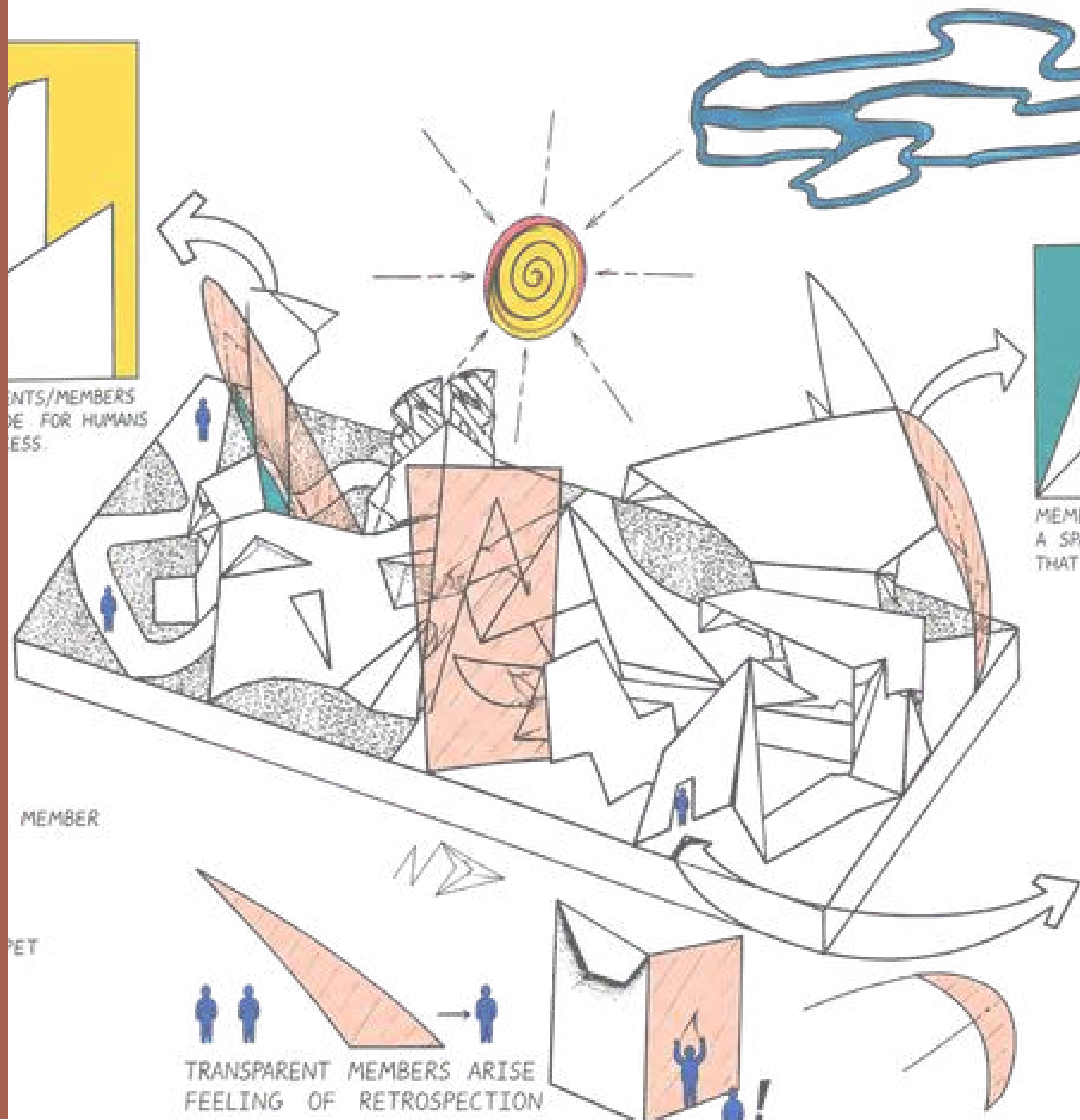
School building, Navi Mumbai



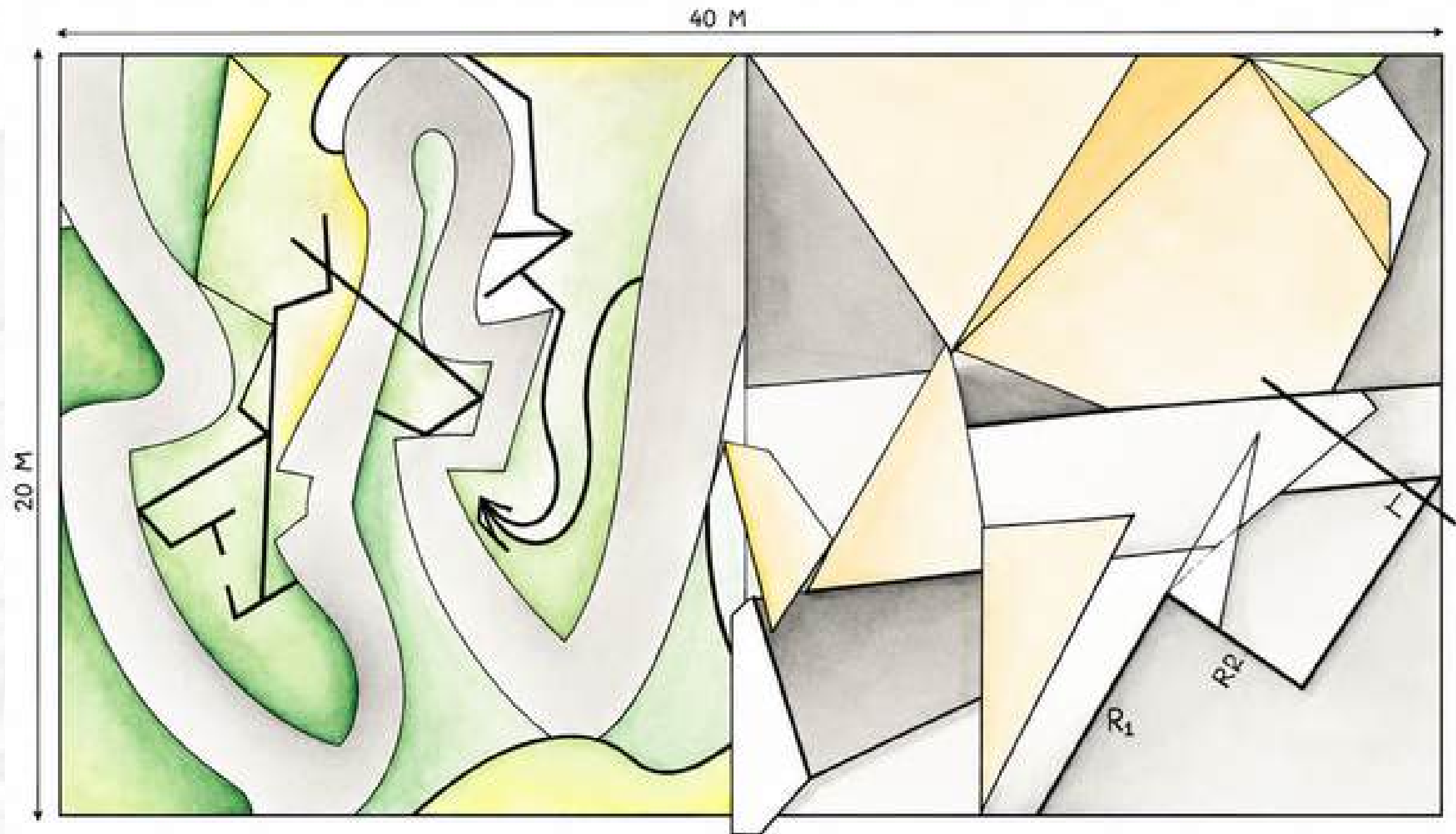
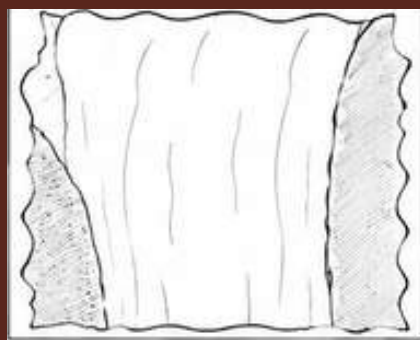
SESTER

1

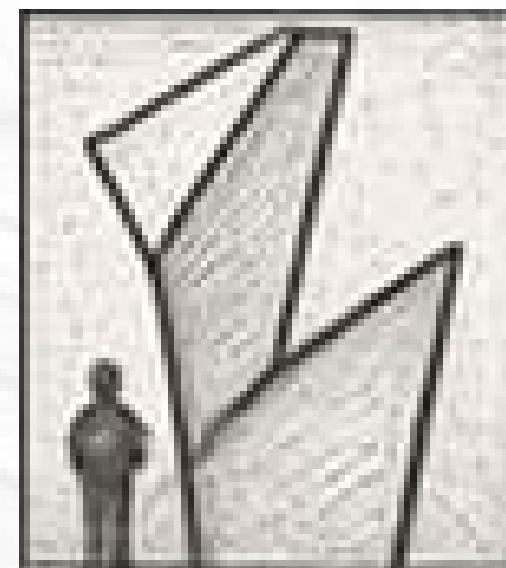
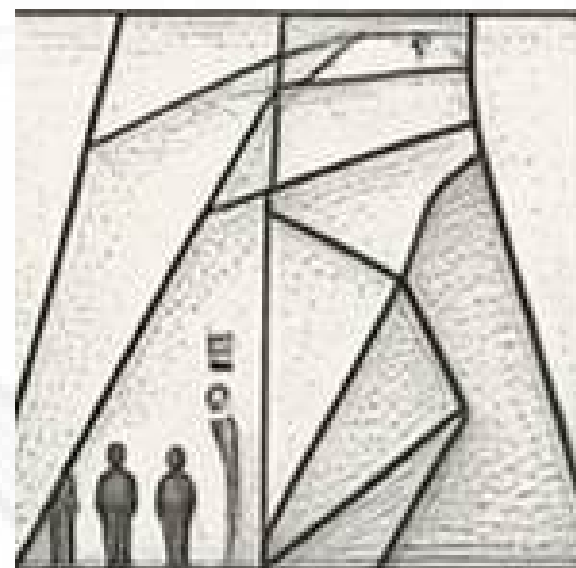
MUSEUM OF
MEMORY
(SCULPTURAL
ACCESSIBLE
BUILDING)

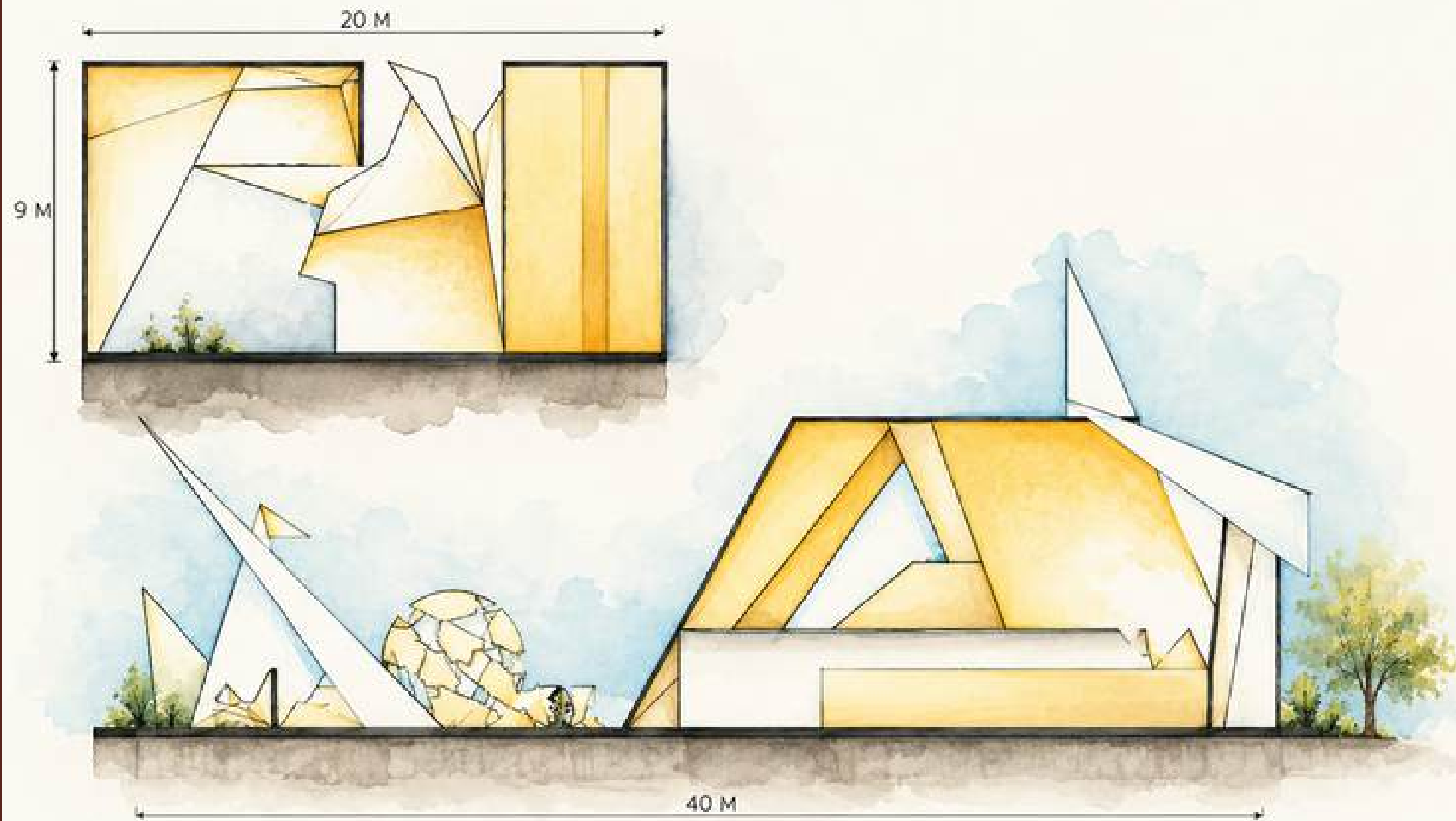
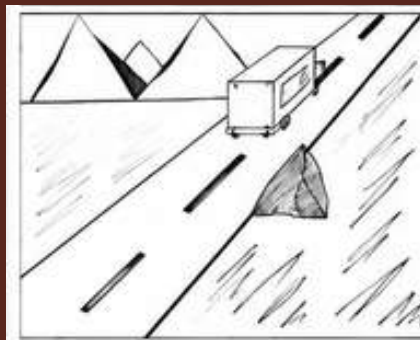
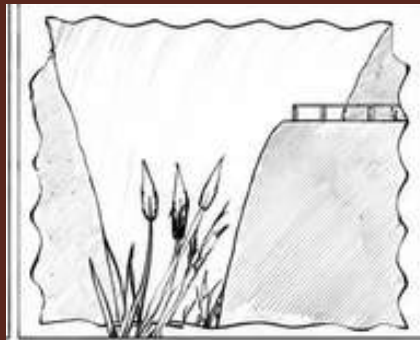
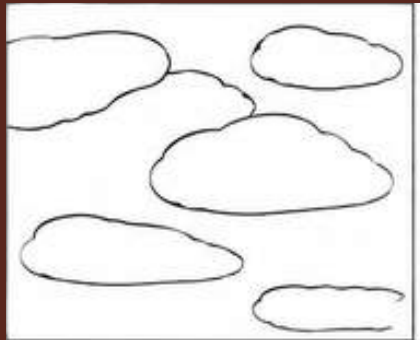


ARCHITECTURAL DESIGN

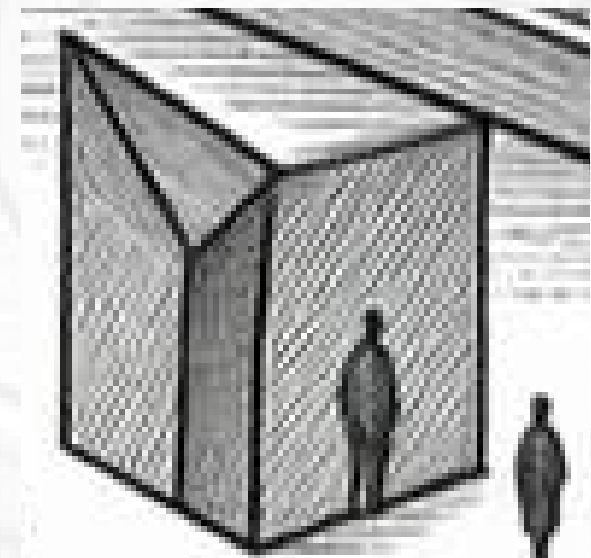
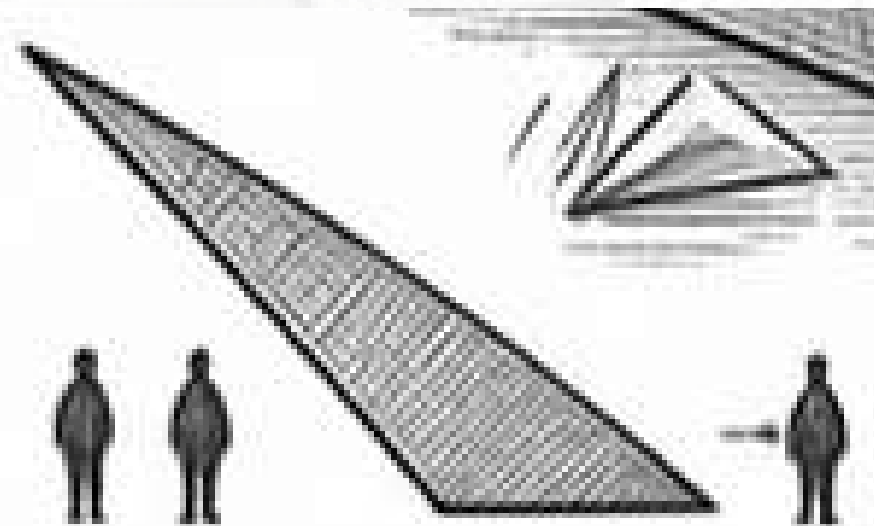
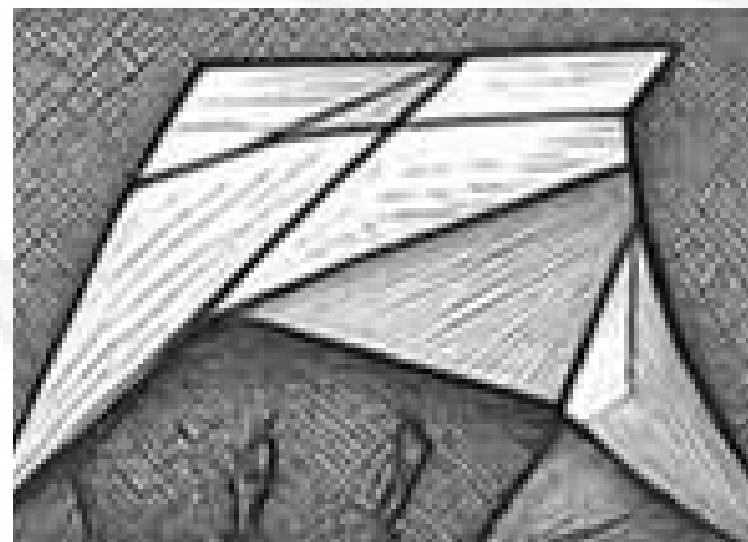


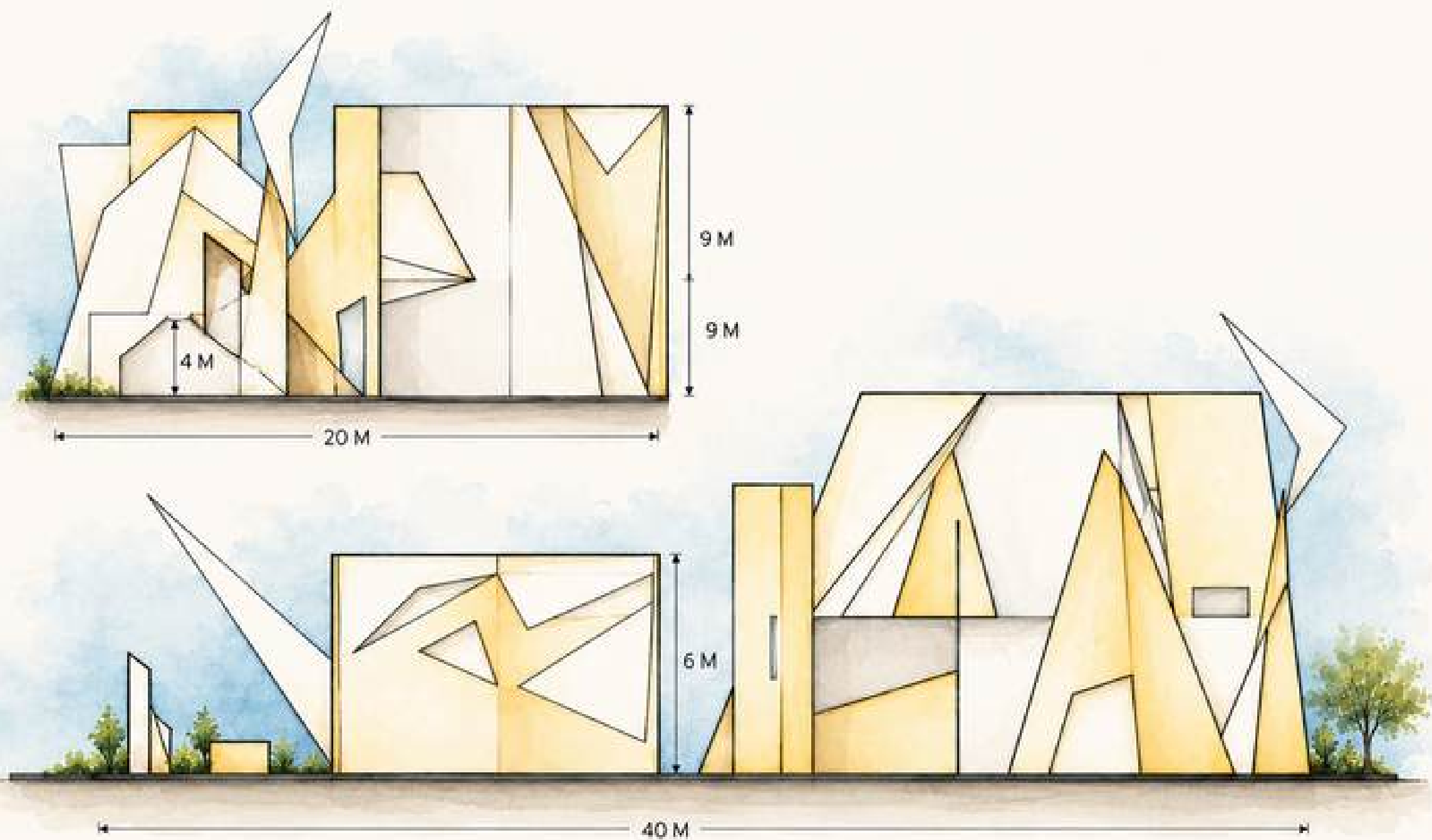
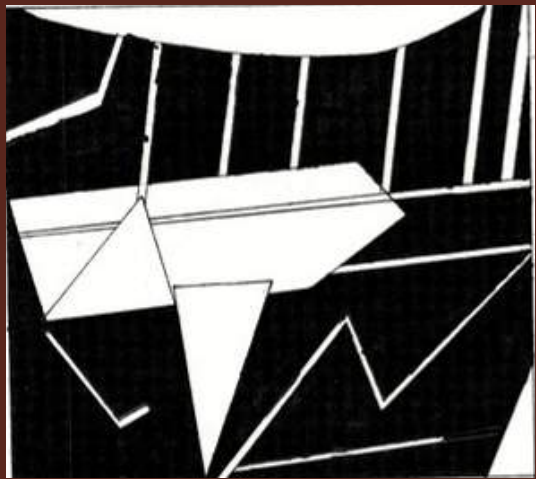
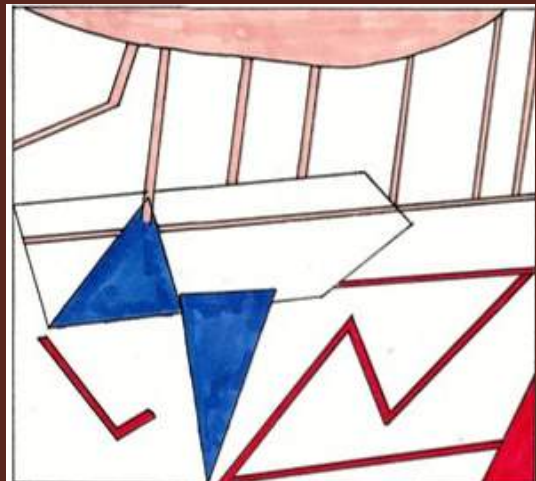
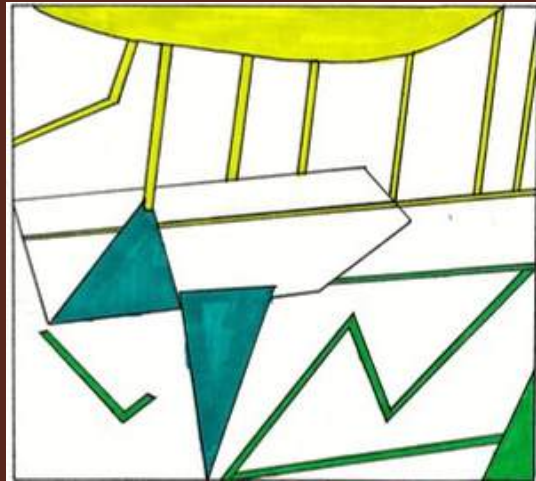
PLAN



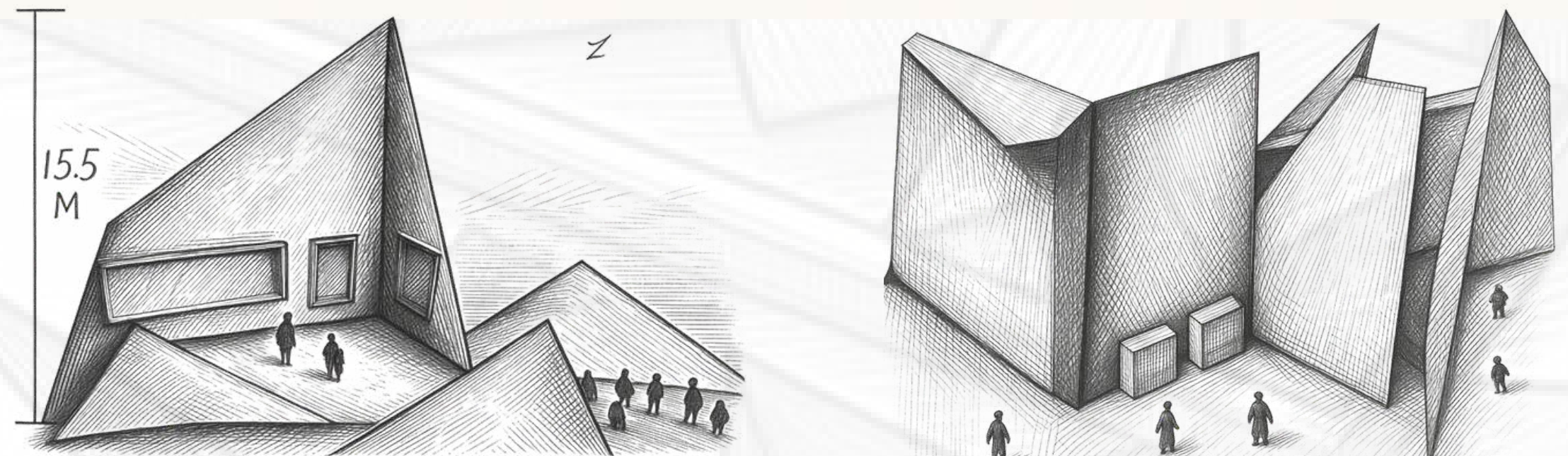


SECTIONS





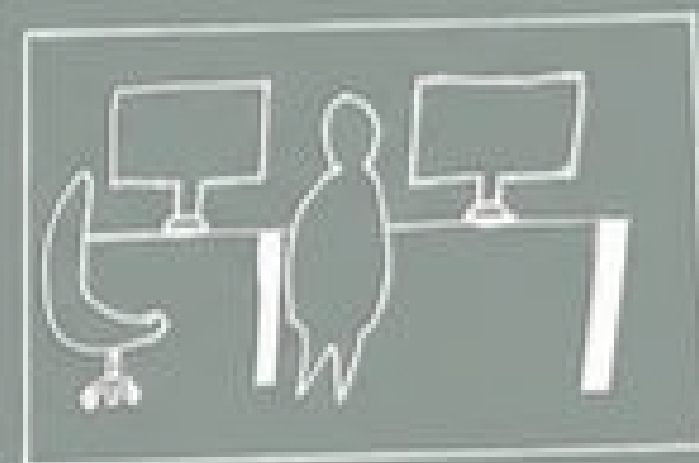
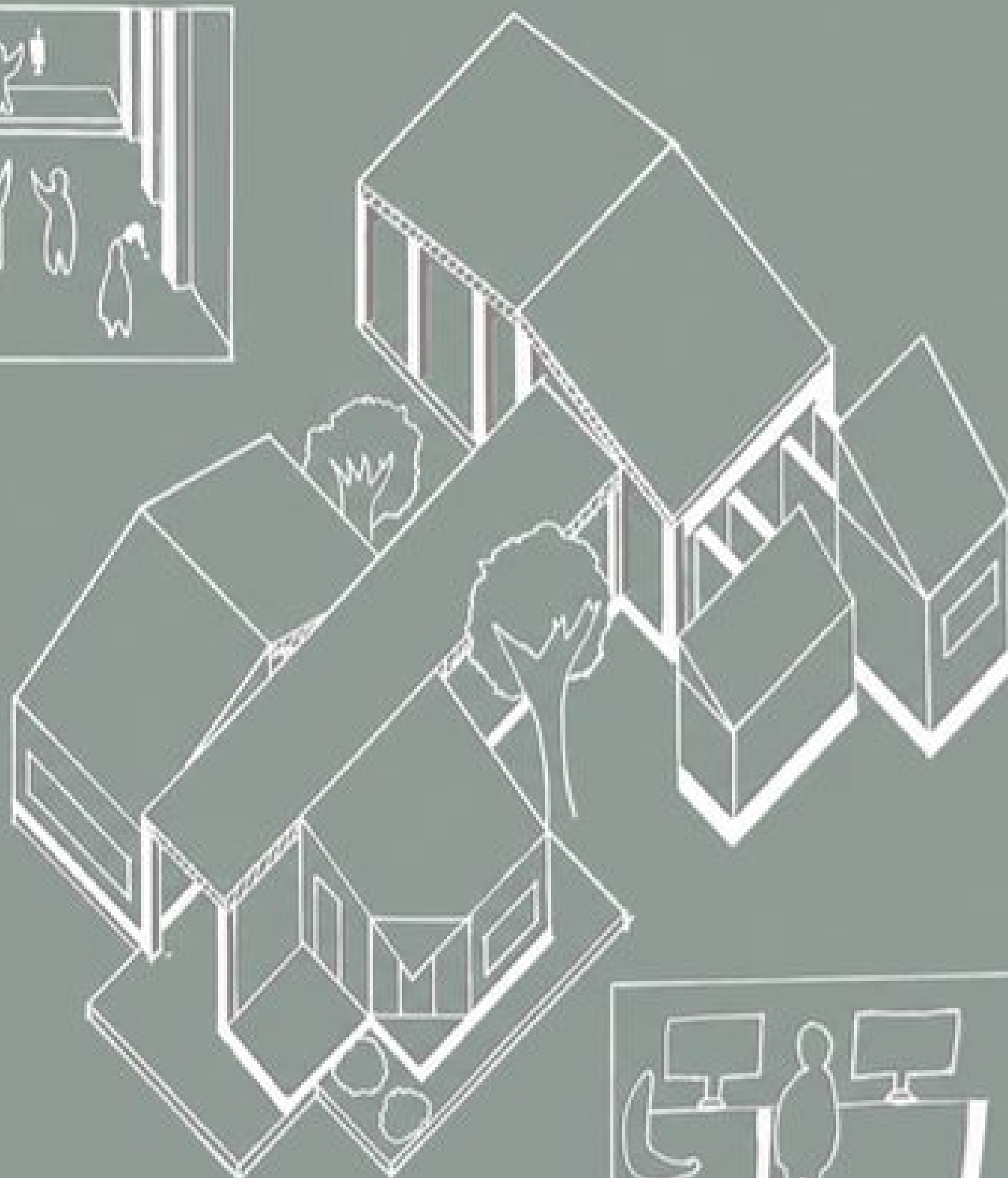
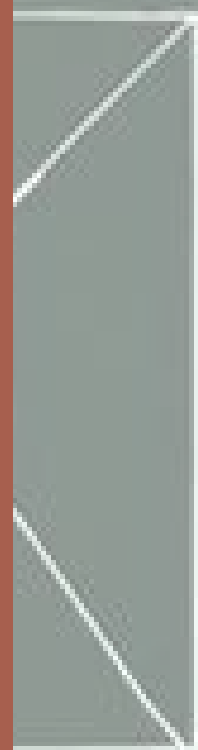
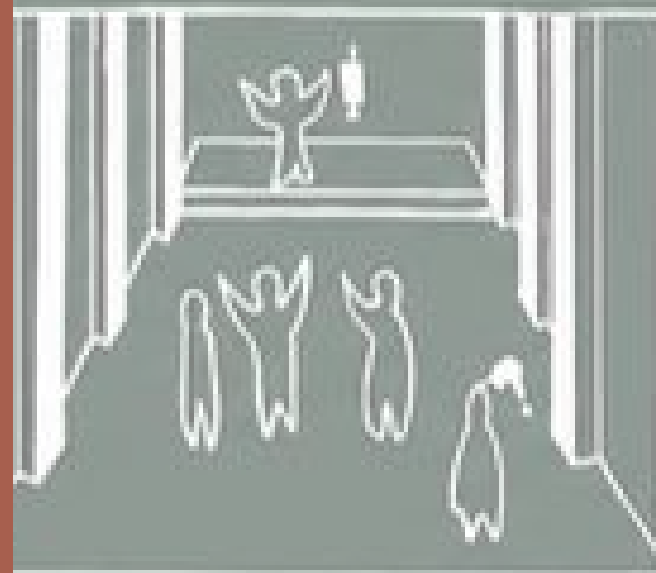
ELEVATIONS



S
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M
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R

2

PLAYCARE
CENTRE



ARCHITECTURAL DESIGN

SWOT ANALYSIS

S (STRENGTH) → ABUNDANCE OF VEGETATION;
CAN BE USED FOR SHADING
OF CANOPY, OFFICES, AIR.

• GOOD WATER SOURCE; PROVIDES
ACCURACY SUPPLY IN NEED.

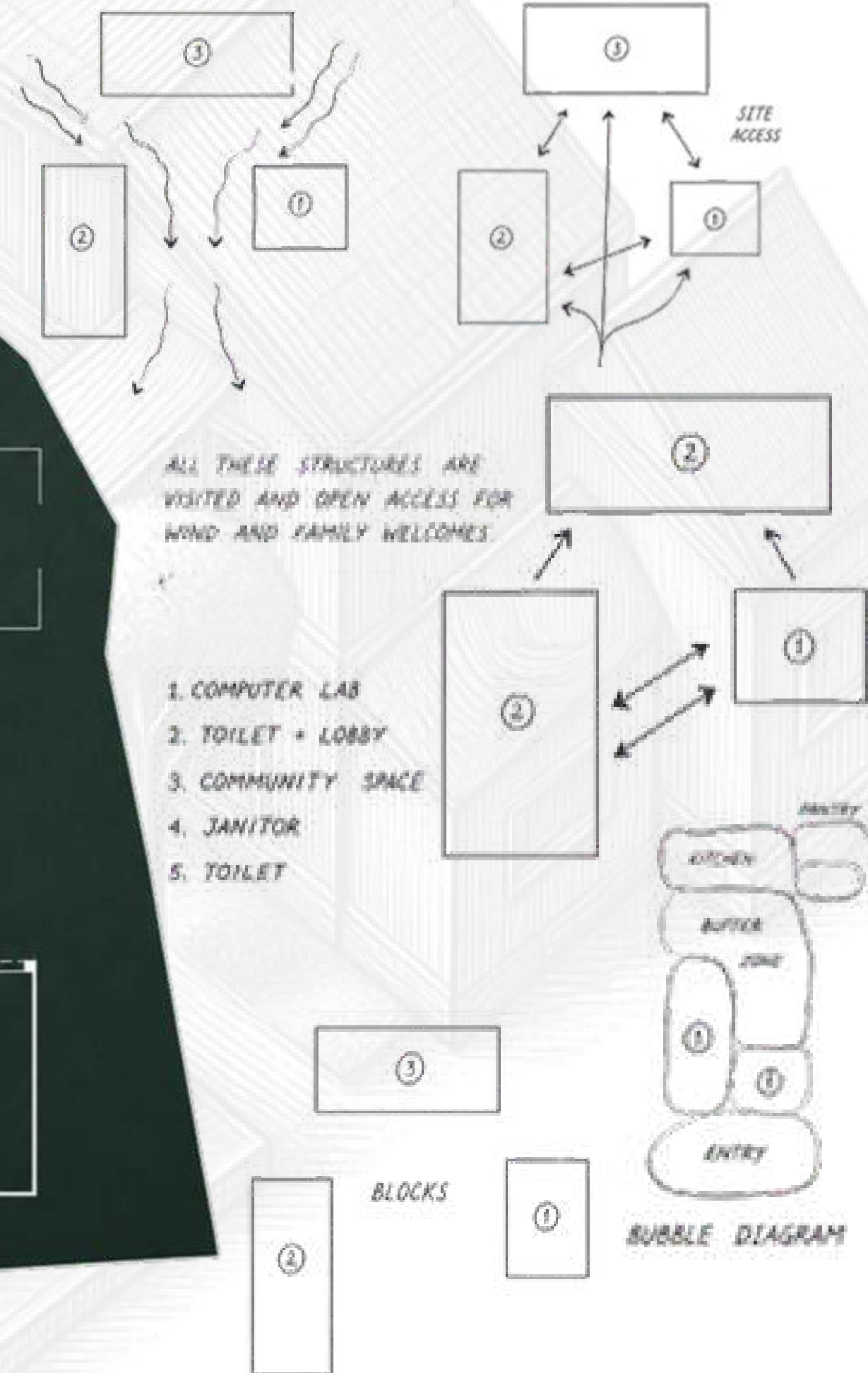
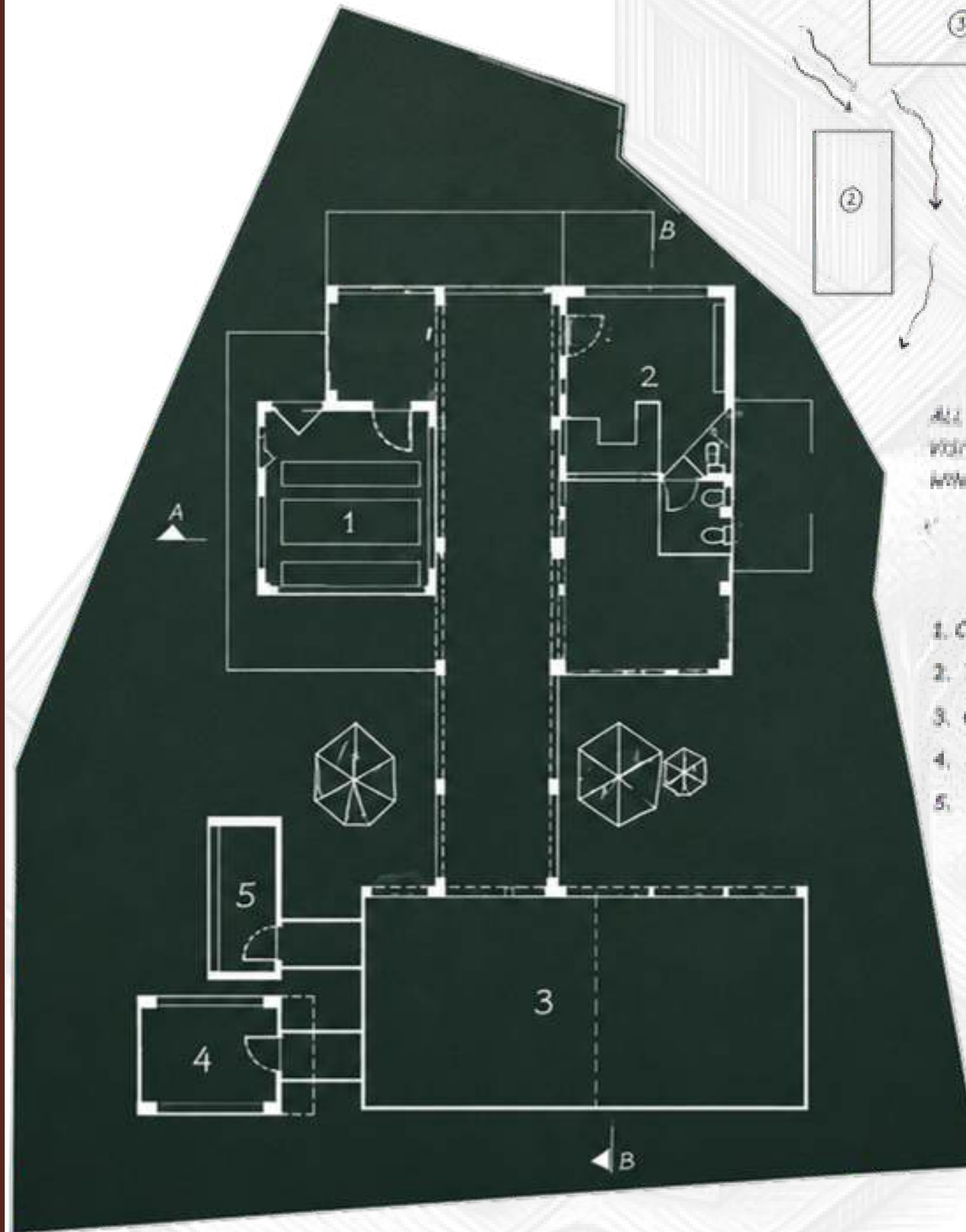
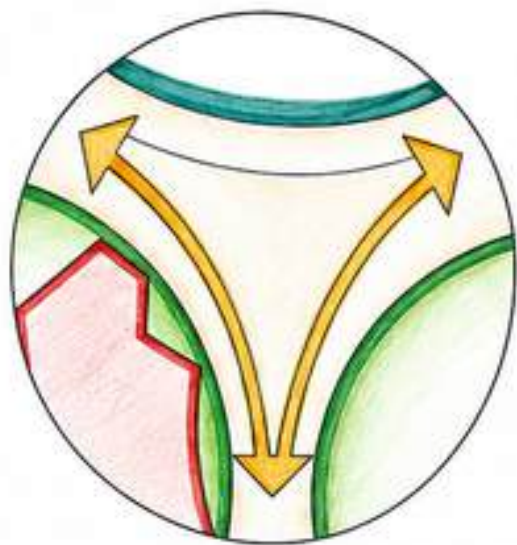


W (WEAKNESS) → EXTREME CLIMATE
(SNOW DURING MONSOON), CAN BE
RESISTED BY INCORPORATING GREEN
SPACES IN TREES.

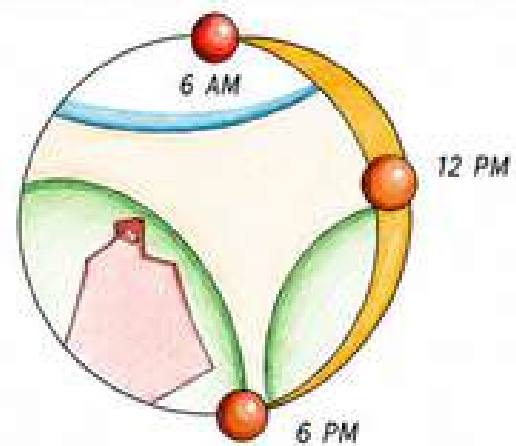


O (OPPORTUNITY) → AGRICULTURAL AND
URBAN DEVELOPER,
DUE TO ABUNDANCE OF
WATER SUPPLY AND ELECTRICITY.

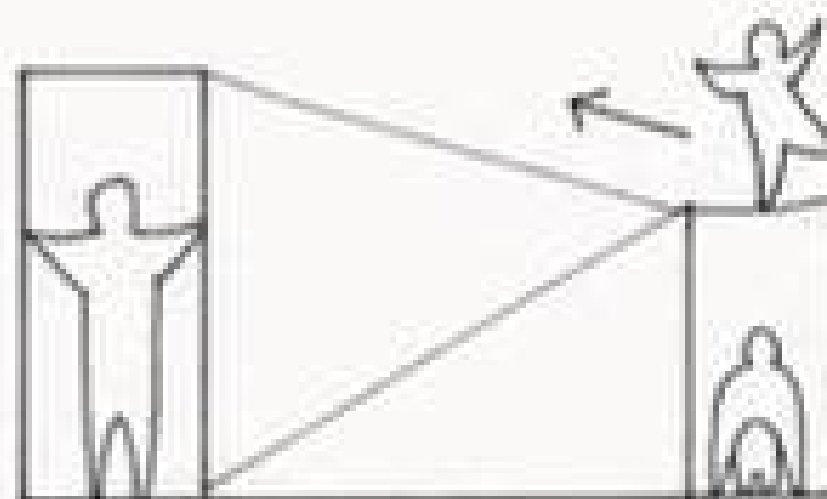
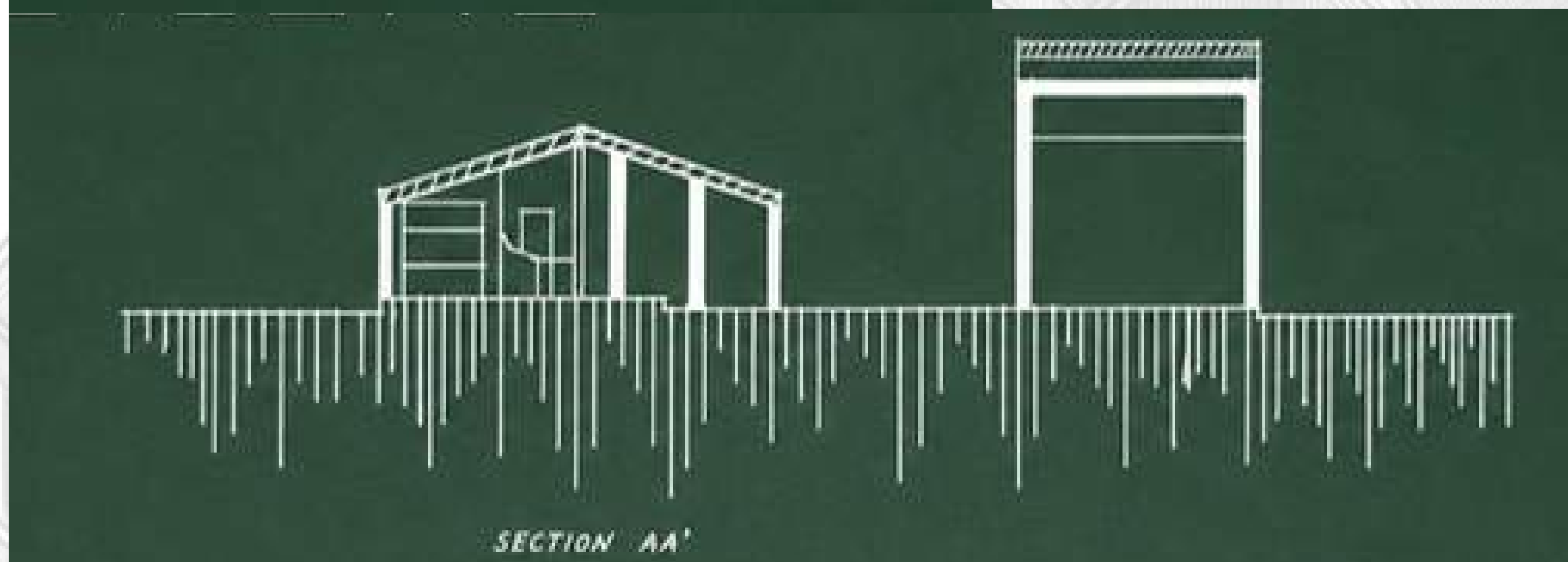
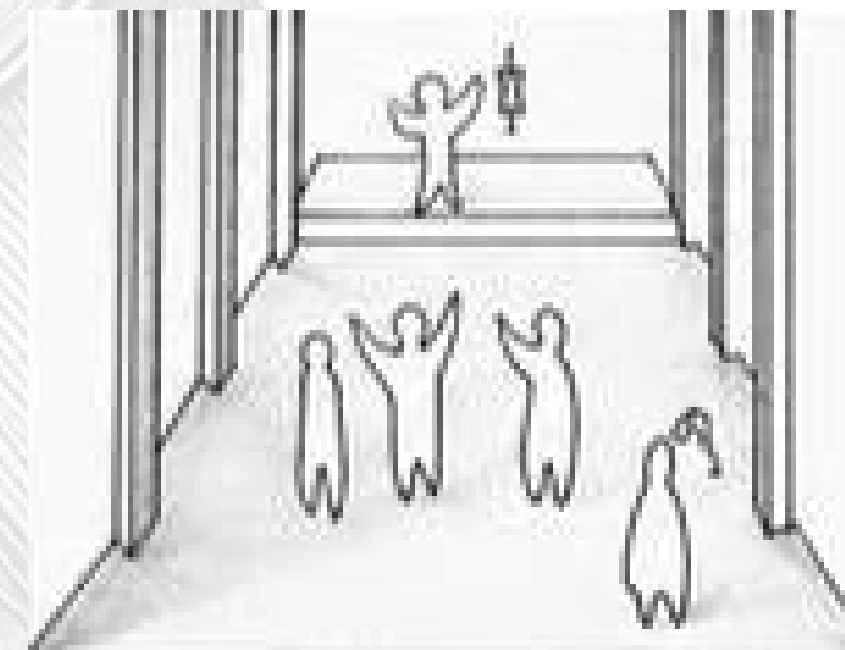
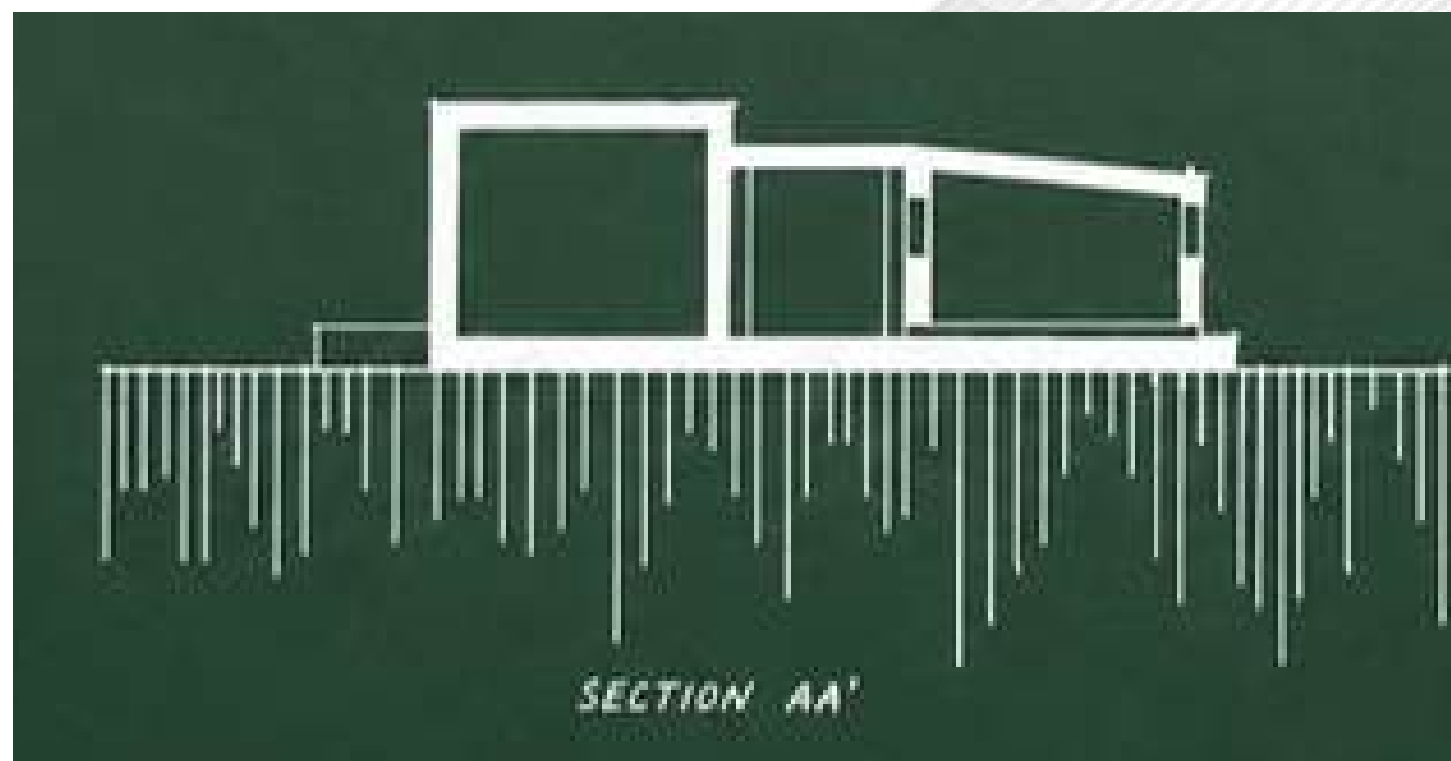
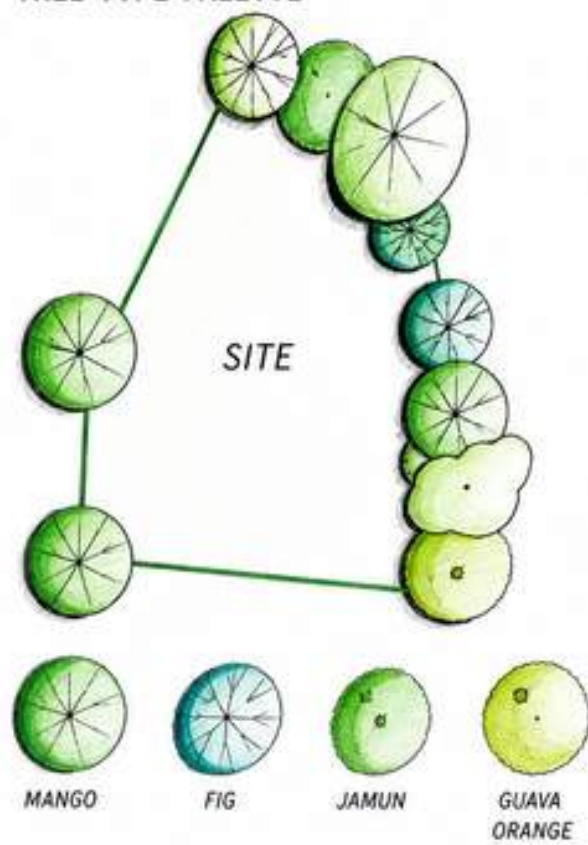
T (THREAT) → ANIMAL FLORA; CAN BE
RESISTED BY INCORPORATING
PLANTS AND LIVING IN DESIGN.

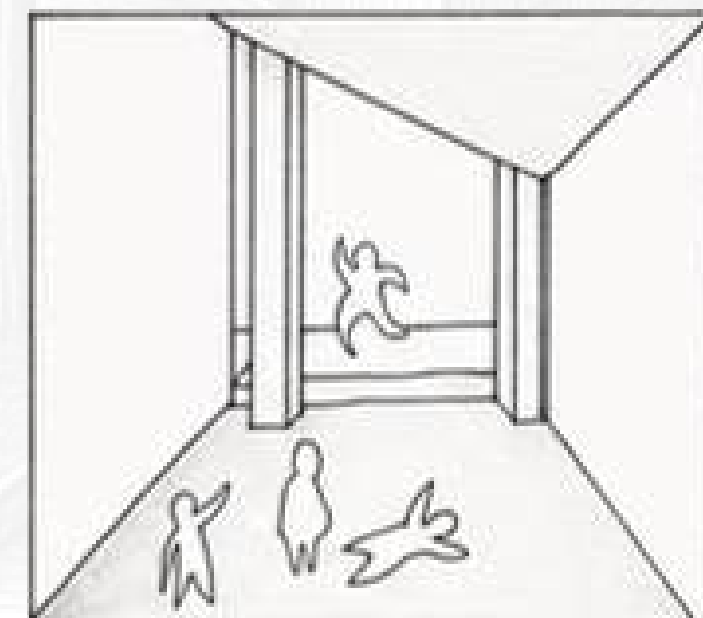
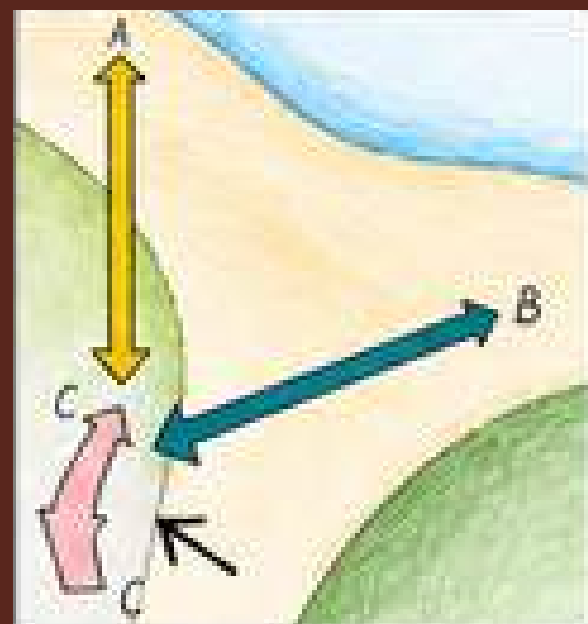
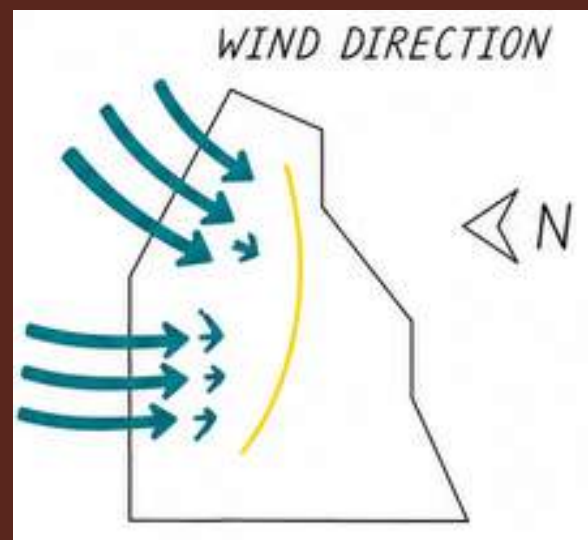
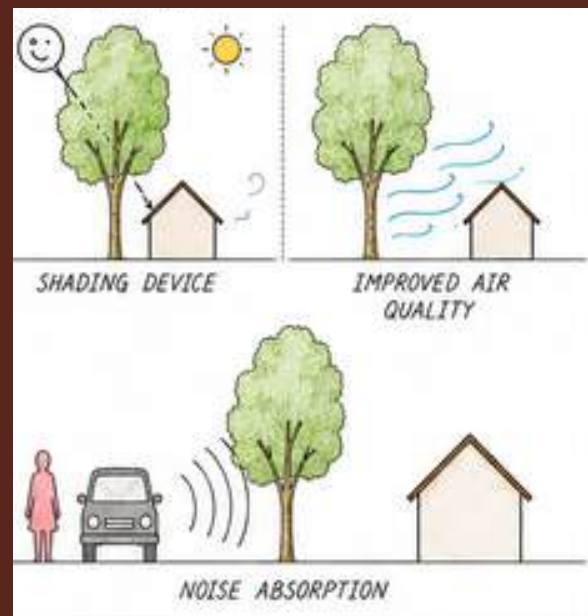


SUN PATH

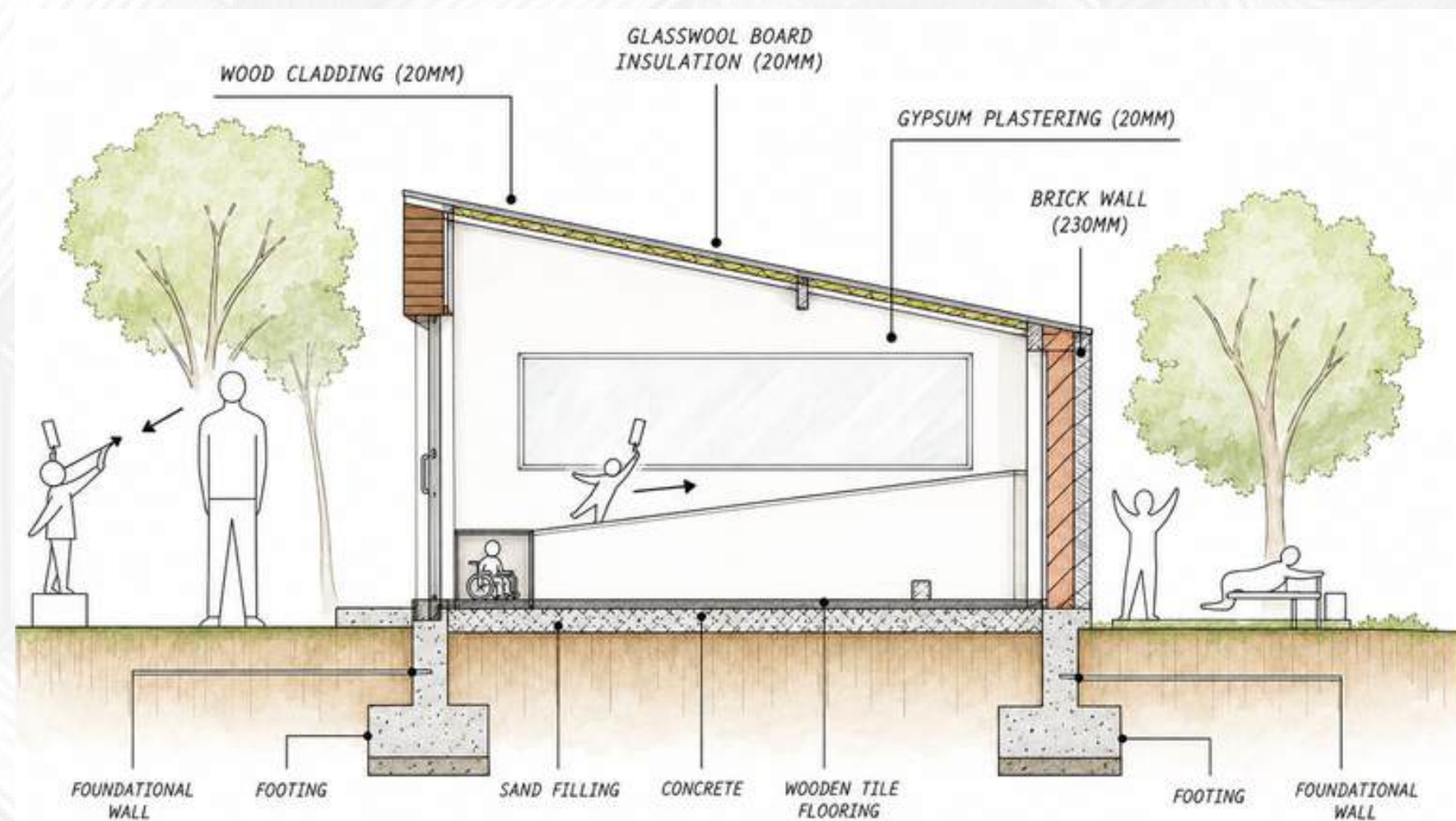
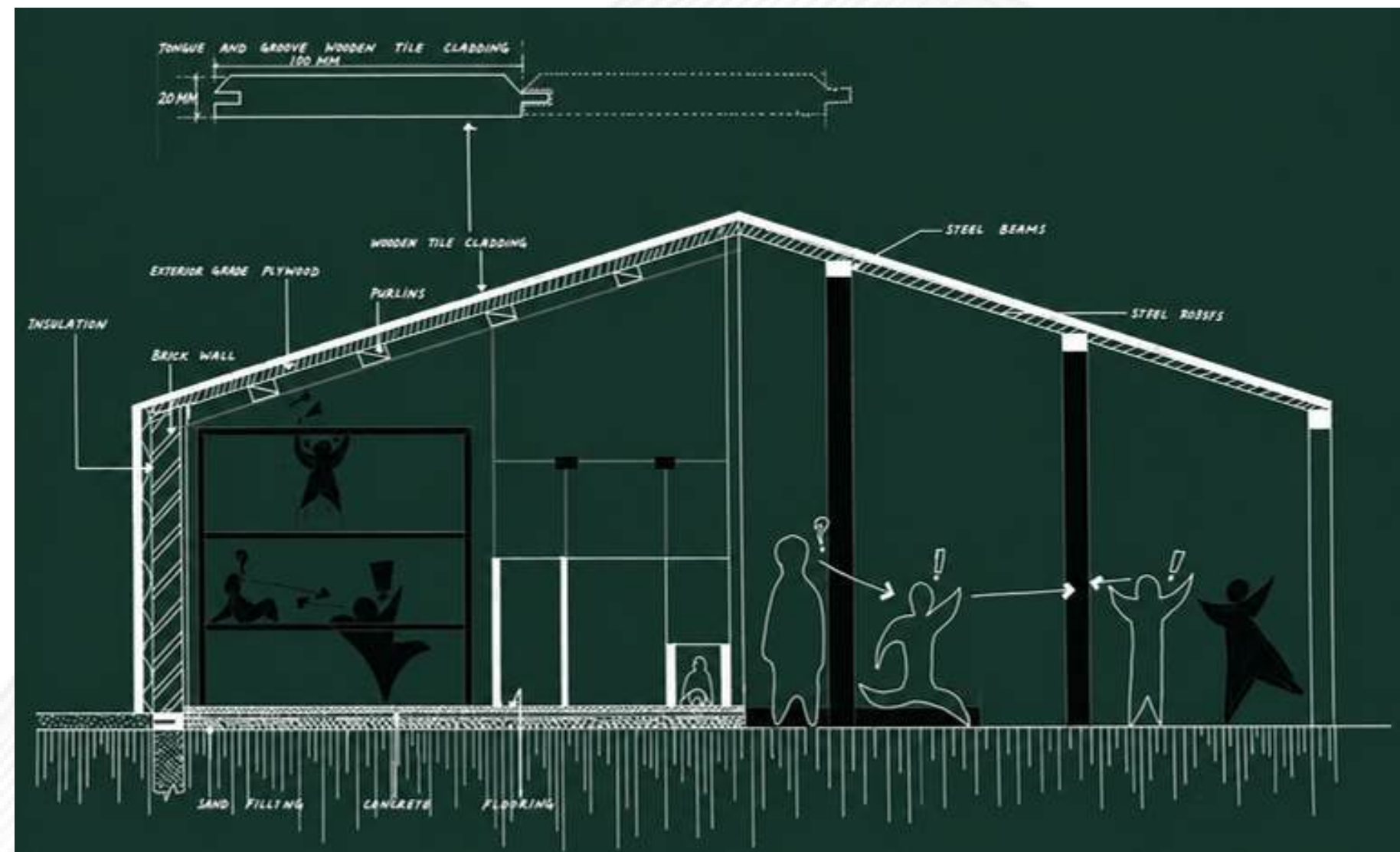
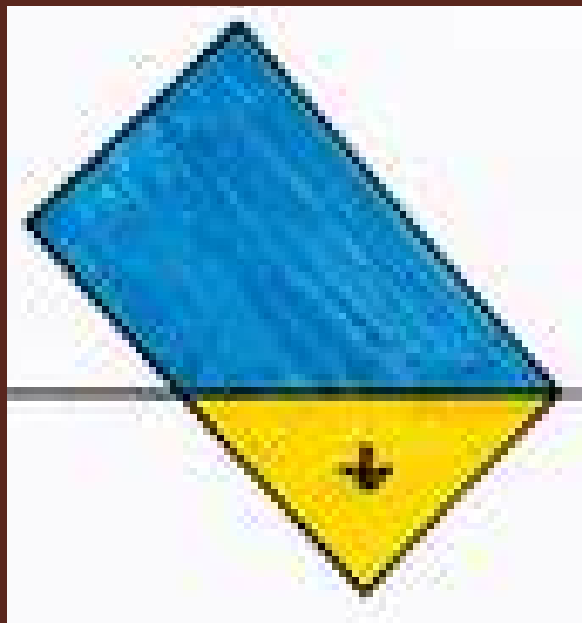
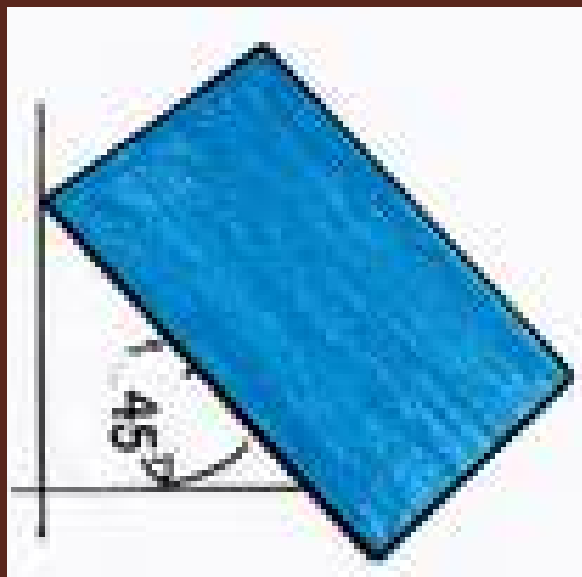
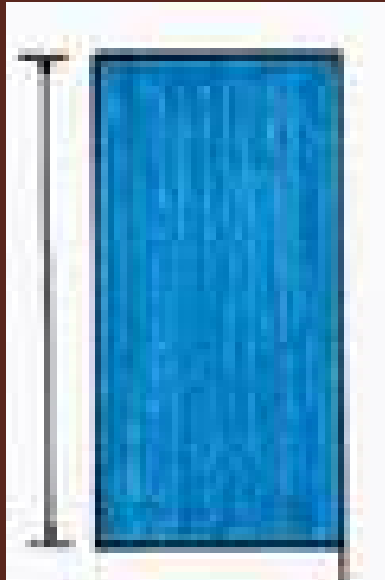


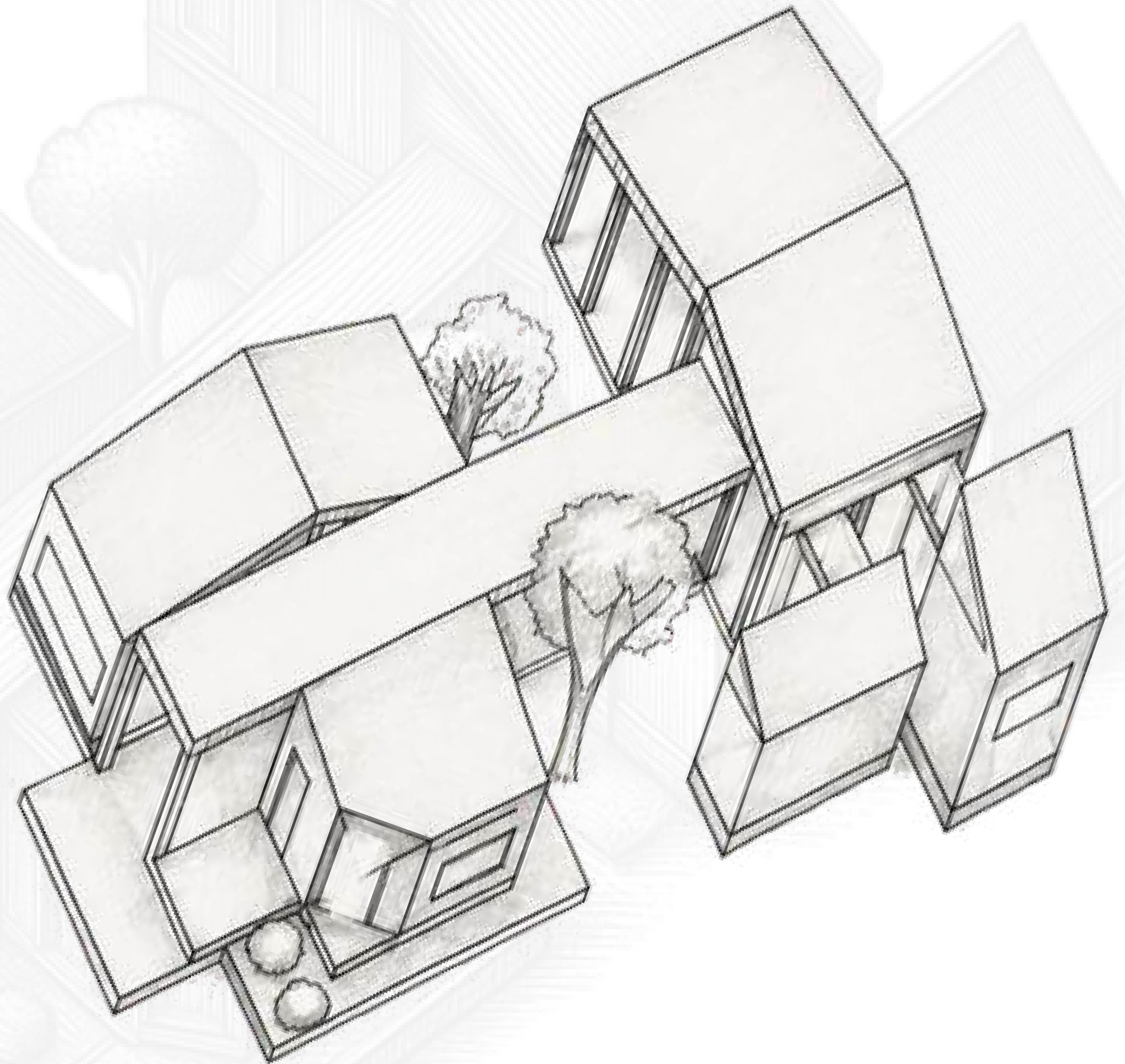
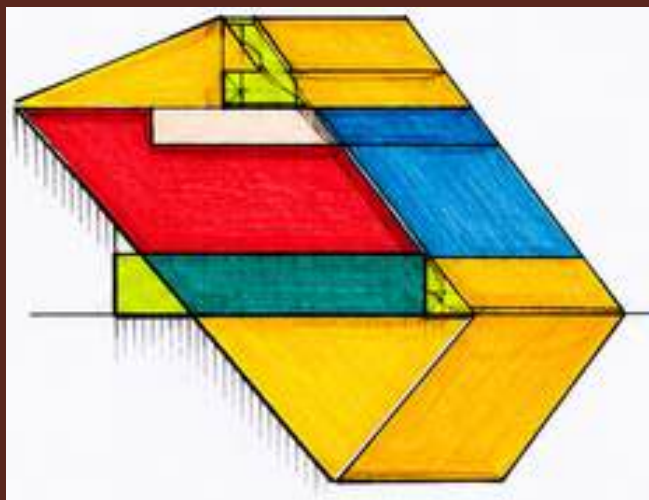
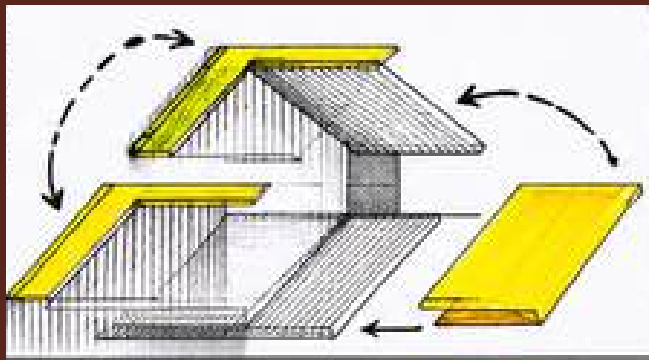
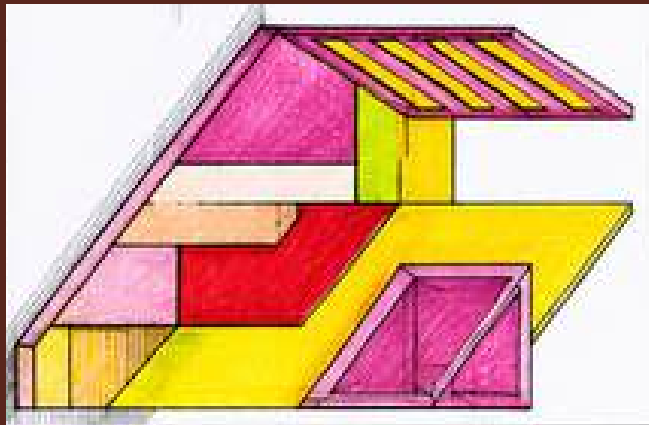
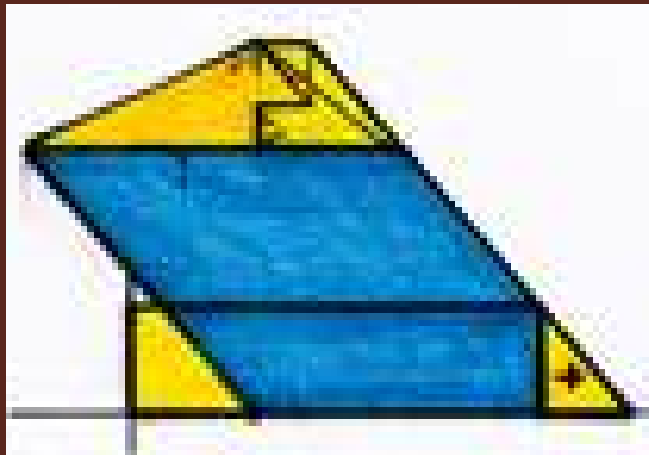
TREE TYPE PALETTE





ALLIED DESIGN

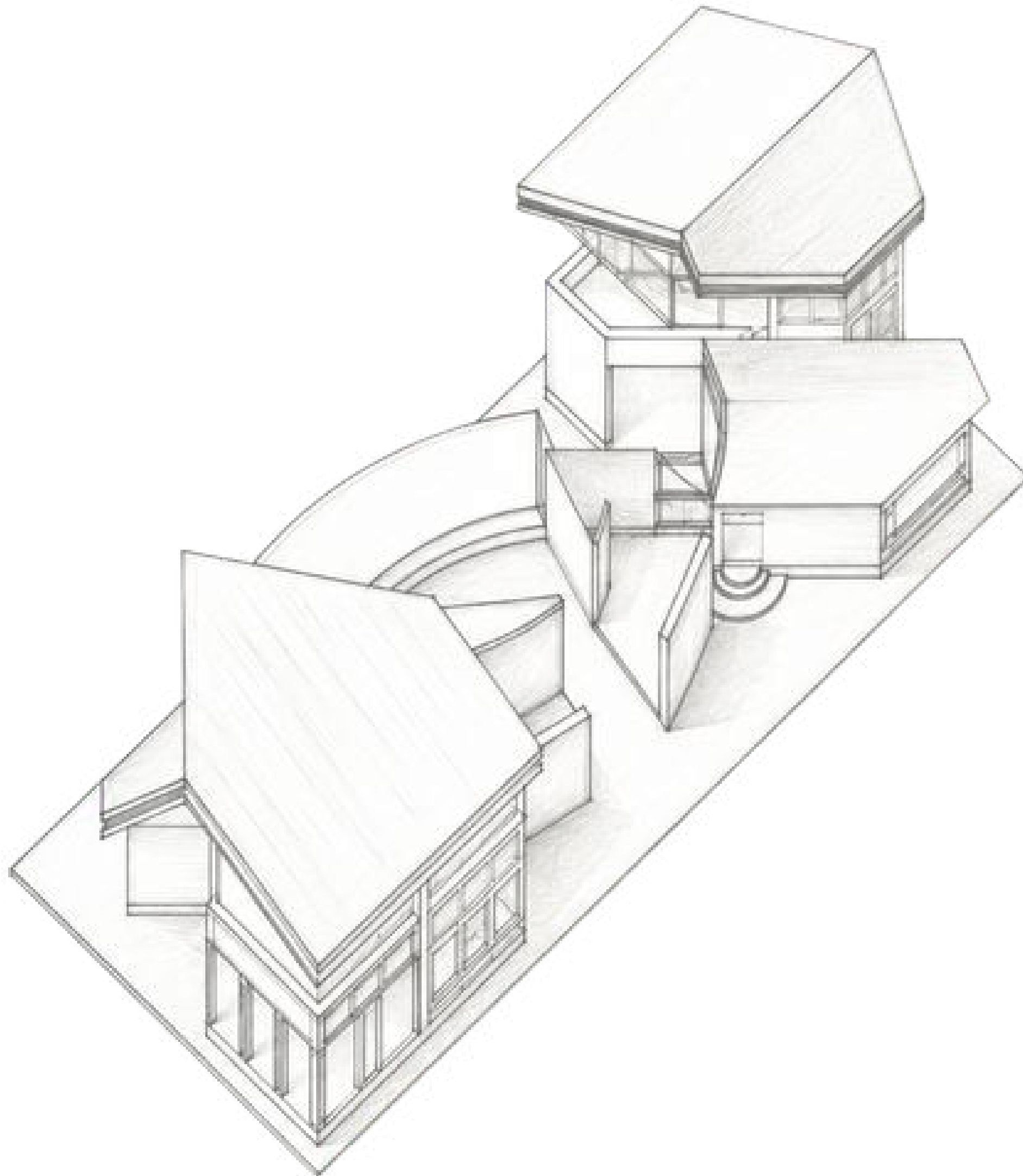




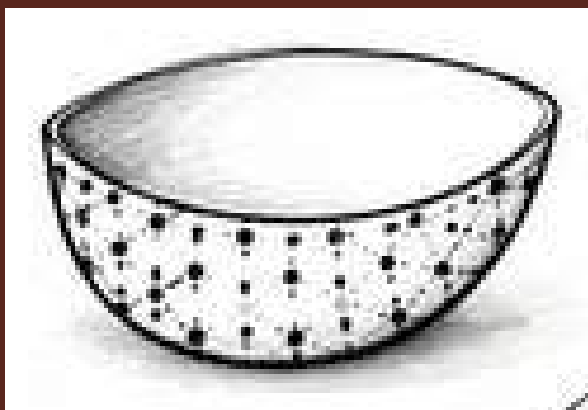
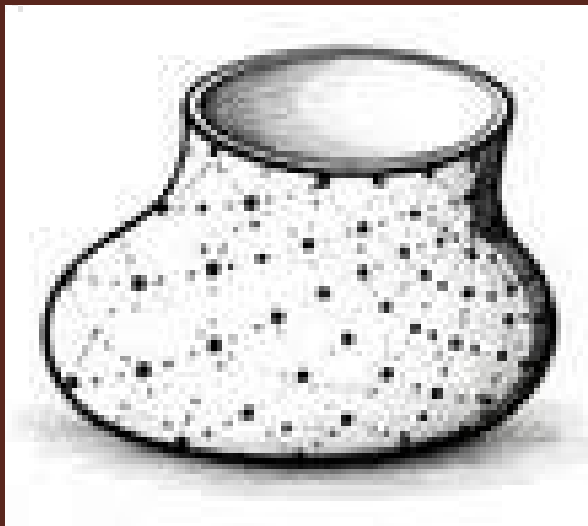
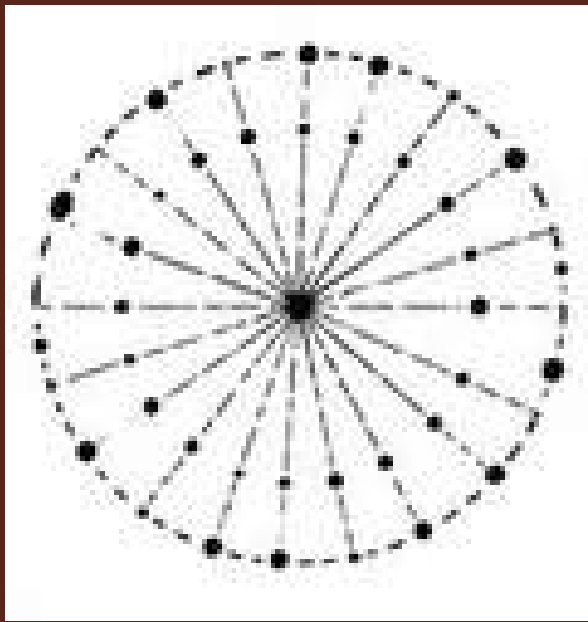
S
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3

SKILL
DEVELOPMENT
CENTRE

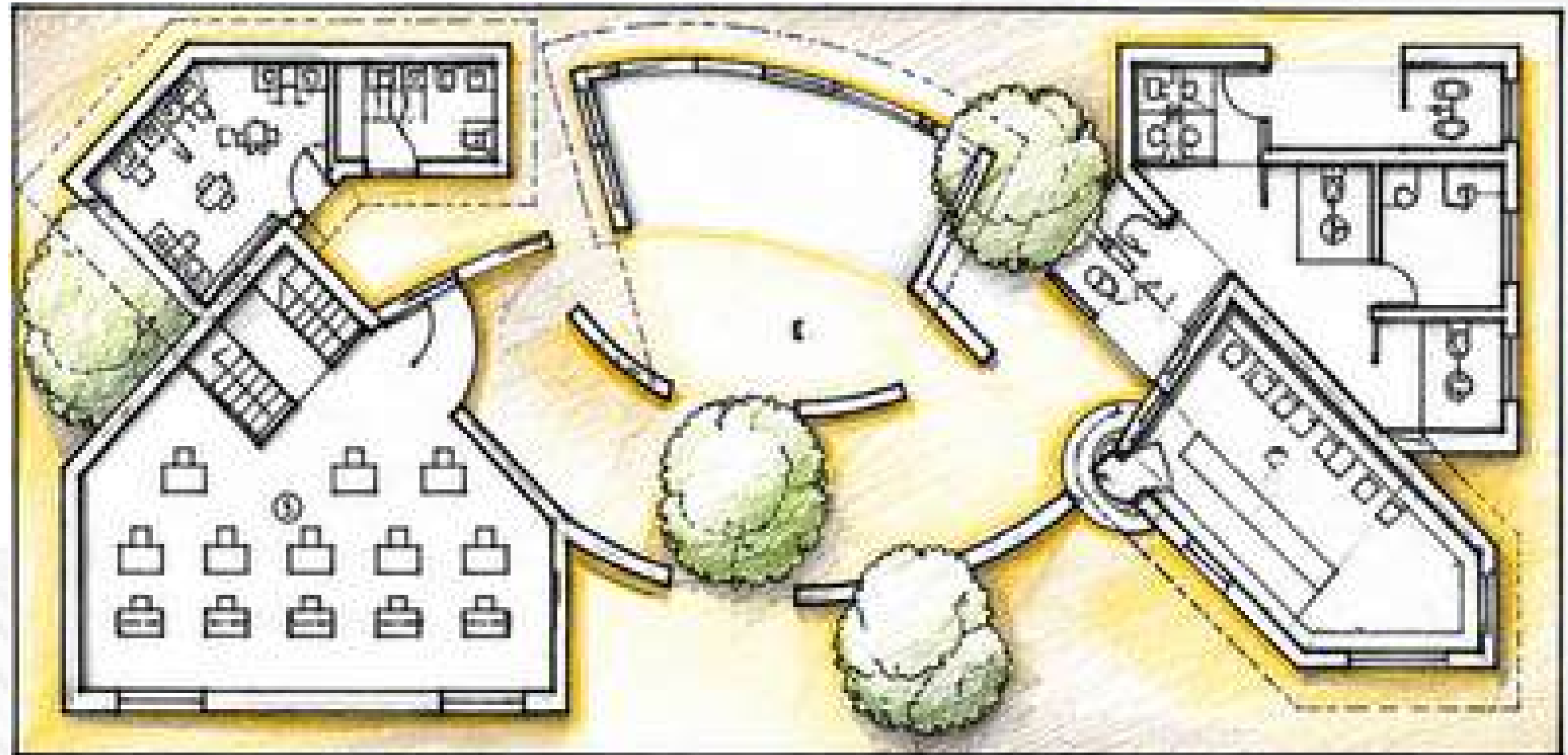


MANDALA ART
SKILL WORKSHOP



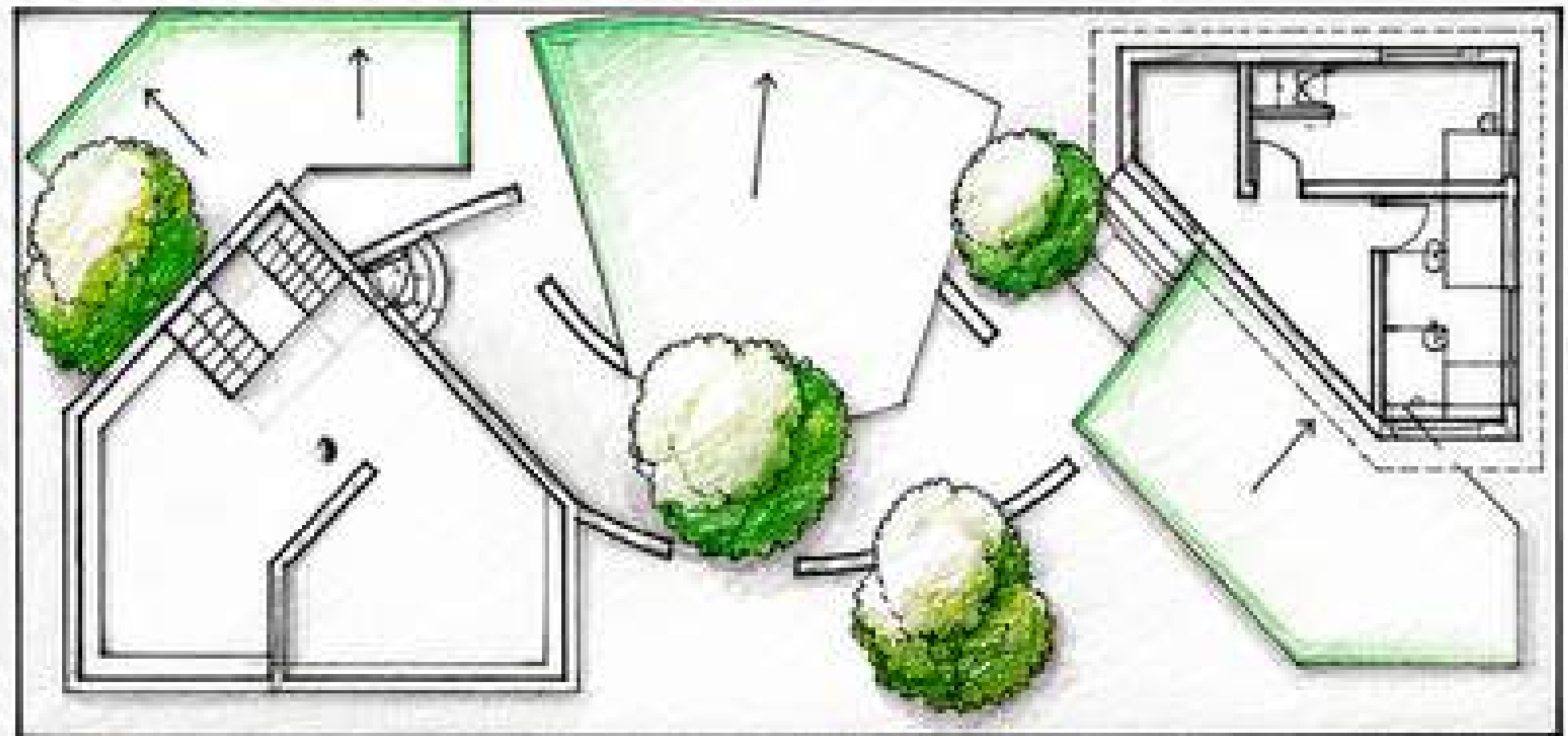
ARCHITECTURAL DESIGN

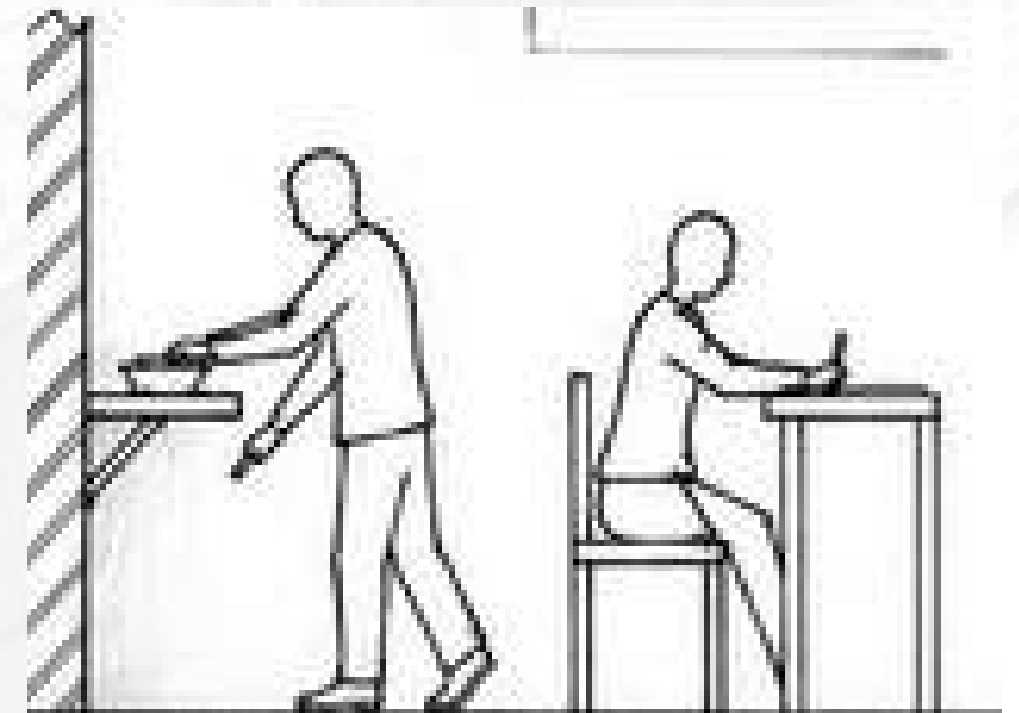
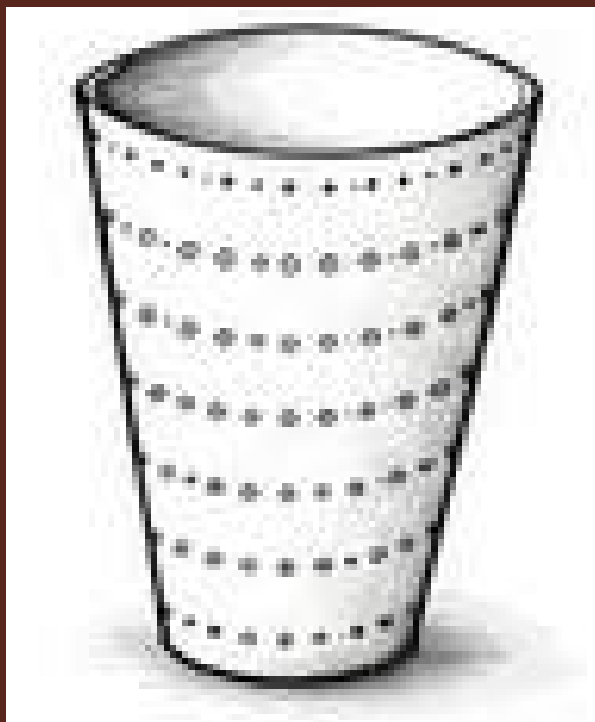
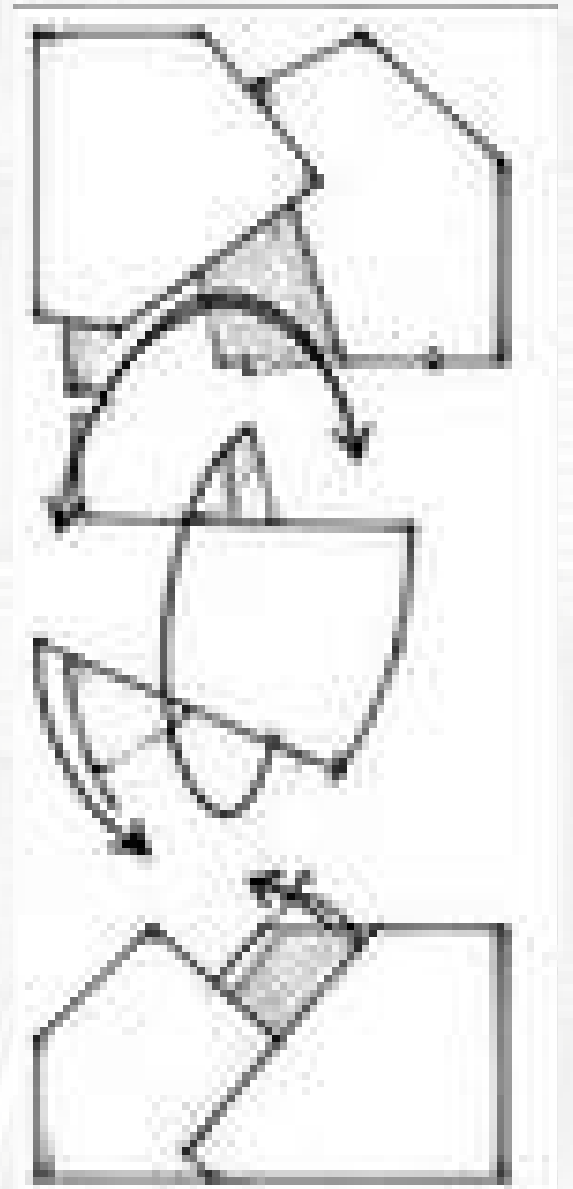
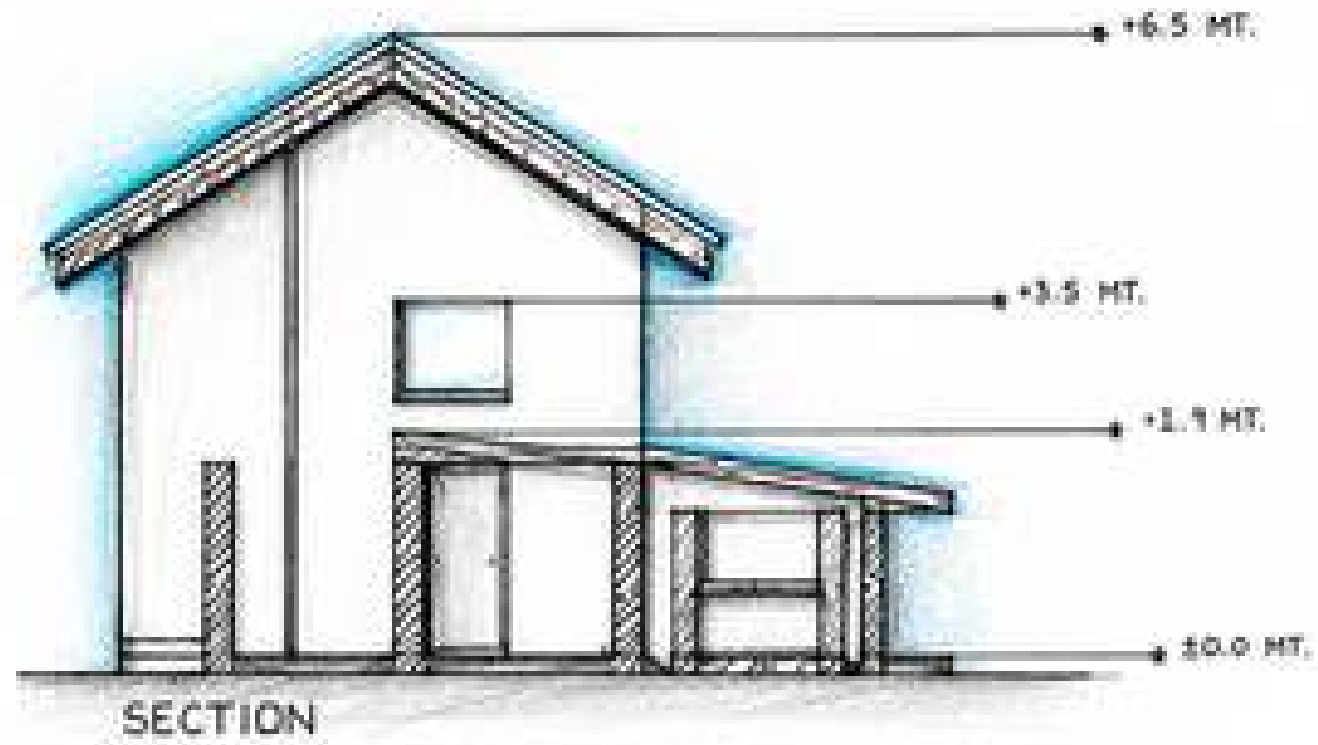
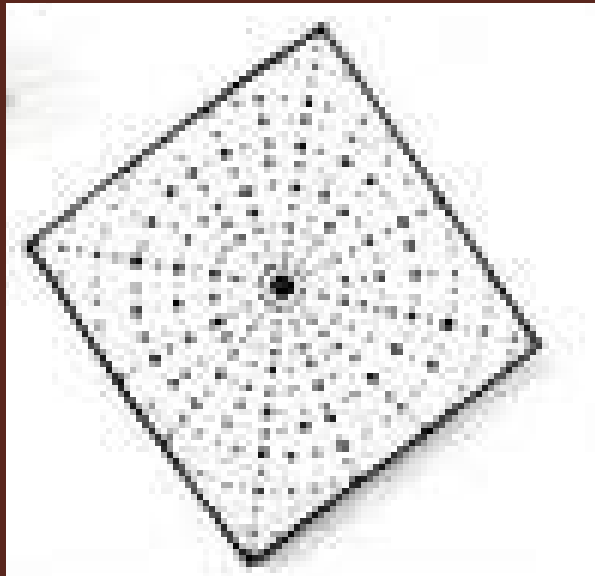
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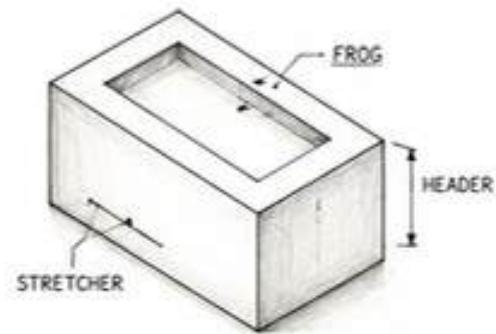
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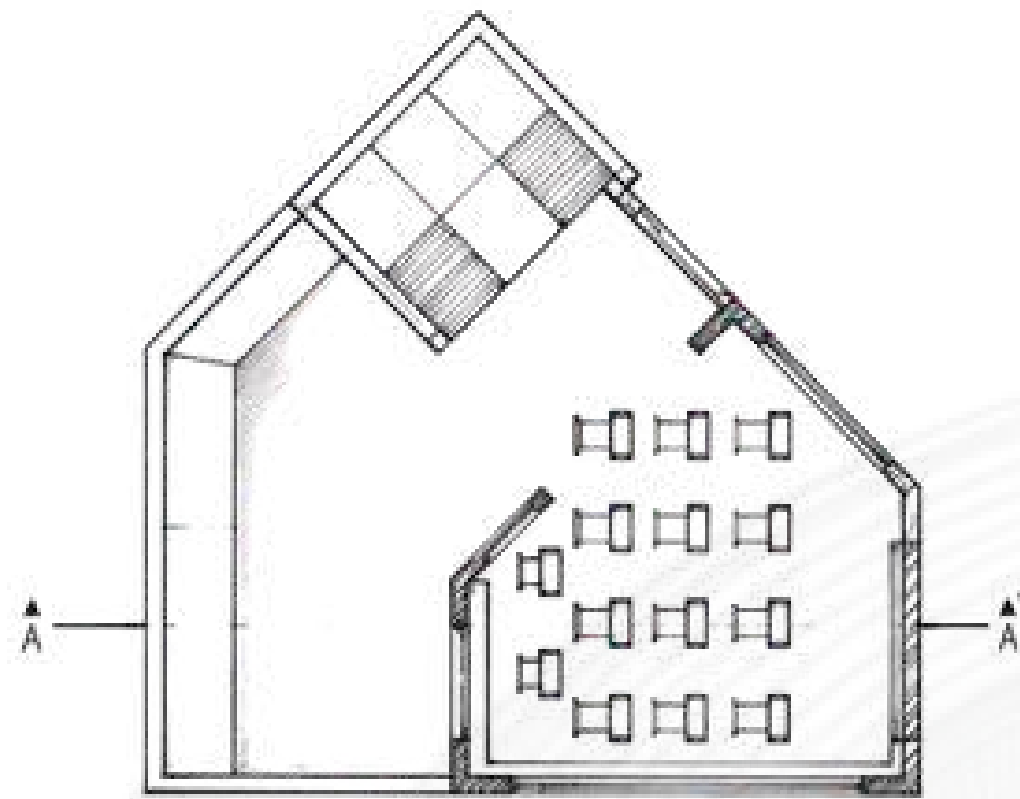
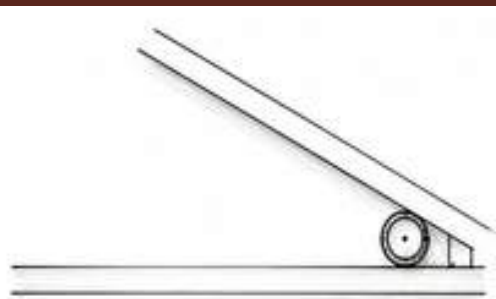
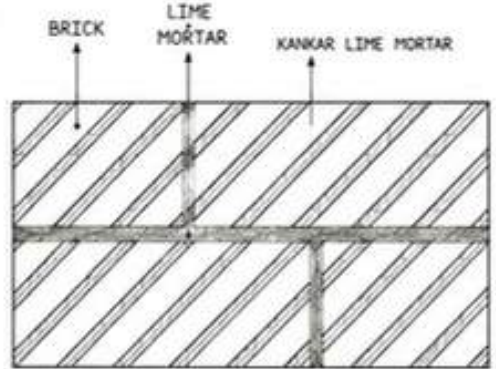




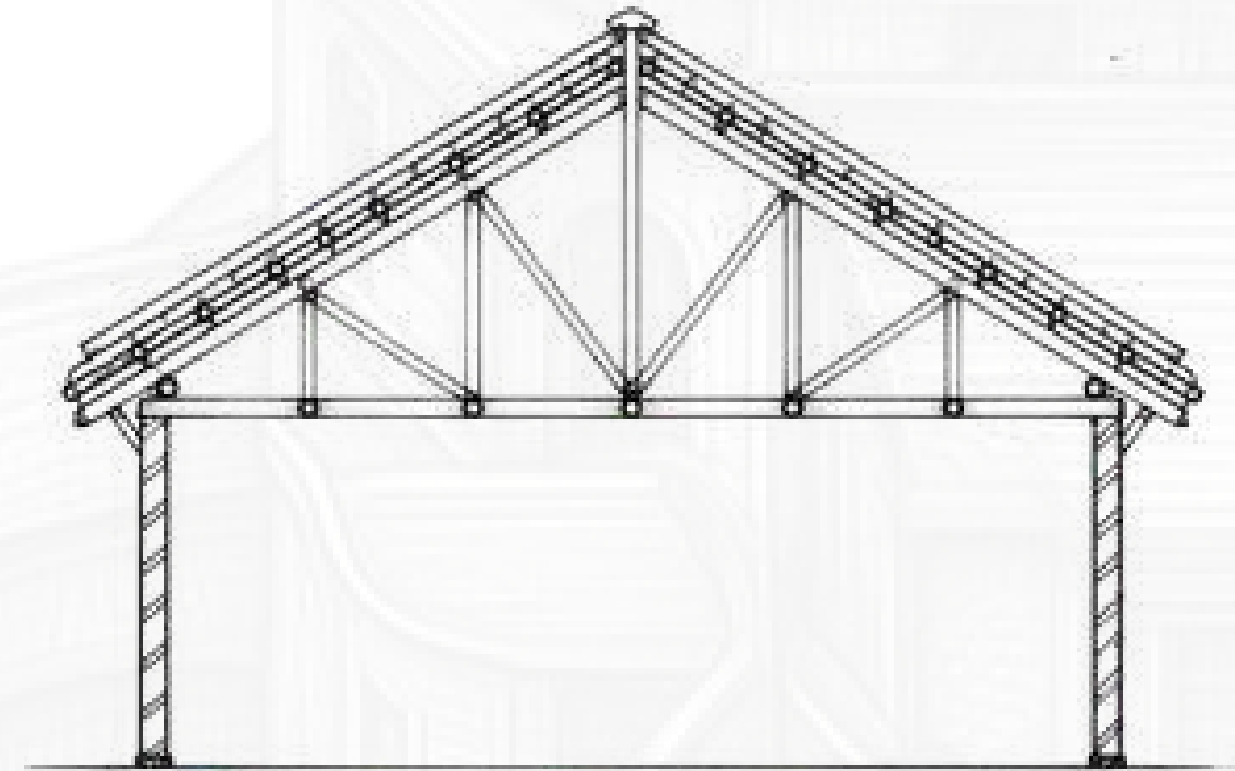
WORKSHOP DETAILS



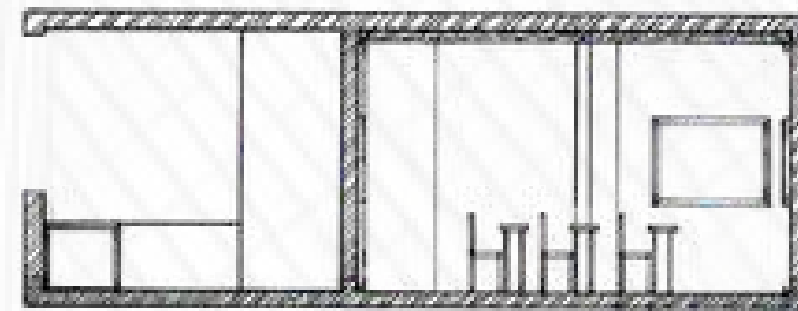
INDIAN
TRADITIONAL
BRICK



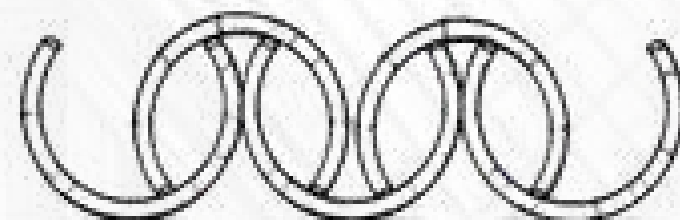
PLAN



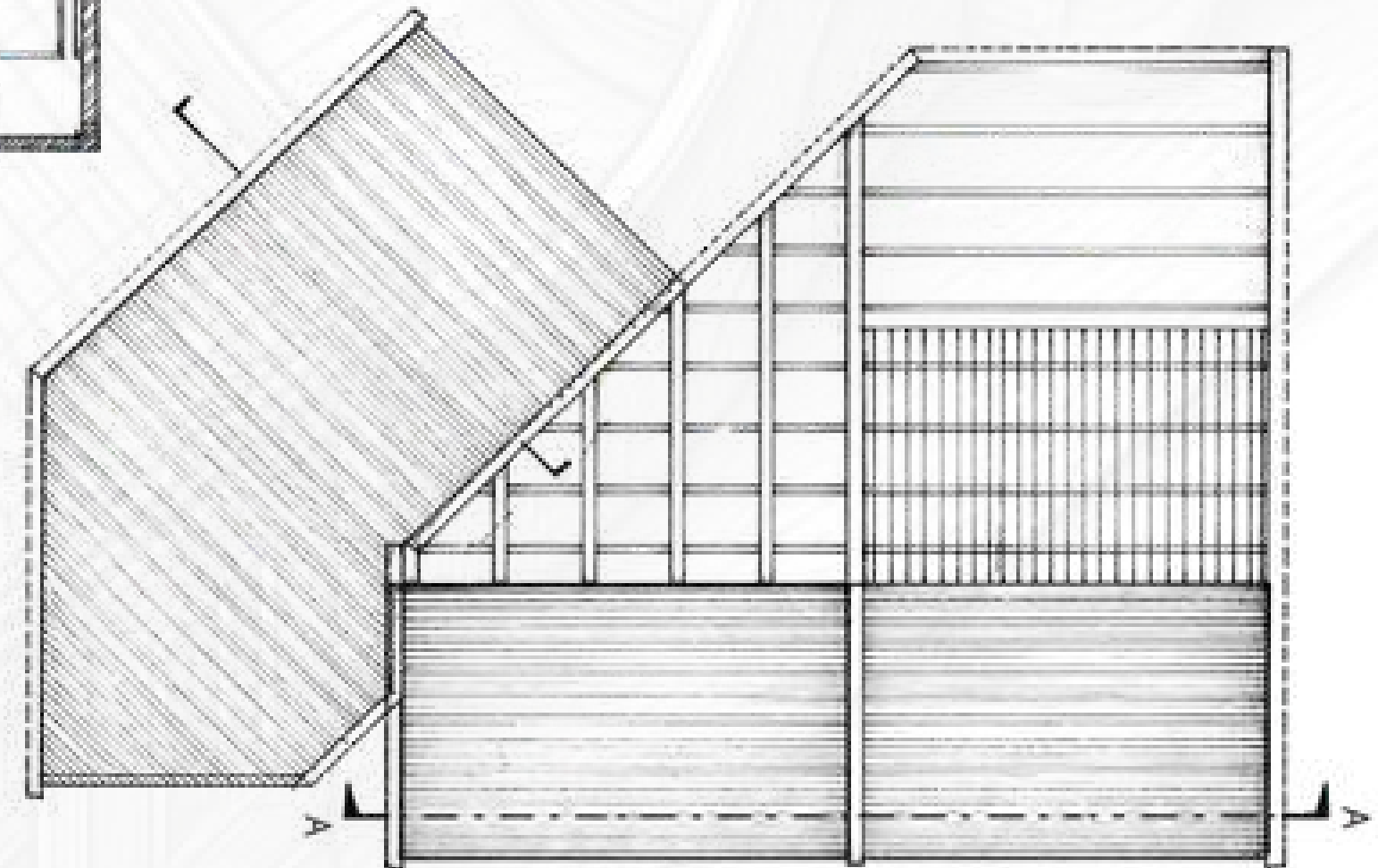
SECTION AA'



SECTION



HALVED BAMBOO ROOF



S
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4

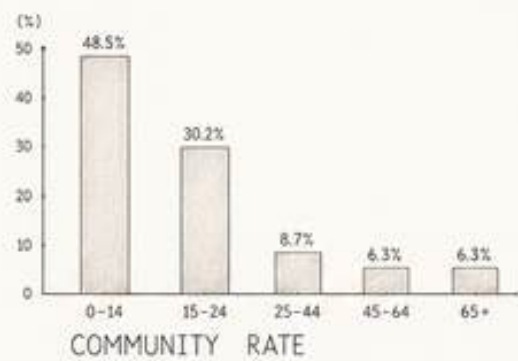
COMMUNITY
HOUSING



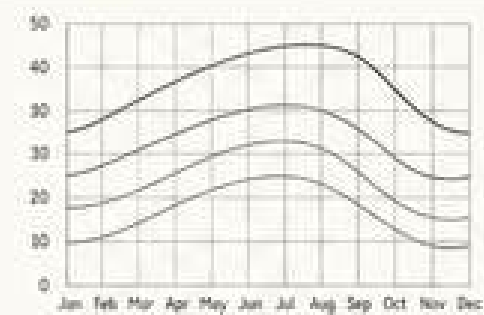
ARCHITECTURAL DESIGN



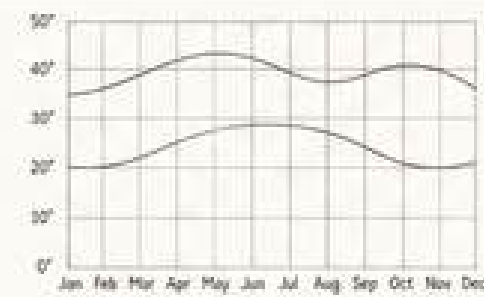
TEXTURA MAPPING



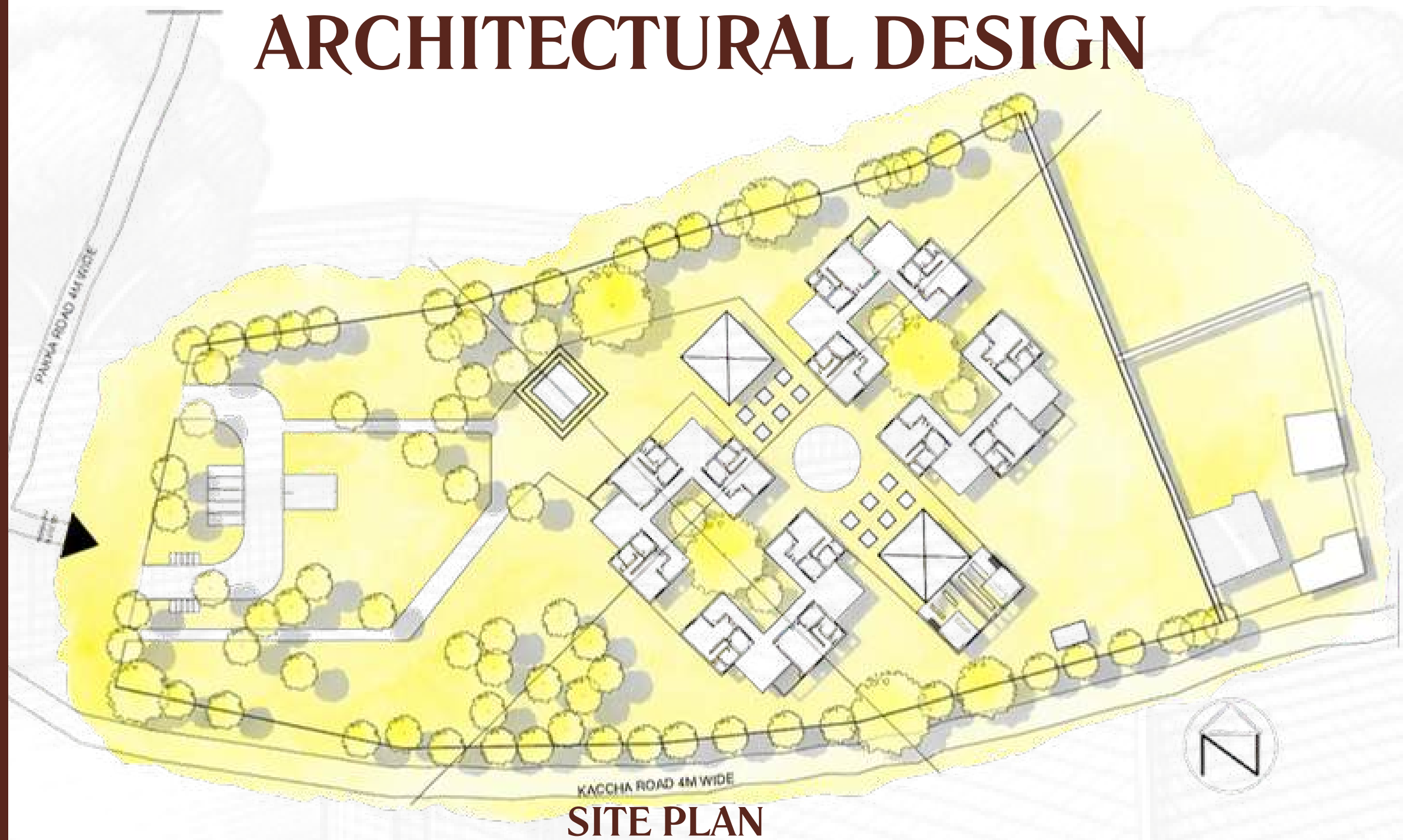
COMMUNITY RATE



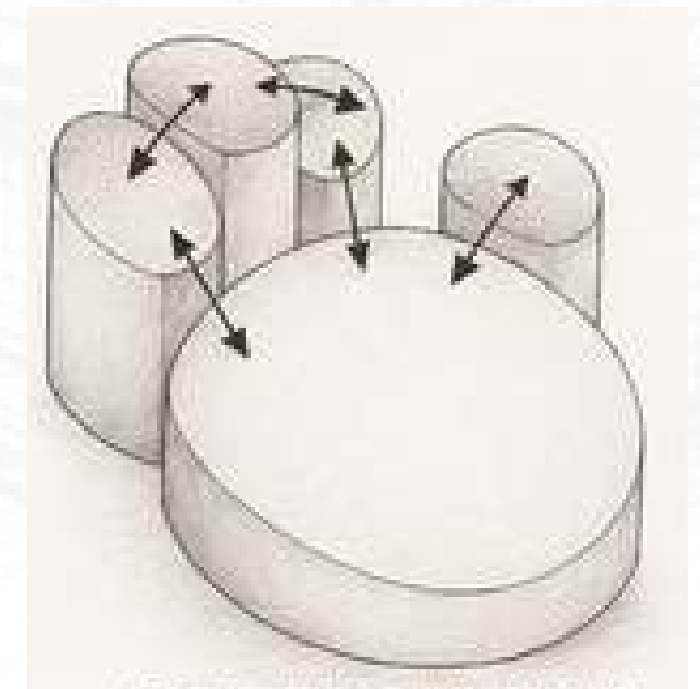
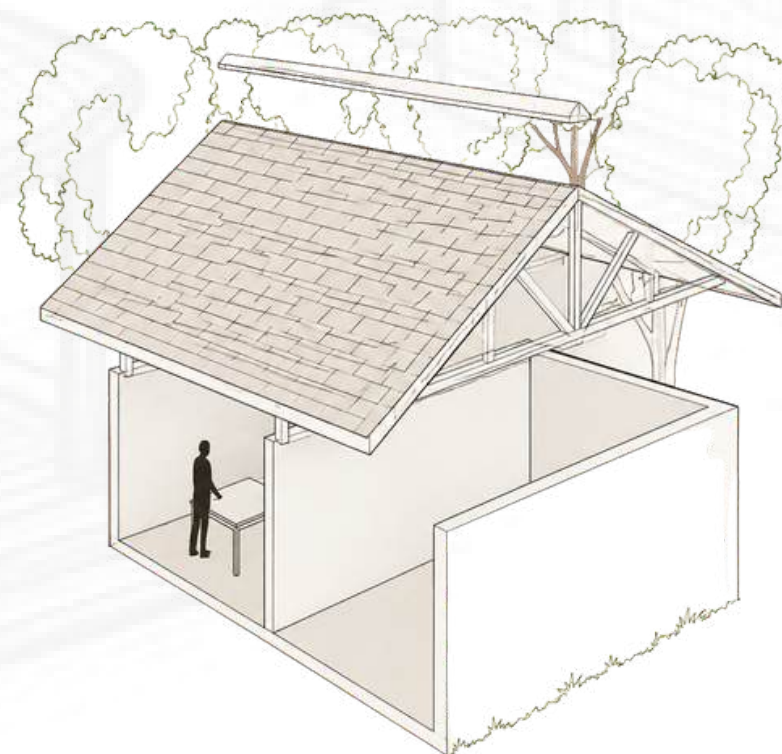
WIND SPEED (Km/H)

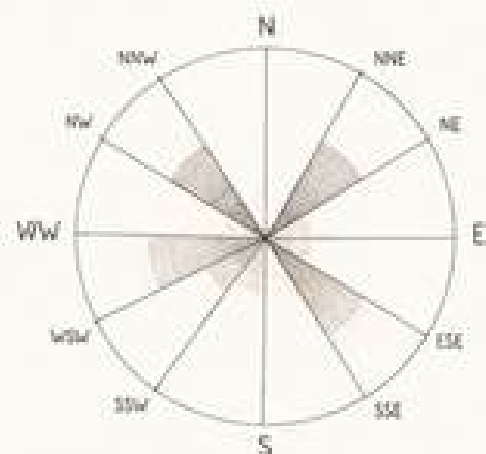


HIGH AND LOW TEMPERATURE

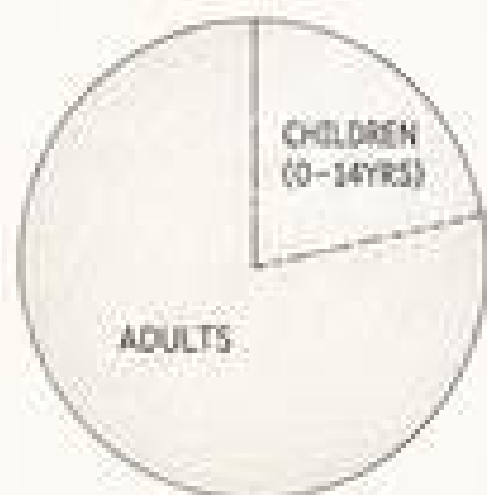


SITE PLAN

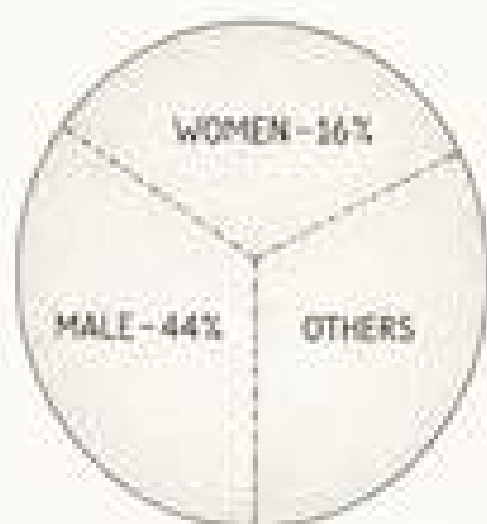




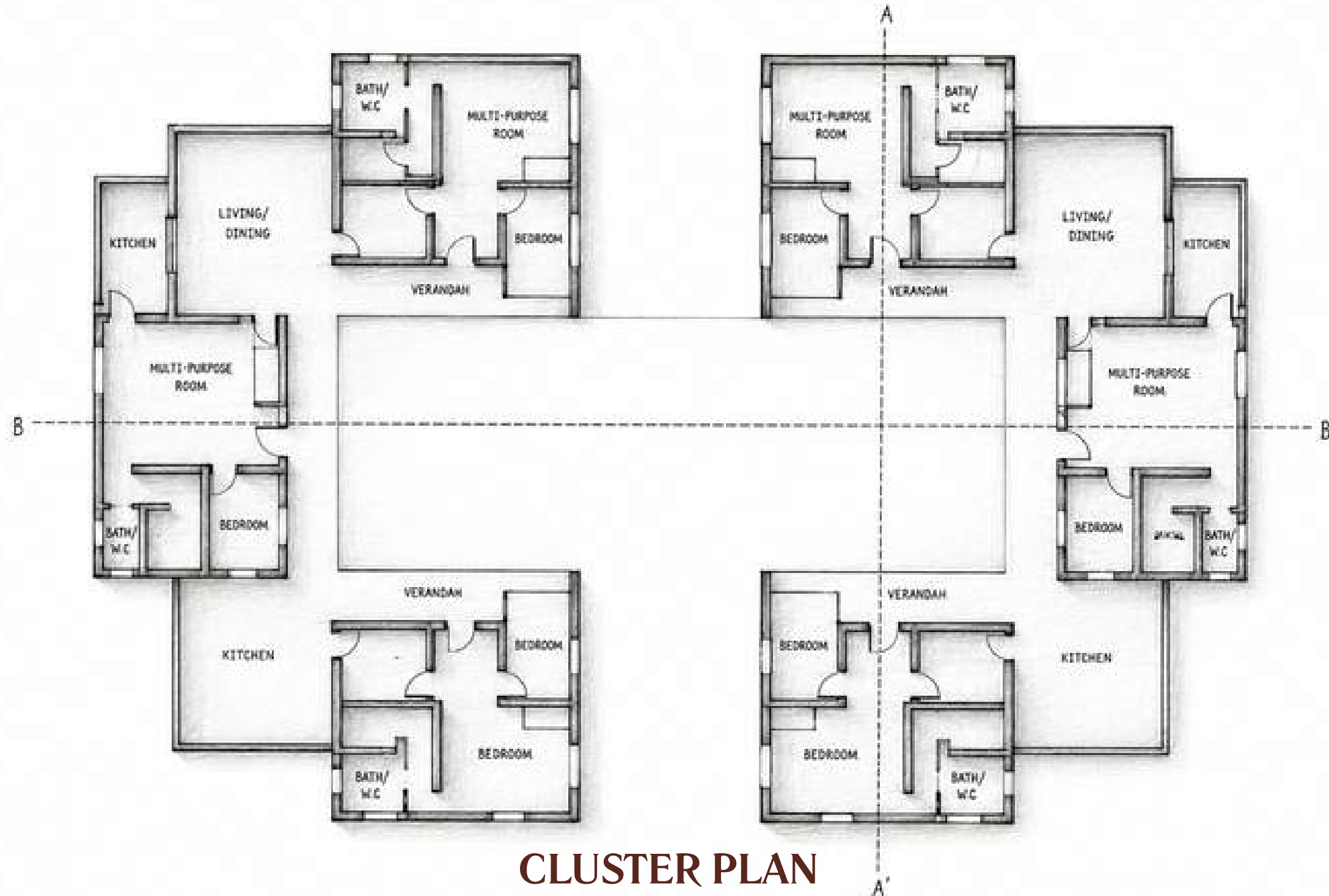
PREVAILING WIND ROSE



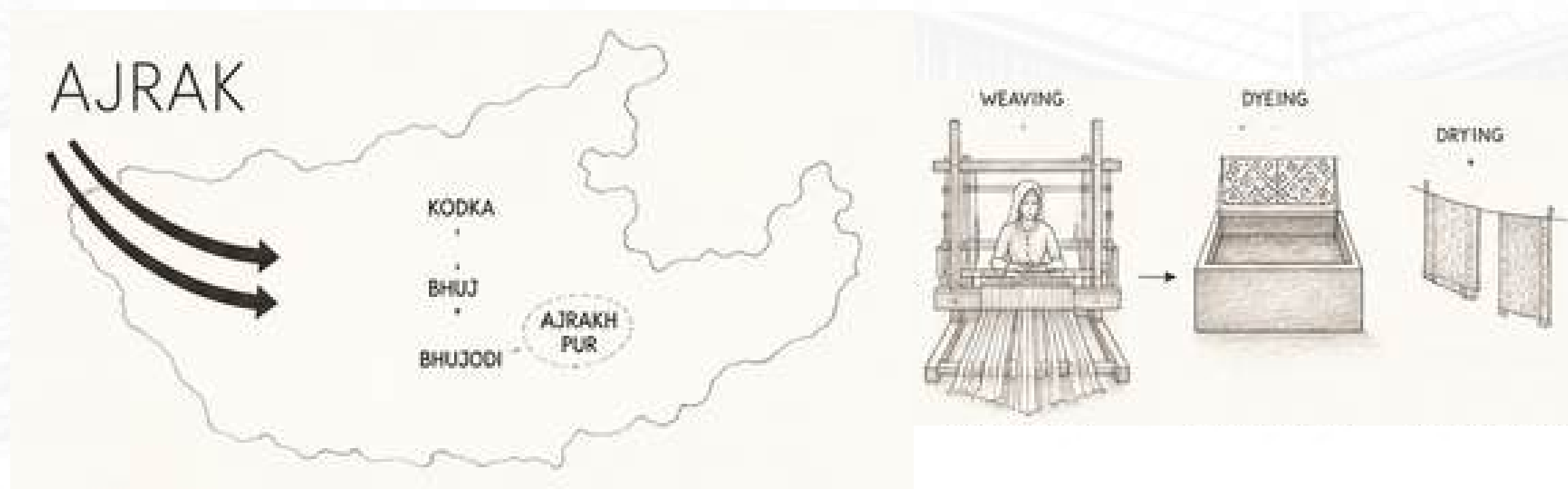
AGE RATIO



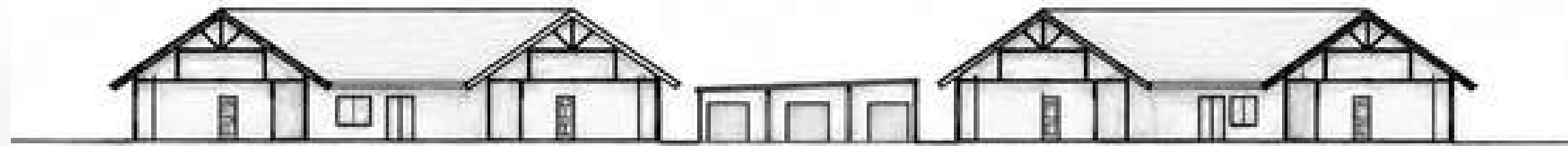
LITERACY RATE



CLUSTER PLAN



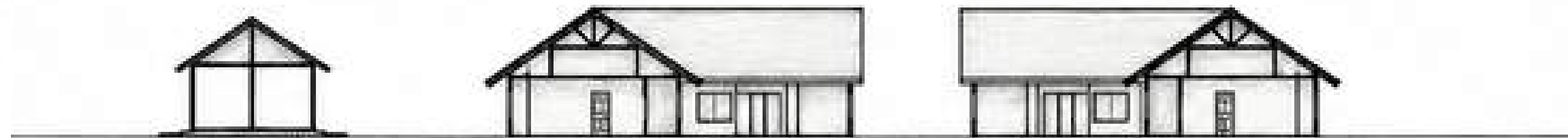
WEAVER



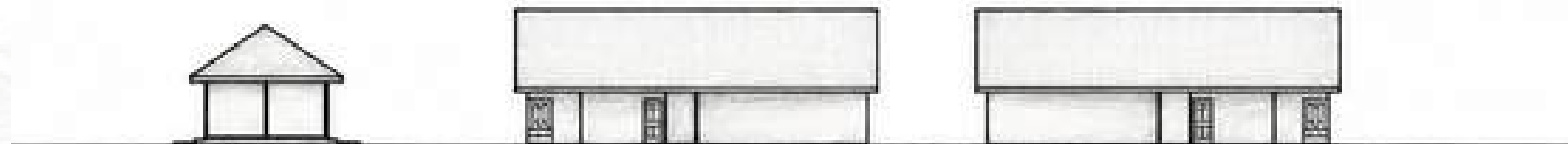
SECTION AA'



NORTH-WEST SIDE ELEVATION

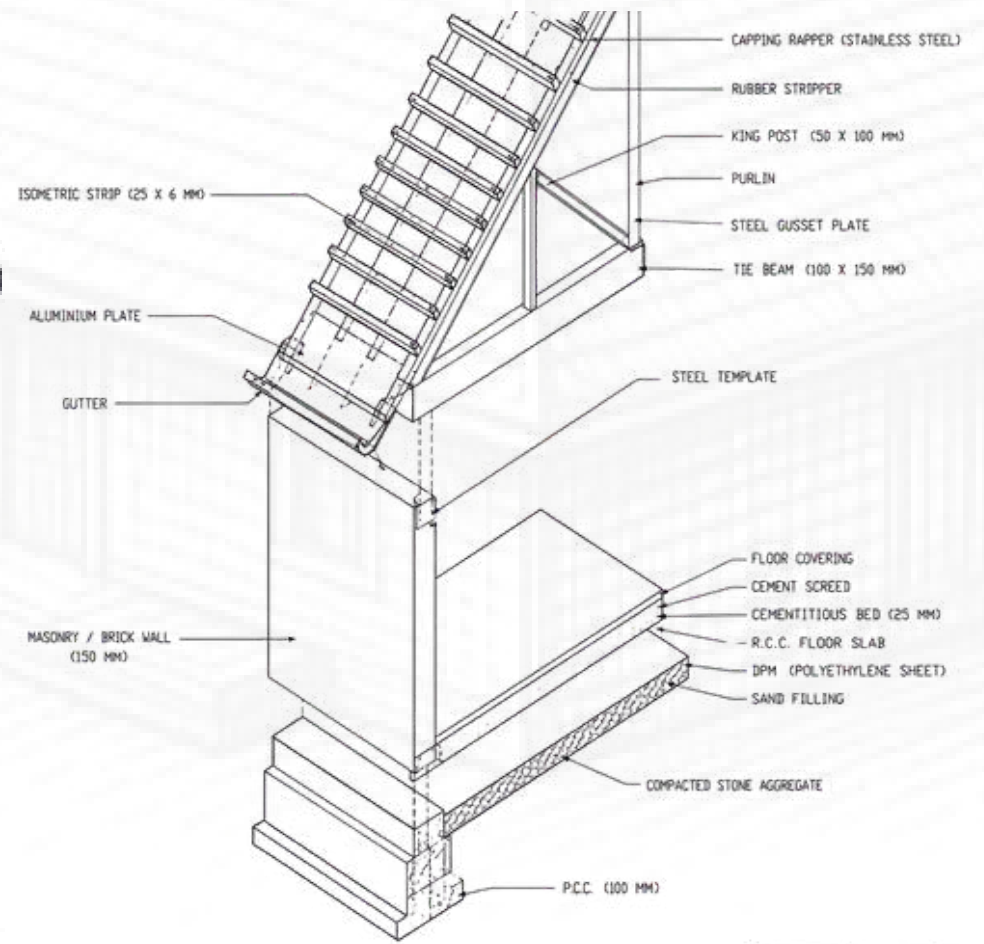
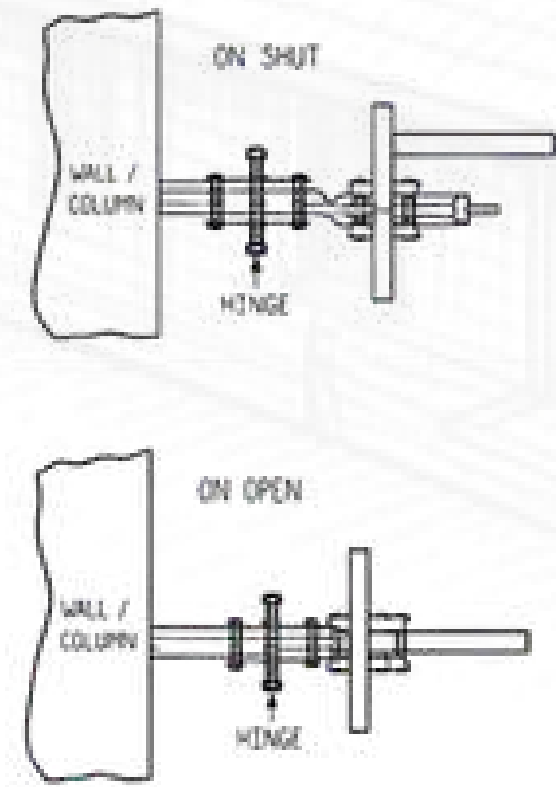
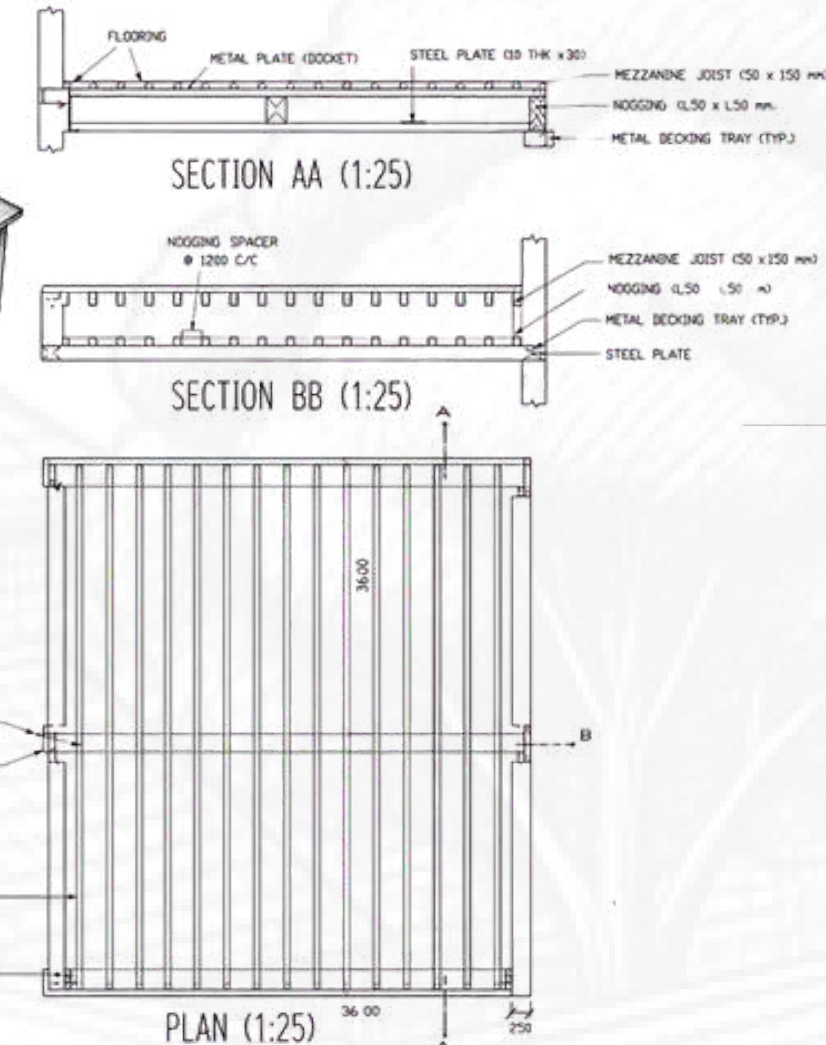
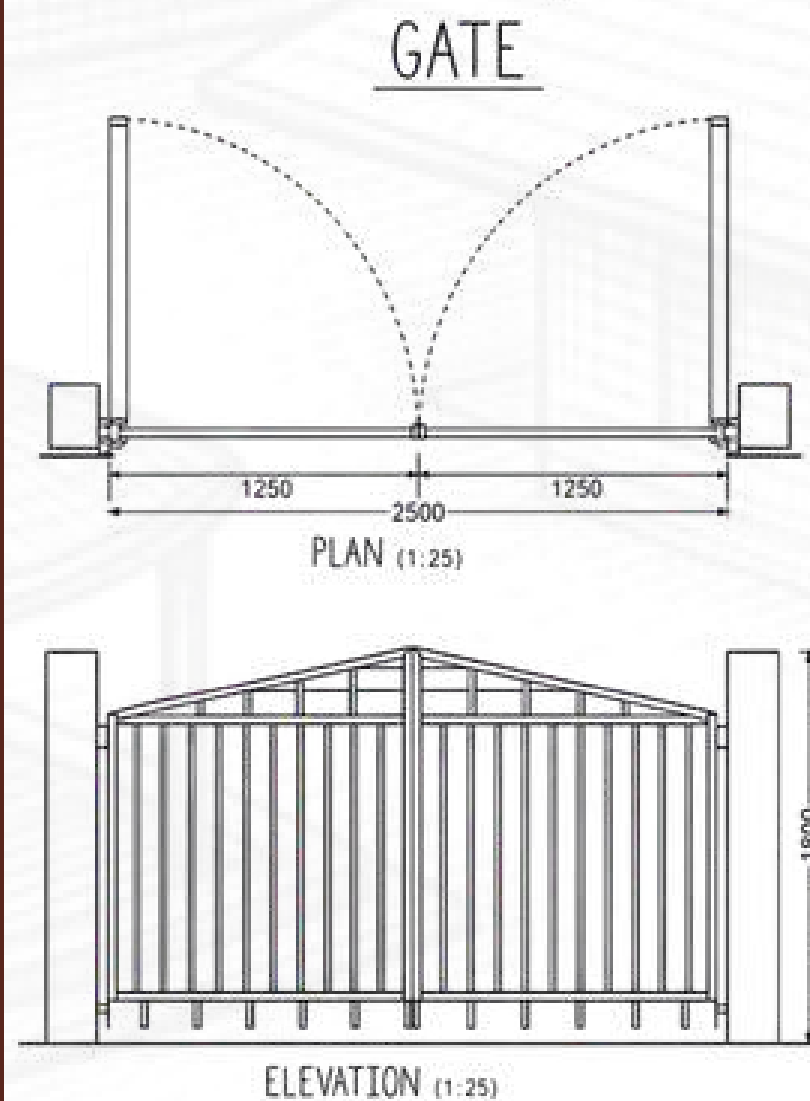
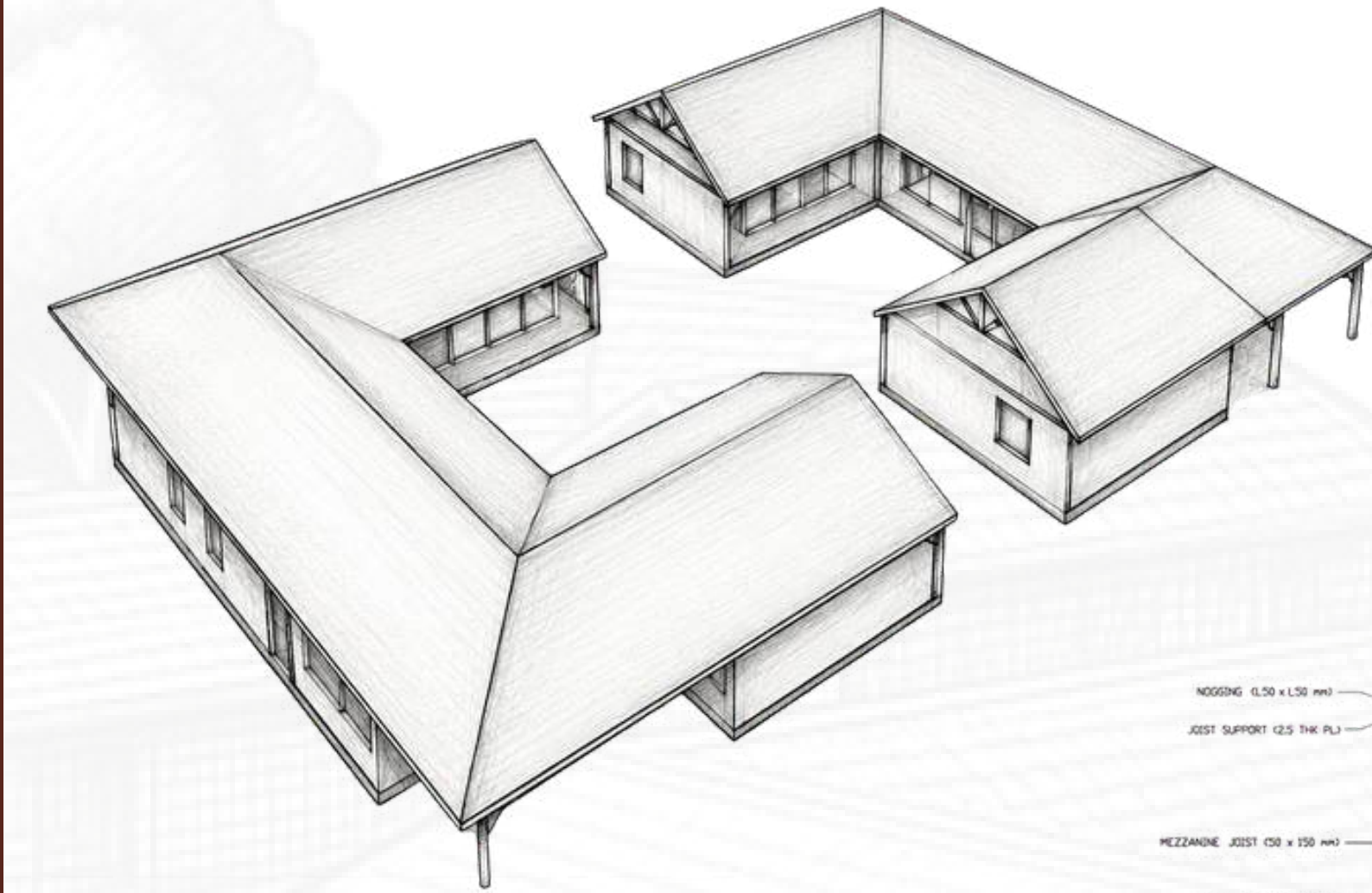
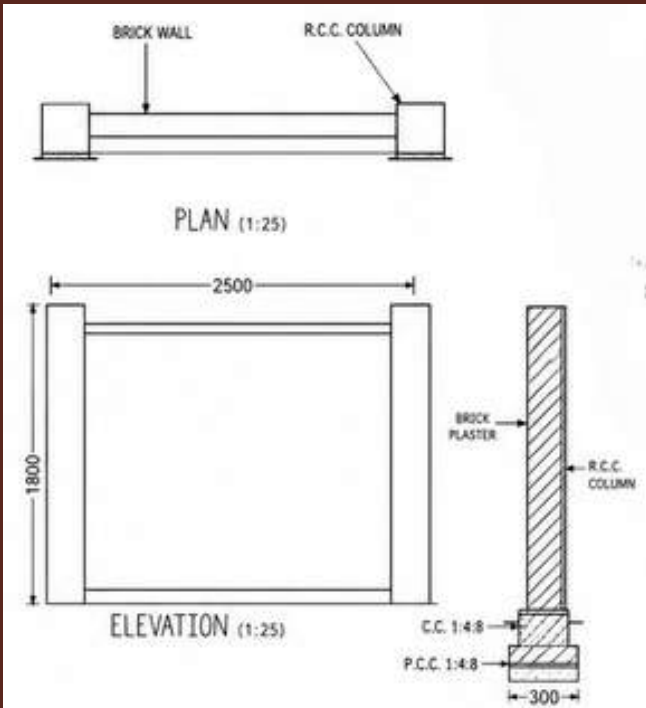
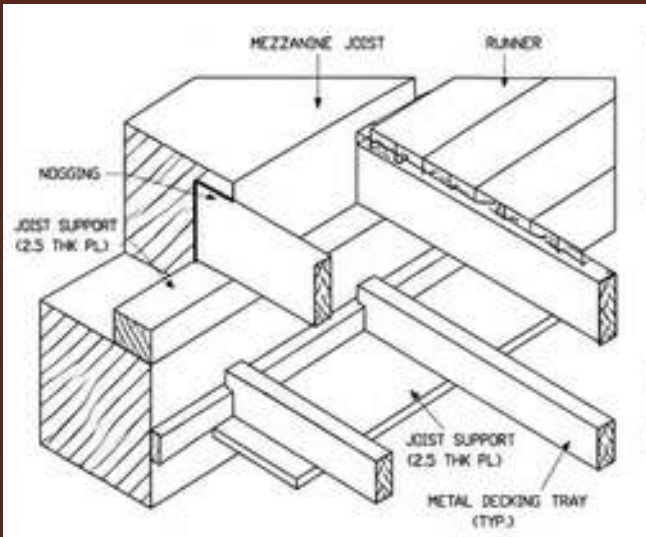
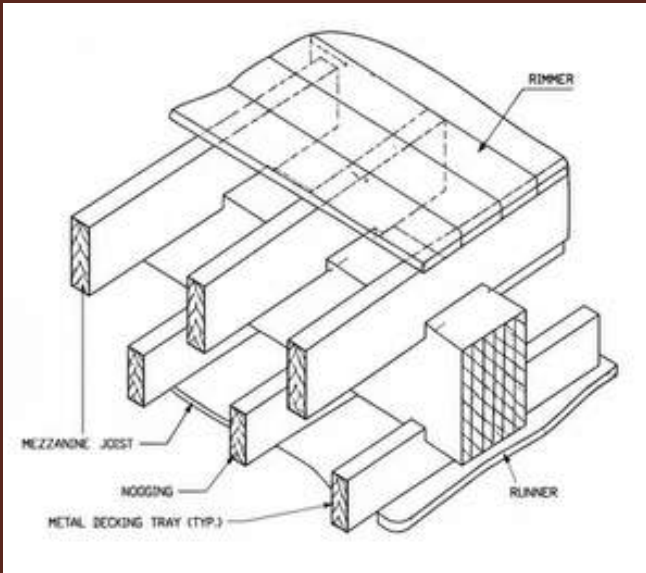


SECTION BB'



SOUTH-WEST SIDE ELEVATION

CONSTRUCTION DETAILS



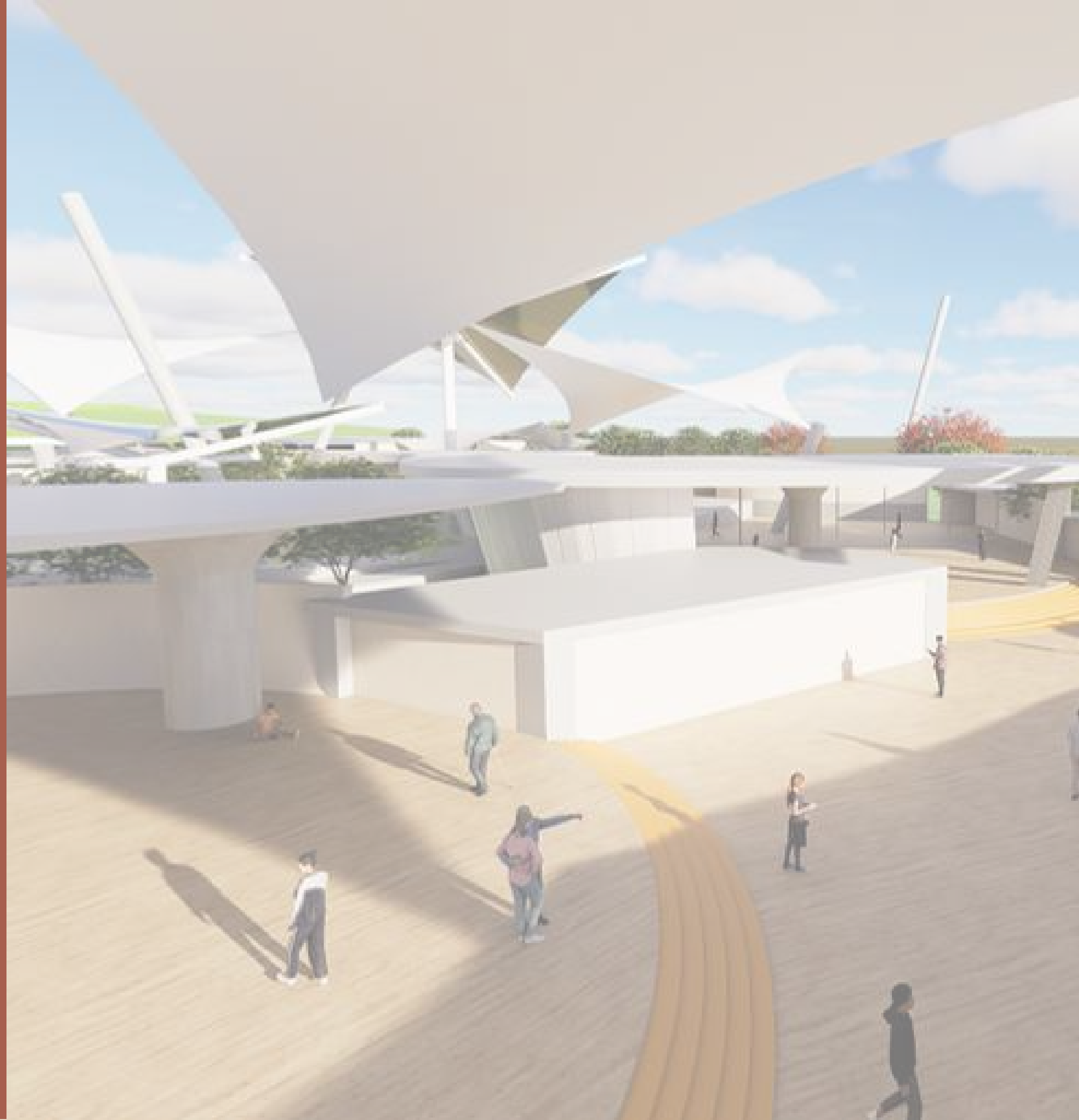


VIEWS

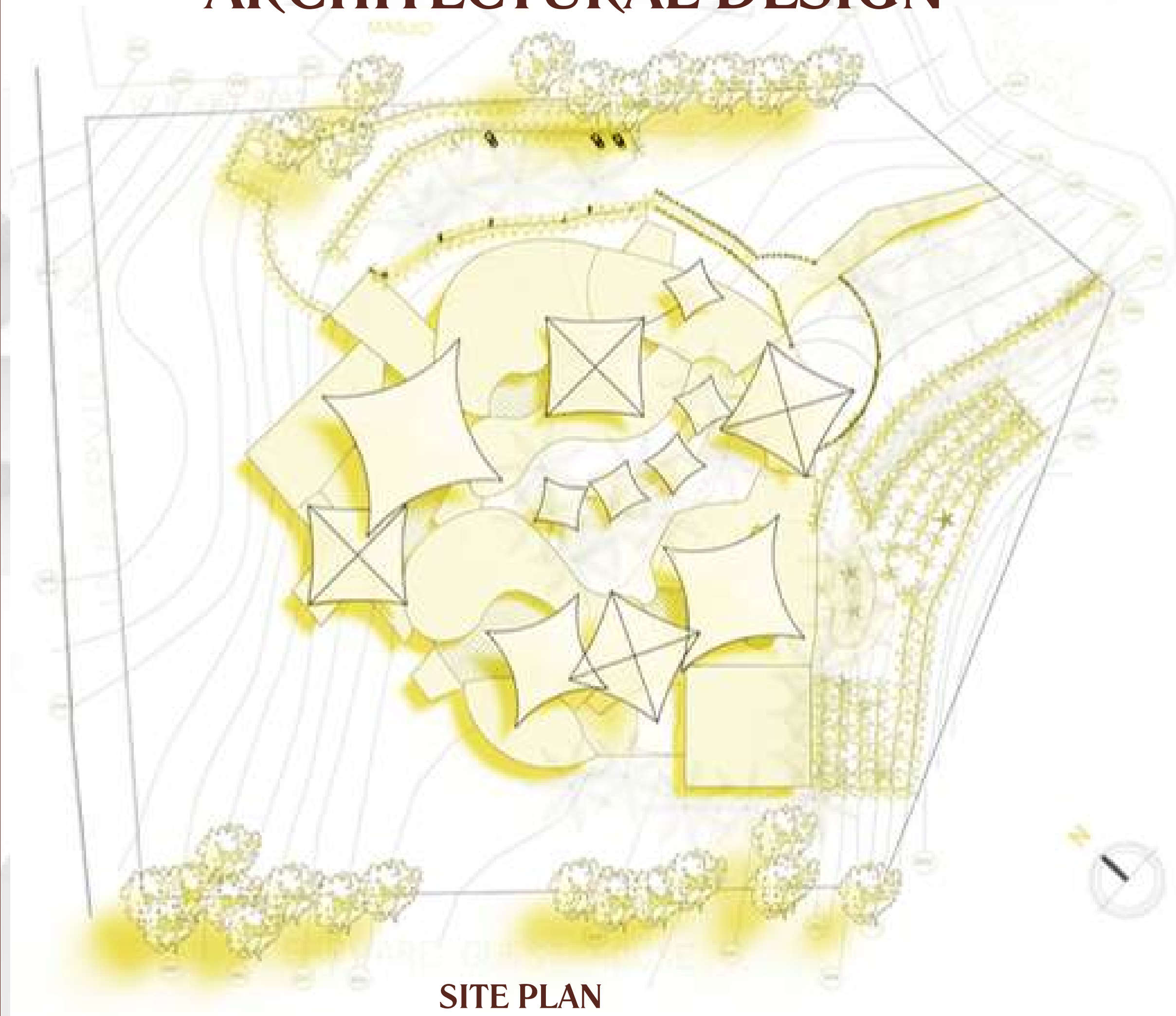
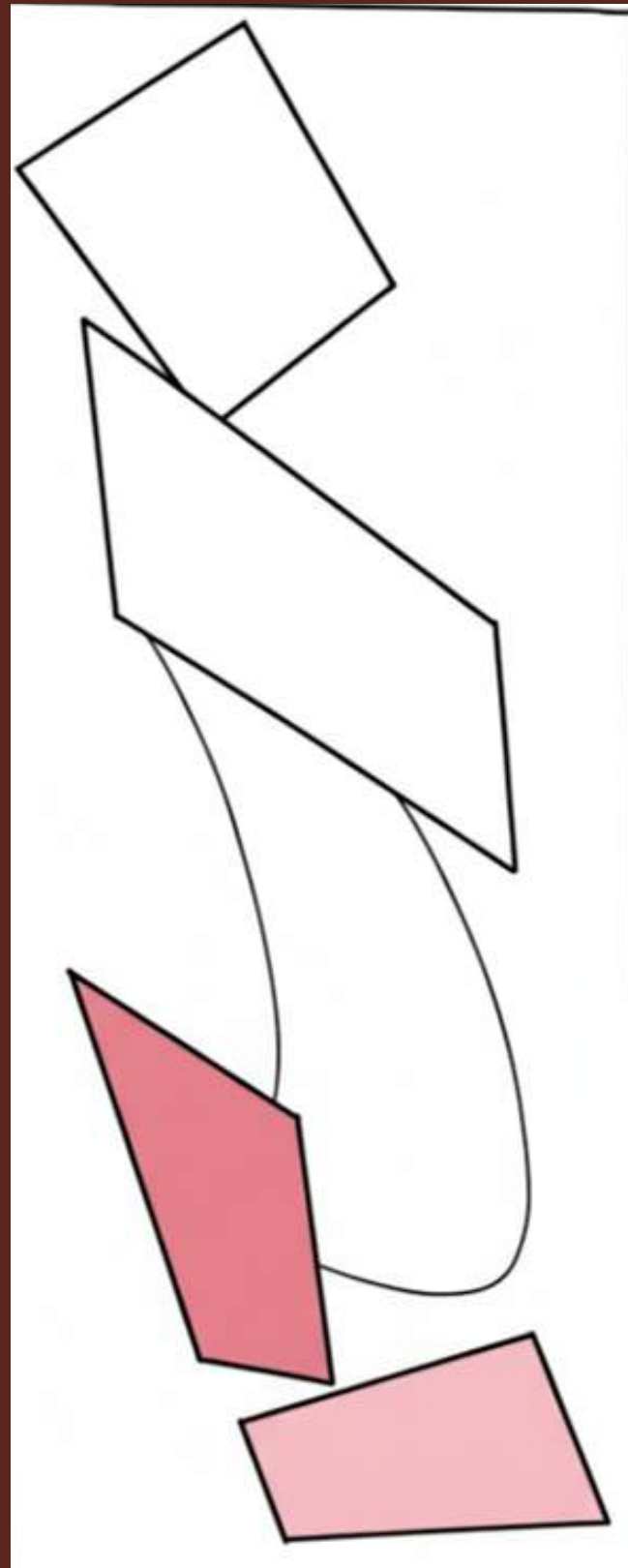


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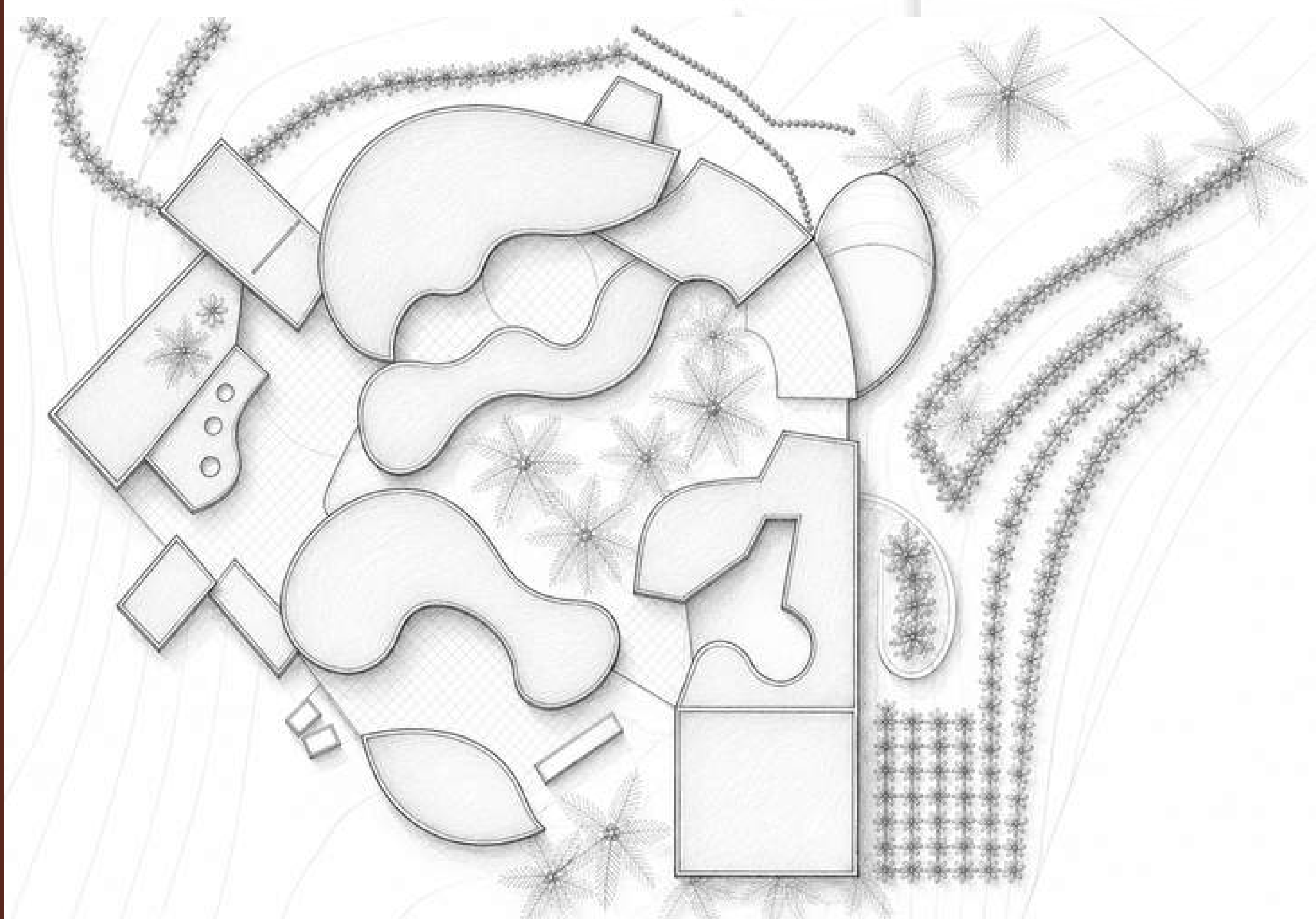
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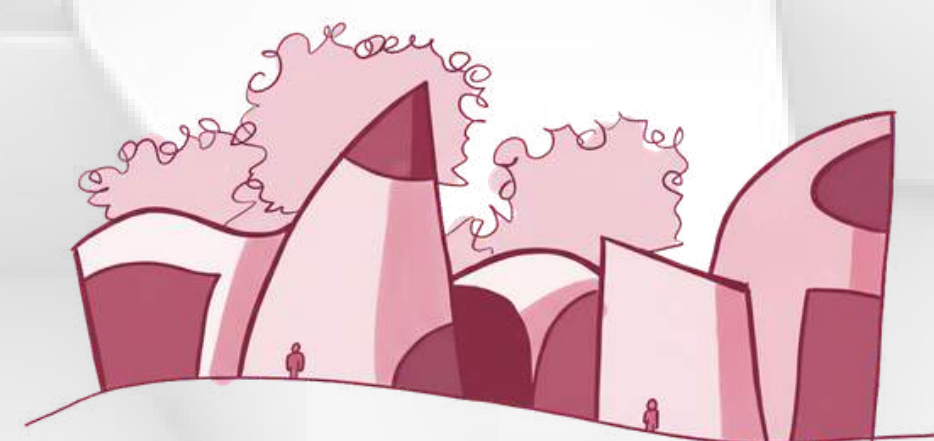
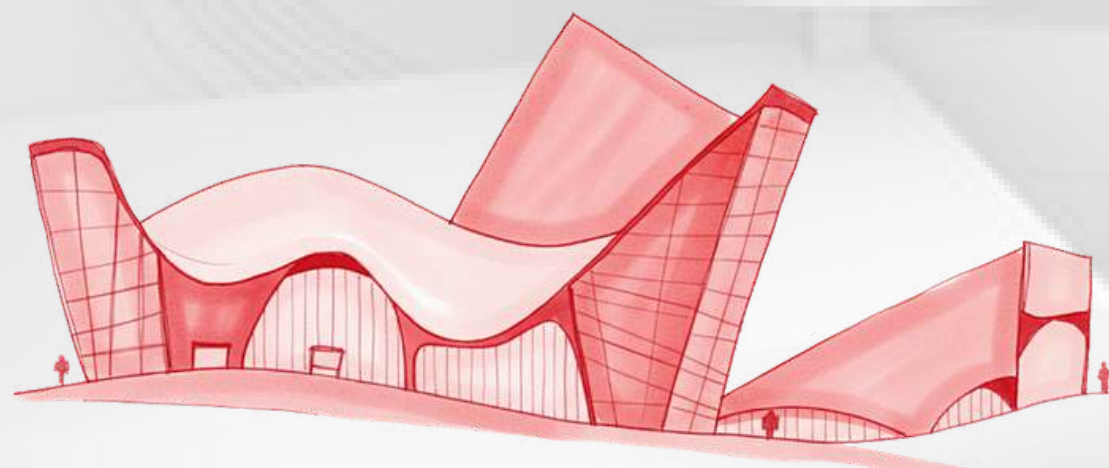
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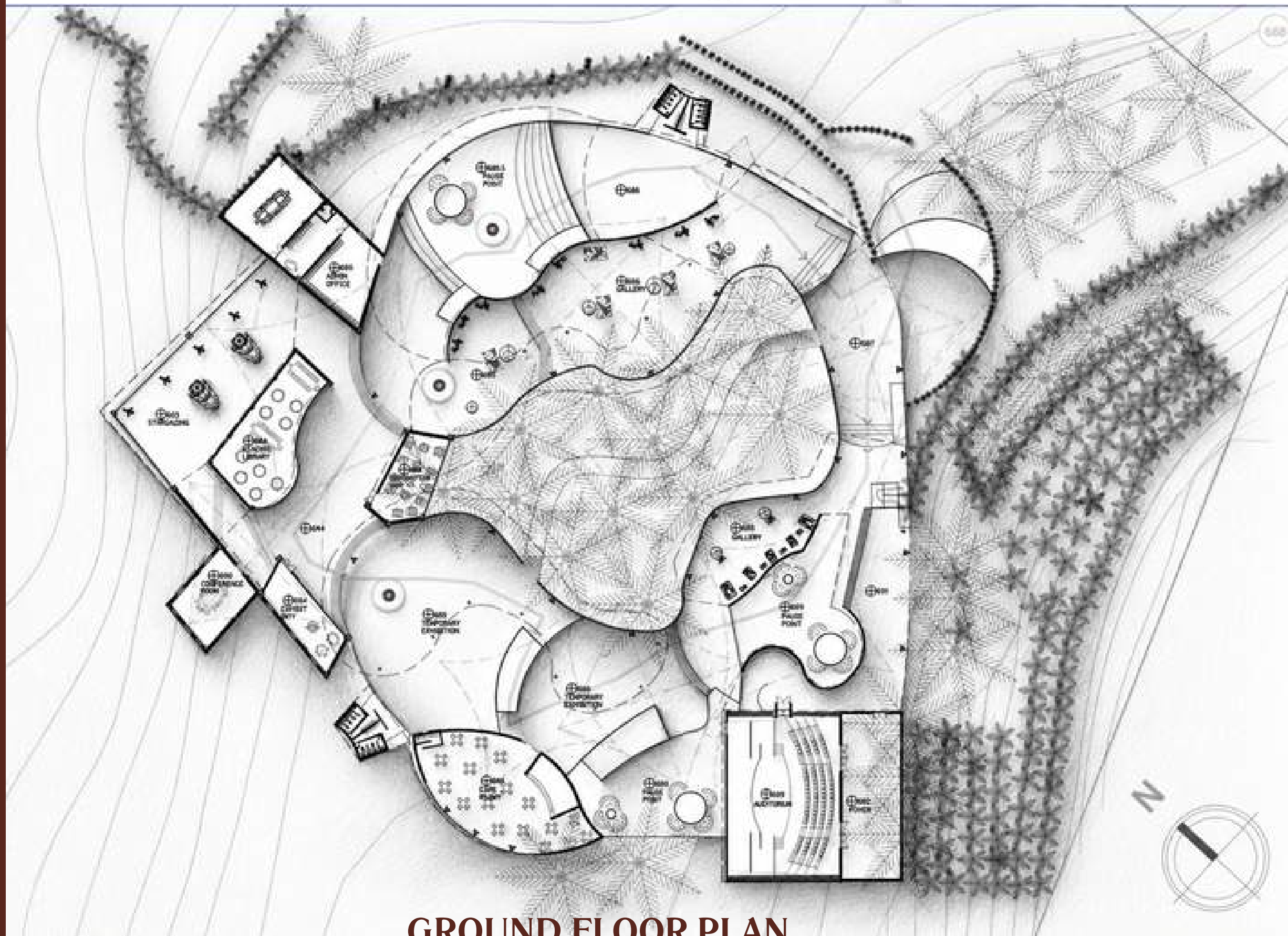
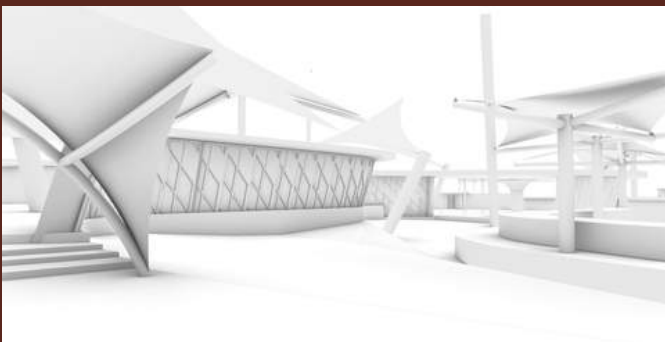
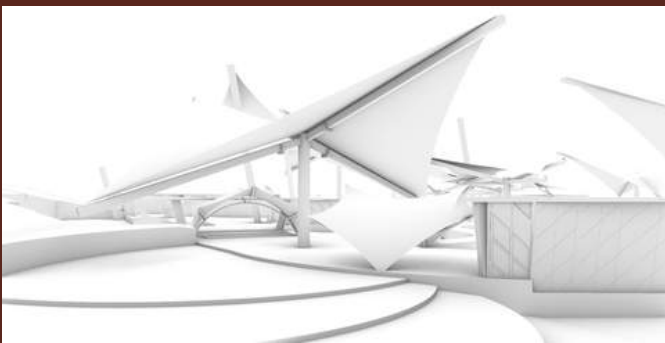
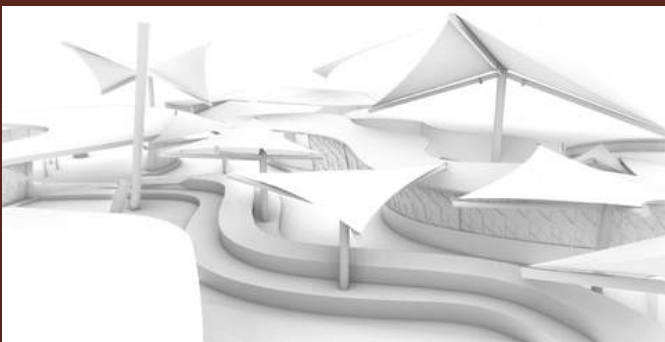


SITE PLAN

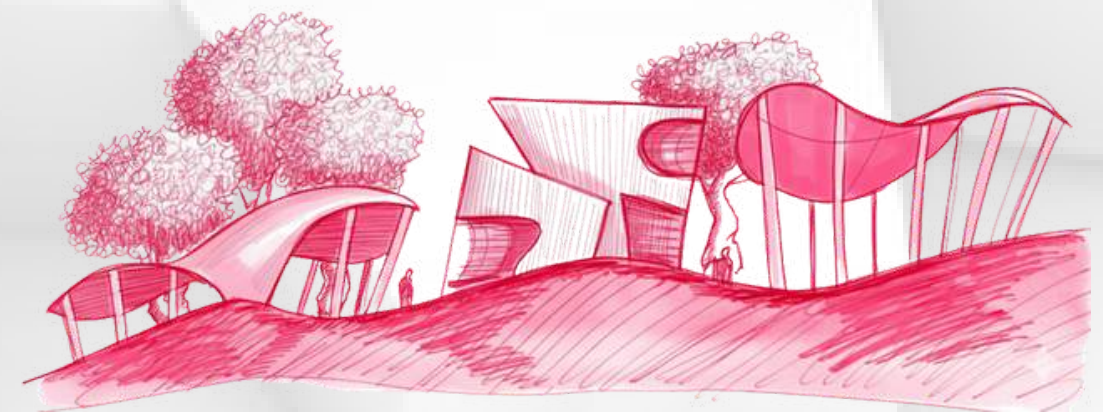


FIRST ROOF LAYER PLAN

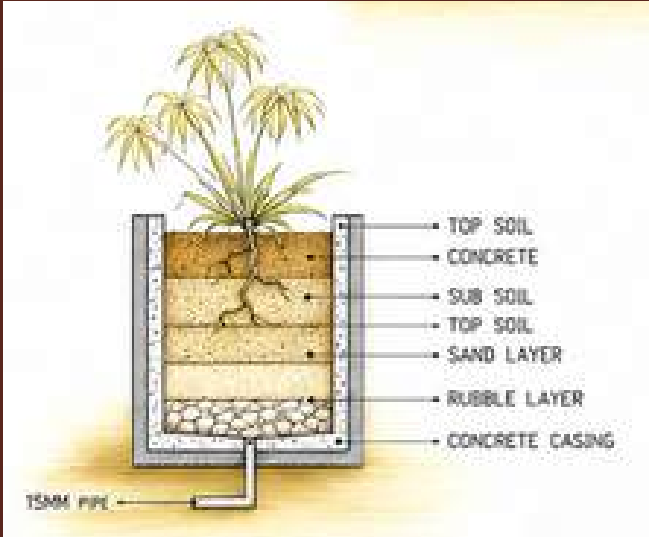




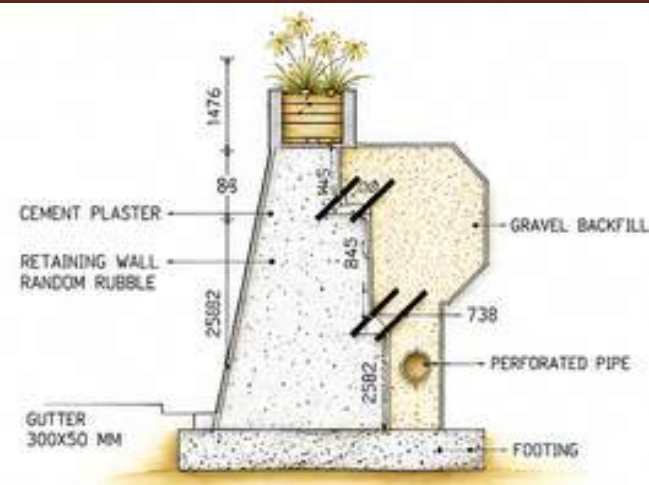
GROUND FLOOR PLAN



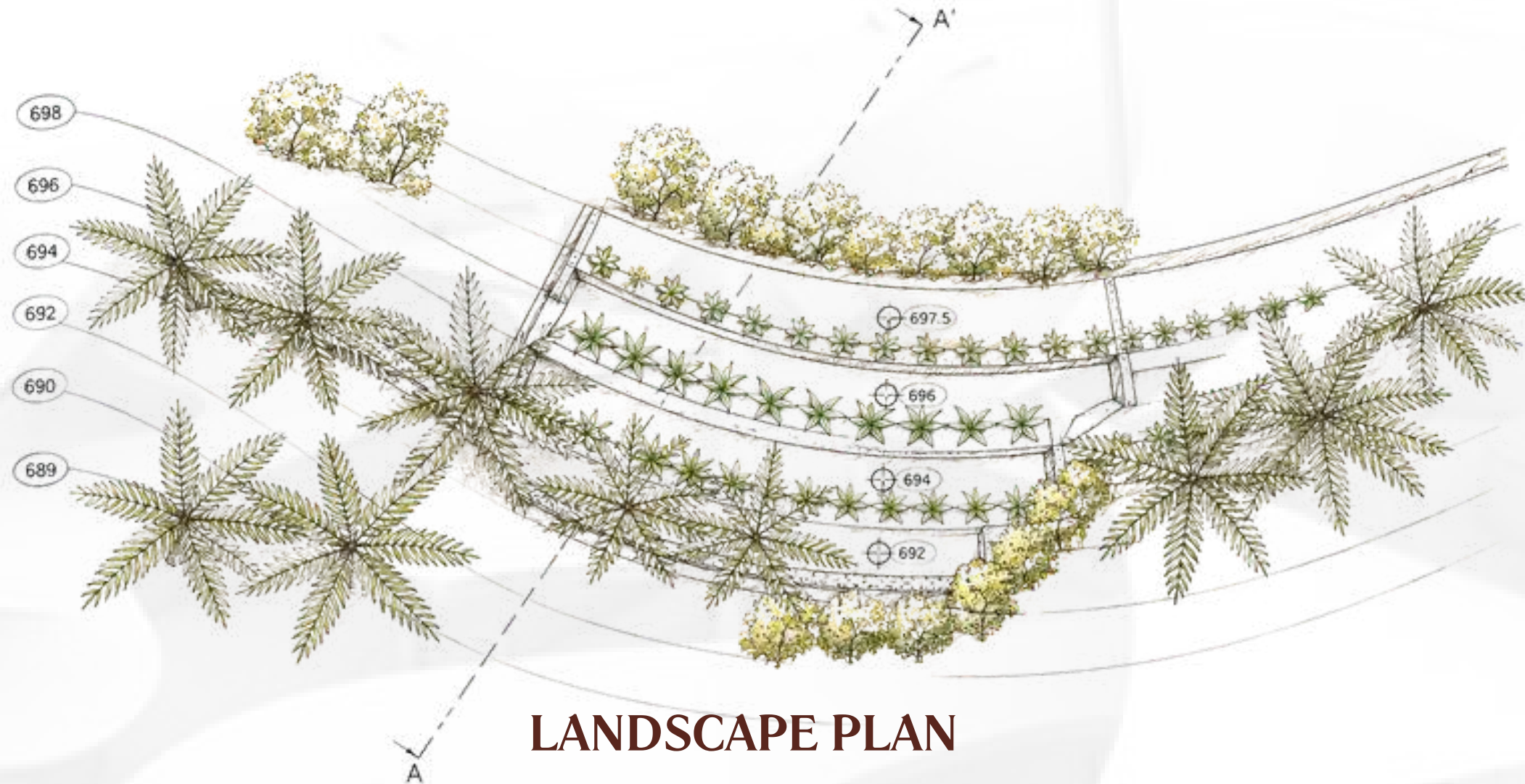
LANDSCAPE AREA



PLANTER DETAIL



RETAINING WALL DETAIL

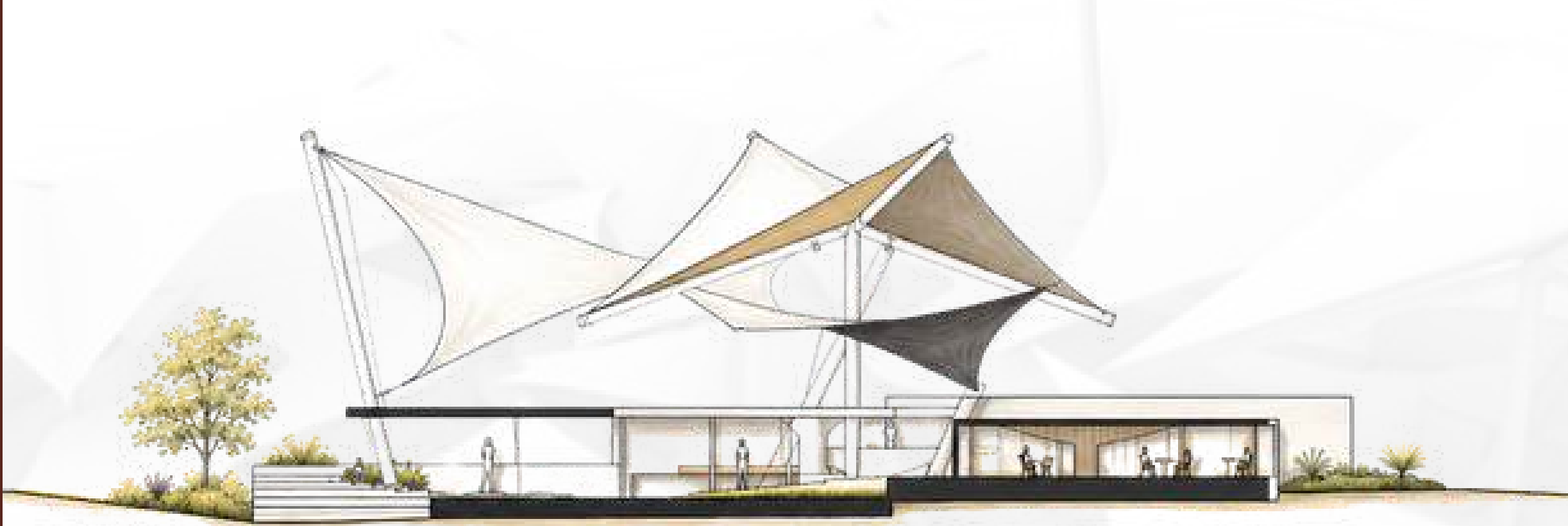
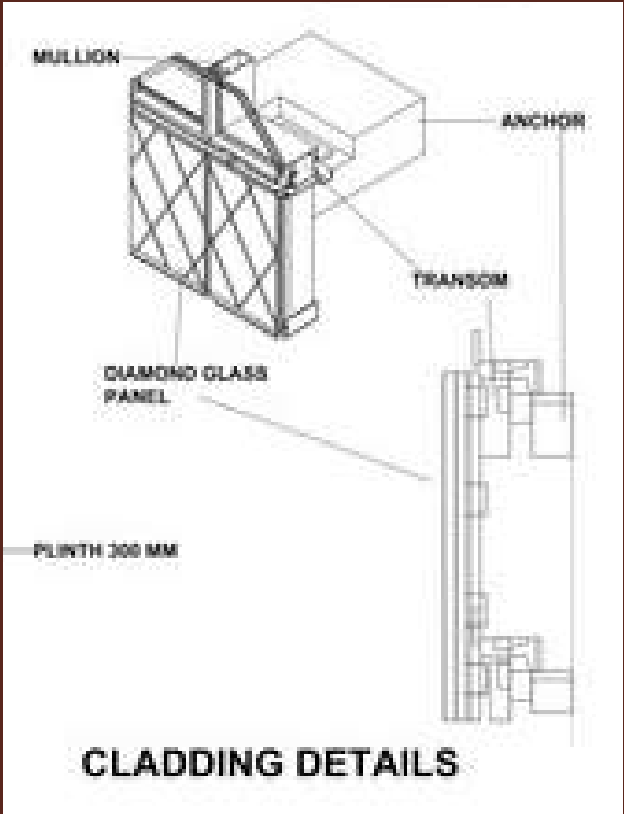
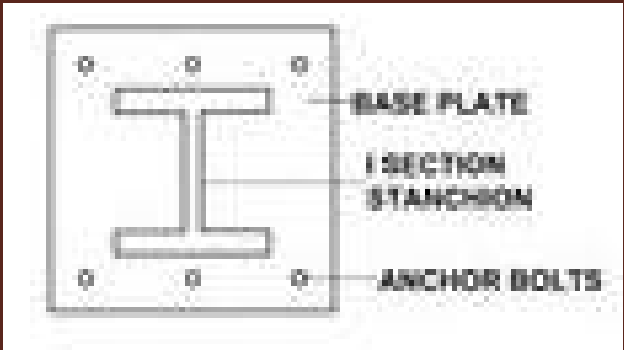
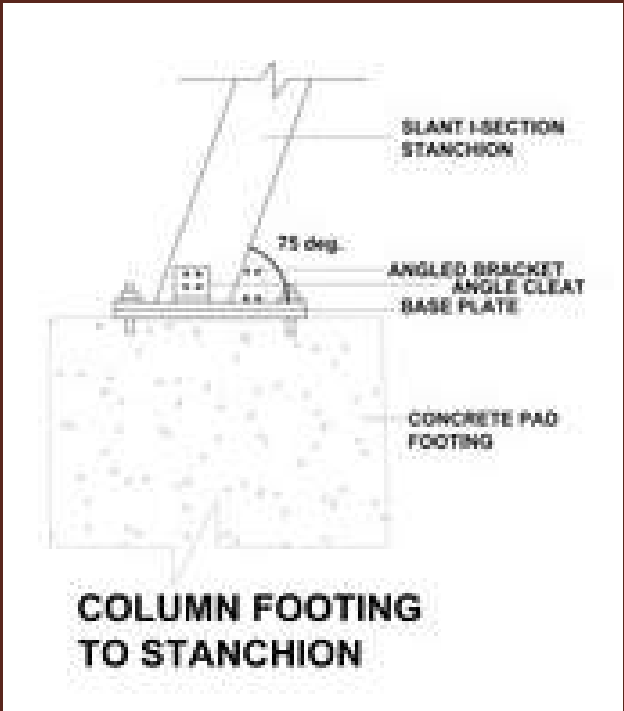


LANDSCAPE PLAN

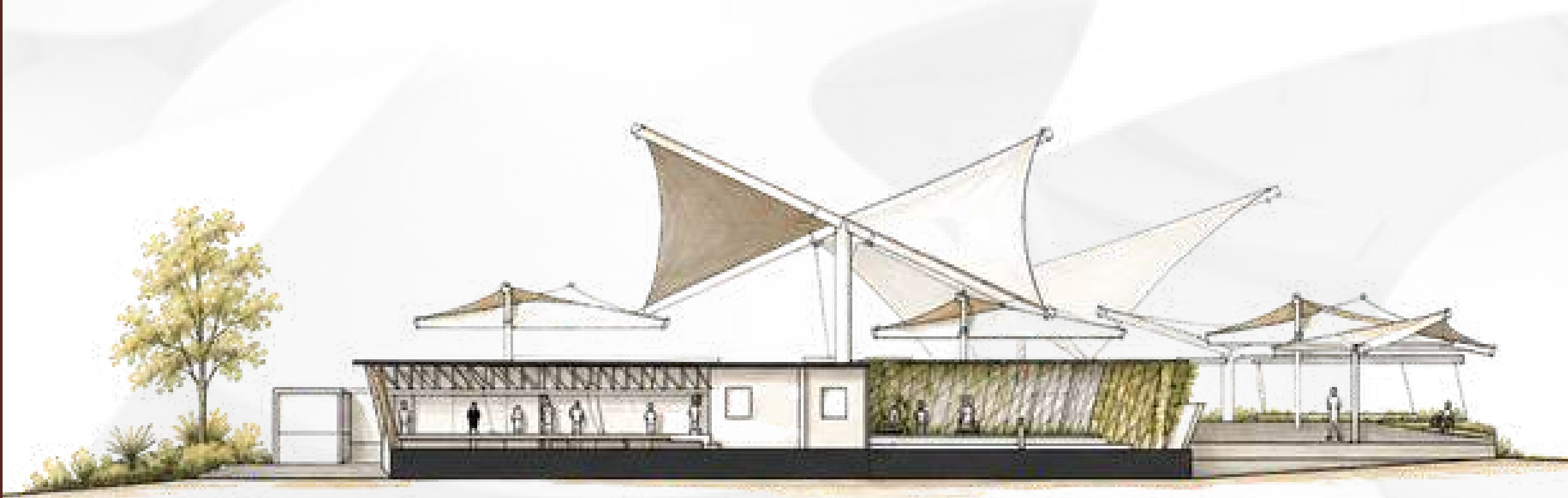


LANDSCAPE SECTION

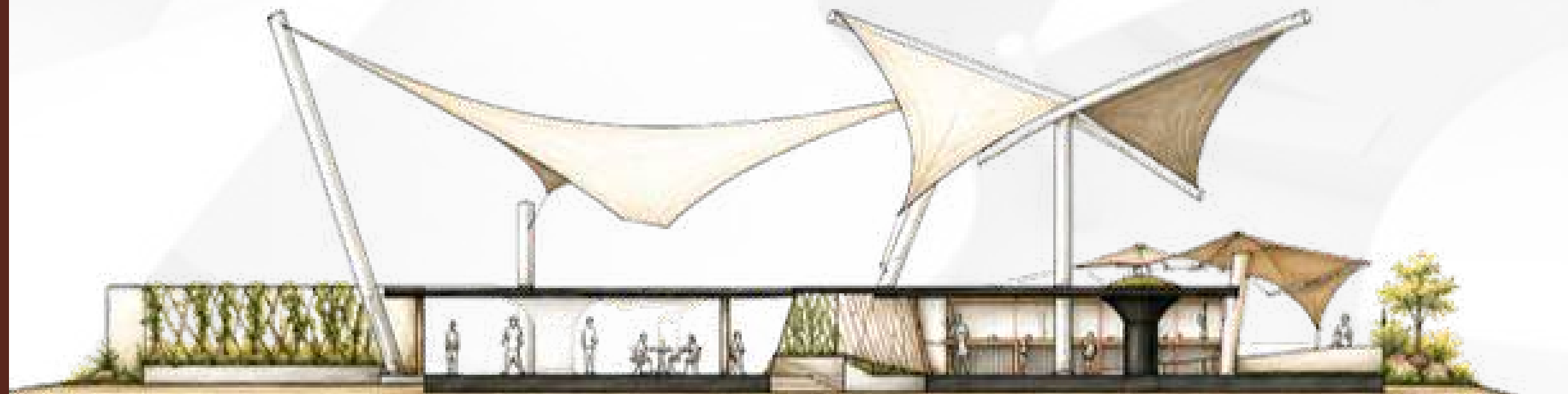
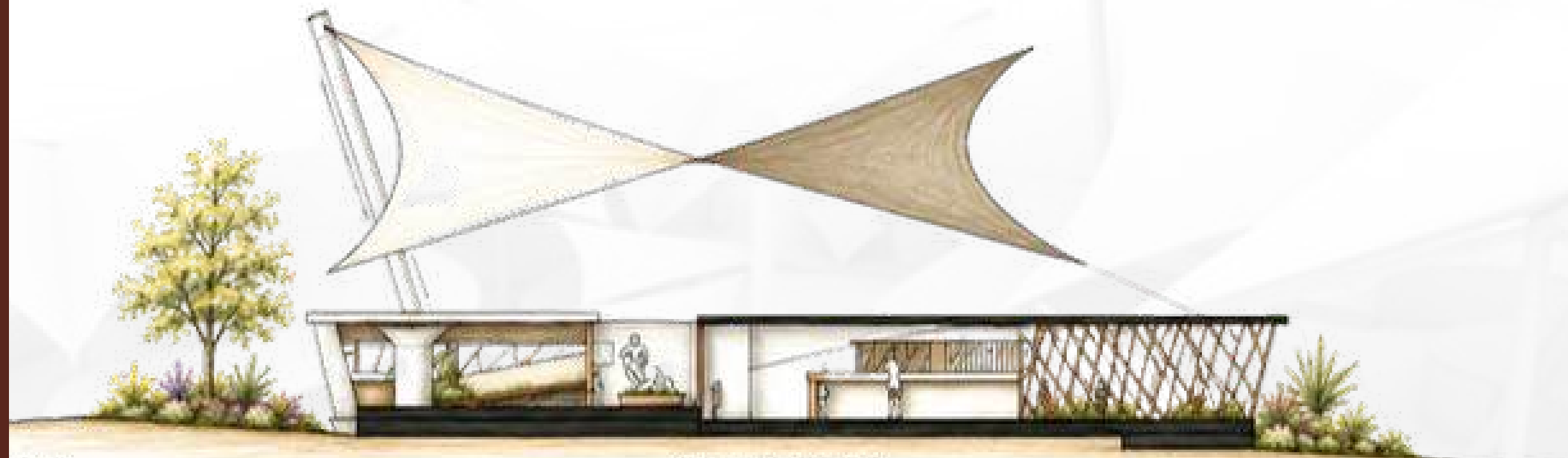
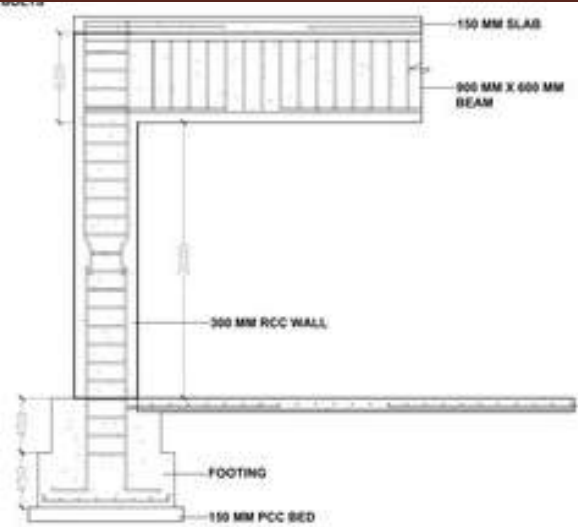
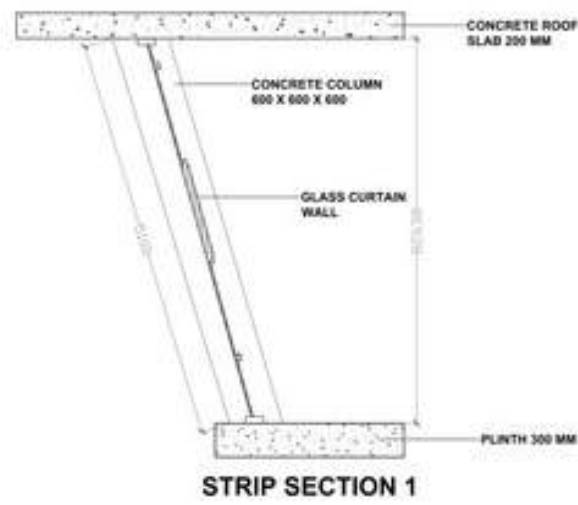
CONSTRUCTION
DETAILS

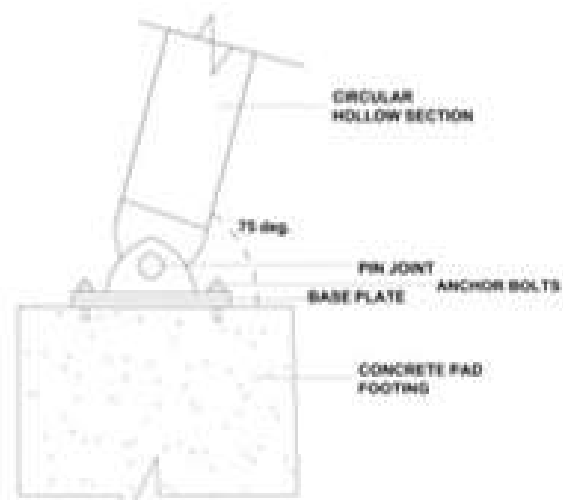


SECTION AA'

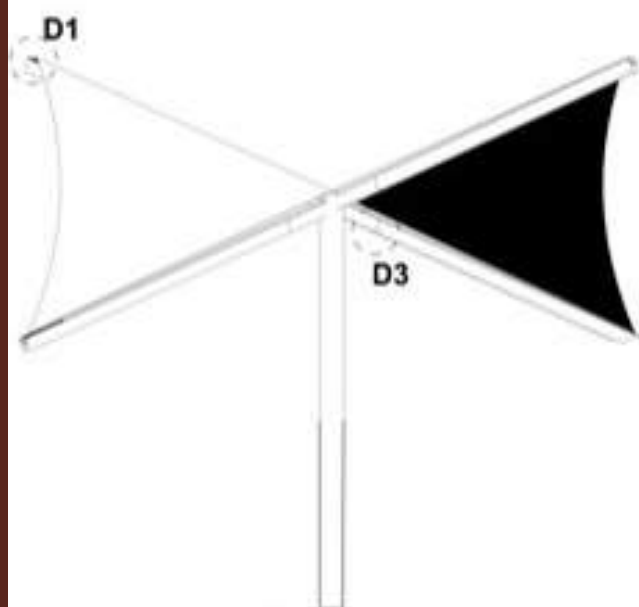


SECTION BB'

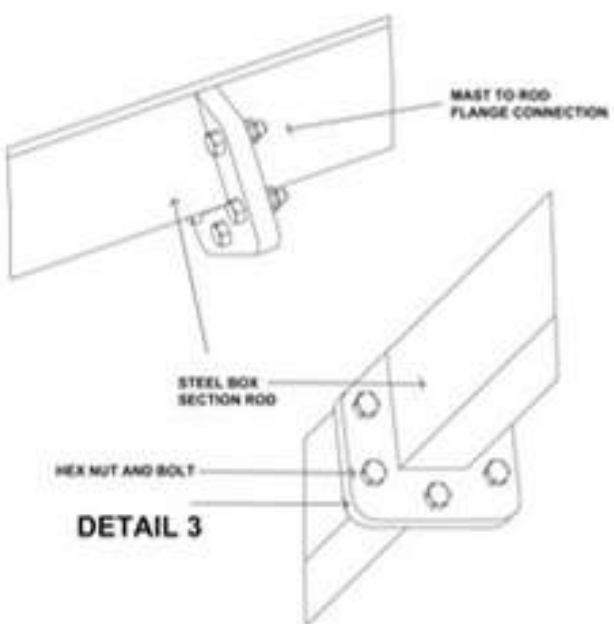




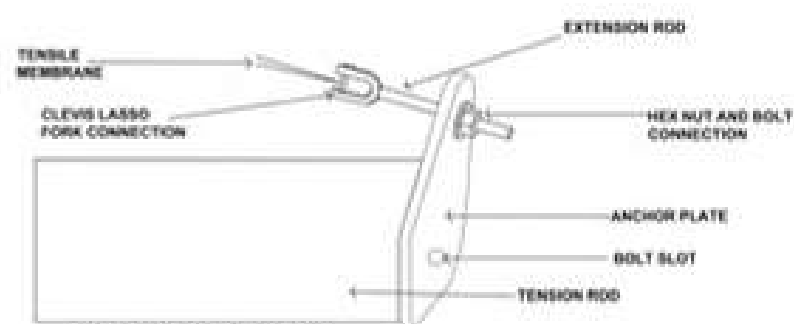
COLUMN TO FOOTING DETAIL 2



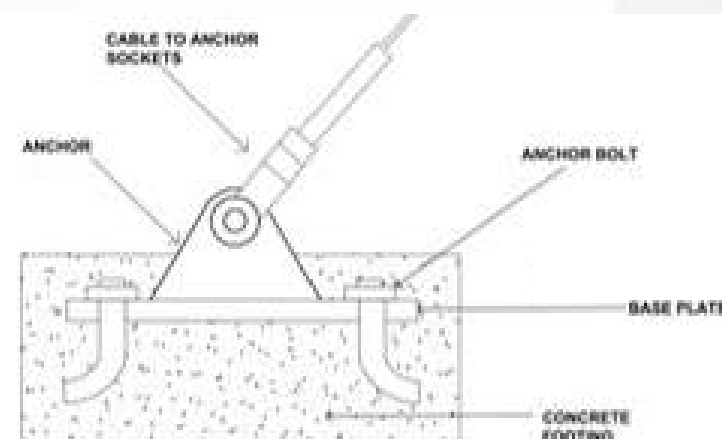
SINGLE POINT HYPAR ROOF



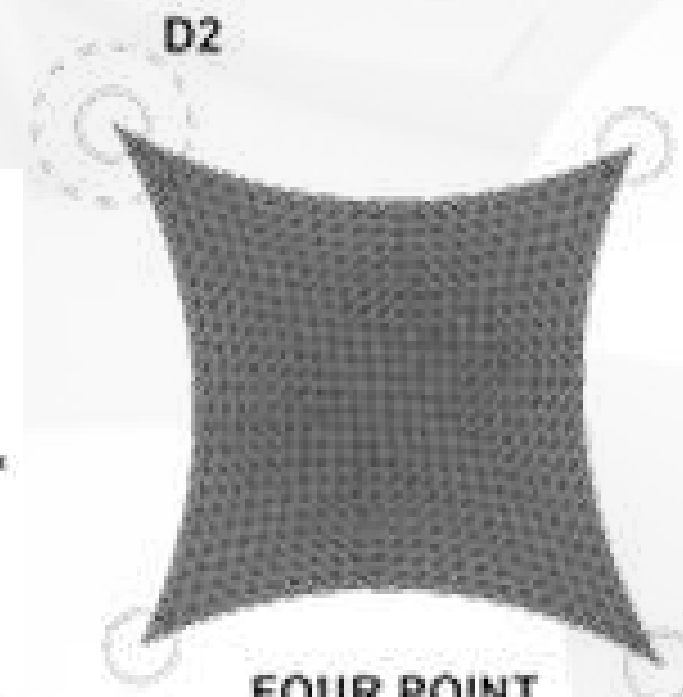
DETAIL 3



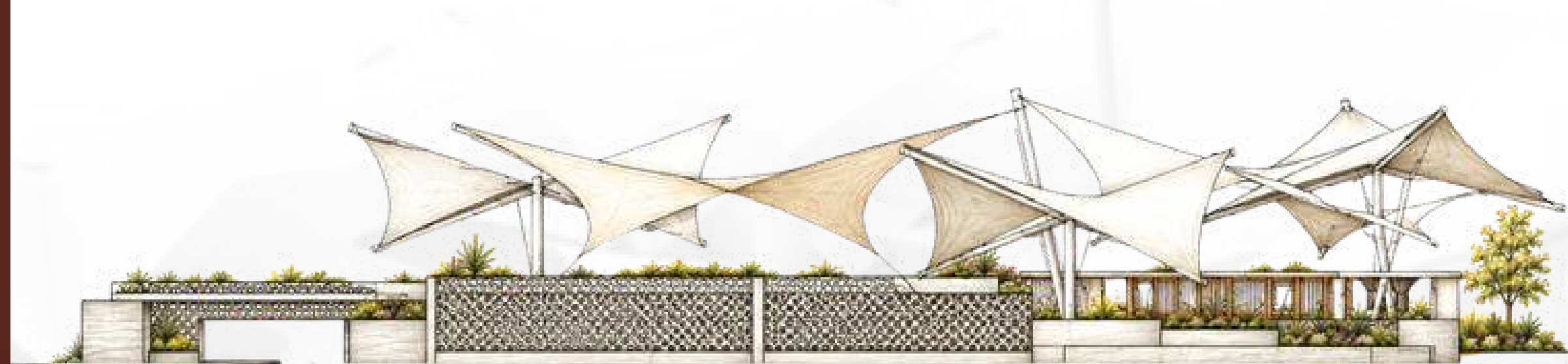
TENSION ROD TO MEMBRANE DETAIL 1



GROUND ANCHOR DETAIL



FOUR POINT HYPAR ROOF

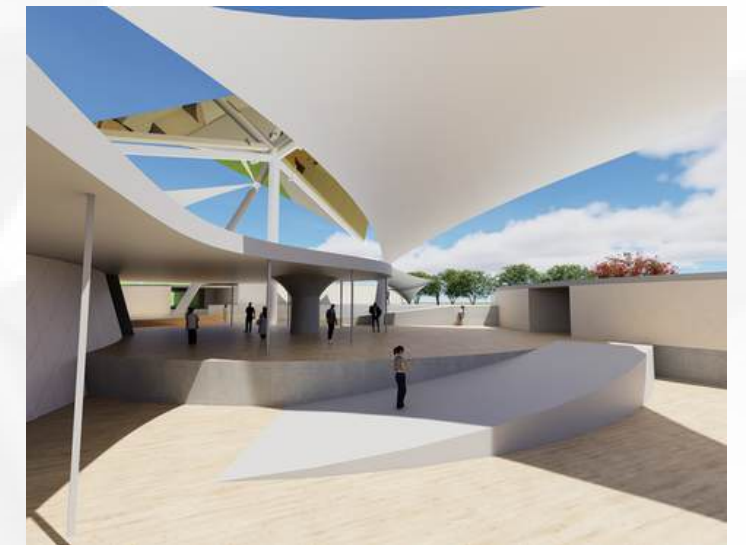
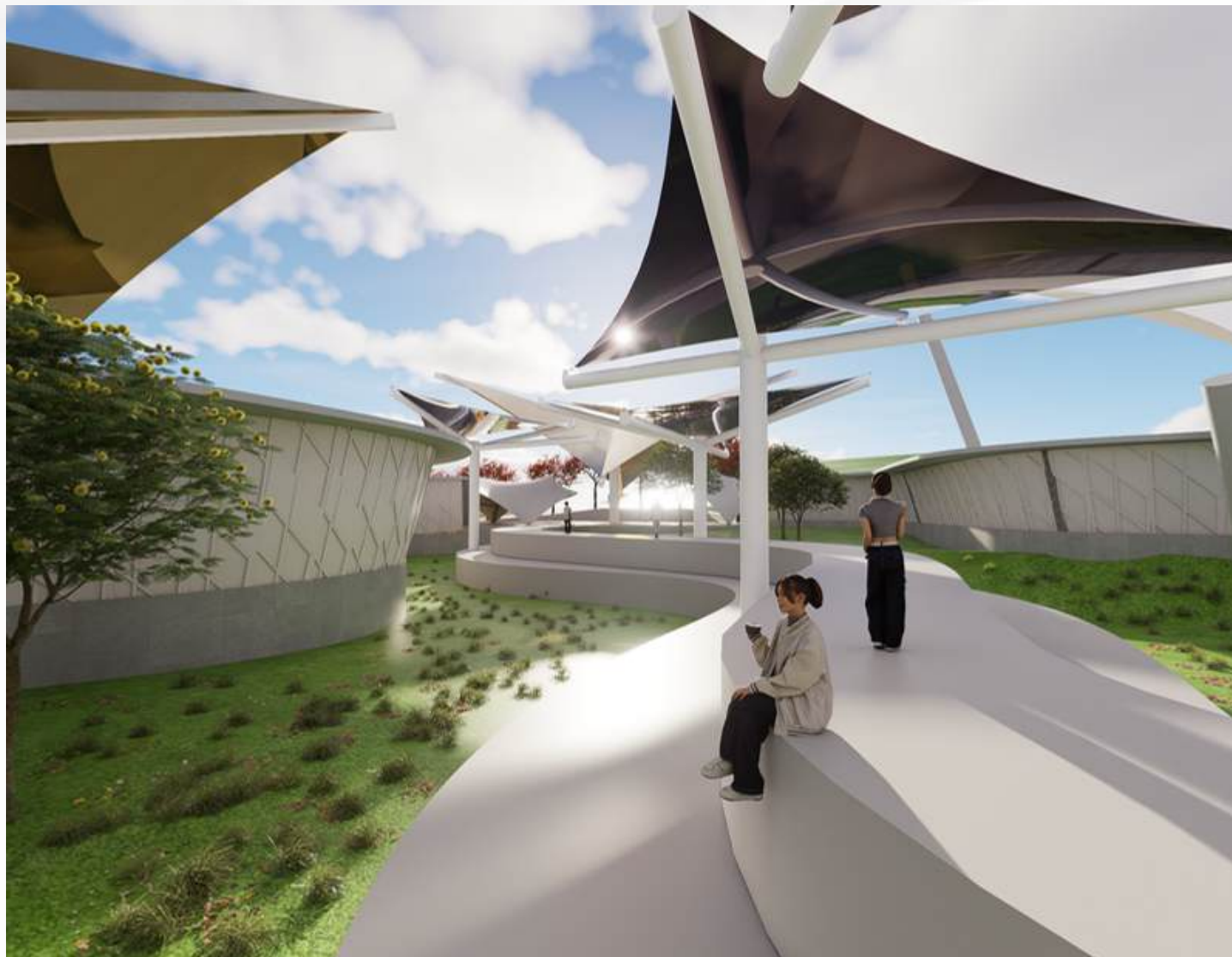
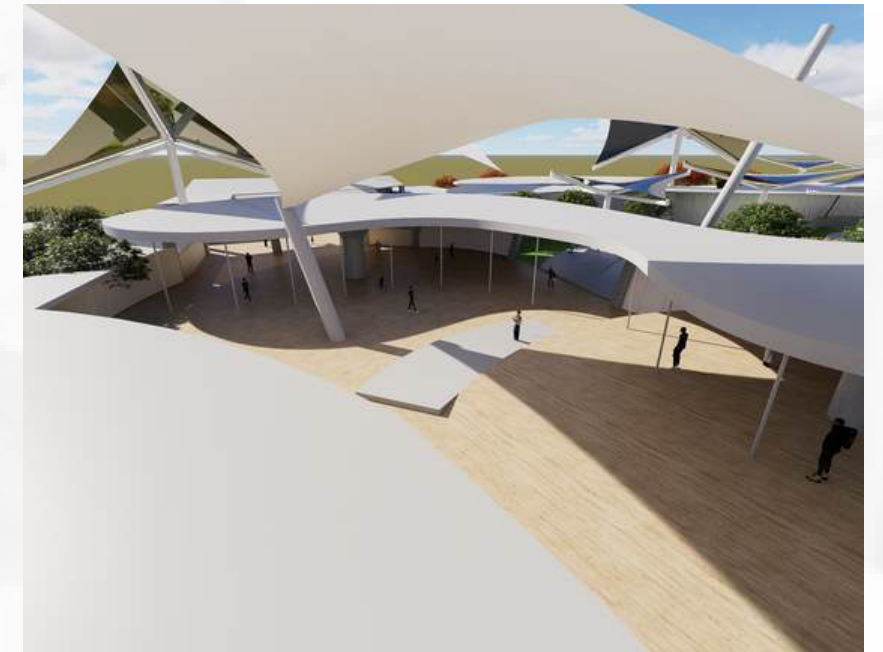


ELEVATION 1



ELEVATION 2

VIEWS



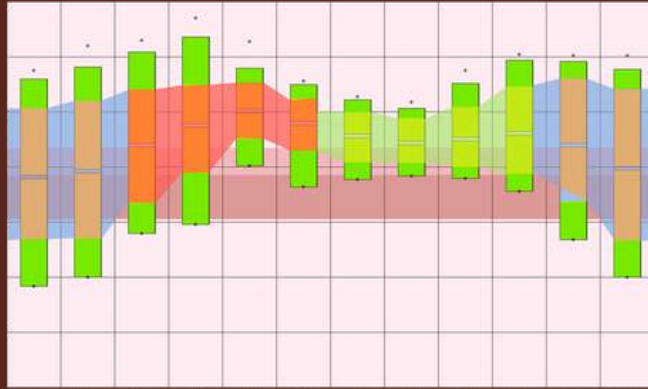
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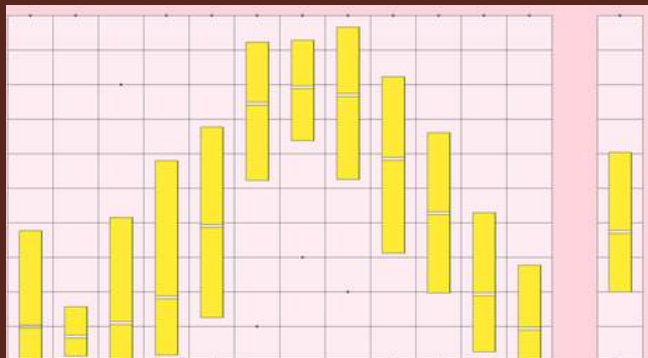
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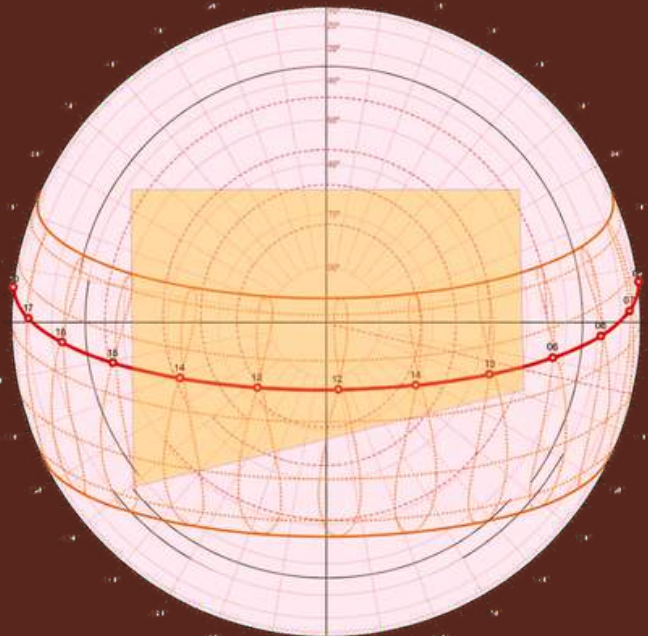
ARCHITECTURAL DESIGN



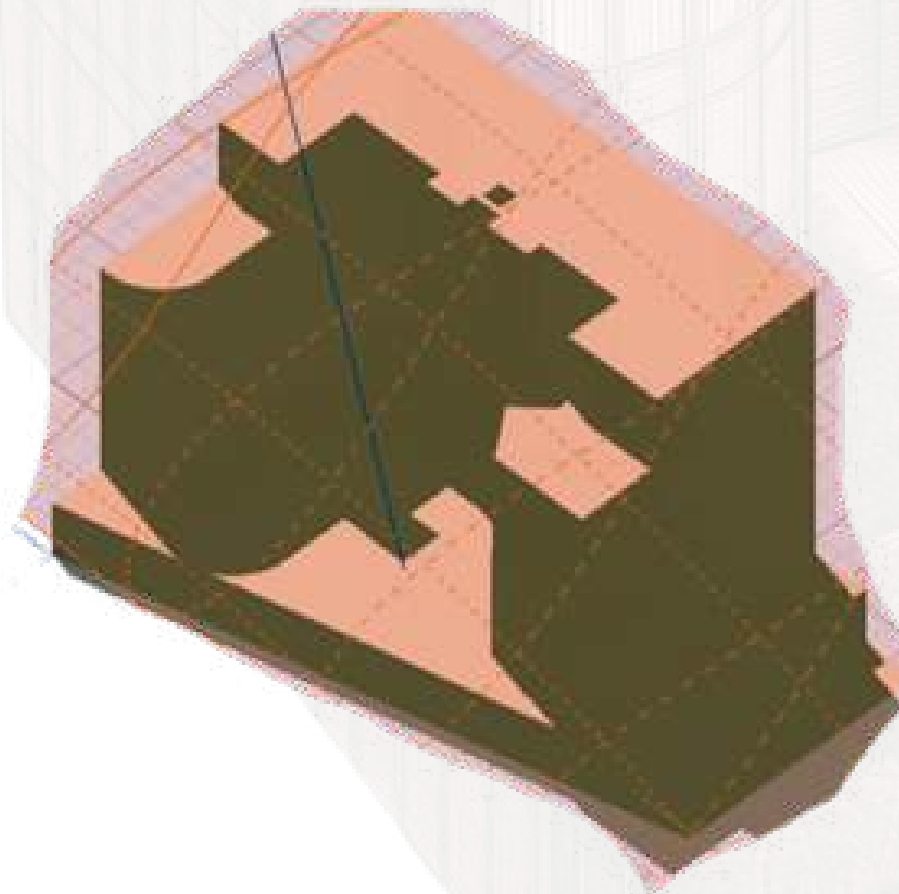
TEMPERATURE



SKY COVER RANGE

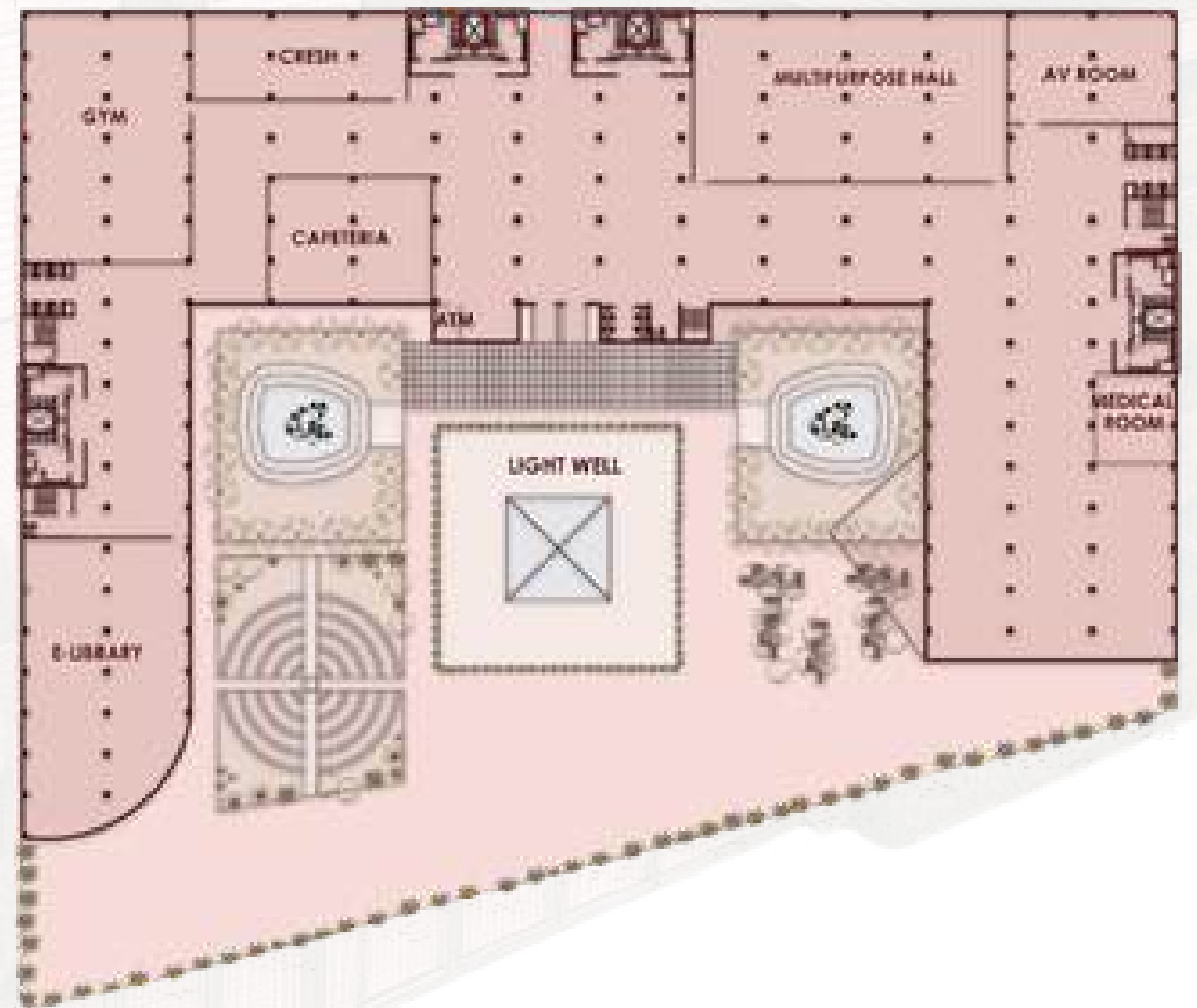


SUN PATH DIAGRAM

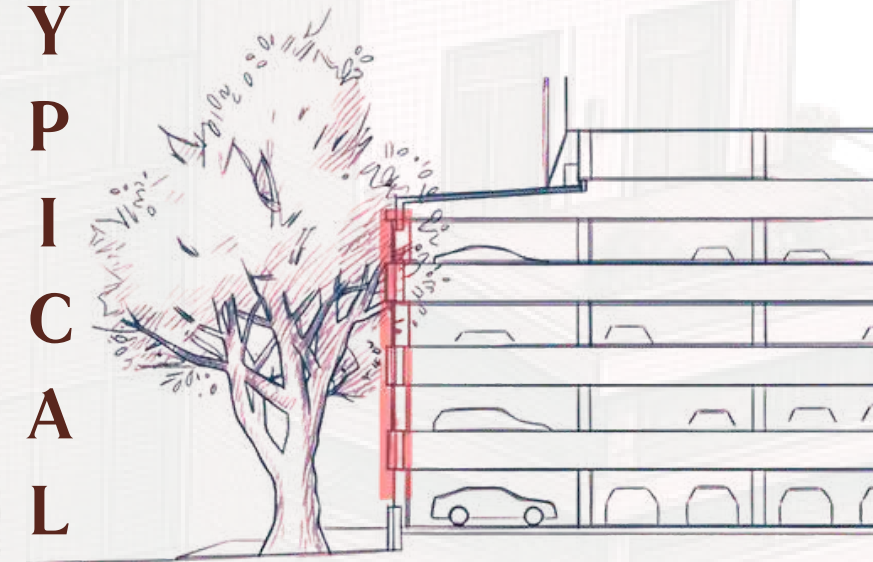


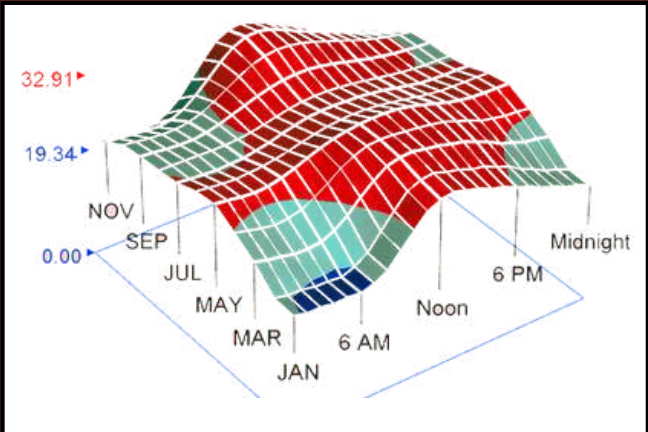
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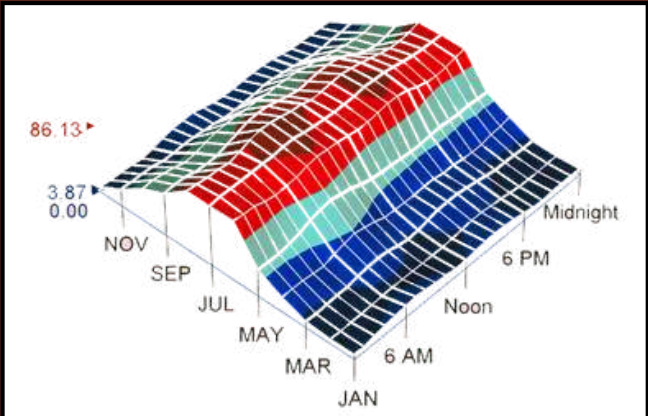


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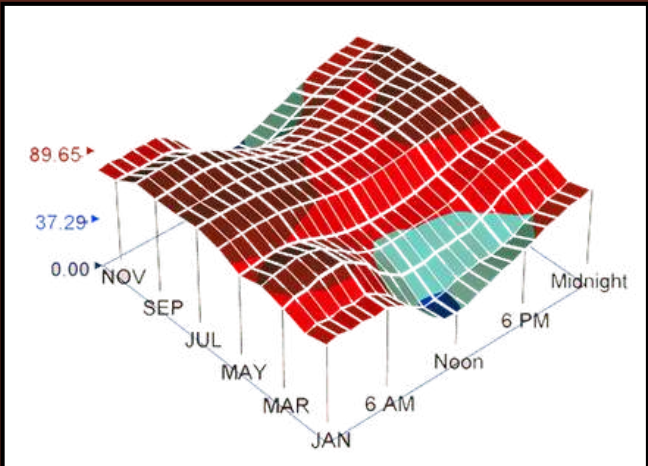




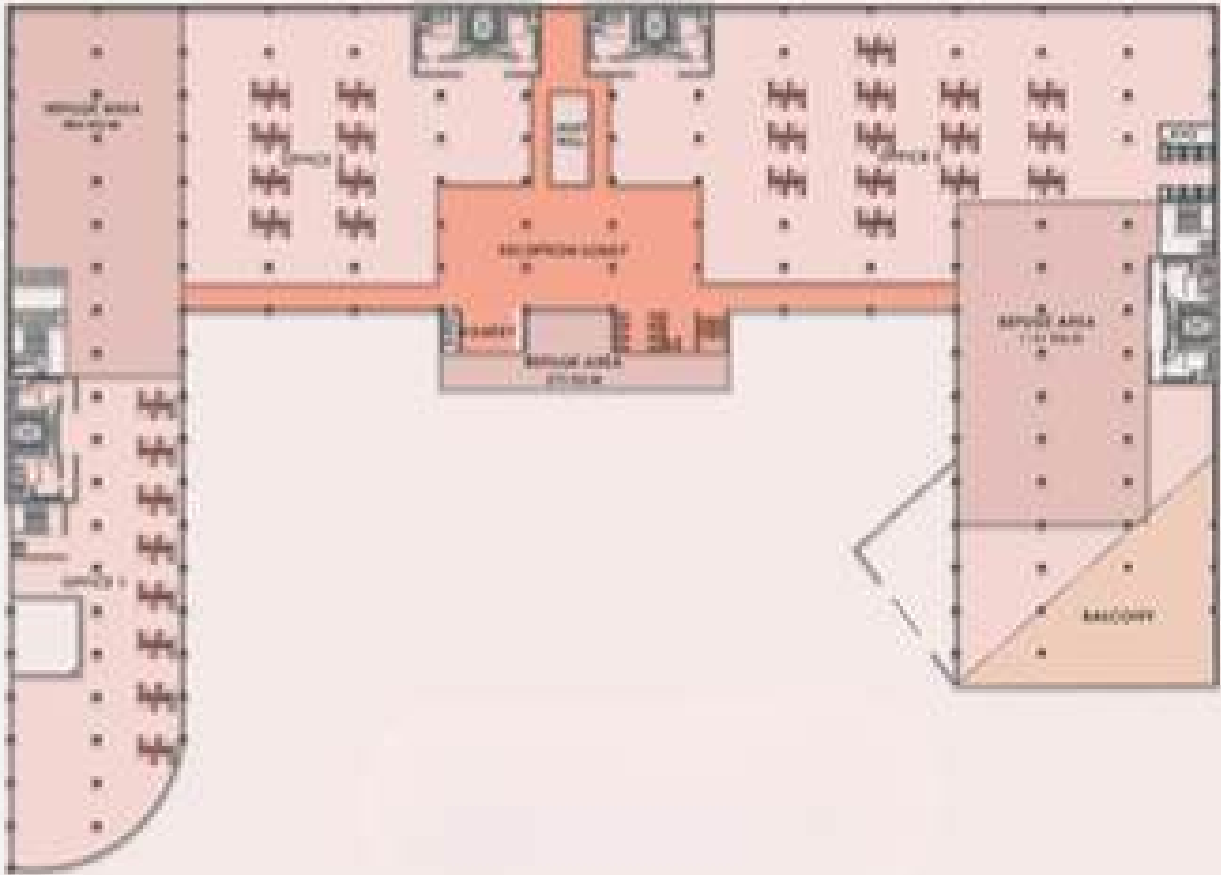
DRY BULB TEMPERATURE



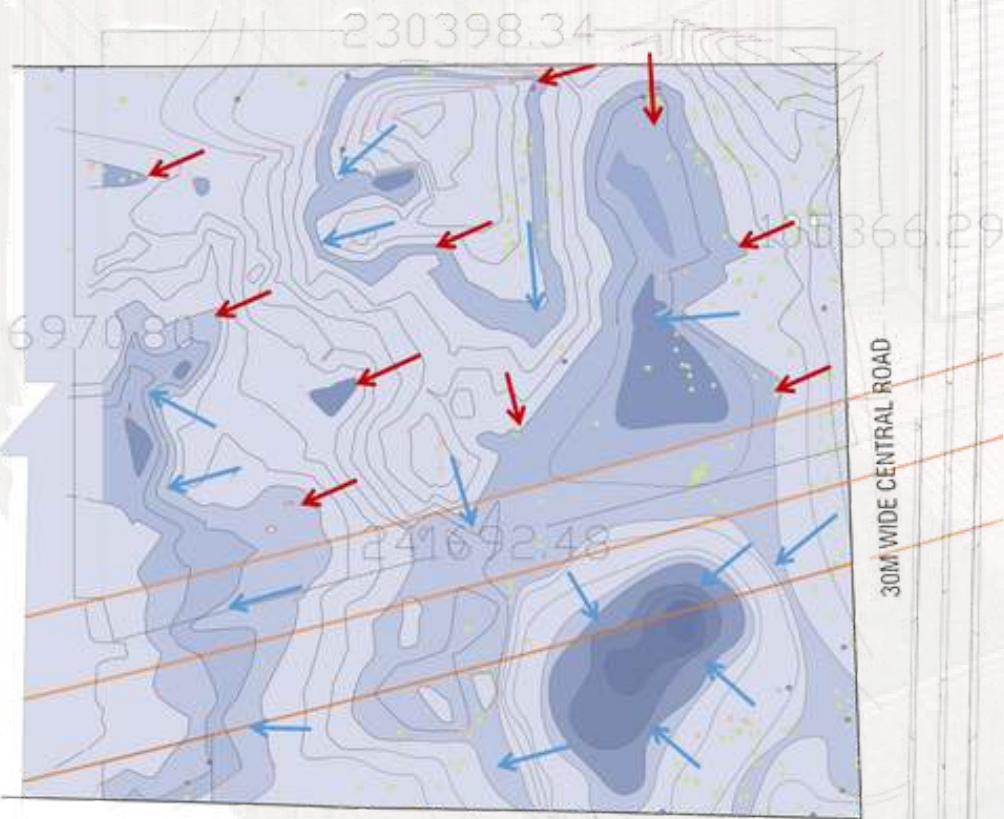
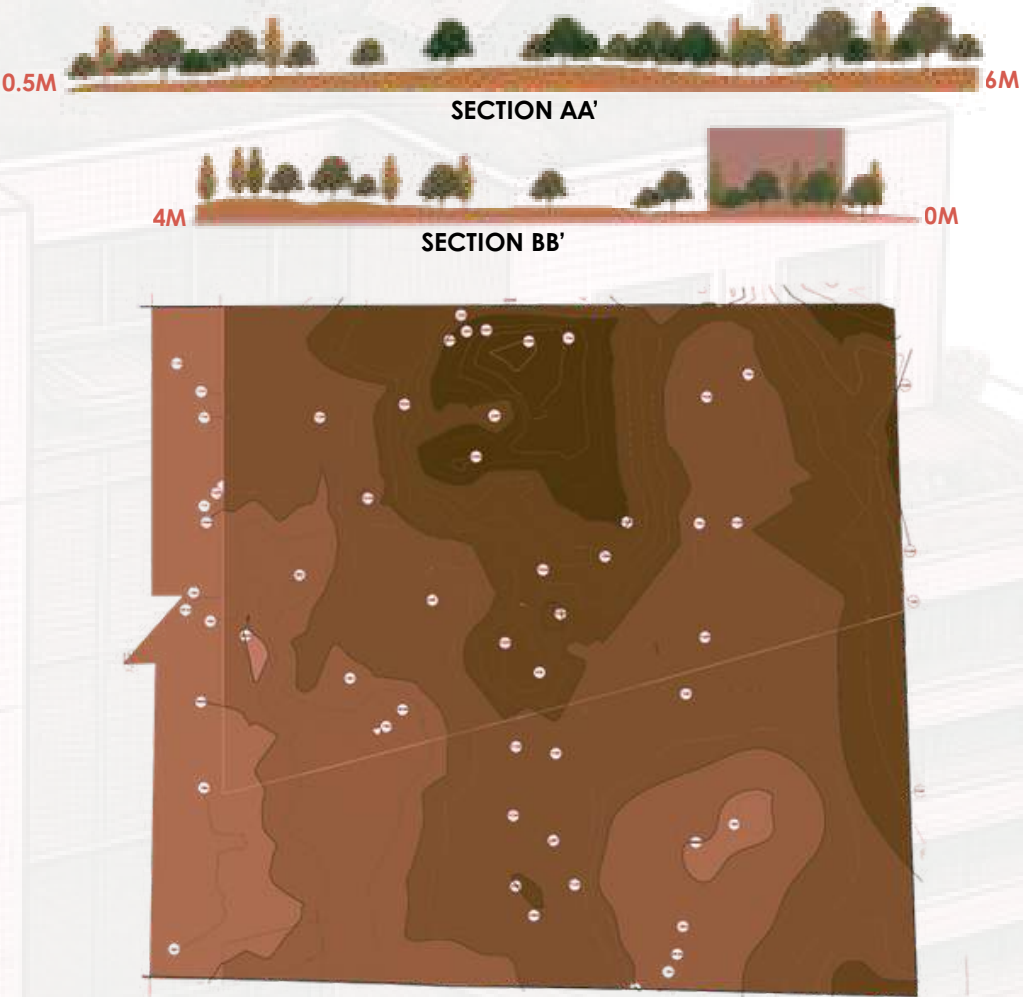
SKY COVER



RELATIVE HUMIDITY

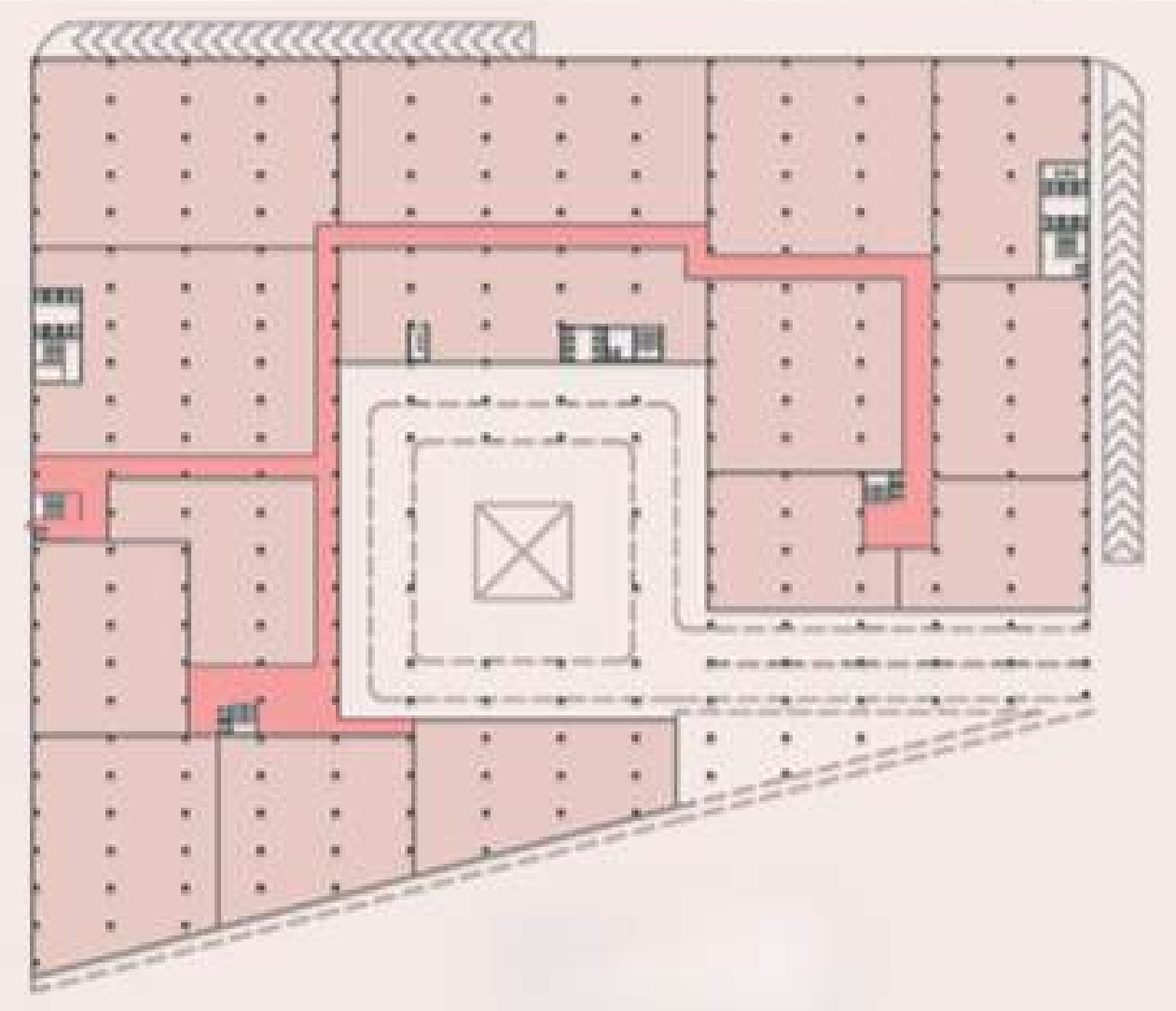


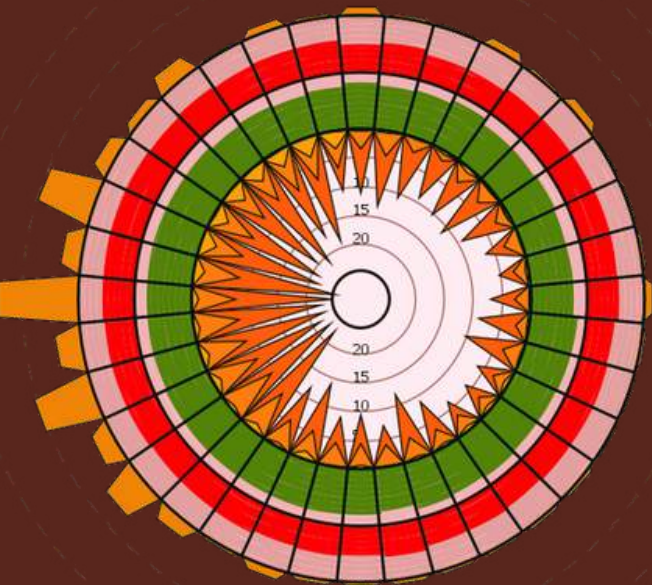
REFUGE PLAN



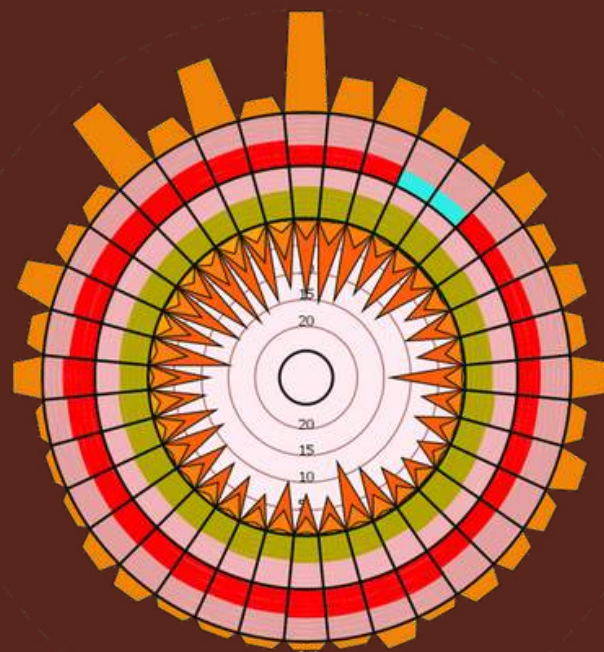
HYDROLOGY

RETAIL FLOOR

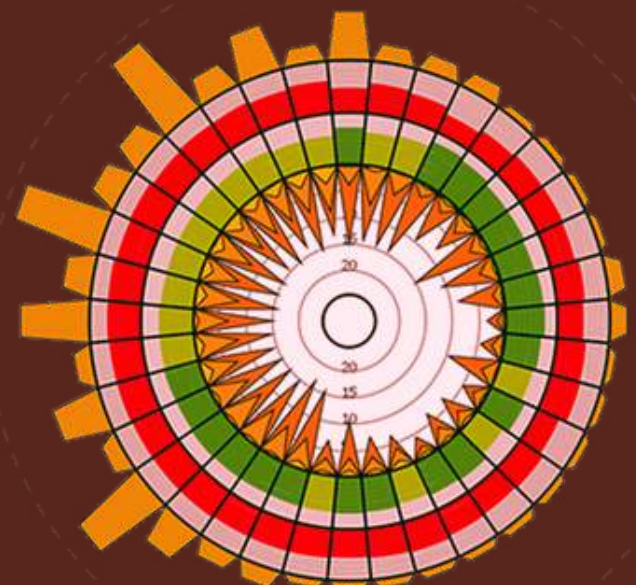




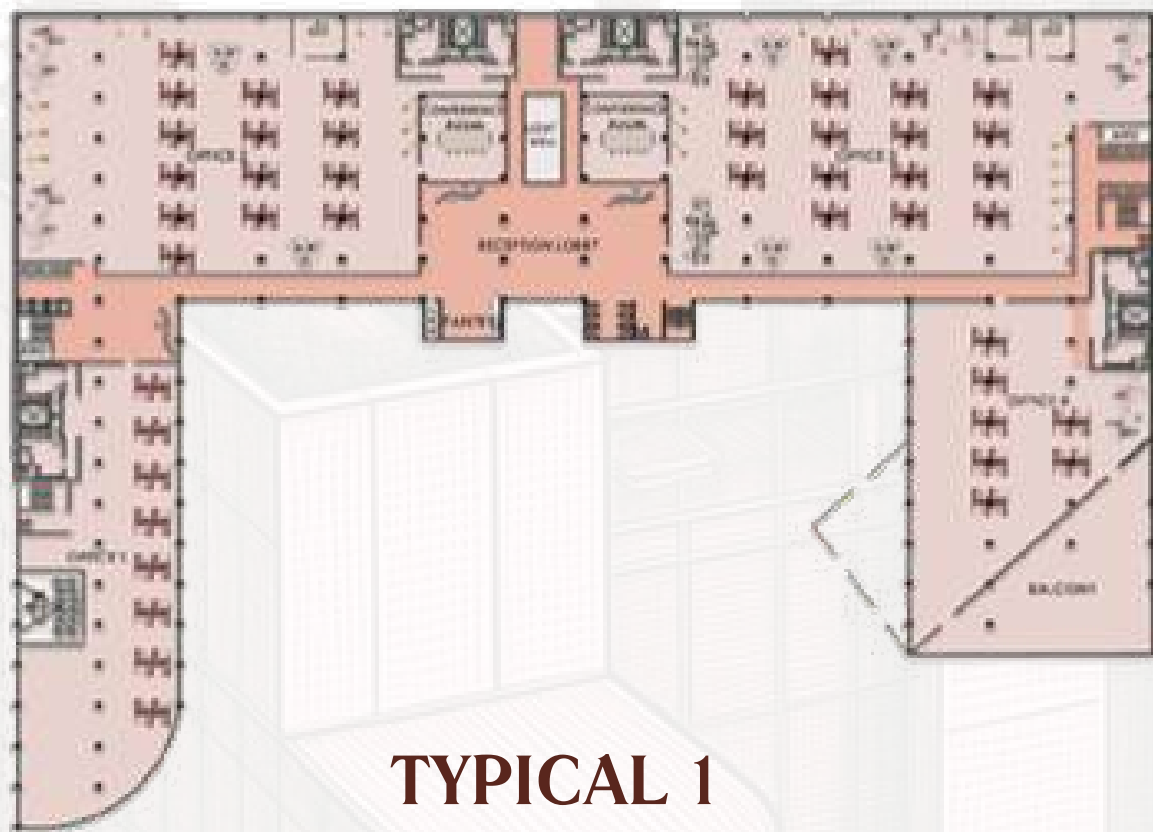
JUNE-OCT WINDS



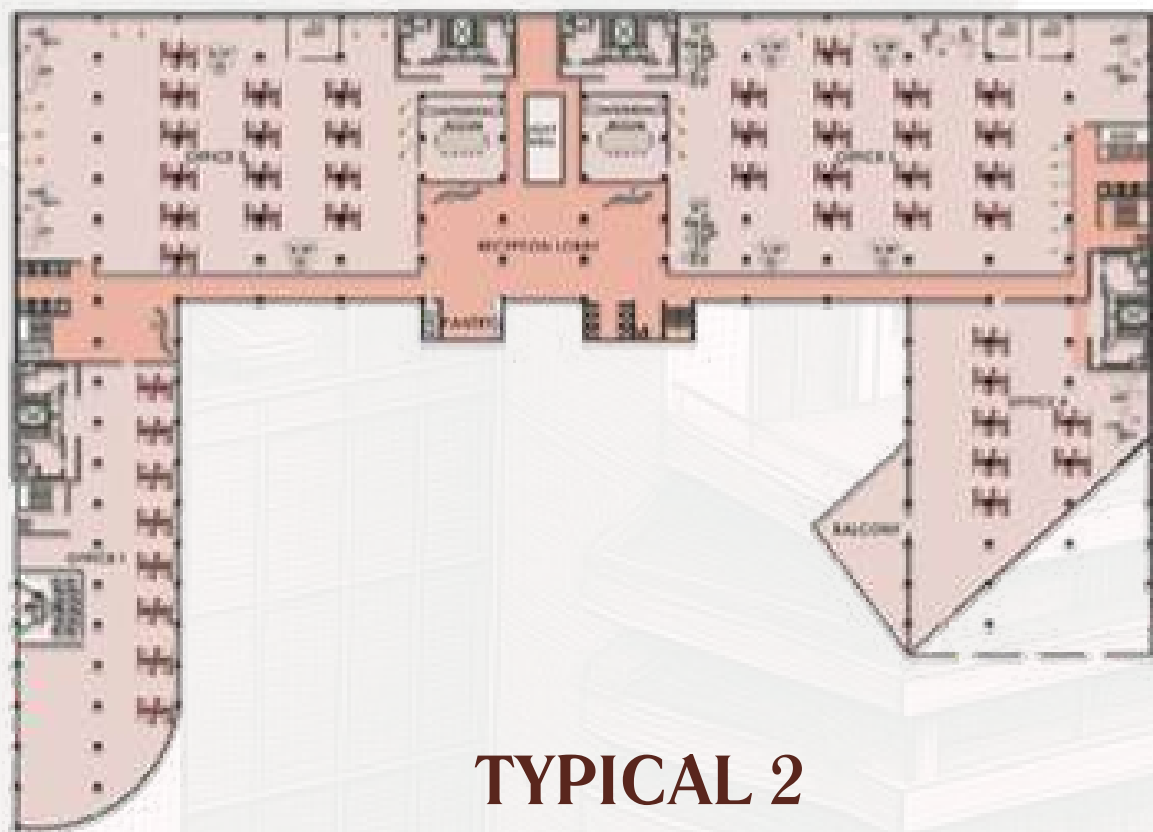
NOV-FEB WINDS



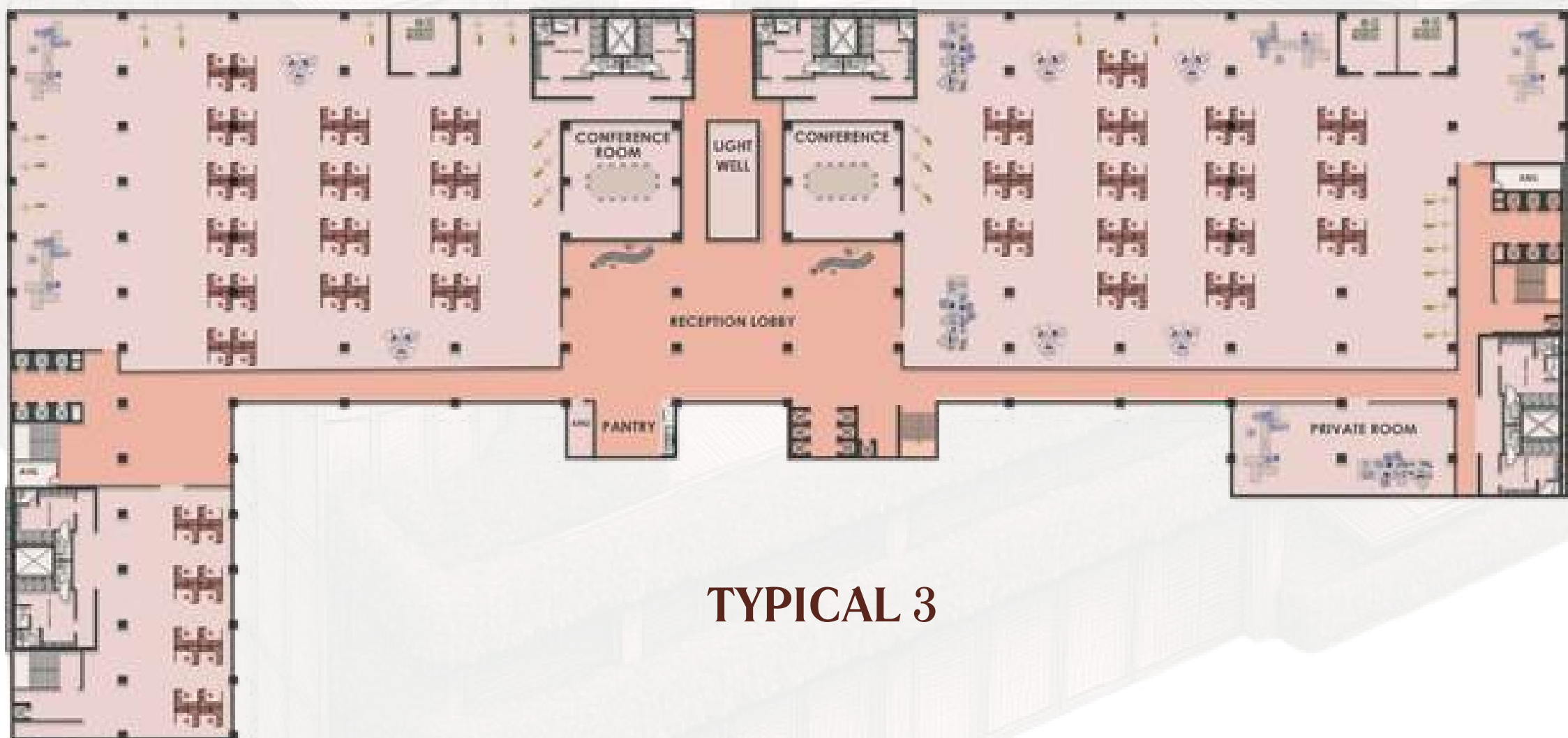
MARCH-MAY WINDS



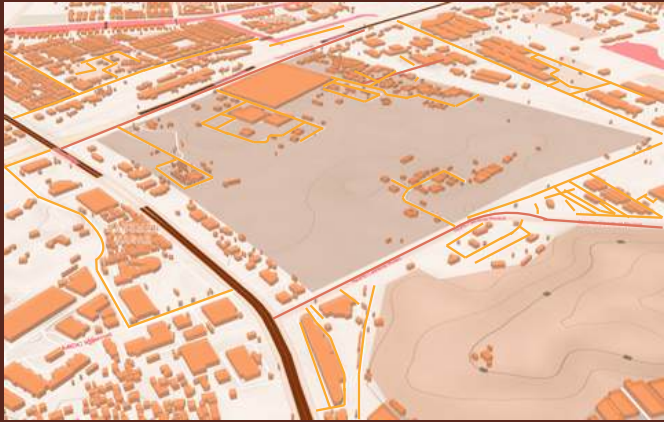
TYPICAL 1



TYPICAL 2



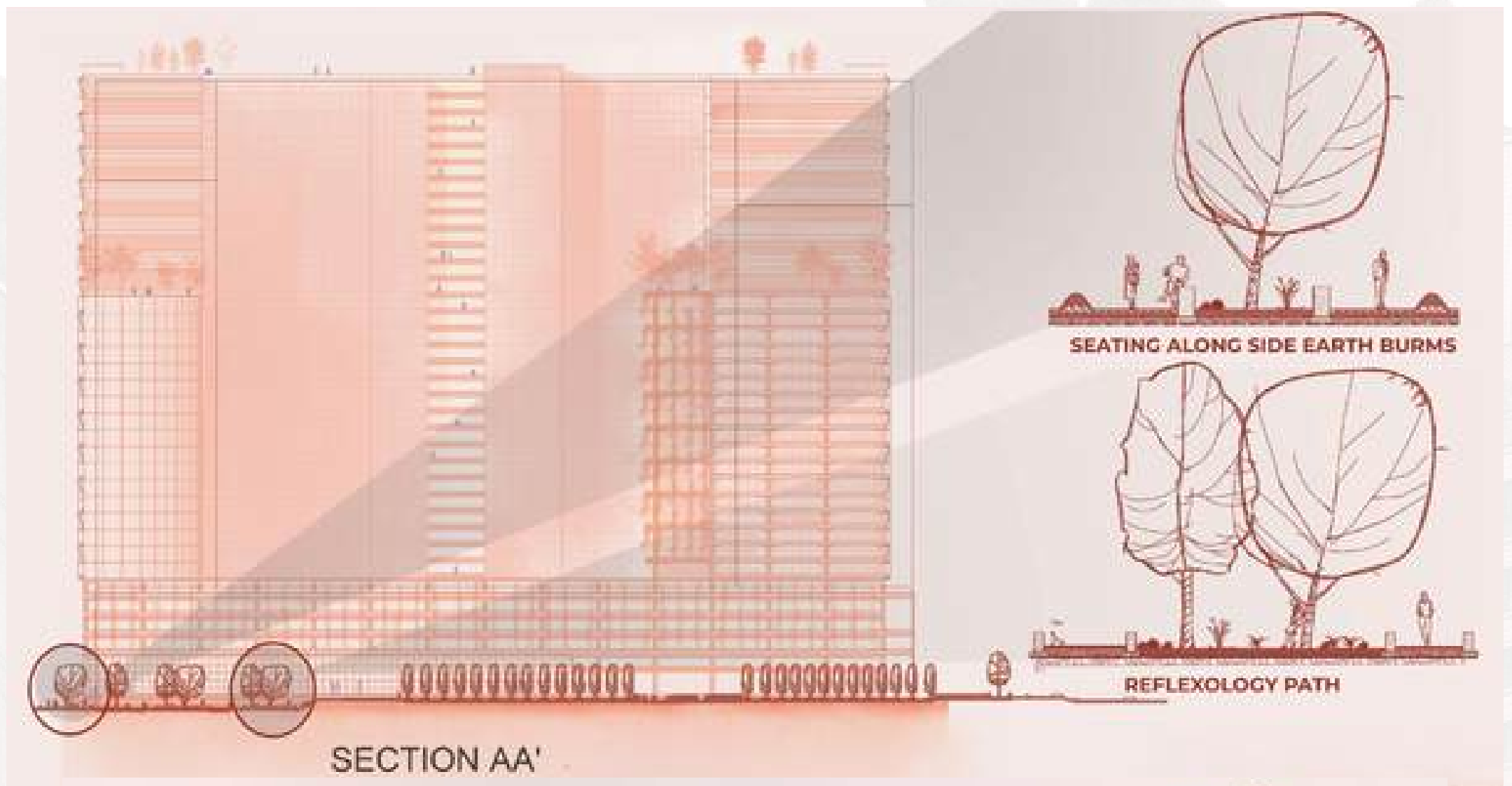
TYPICAL 3

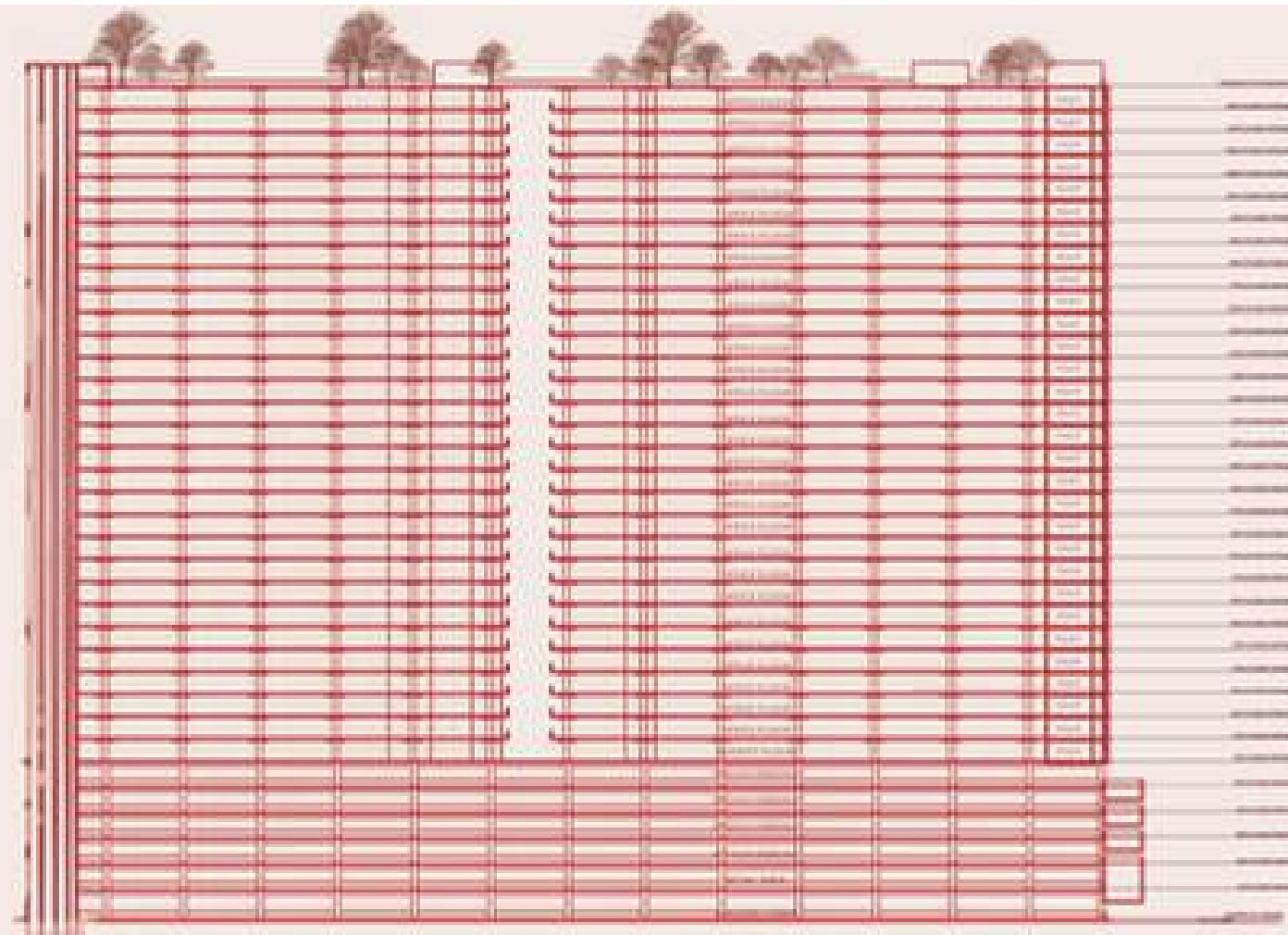
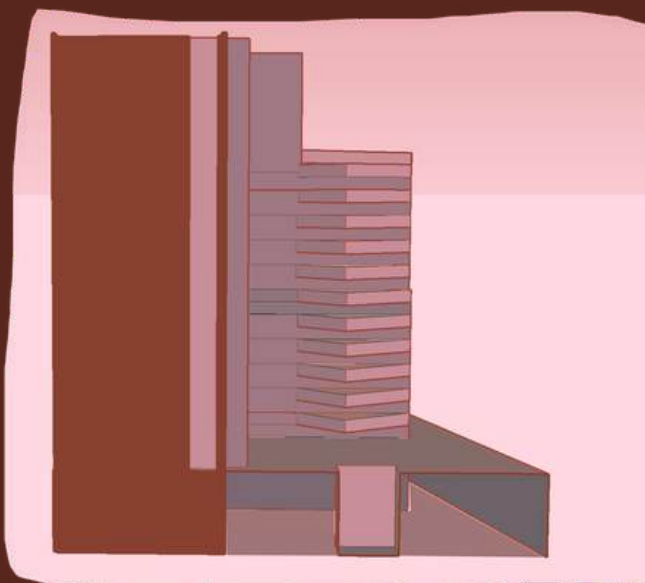


LEGEND PRIMARY ROADS
SECONDARY ROADS TERTIARY ROADS
CIRCULATION



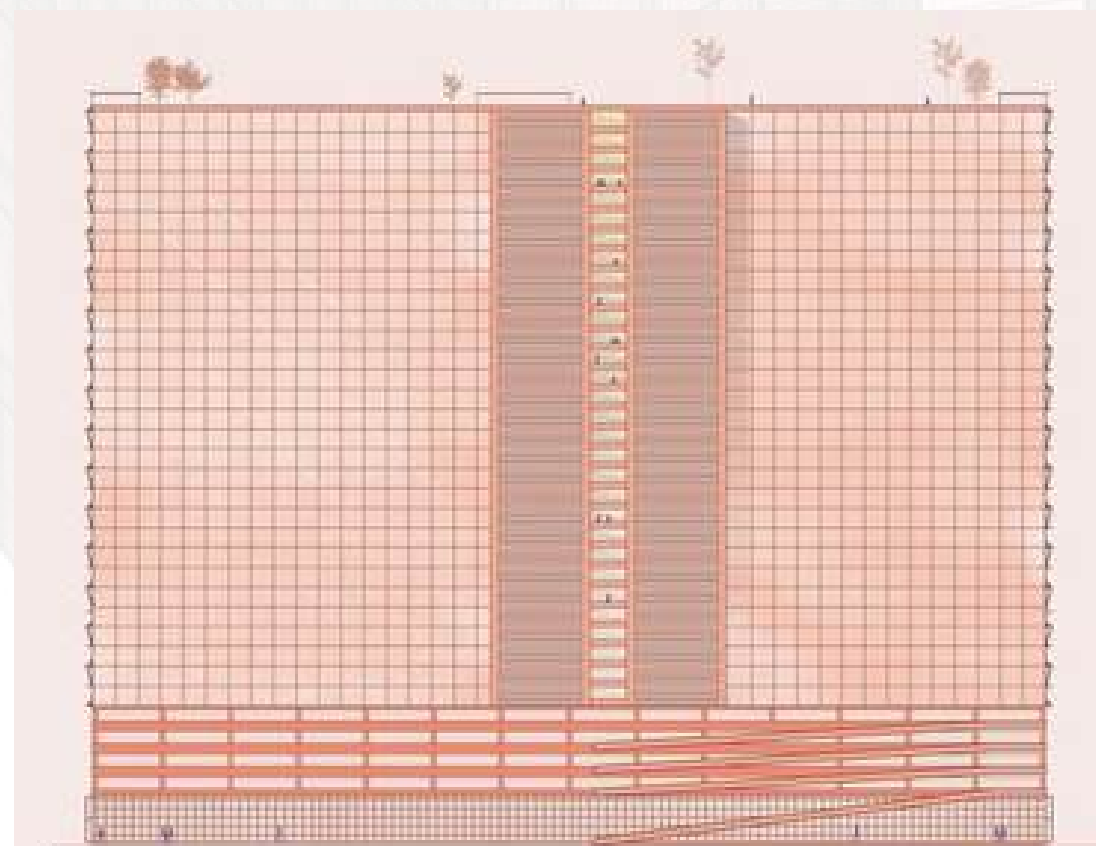
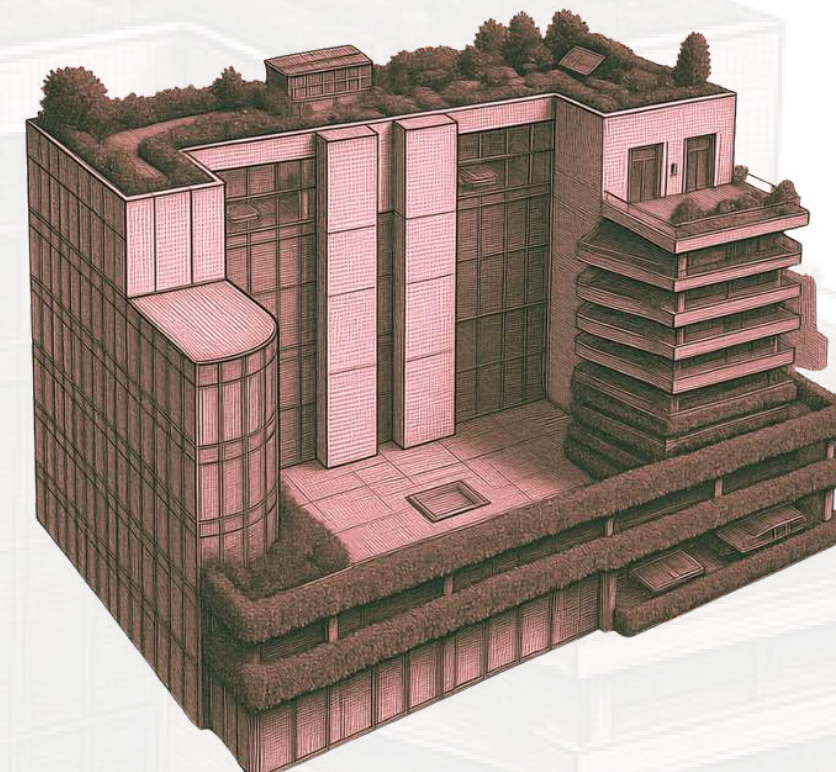
LEGEND SITE INDUSTRIAL BUILDINGS
RESIDENTIAL BUILDINGS COMMERCIAL BUILDINGS
HOTELS GHANSOLI RAILWAY STATION
TPOLOGY



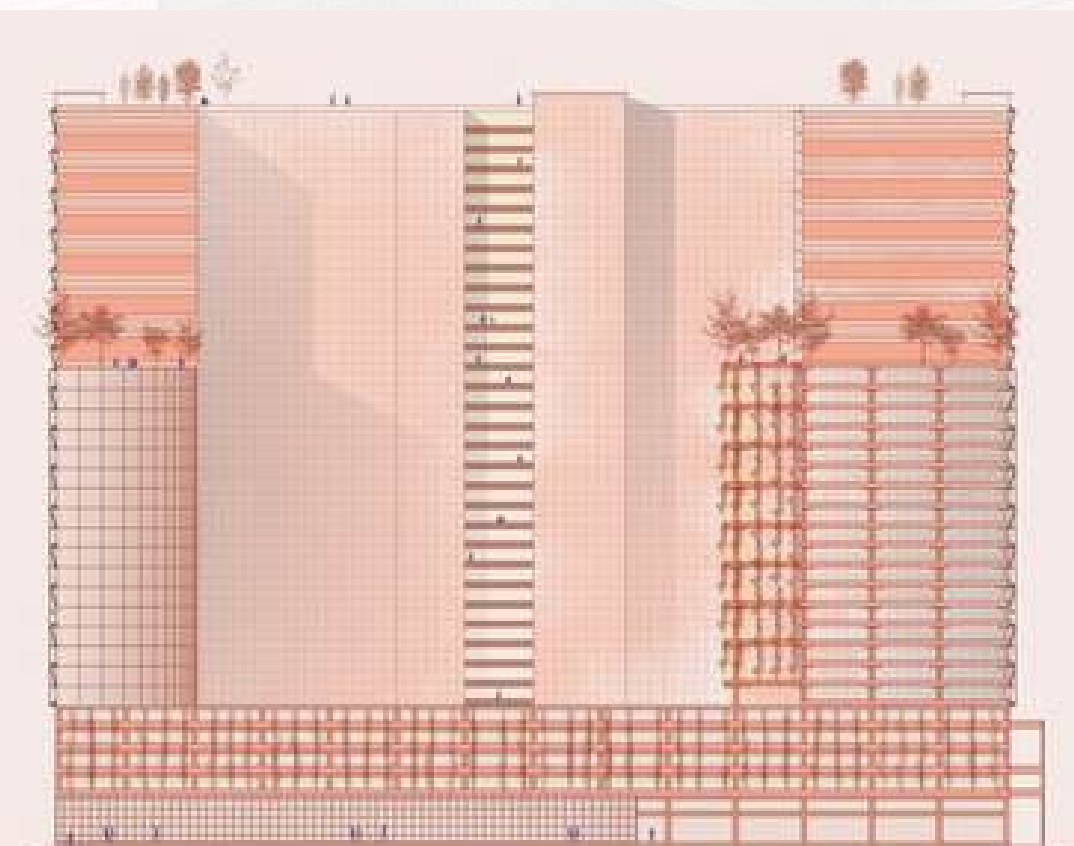


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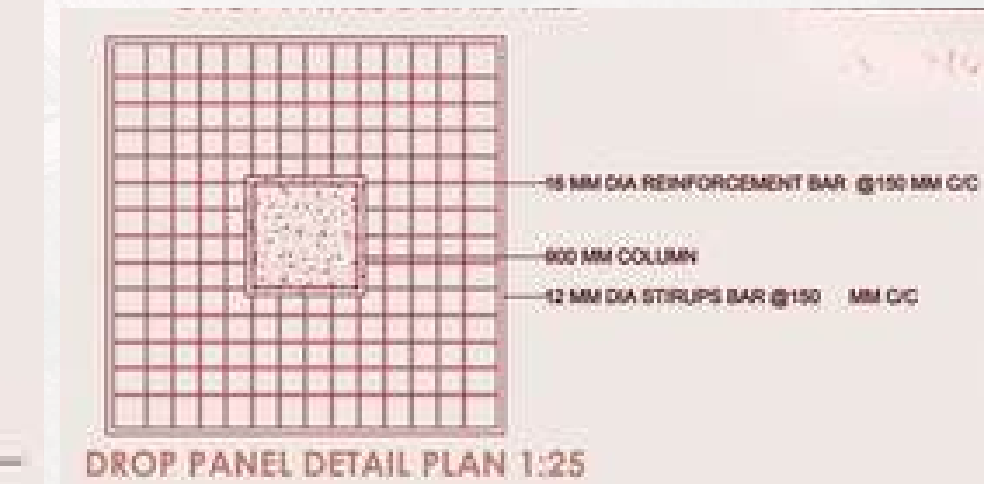
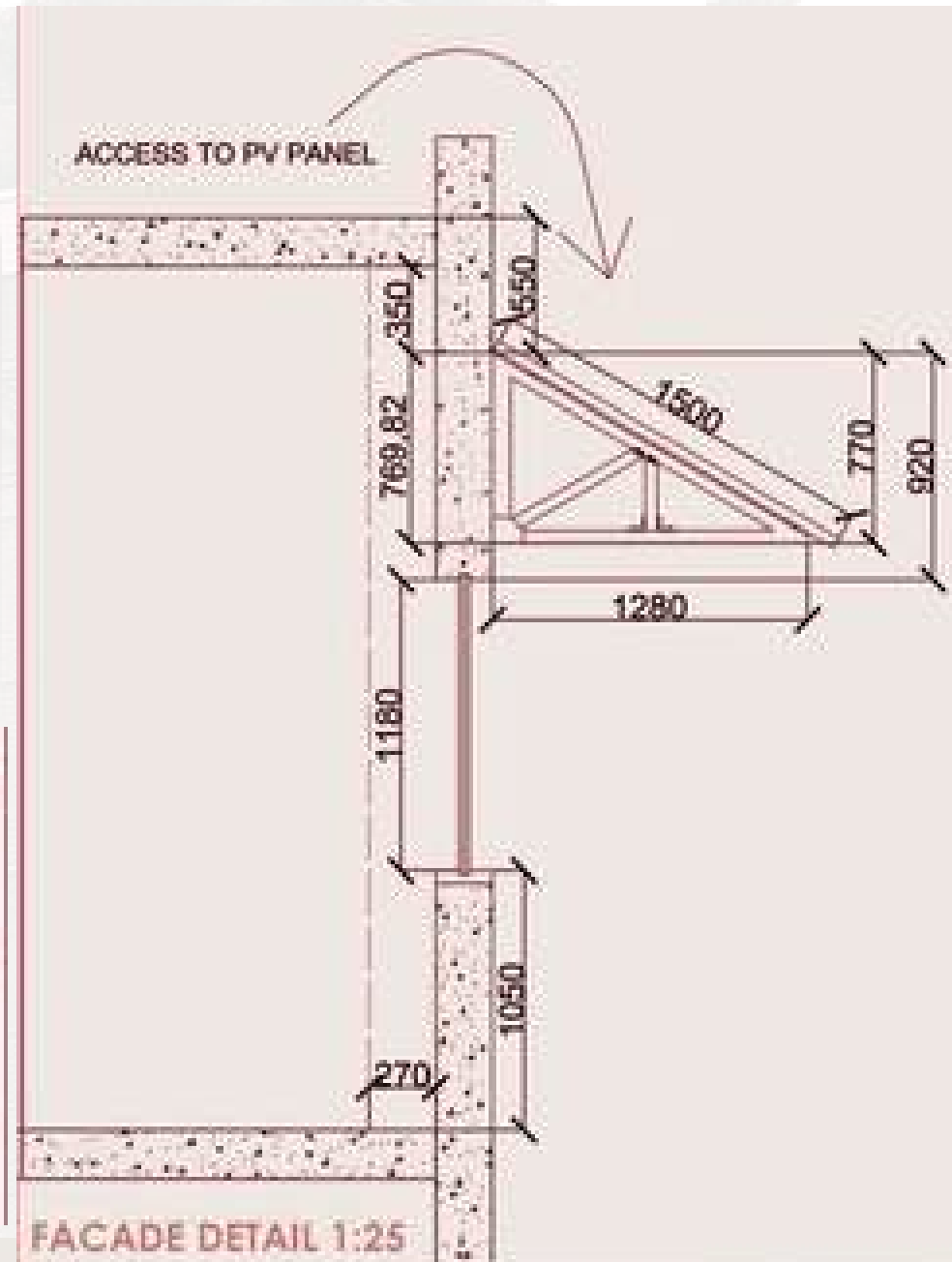
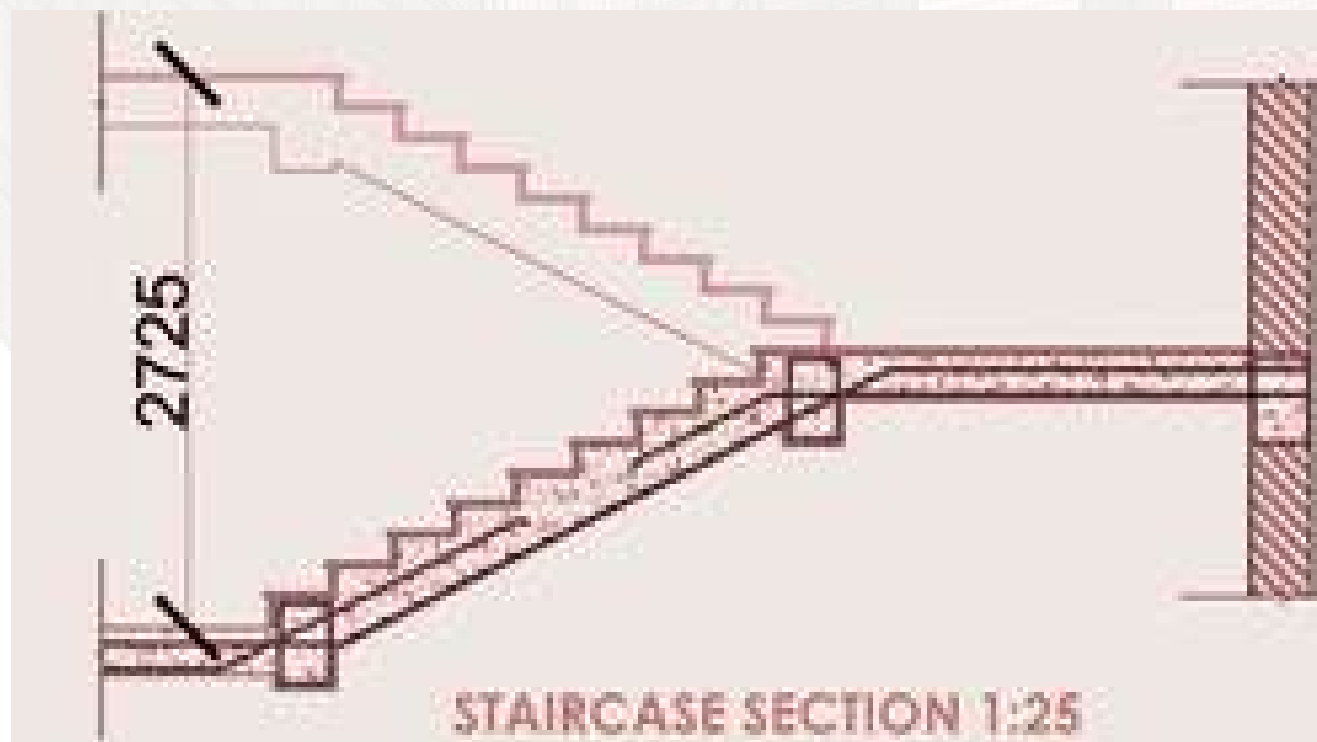
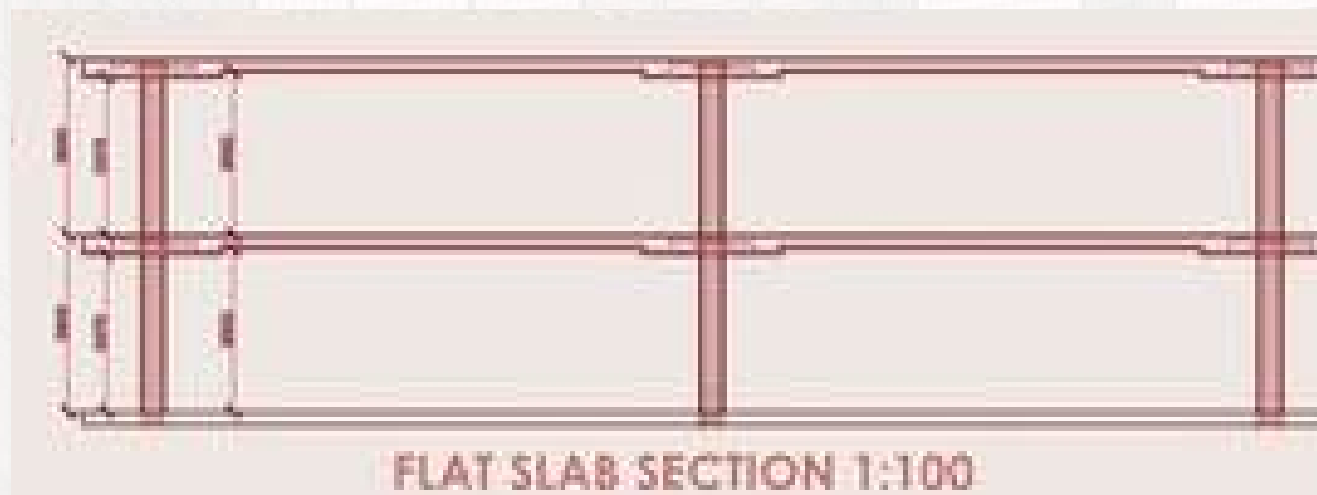
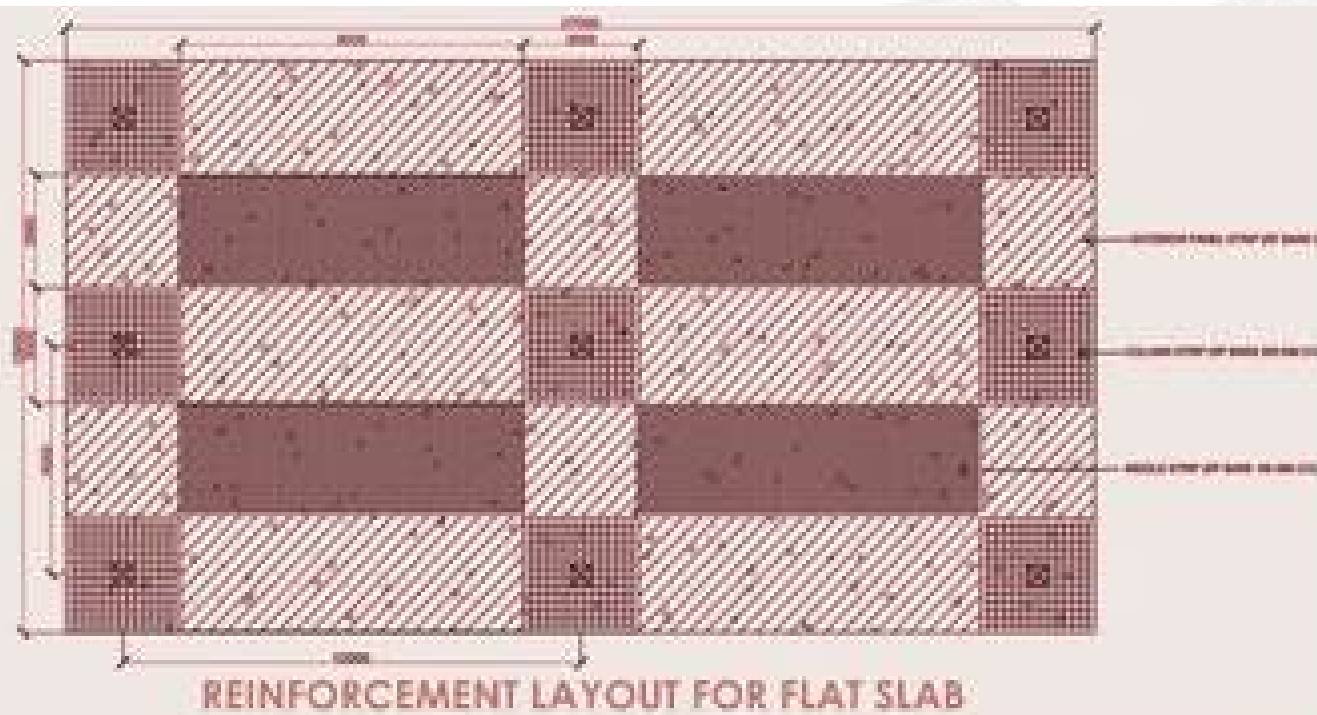
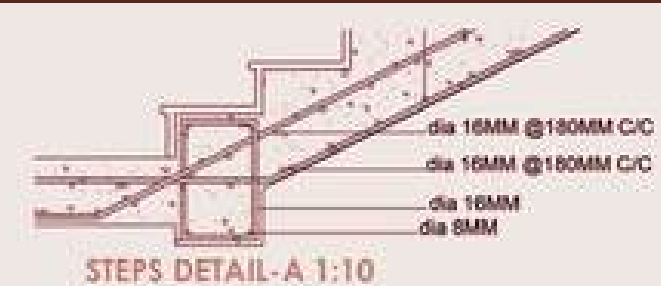
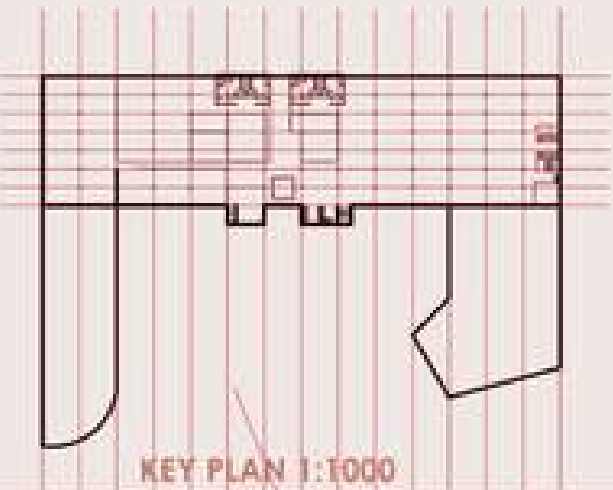


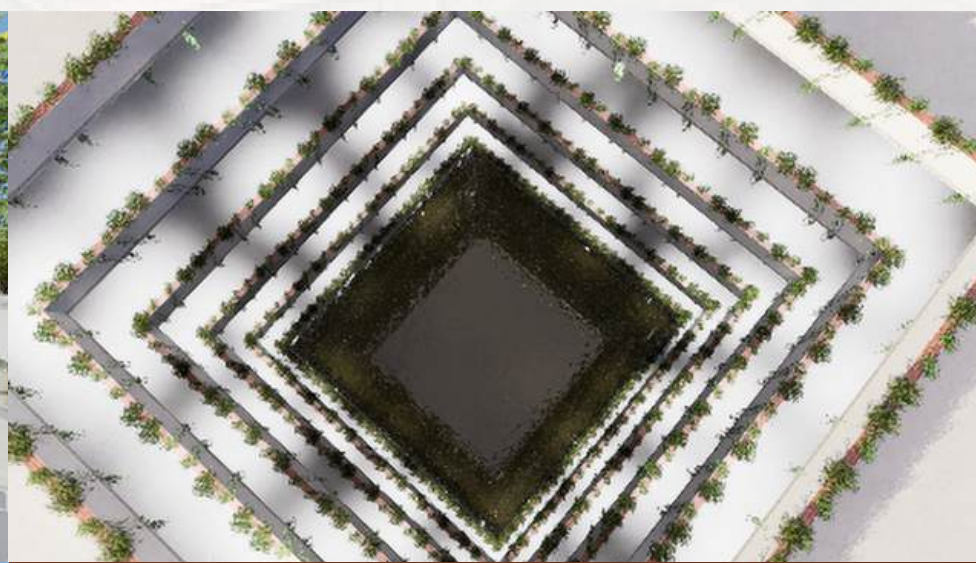
SOUTH-EAST ELEVATION



NORTH-WEST ELEVATION

CONSTRUCTION DETAILS





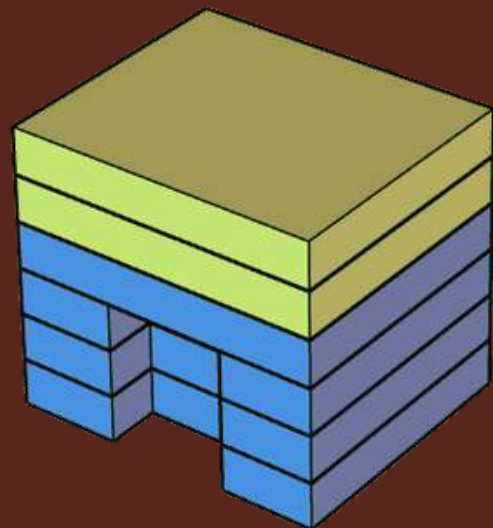
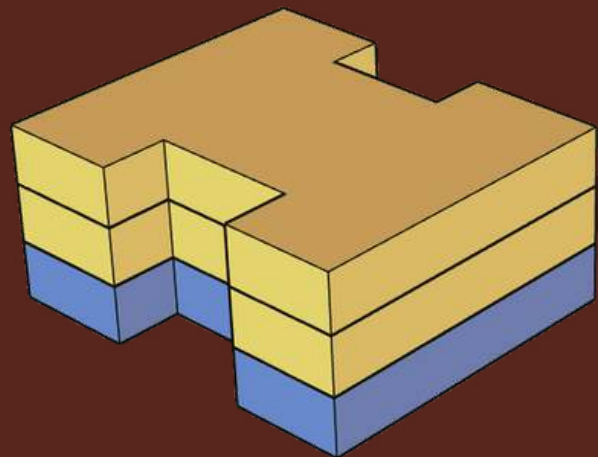
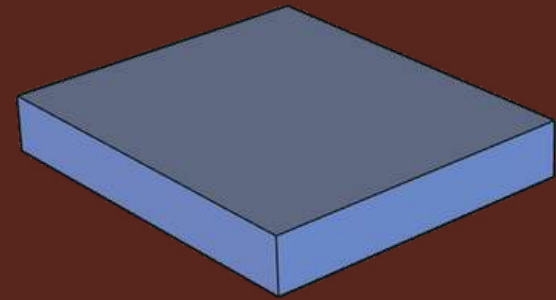
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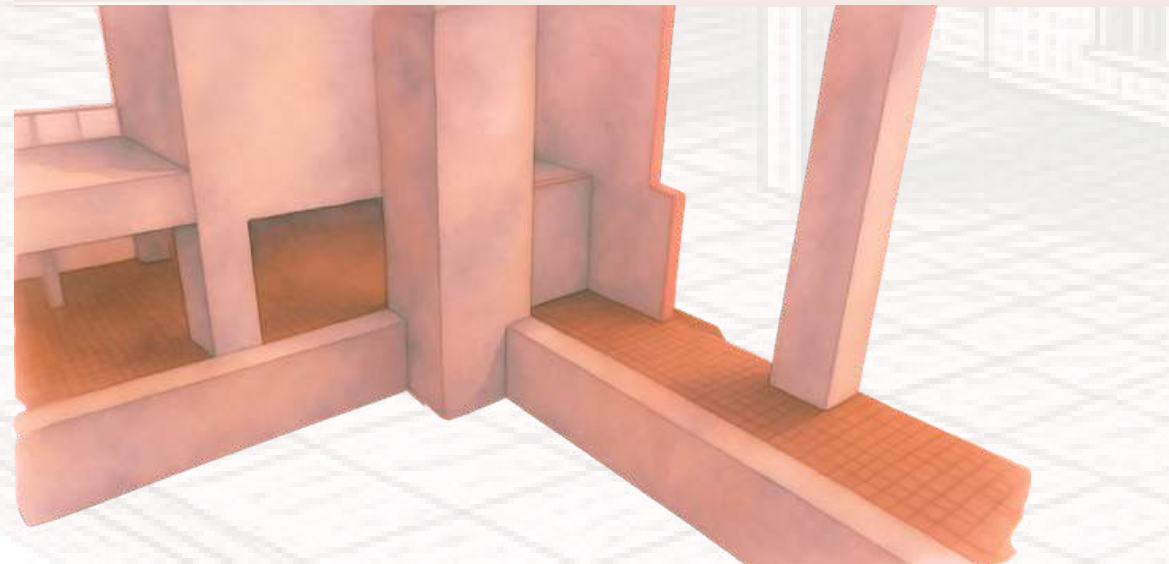
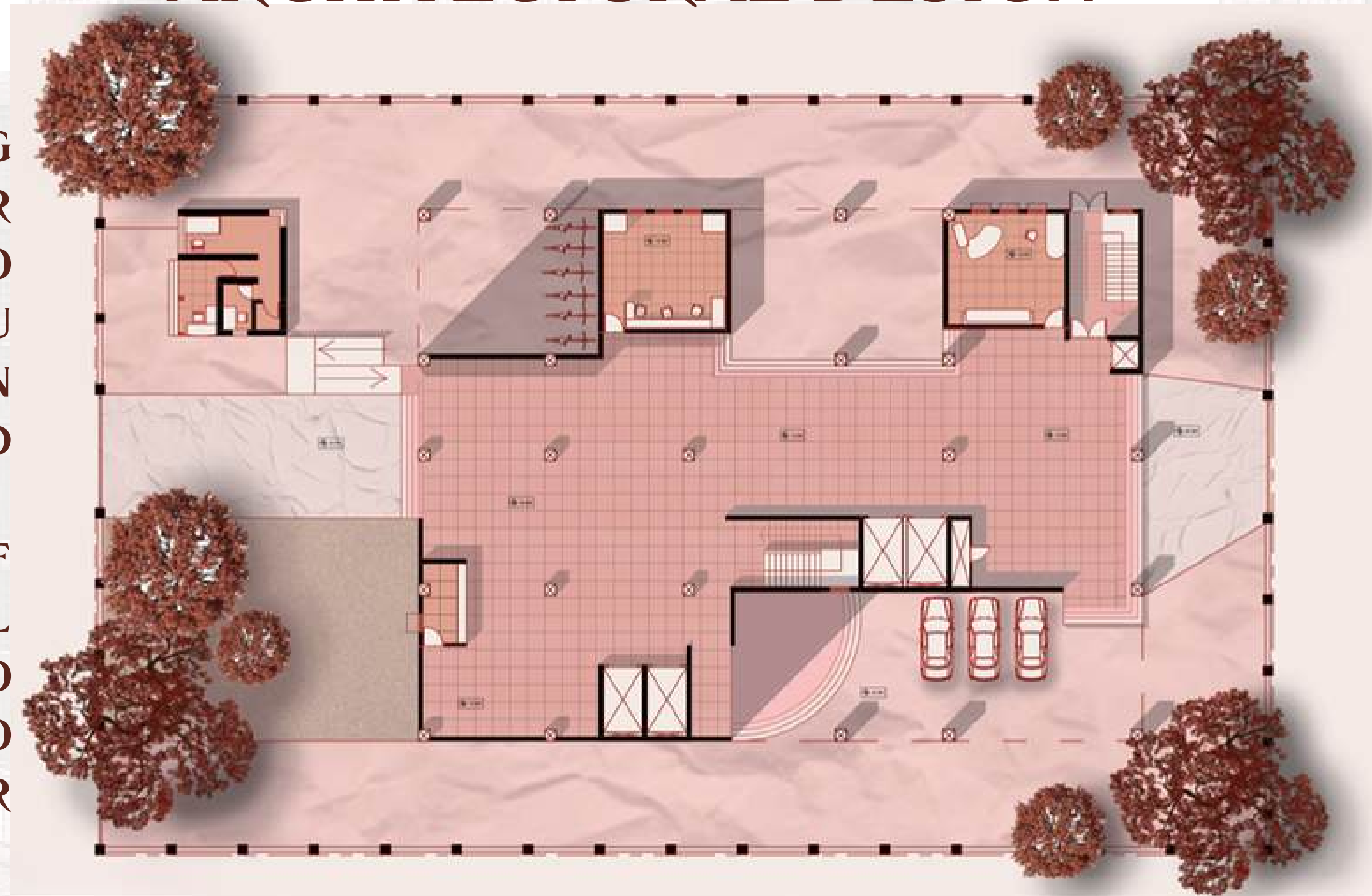


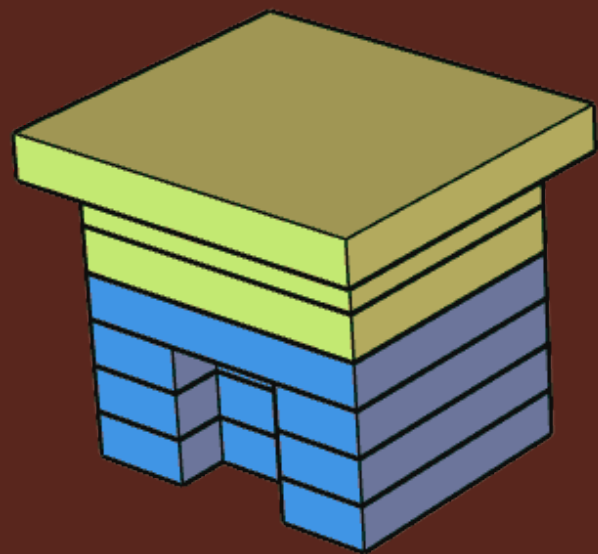
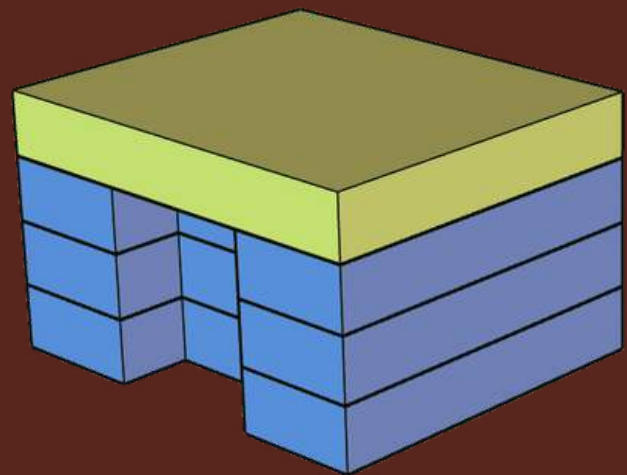
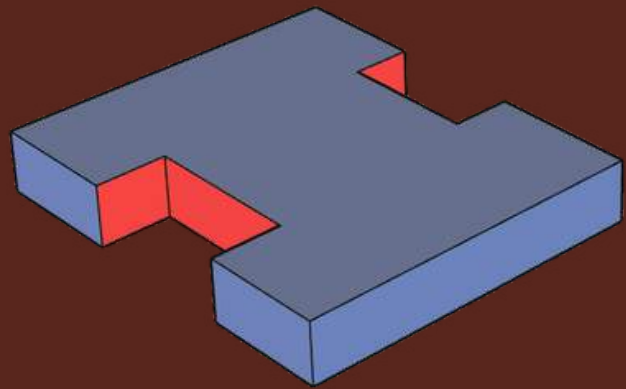
FORM DEVELOPMENT



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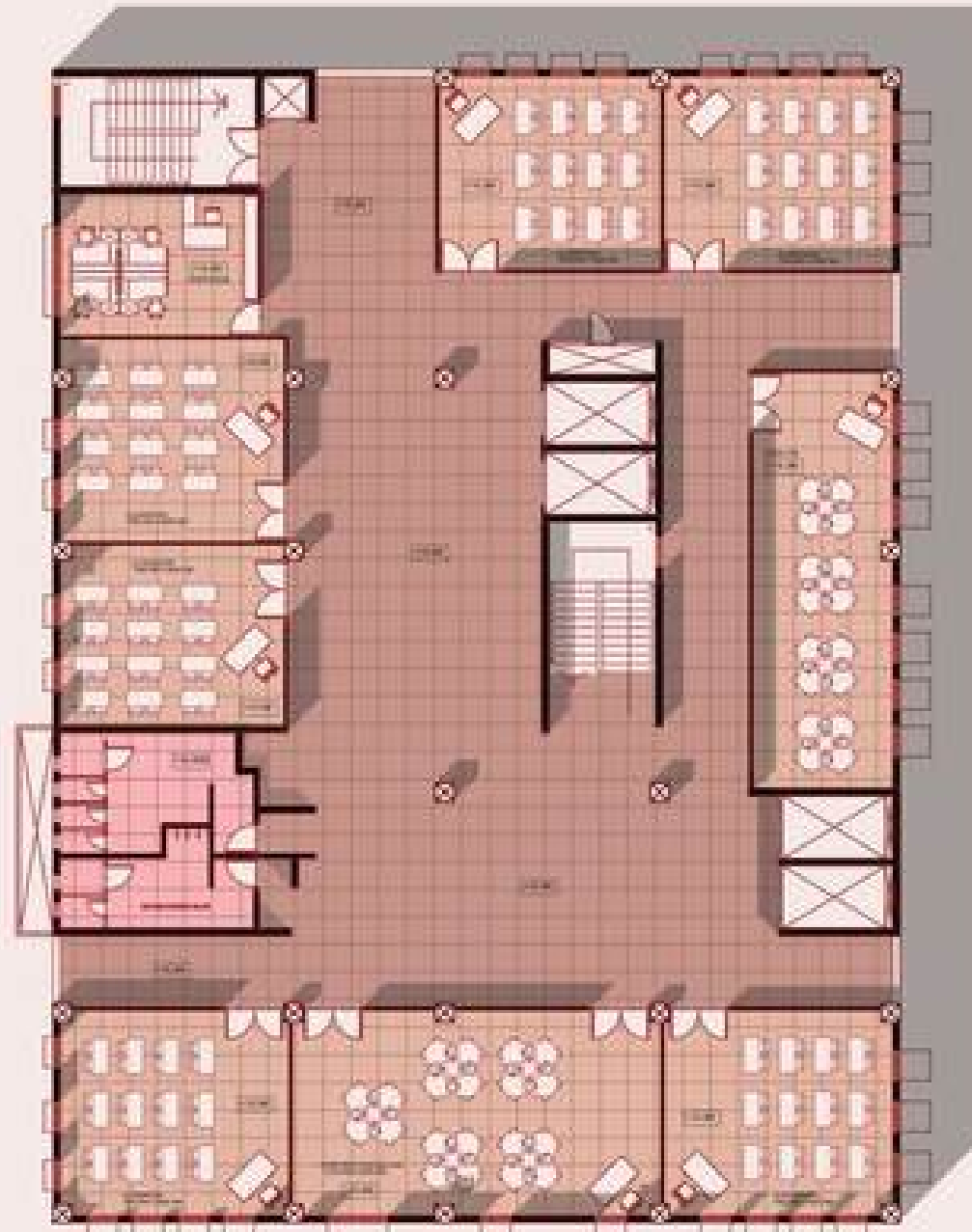
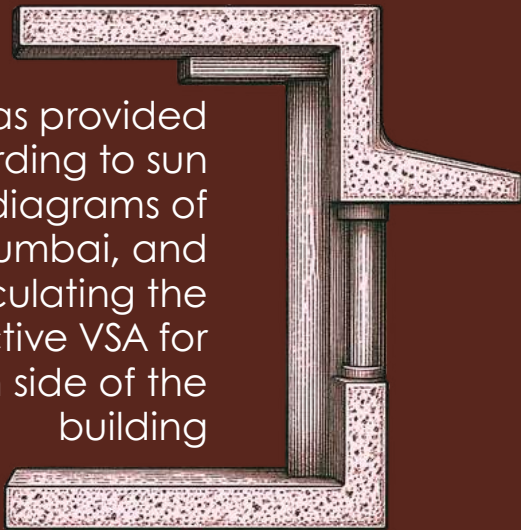
FIRST FLOOR PLAN



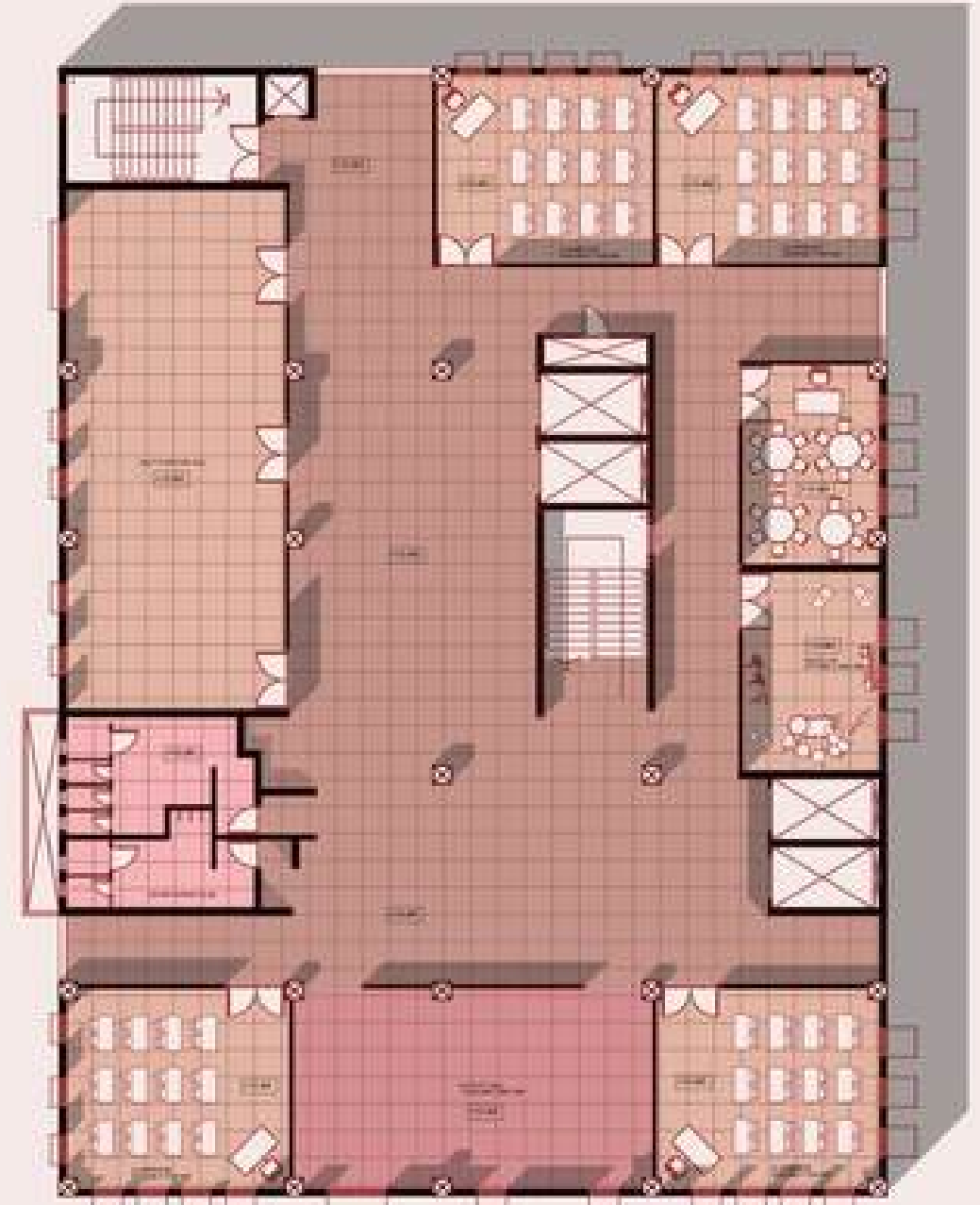
SECOND FLOOR PLAN



Chajjas provided according to sun path diagrams of navi mumbai, and calculating the repective VSA for each side of the building

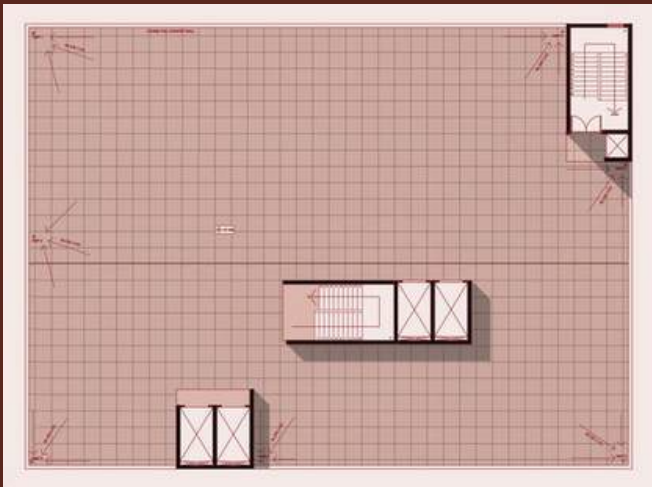
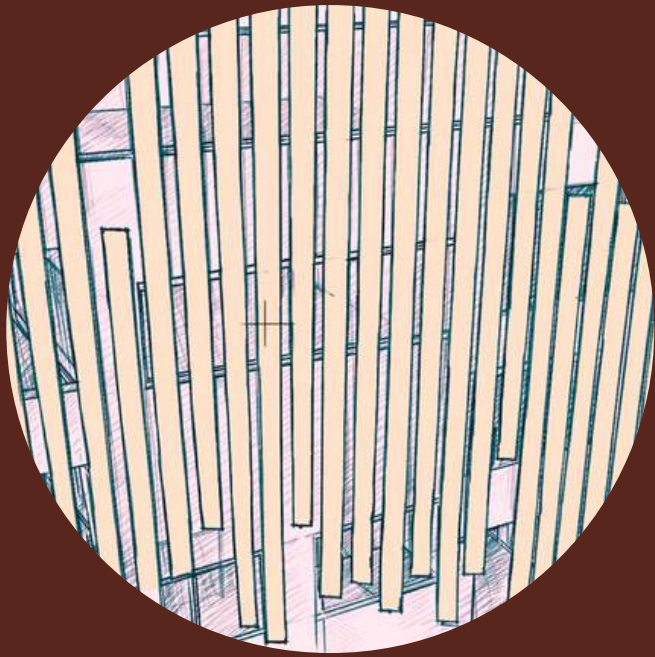


THIRD FLOOR PLAN

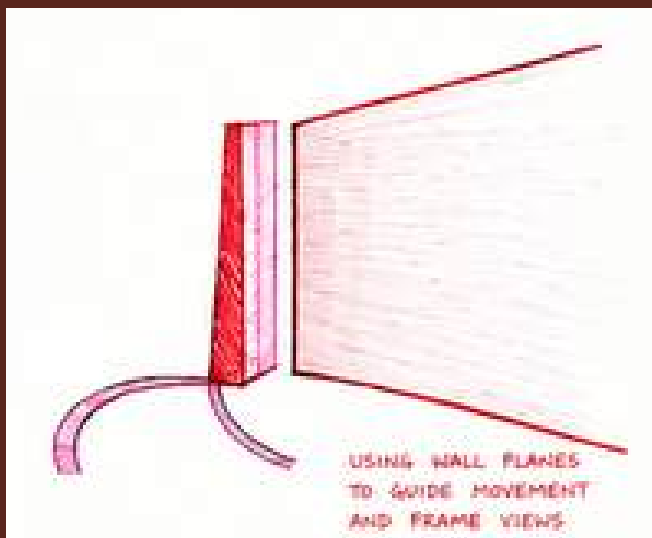


FOURTH FLOOR PLAN

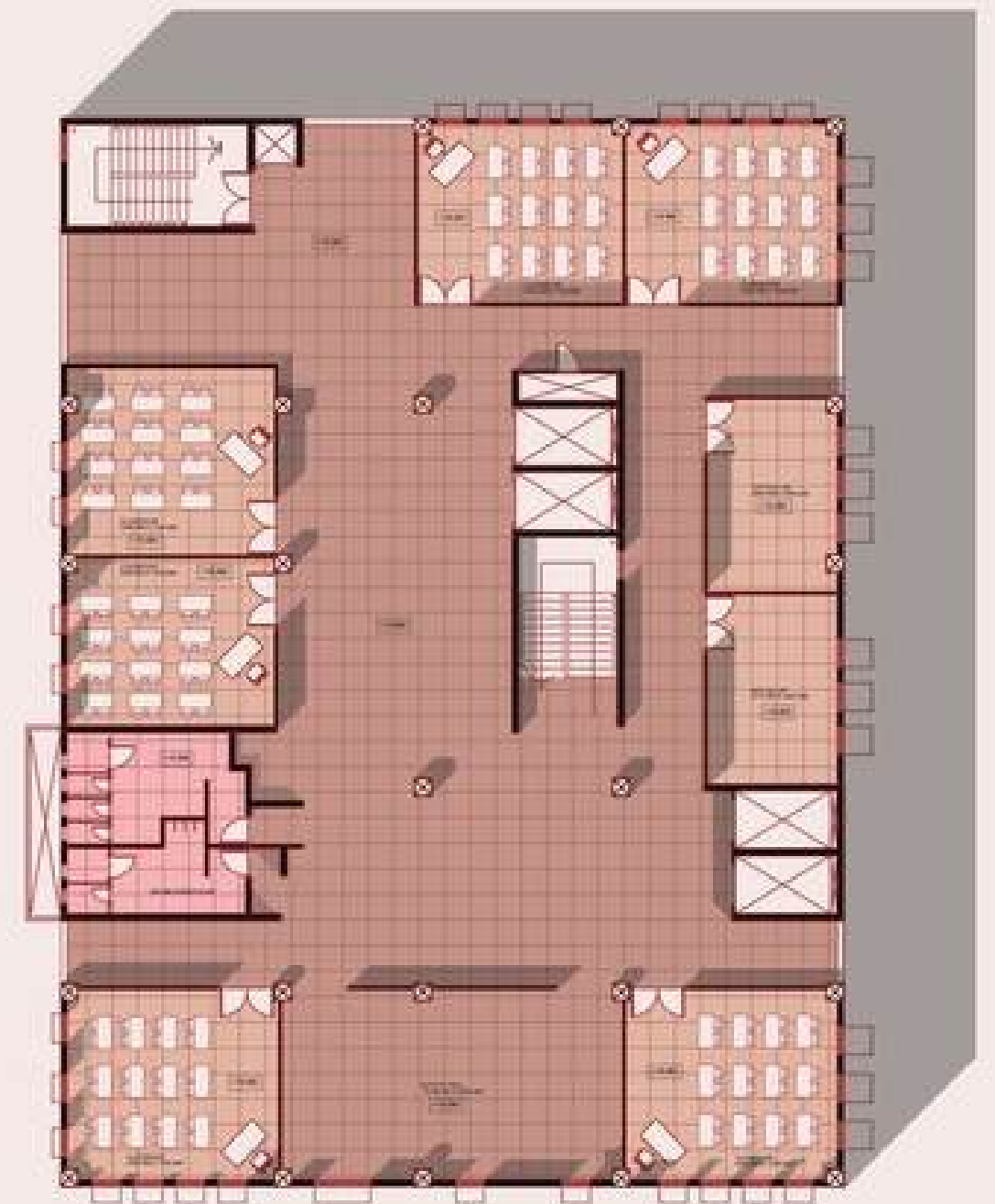




TERRACE FLOOR PLAN



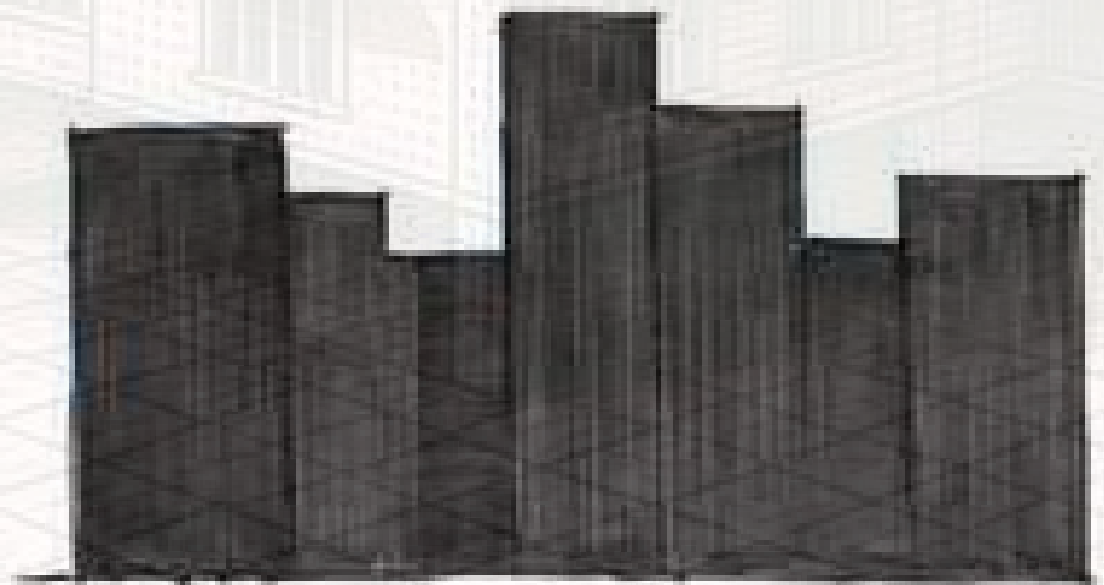
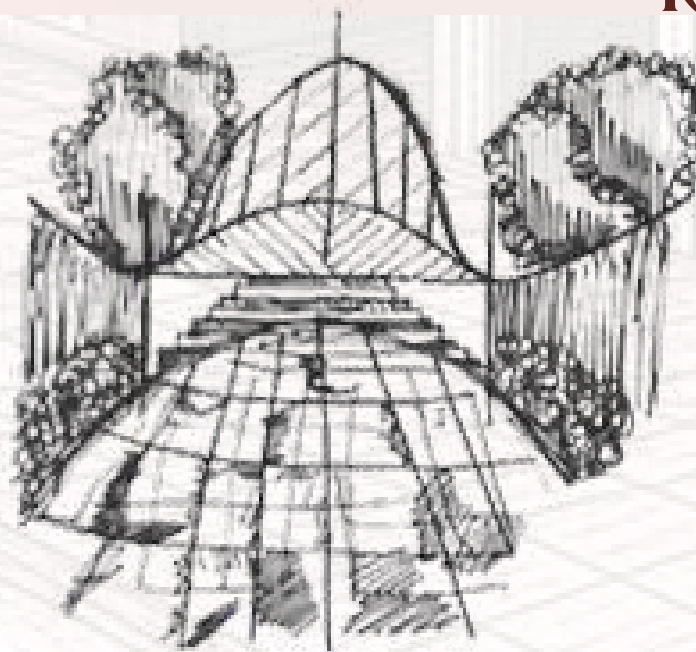
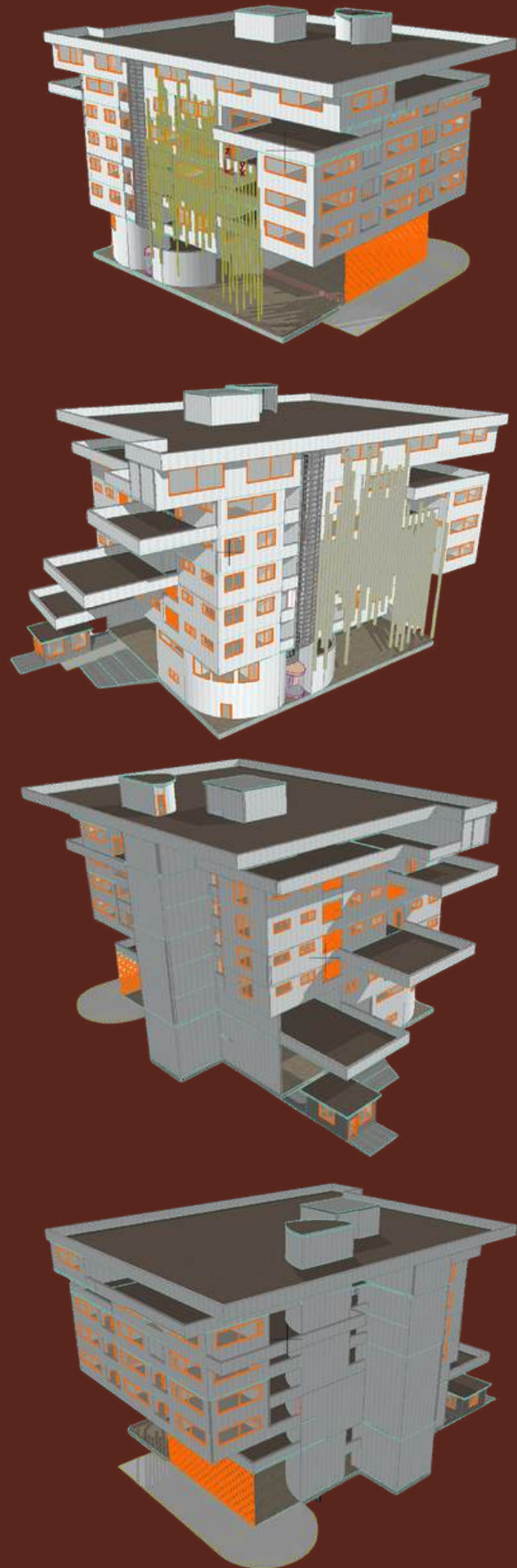
FIFTH FLOOR PLAN

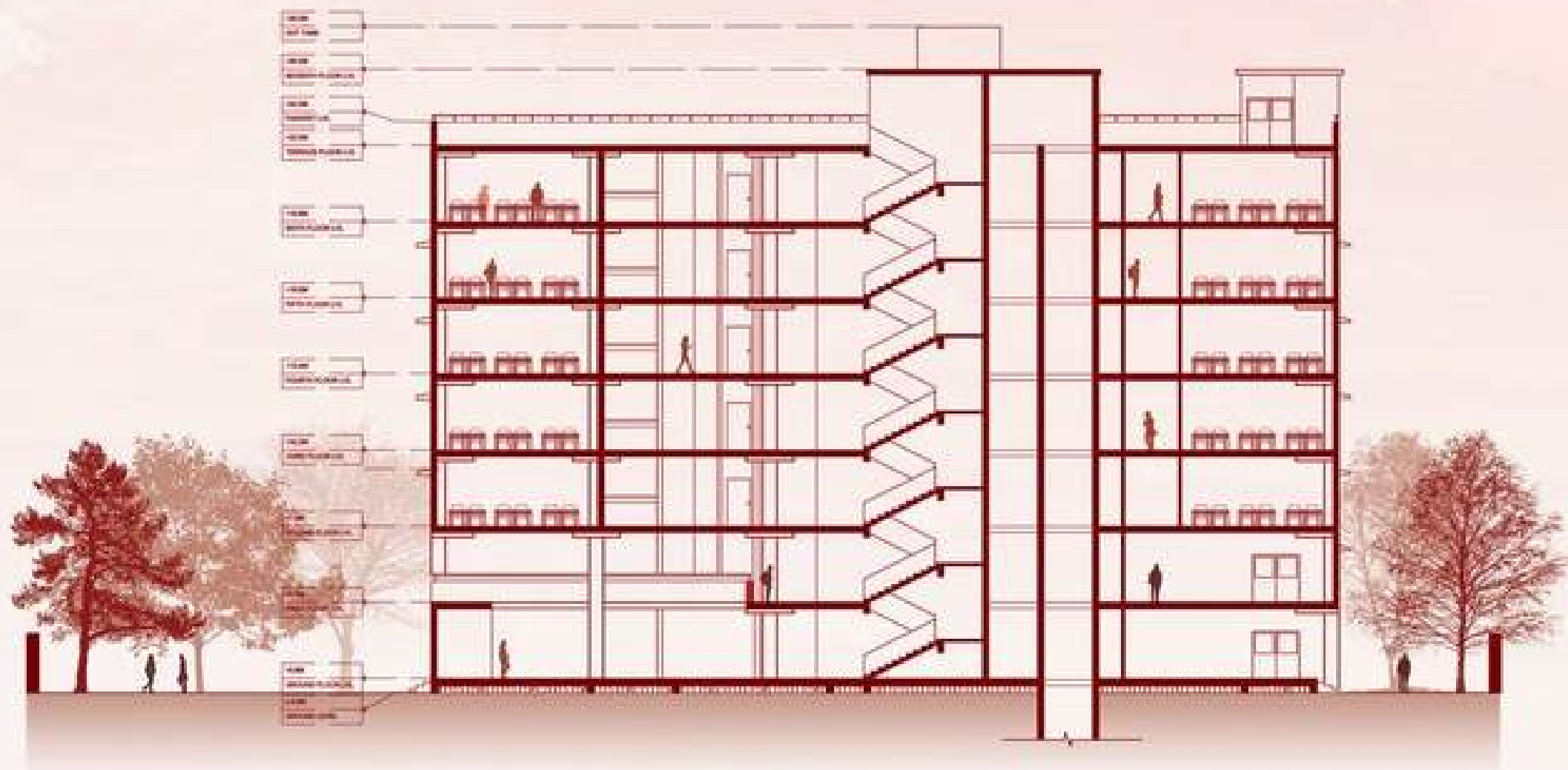
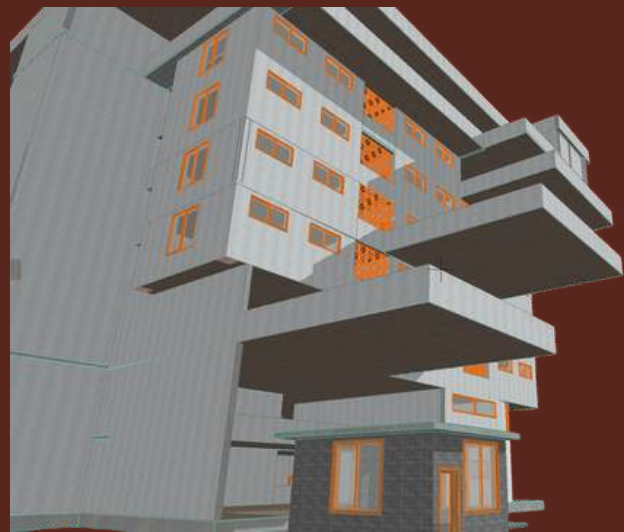
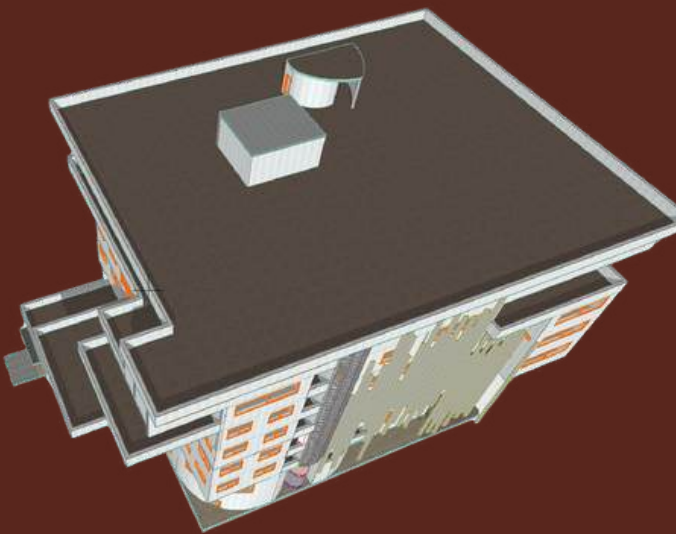
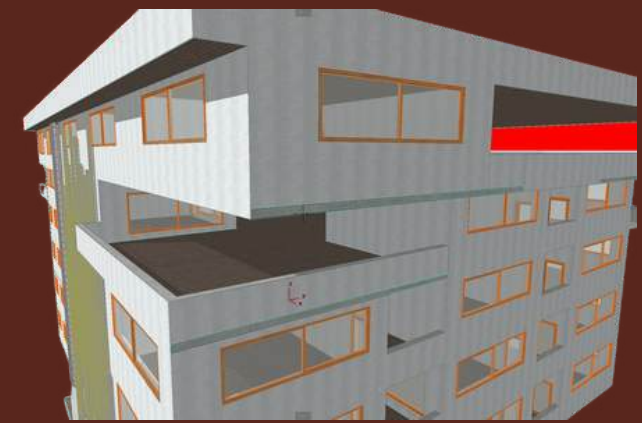


SIXTH FLOOR PLAN



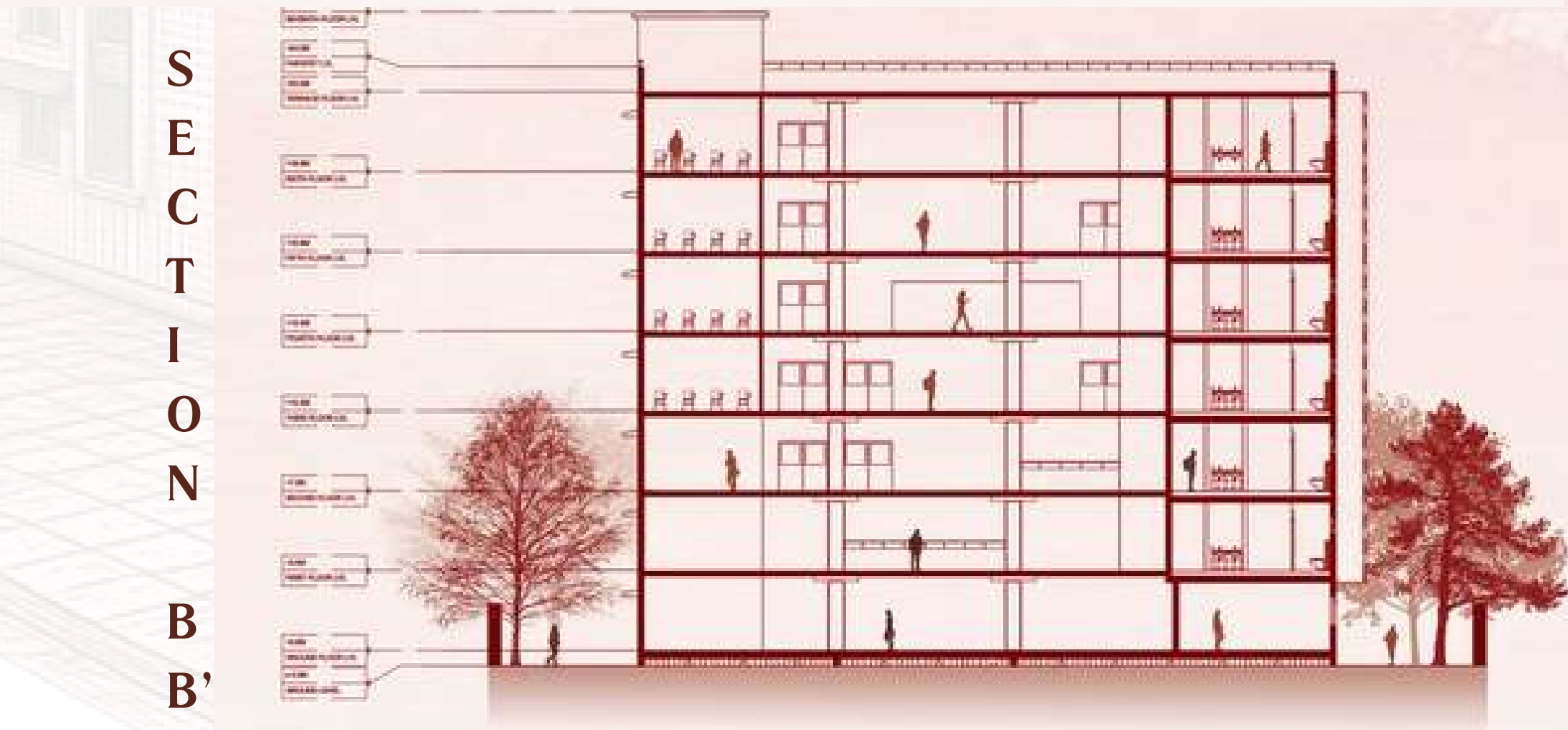
CONCEPTUAL DESIGN VIEWS





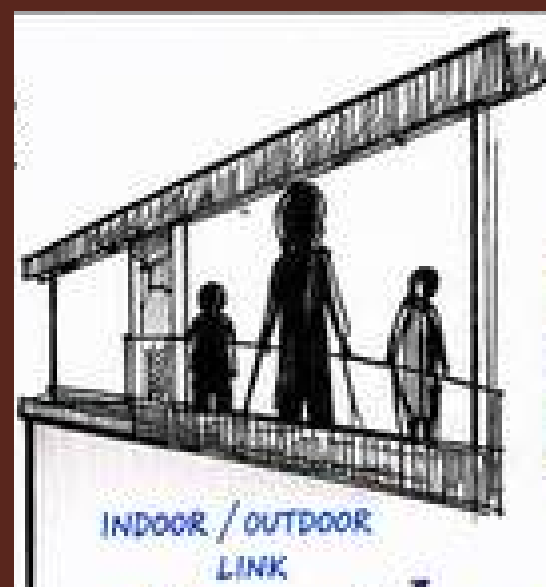
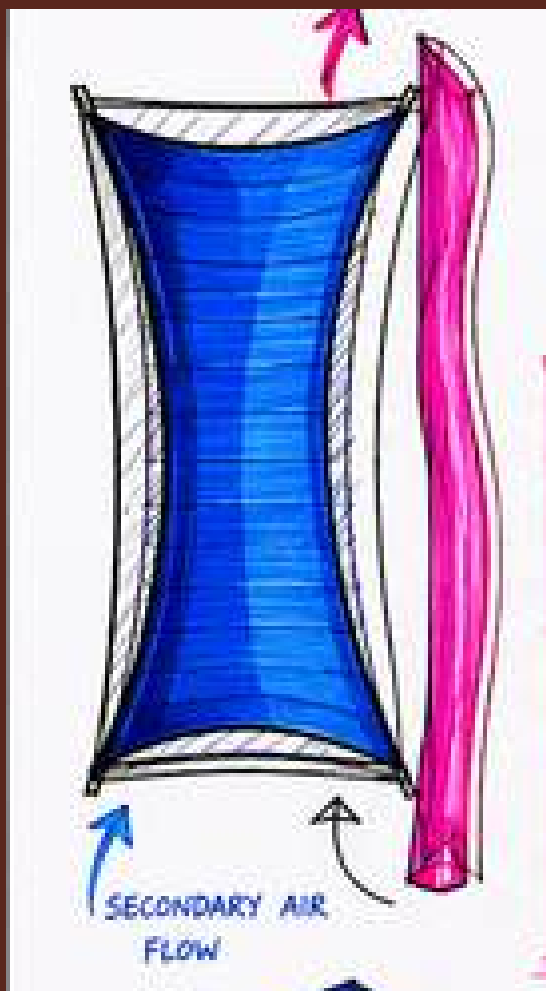
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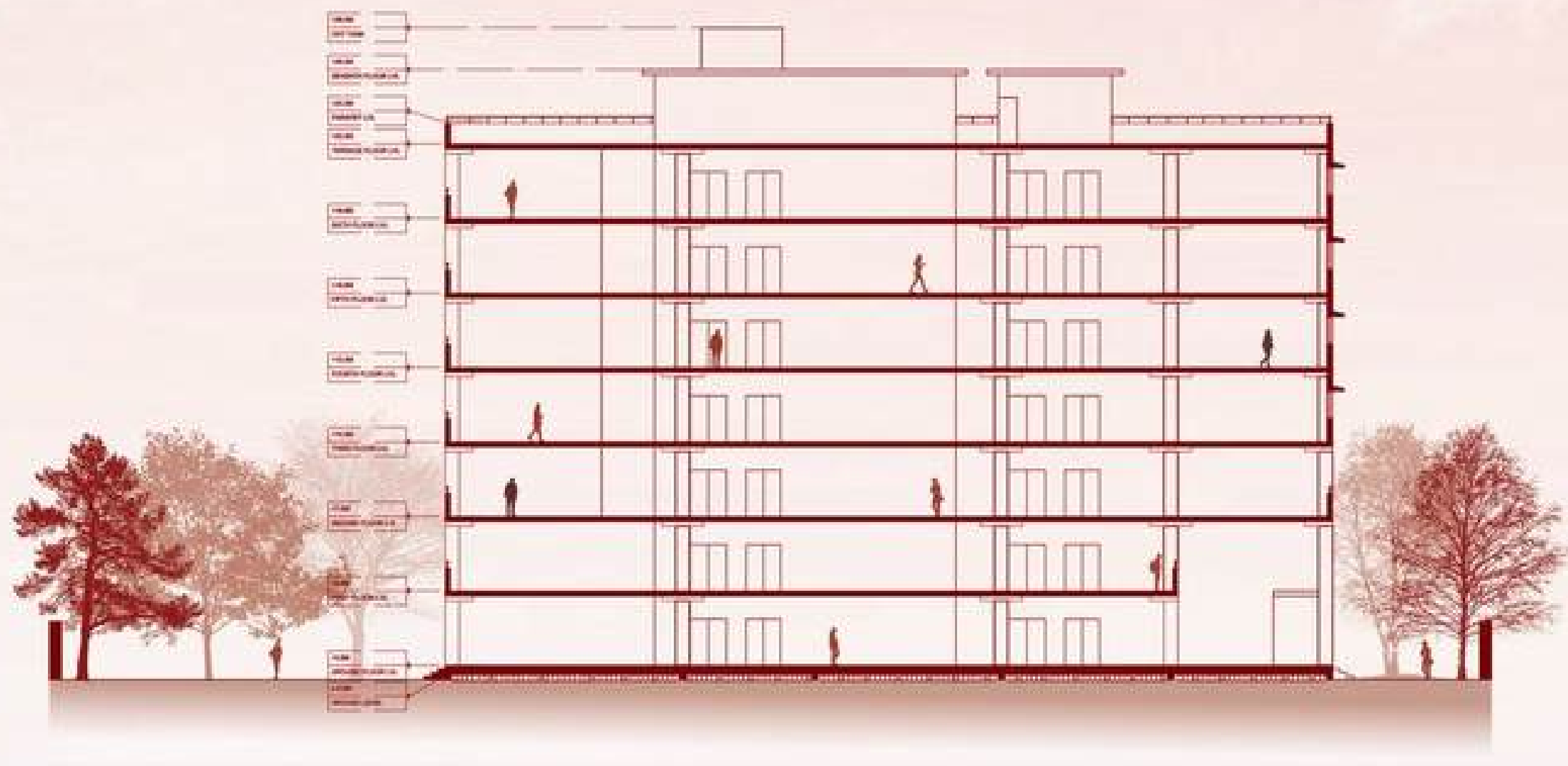


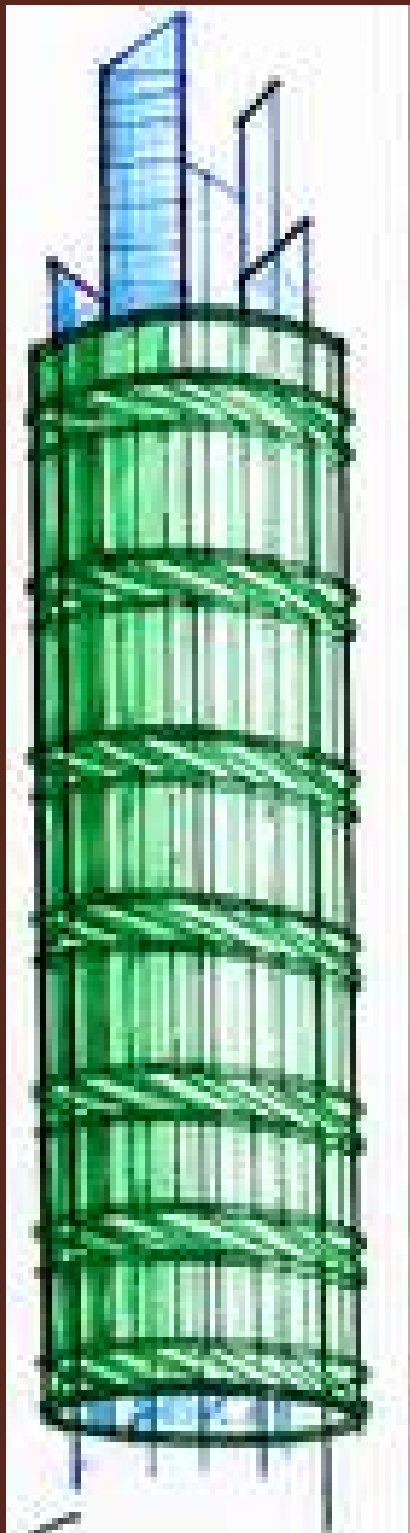
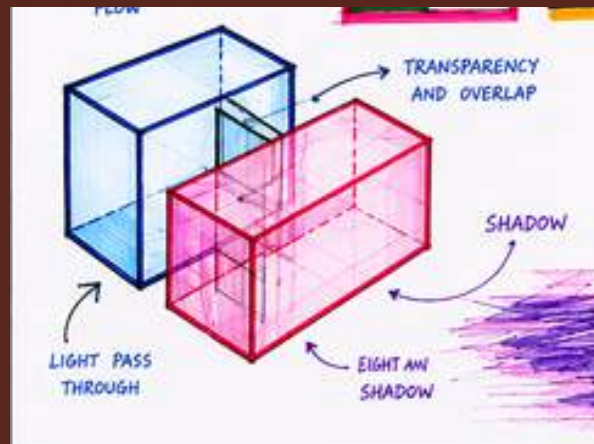
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SECTION D-D'

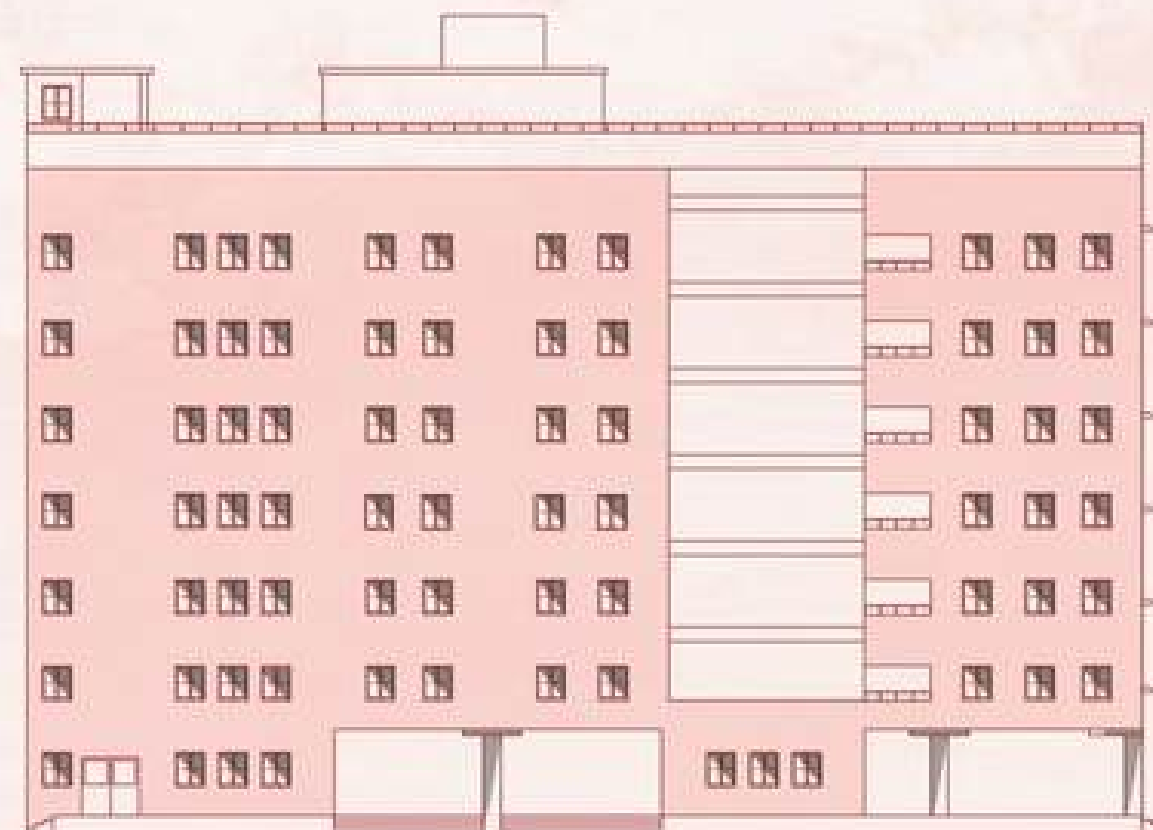




NORTHSIDE ELEVATION



WESTSIDE ELEVATION



EASTSIDE ELEVATION



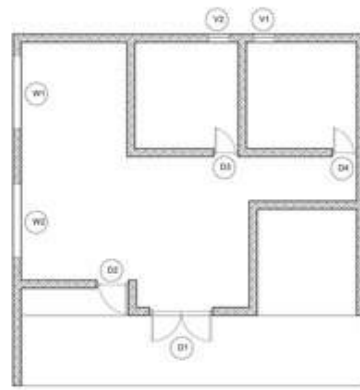
SOUTHSIDE ELEVATION



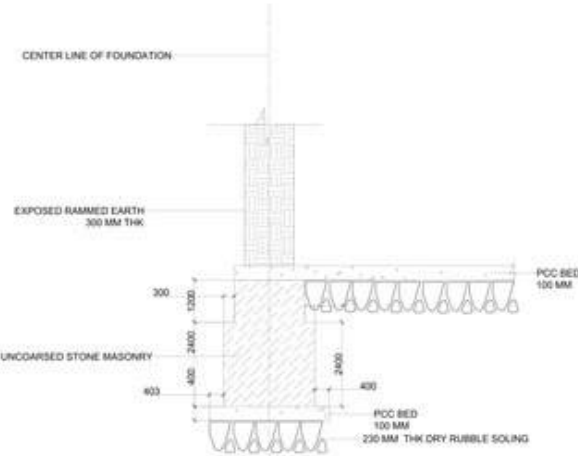
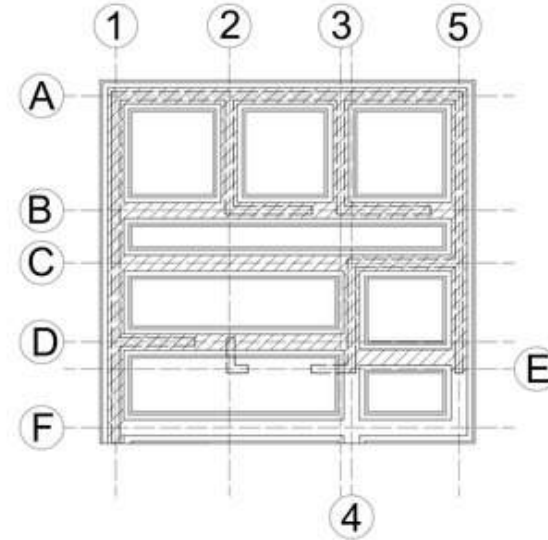
VIEWS



DRAWINGS

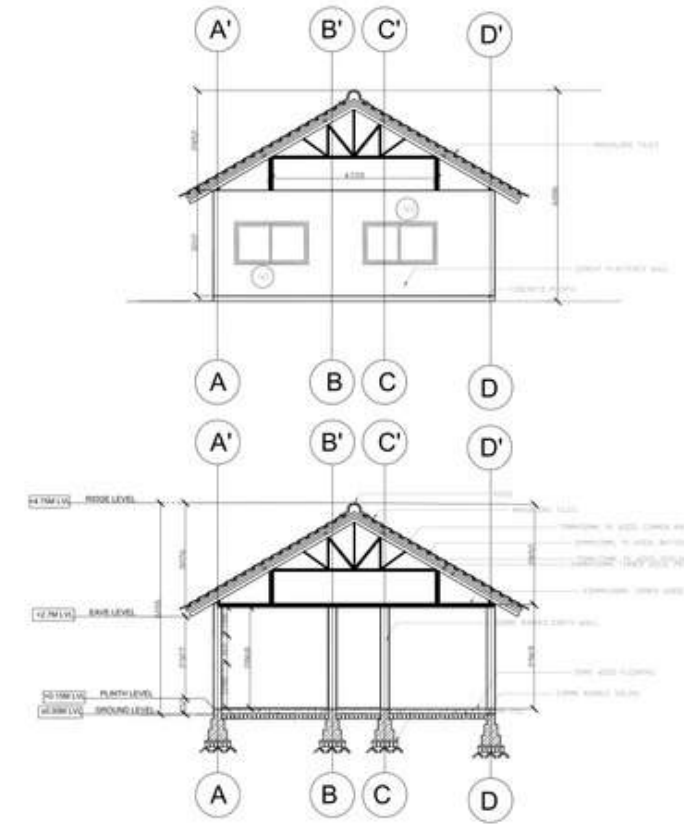
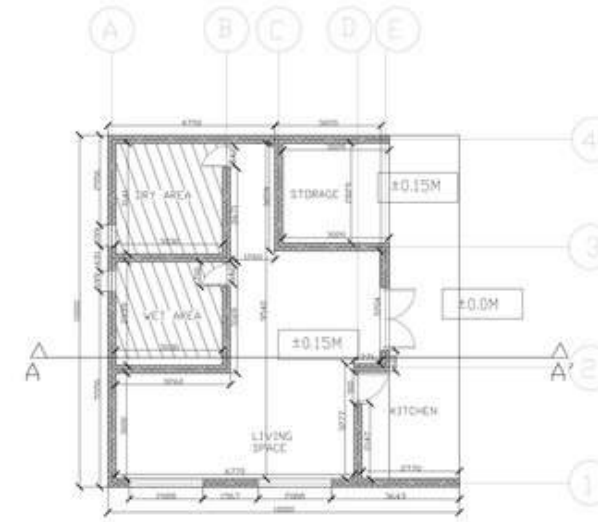


NAME	DIMENSIONS	DESCRIPTION
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D2	1200MMX2000MM	SINGLE LEAF DOUBLE PANELLED TEAK WOOD DOOR
D3	900MMX1800MM	SINGLE LEAF DOUBLE PANELLED TEAK WOOD DOOR
D4	900MMX1800MM	SINGLE LEAF DOUBLE PANELLED TEAK WOOD DOOR
W1	2000MMX1800MM	DOUBLE SHUTTER SINGLE PANELLED TEAK WOOD WINDOW
W2	2000MMX1800MM	DOUBLE SHUTTER FOUR PANELLED TEAK WOOD WINDOW
V1	800MMX500MM	LOUVERED WINDOW
V2	800MMX500MM	LOUVERED WINDOW



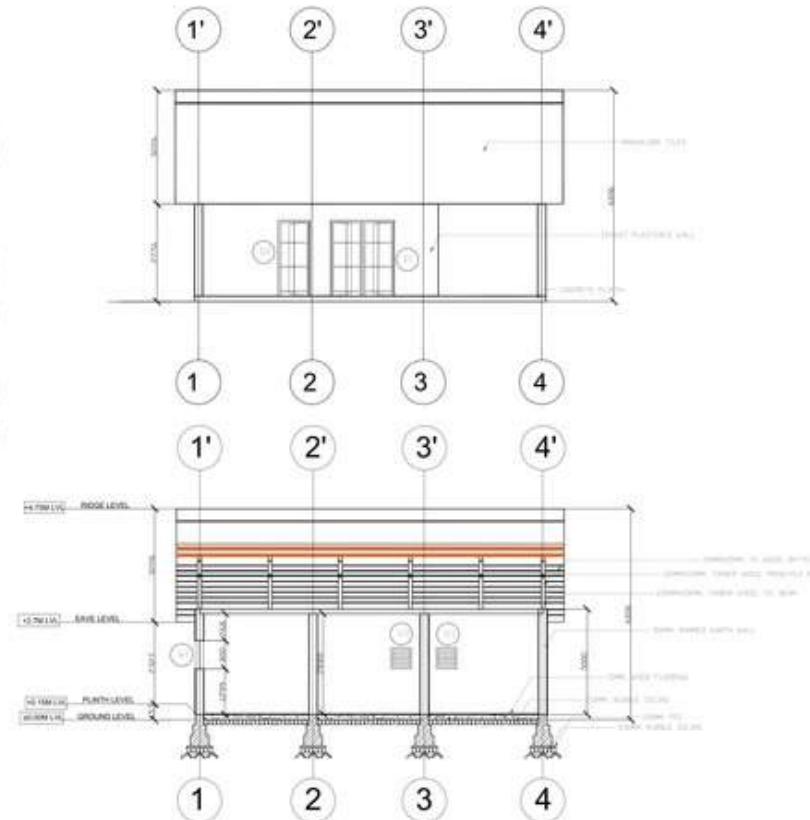
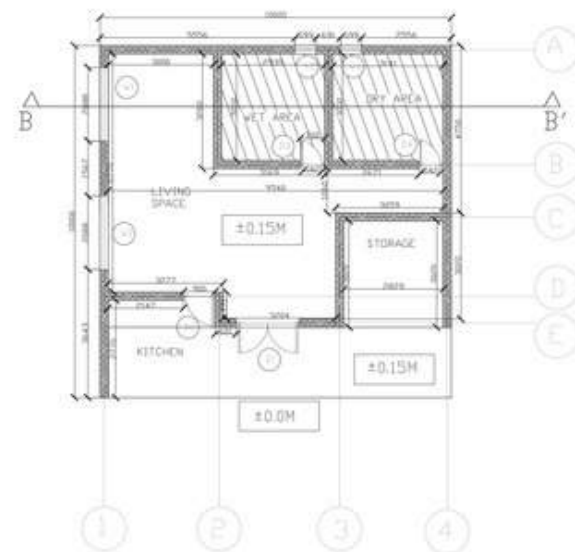
ARCHITECTURAL PLAN, FOUNDATION PLAN SCALE- 1:50

SIGN/STAMP	GRADE	V.E.S.C.O.A.
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		ROLL NO - AR23048 , BATCH-B
		SUBJECT- WD
		YEAR- 2ND , SEM-4



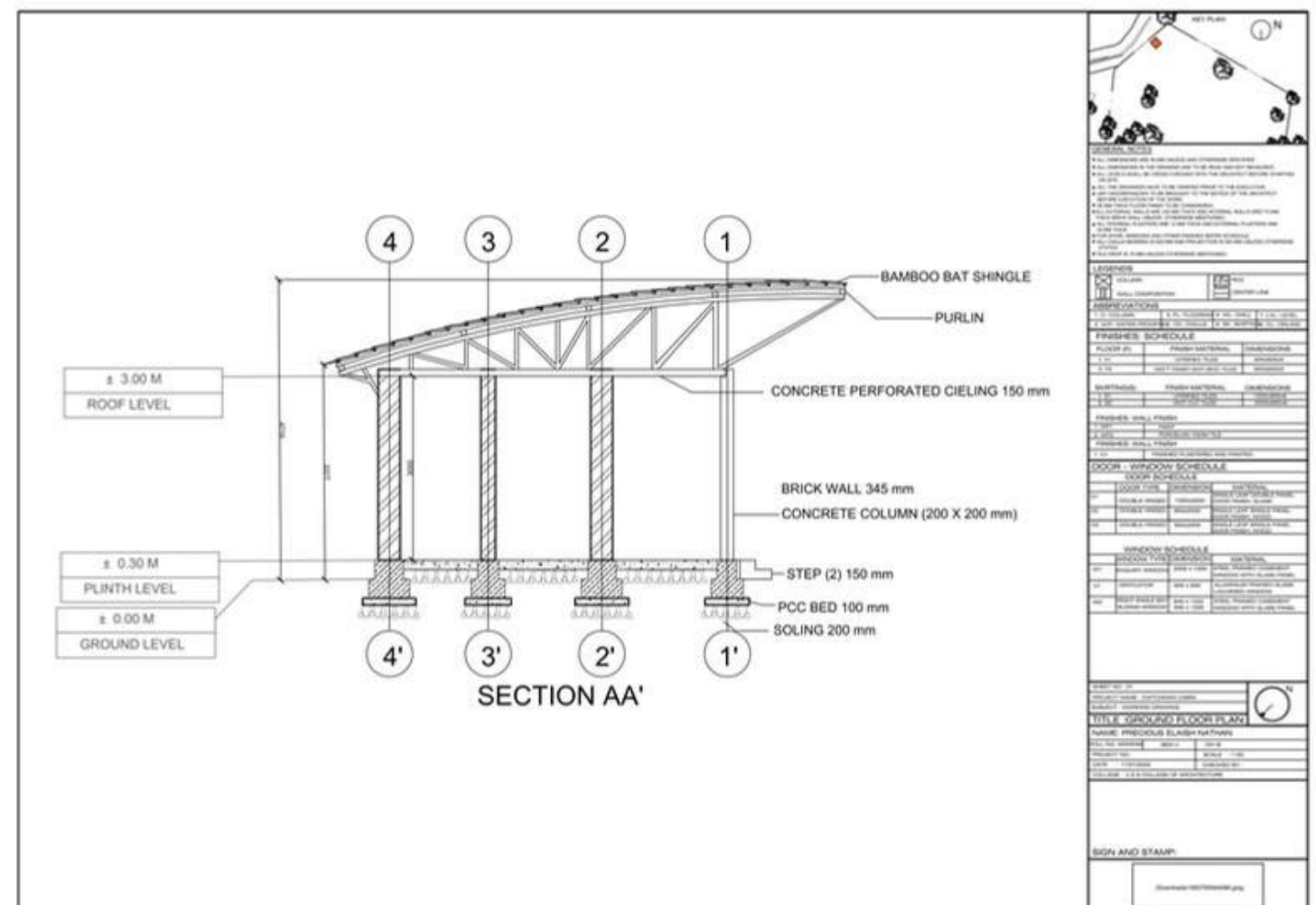
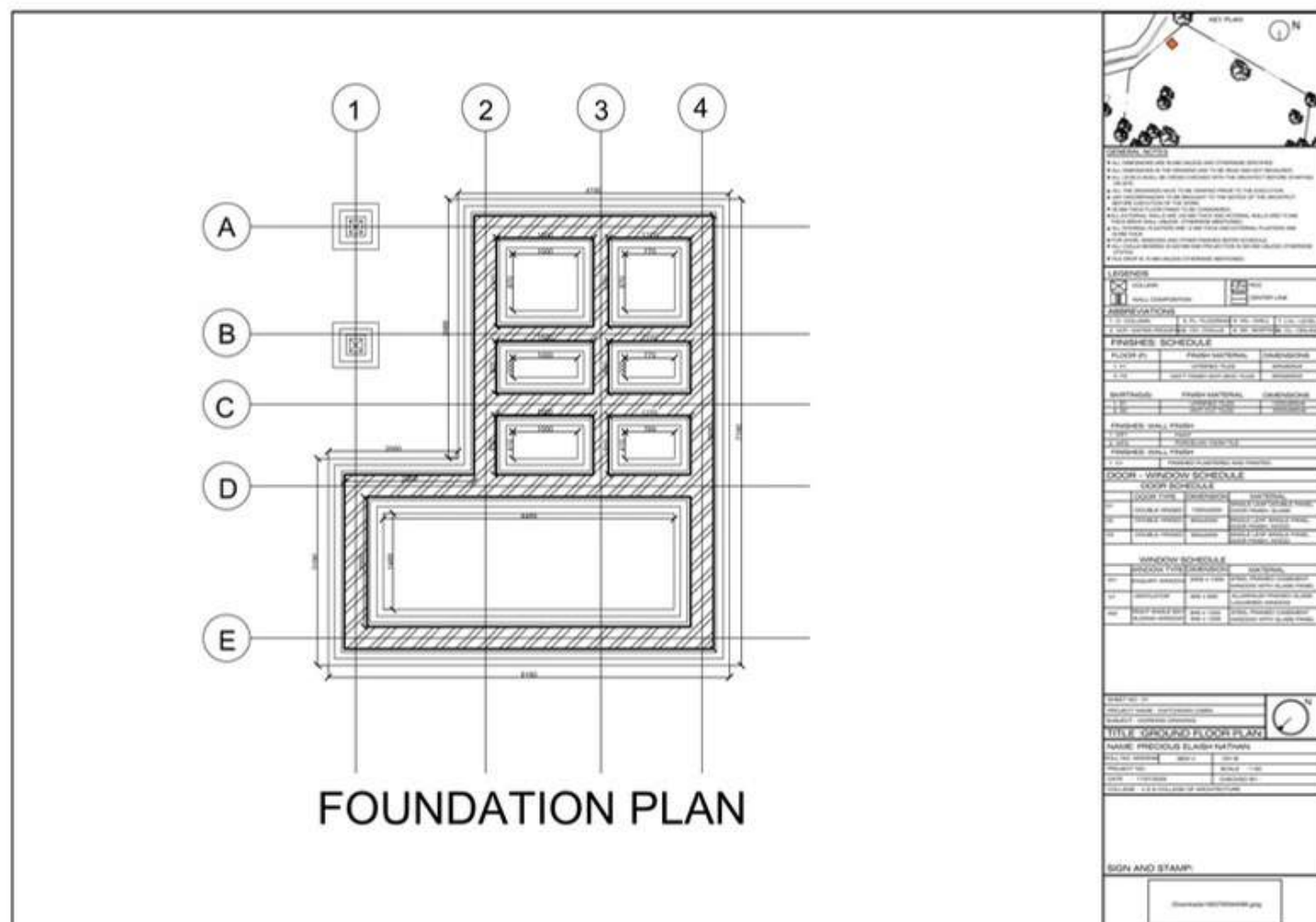
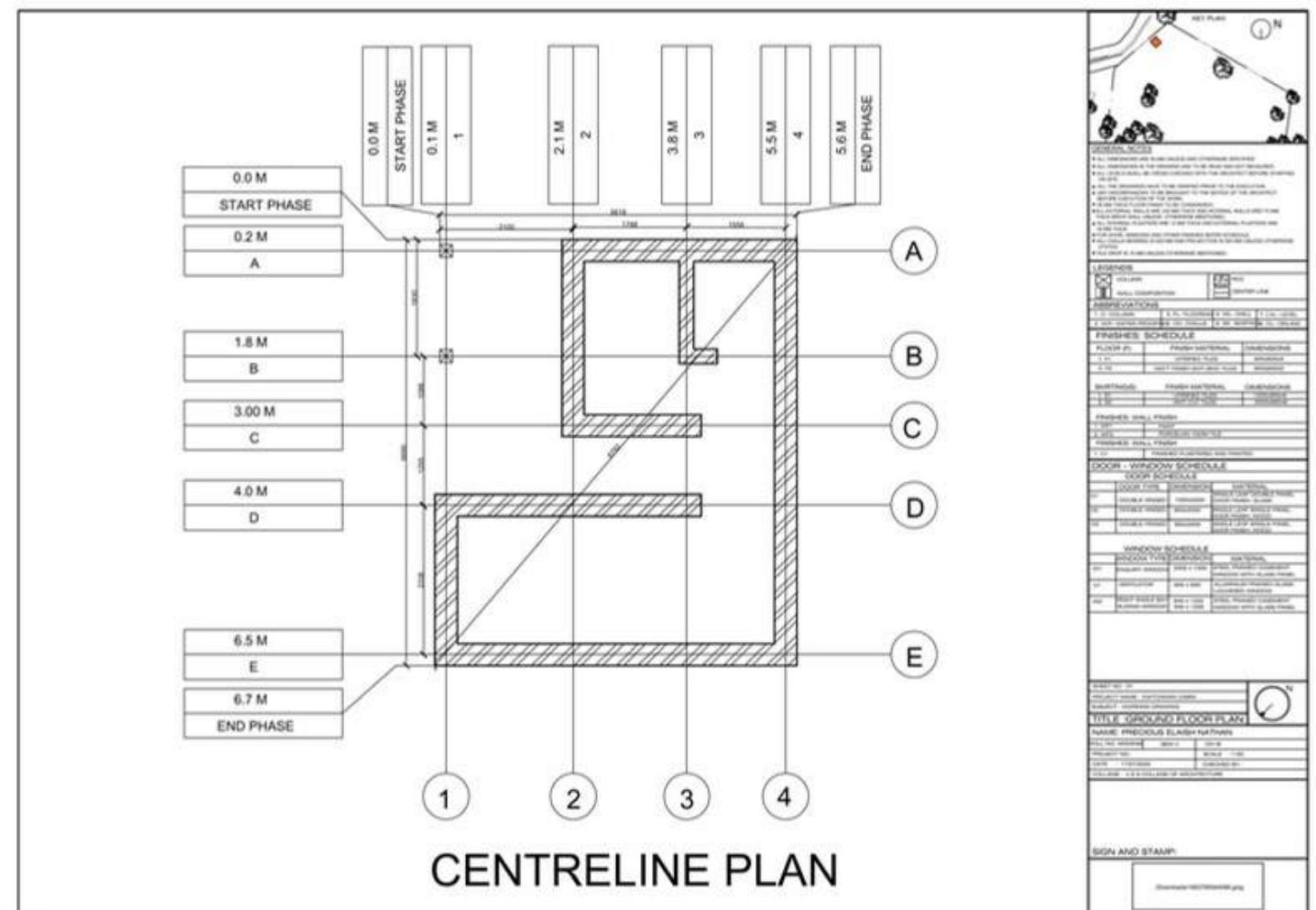
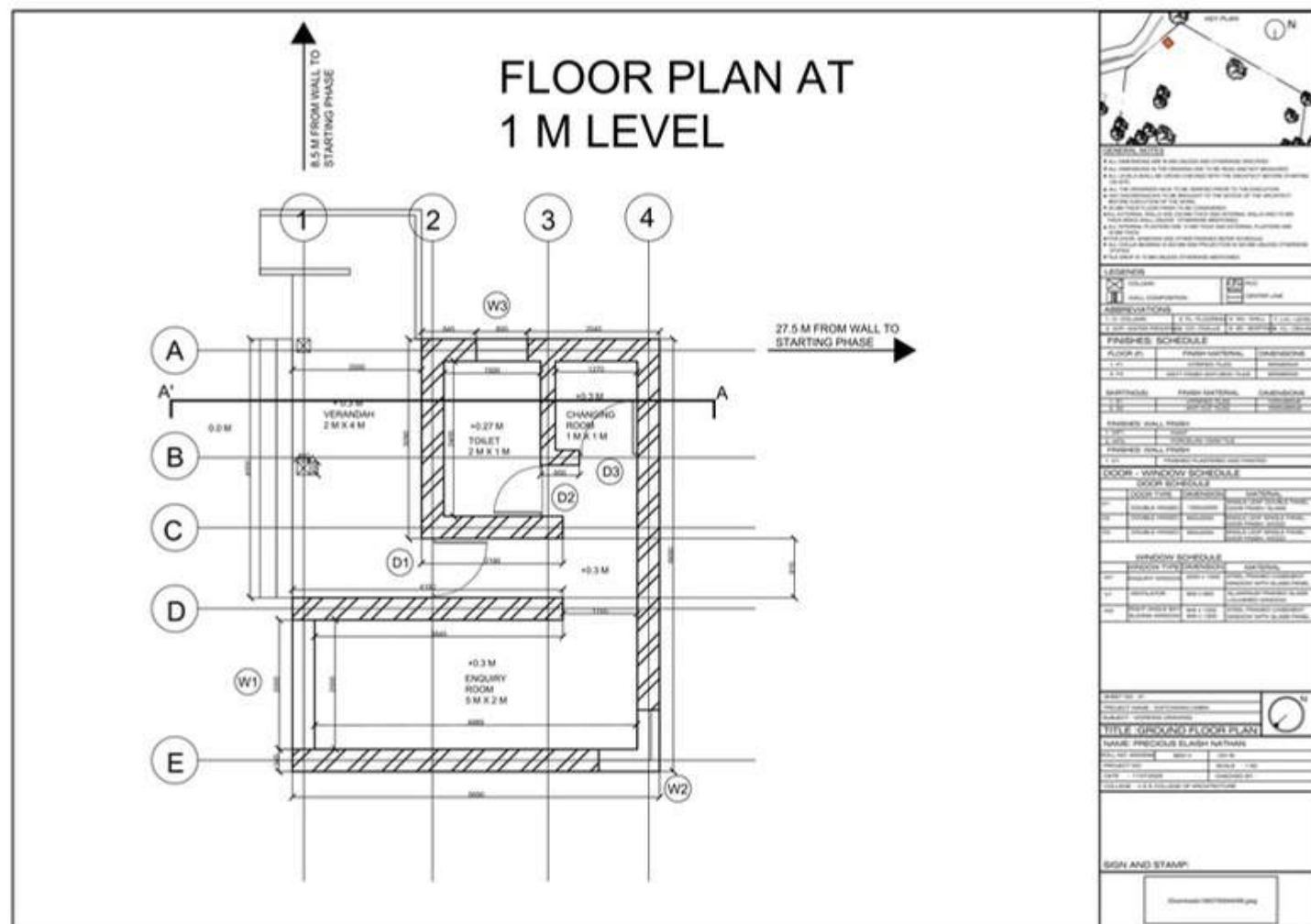
PLAN, SECTION AA', ELEVATION SCALE- 1:50

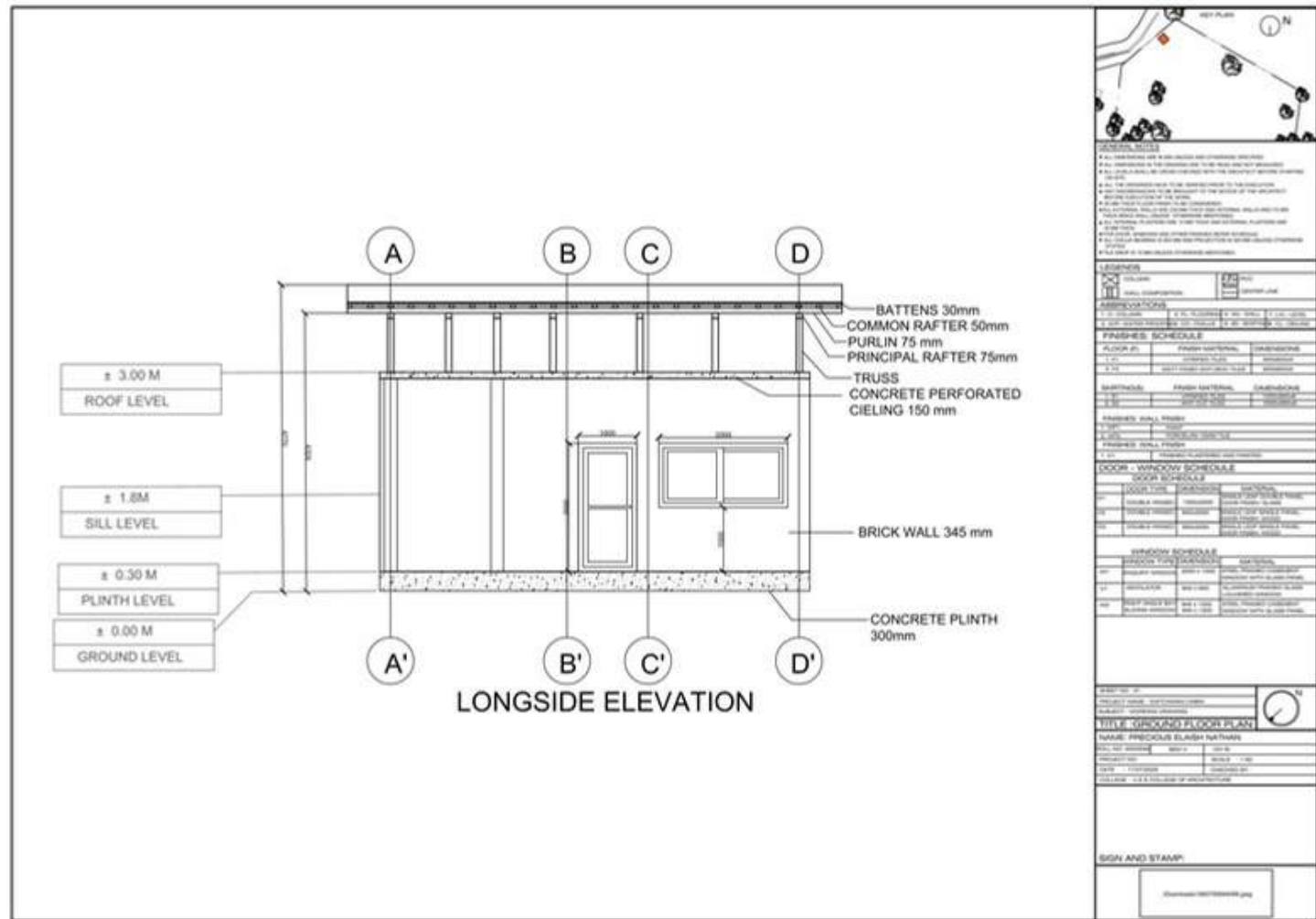
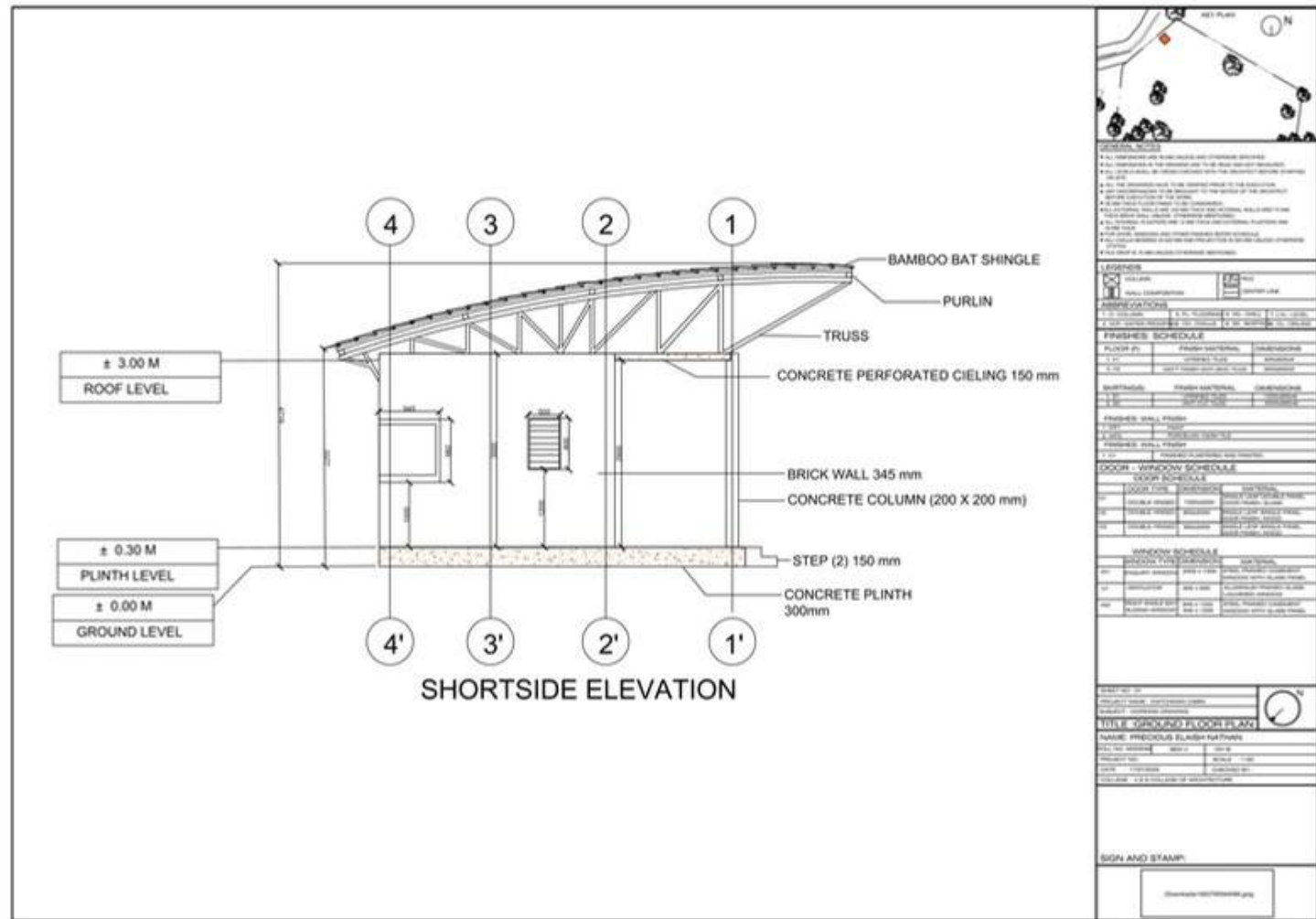
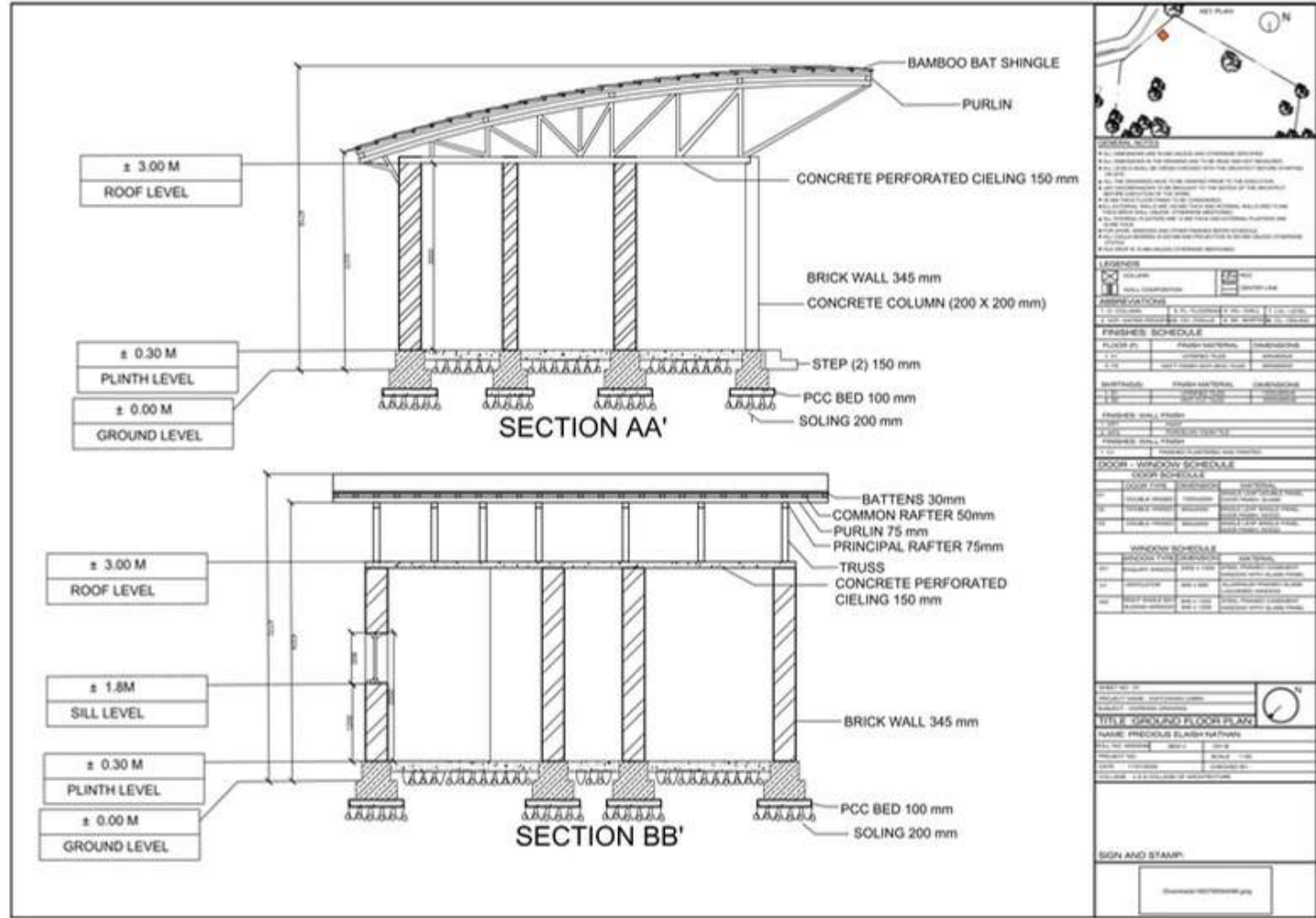
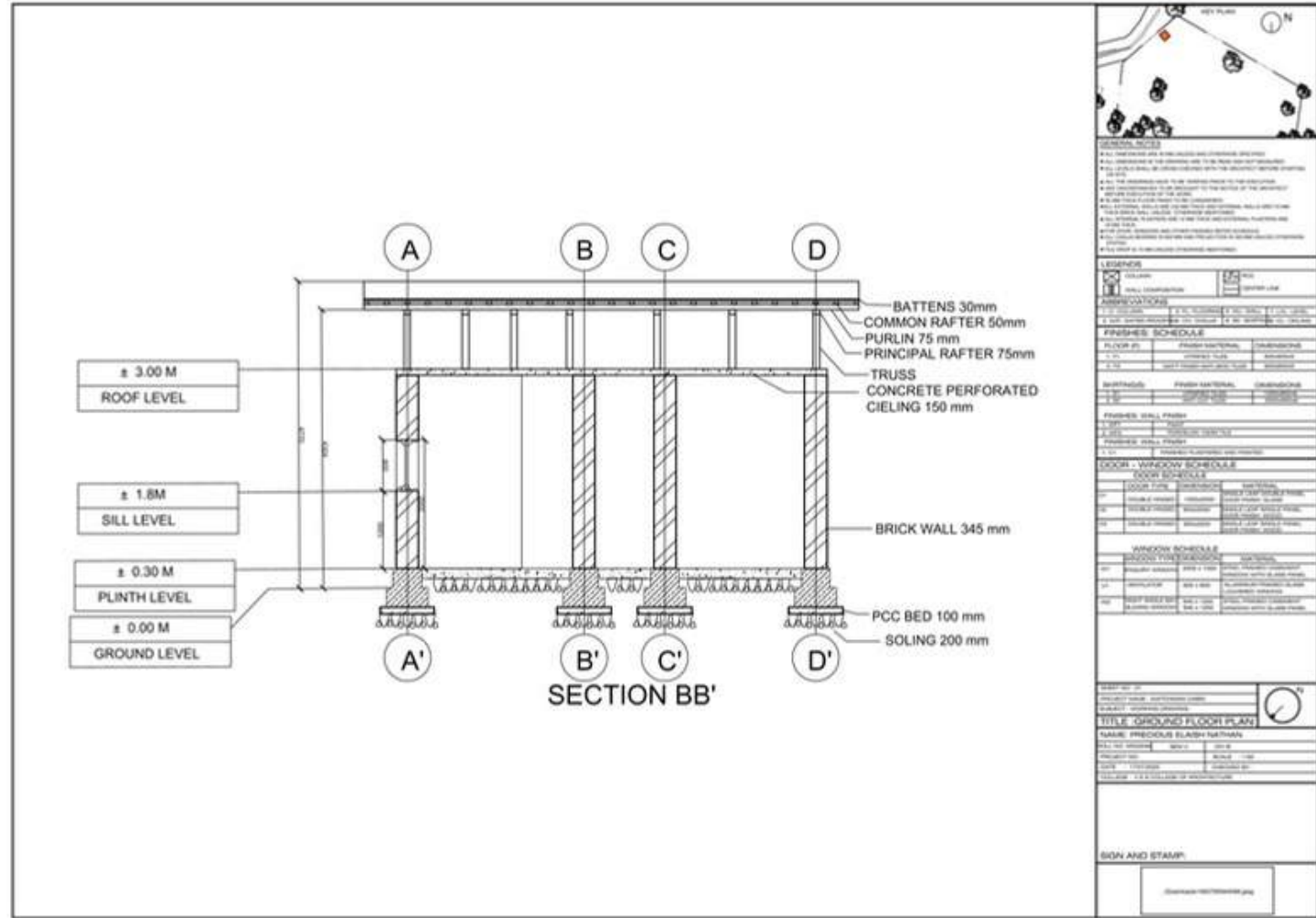
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		ROLL NO - AR23048 , BATCH-B
		SUBJECT- WD
		YEAR- 2ND , SEM-4

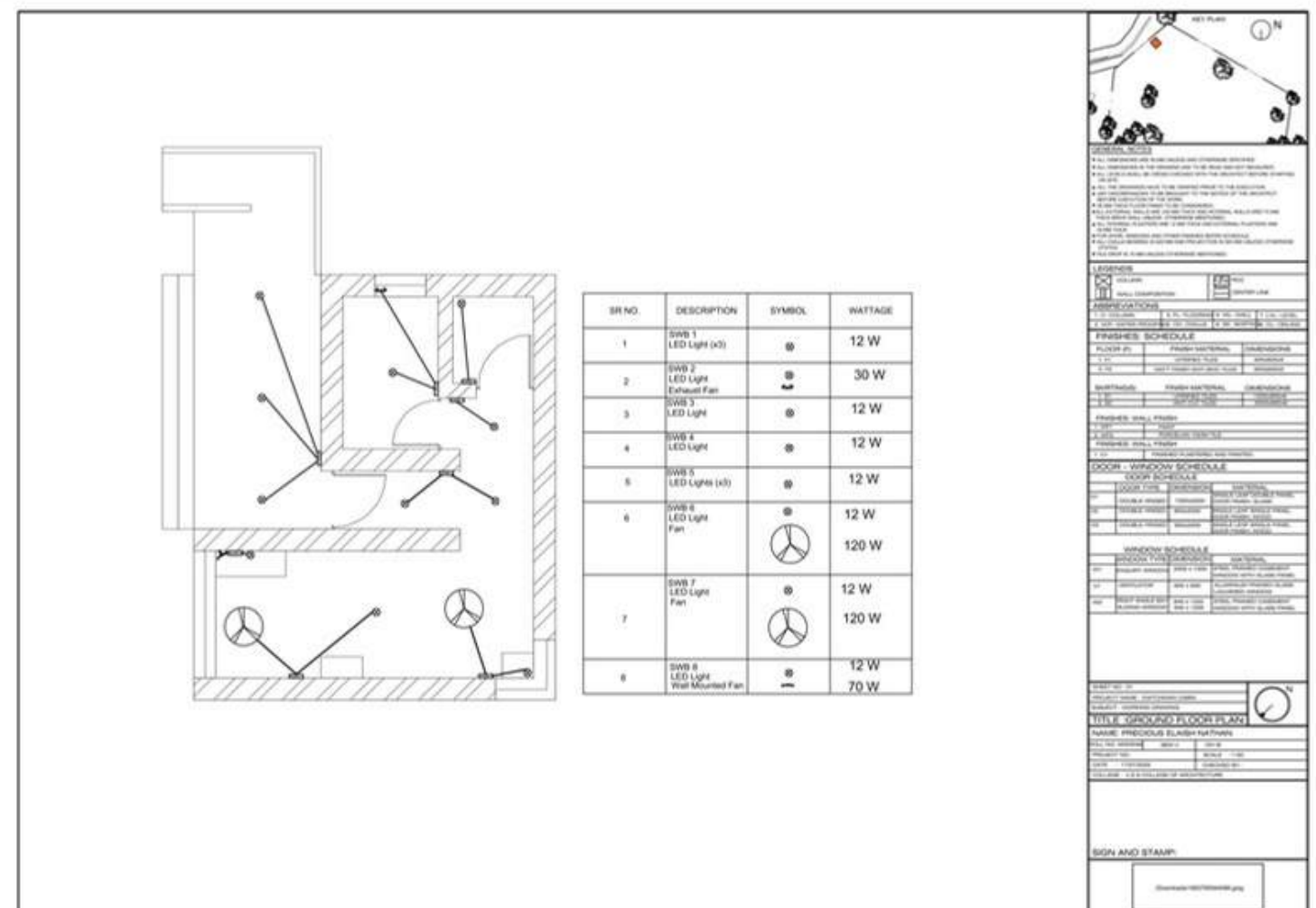
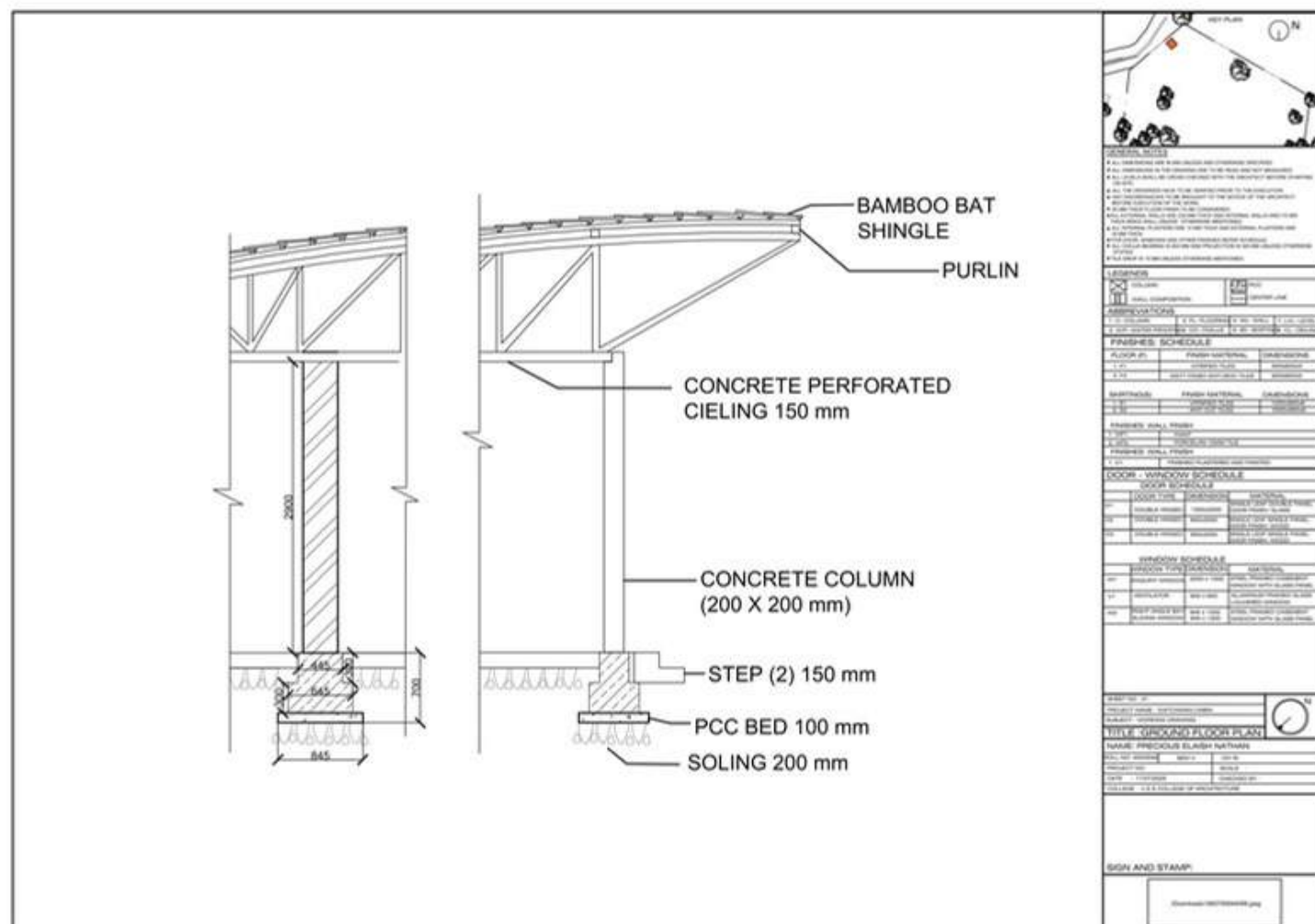
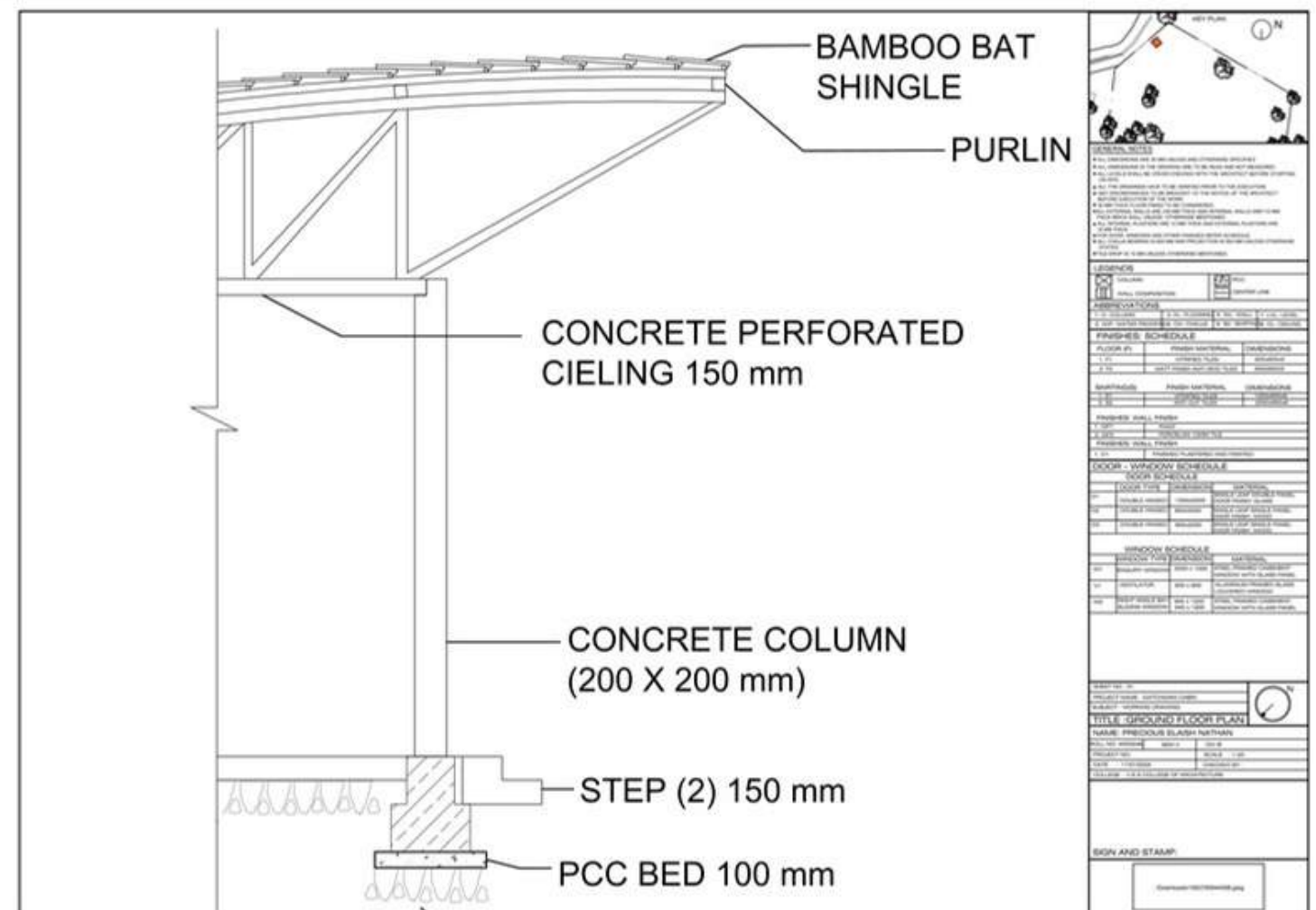
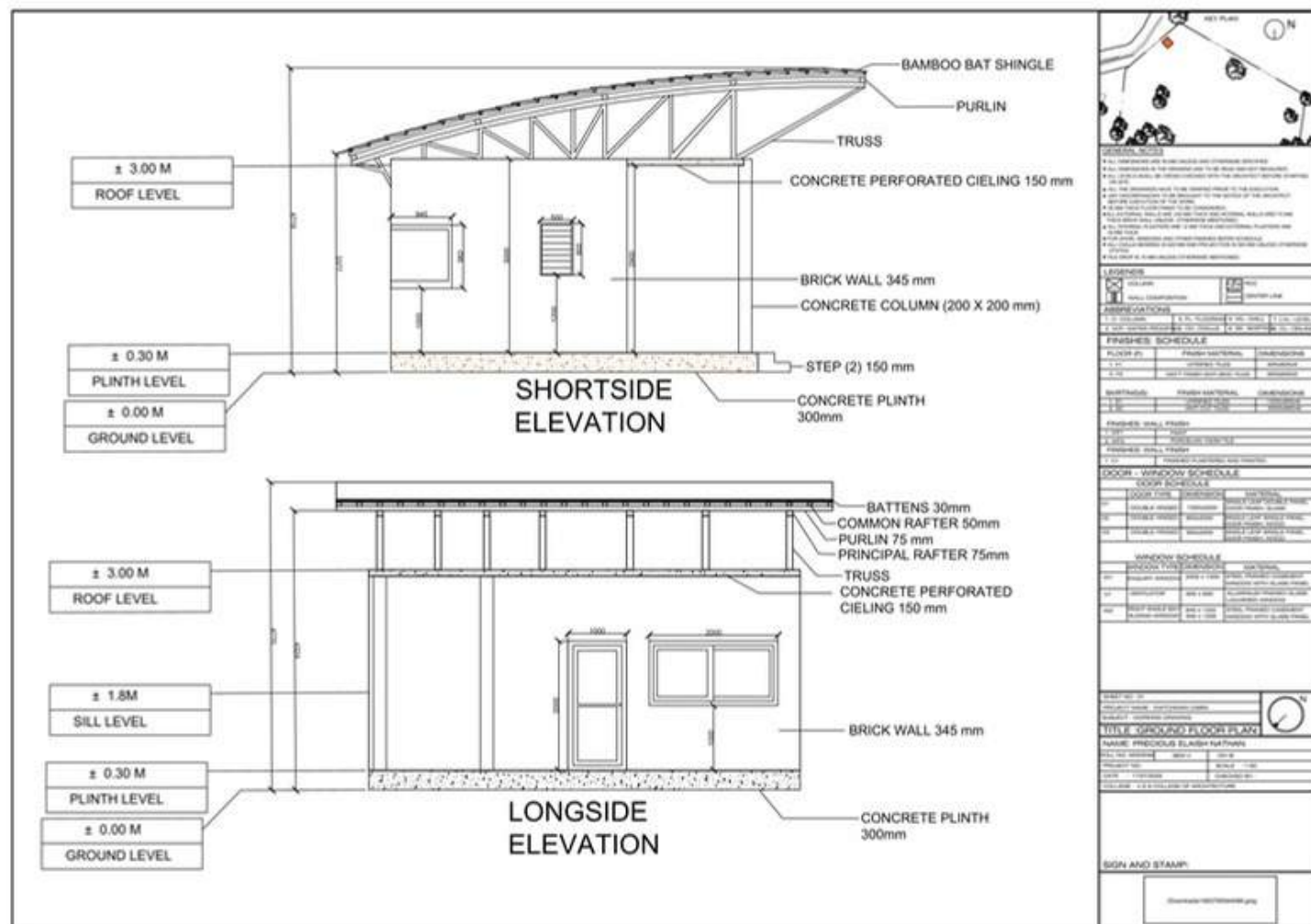


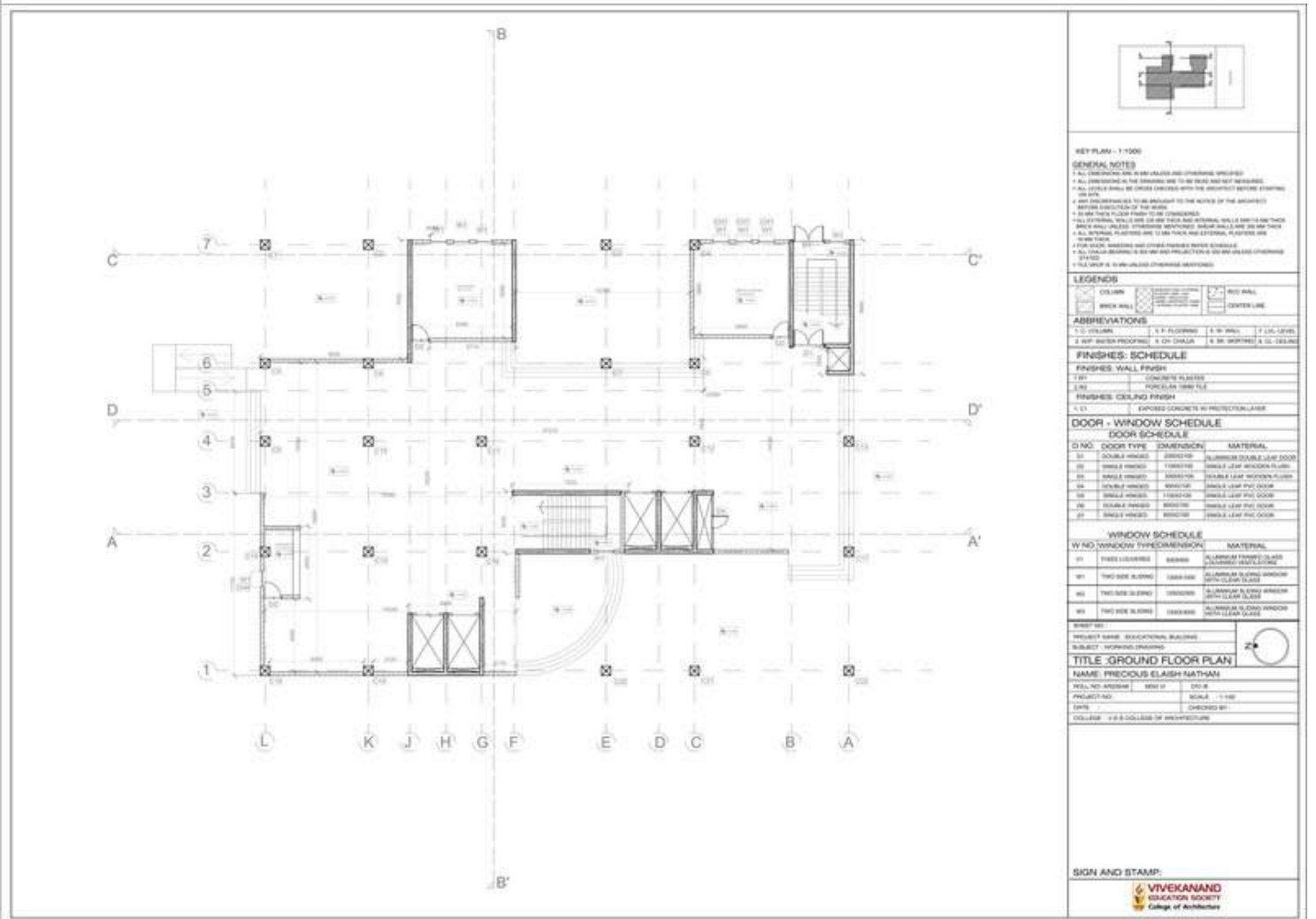
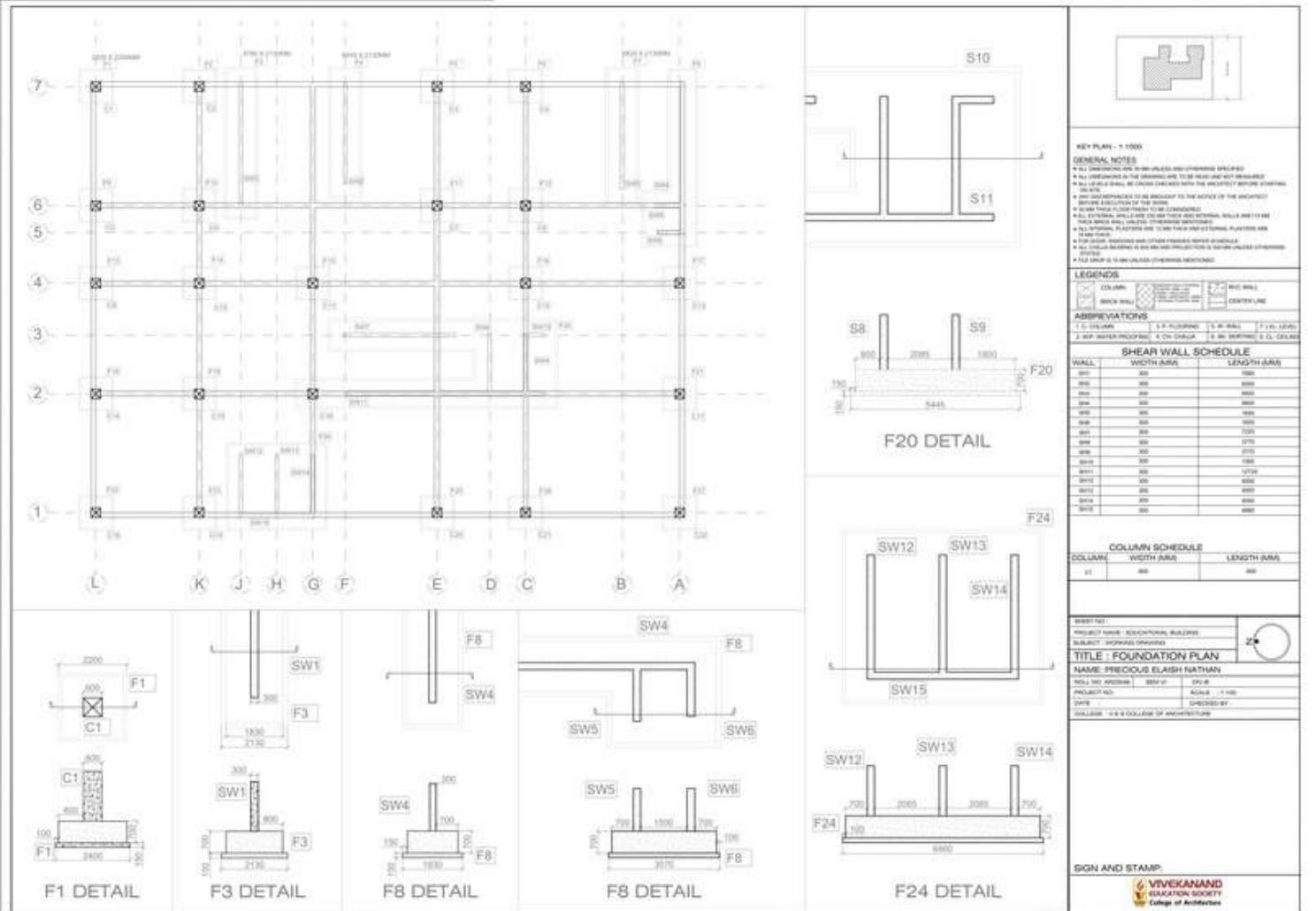
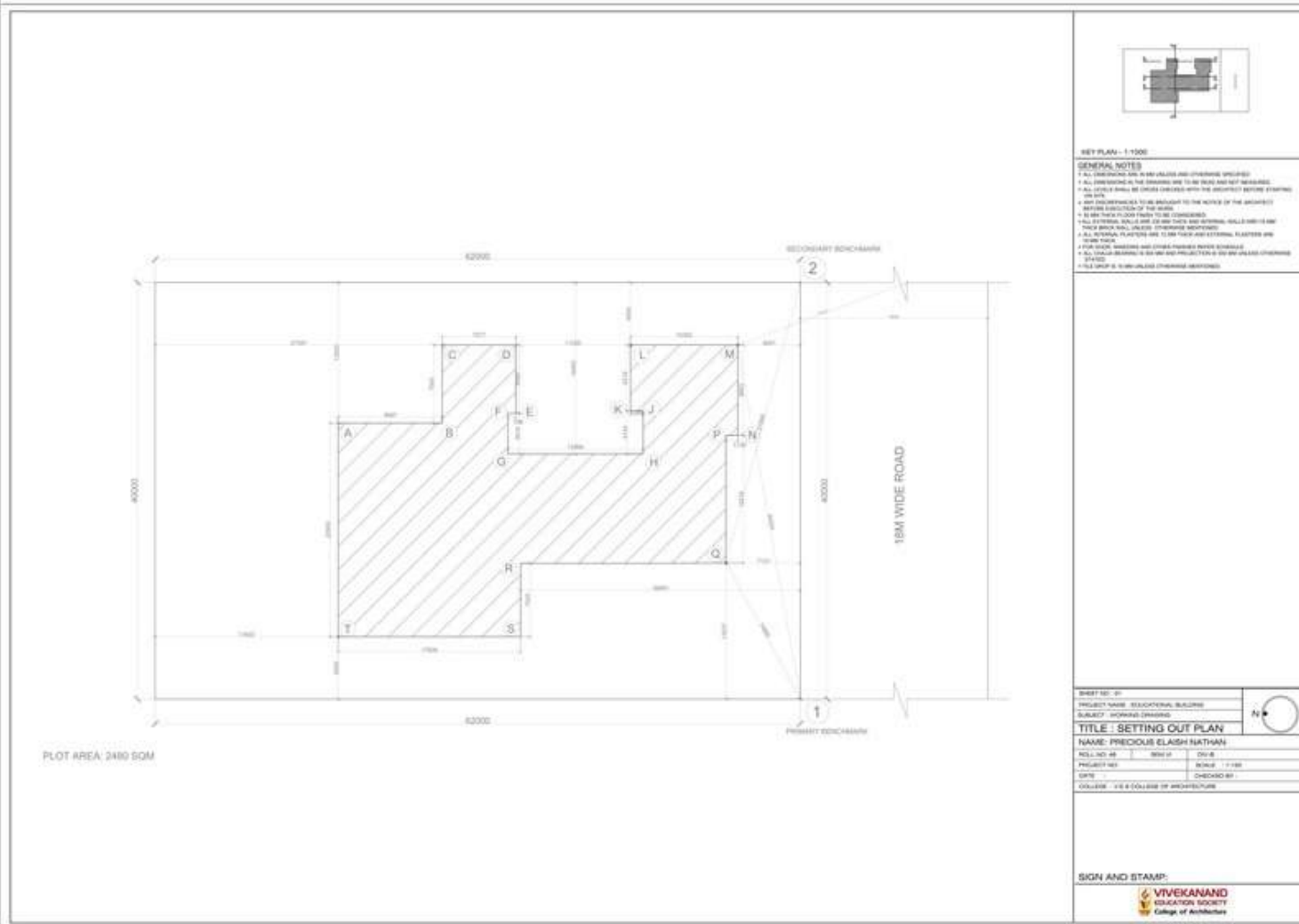
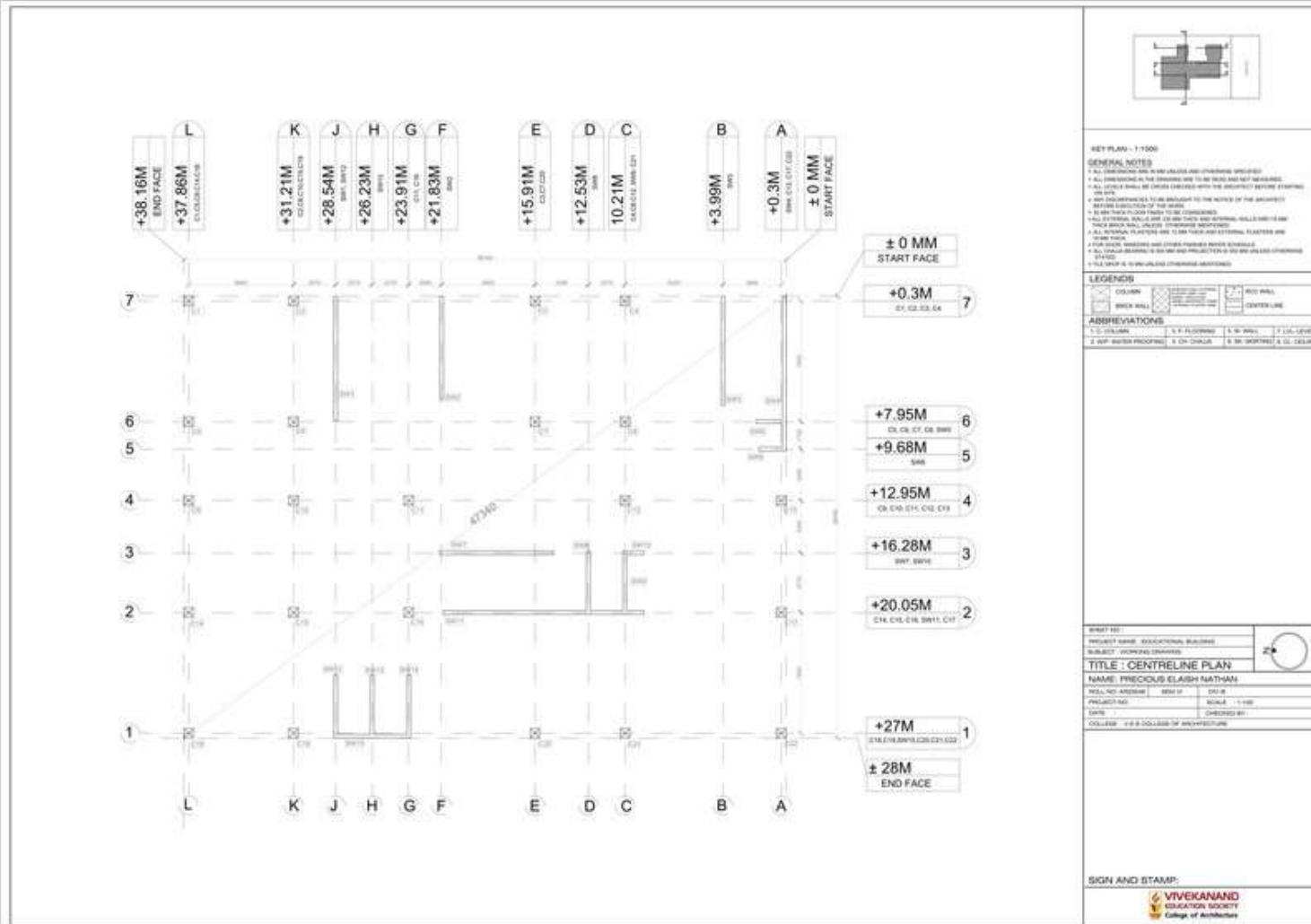
PLAN, SECTION BB', ELEVATION SCALE- 1:50

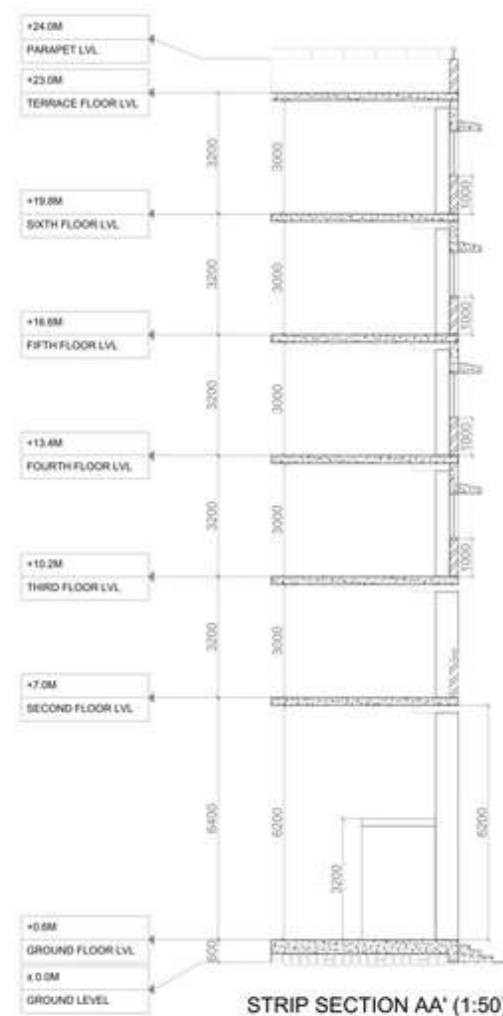
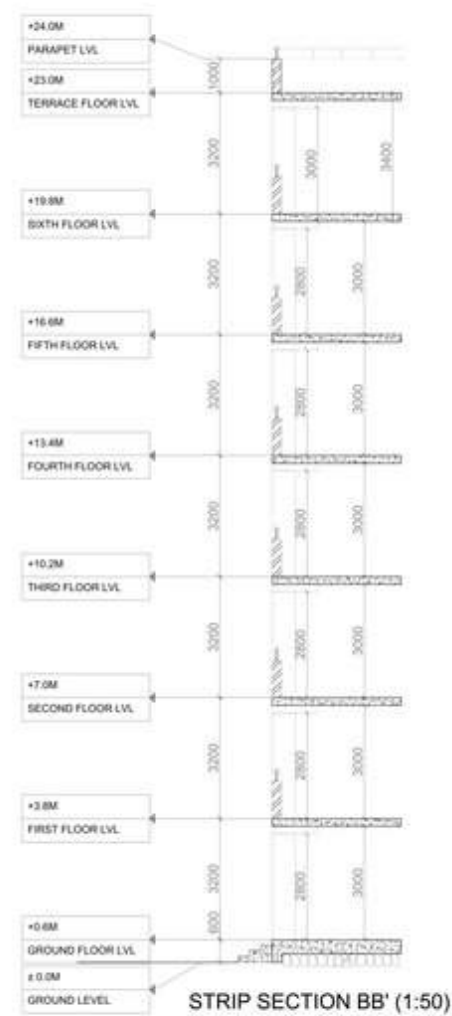
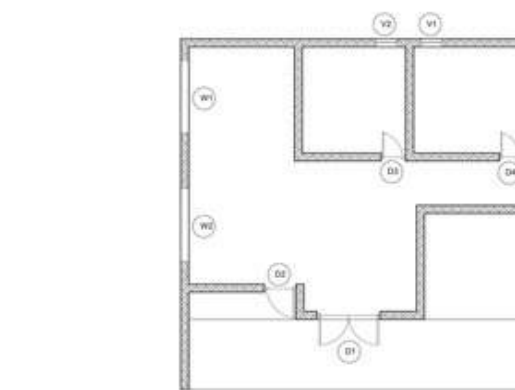
SIGN/STAMP	GRADE	V.E.S.C.O.A.
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		ROLL NO - AR23048 , BATCH-B
		SUBJECT- WD
		YEAR- 2ND , SEM-4



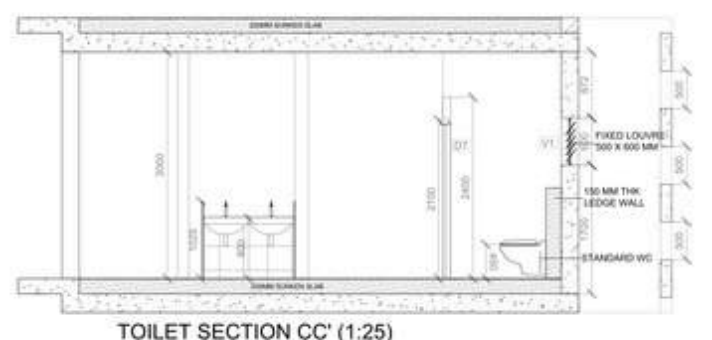
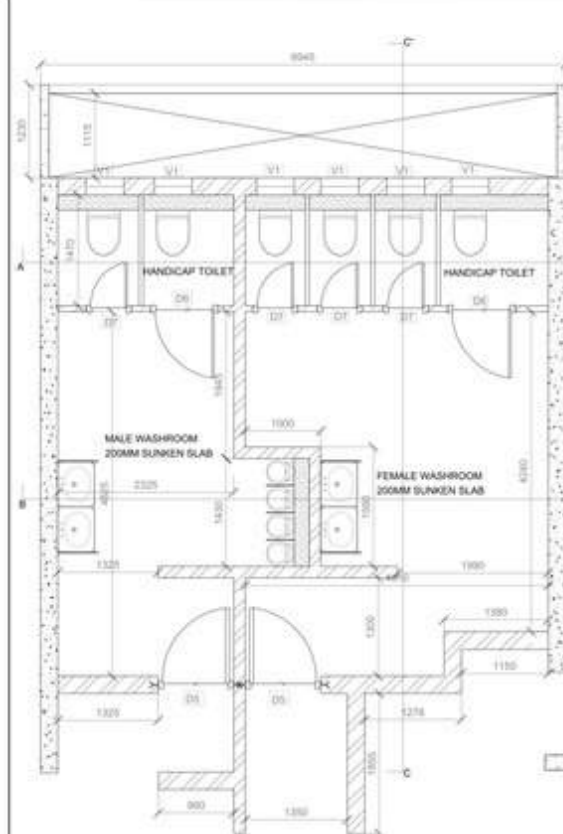
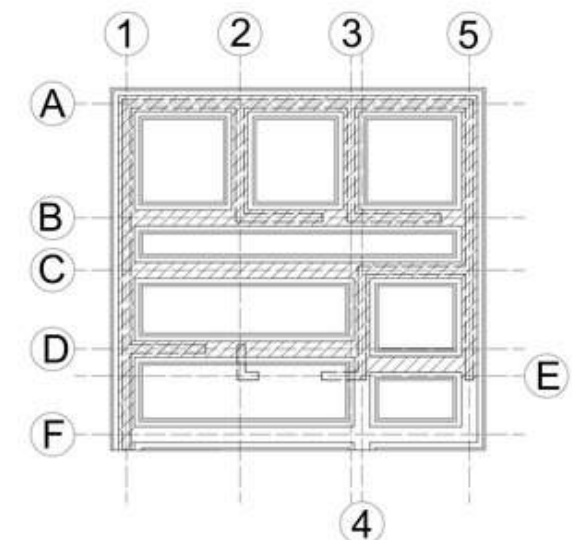






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DOOR WINDOW SHEDULE		
NAME	DIMENSIONS	DESCRIPTION
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D2	1000MMX2000MM	SINGLE LEAF DOUBLE PANELED TEAK WOOD DOOR
D3	900MMX2000MM	SINGLE LEAF DOUBLE PANELED TEAK WOOD DOOR
D4	900MMX2000MM	SINGLE LEAF DOUBLE PANELED TEAK WOOD DOOR
W1	2000MMX1000MM	DOUBLE SHUTTER SINGLE PANELED TEAK WOOD WINDOW
W2	2000MMX1000MM	DOUBLE SHUTTER FOUR PANELED TEAK WOOD WINDOW
V1	600MMX600MM	LOUVERED WINDOW
V2	800MMX800MM	LOUVERED WINDOW

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MISCELLANEOUS





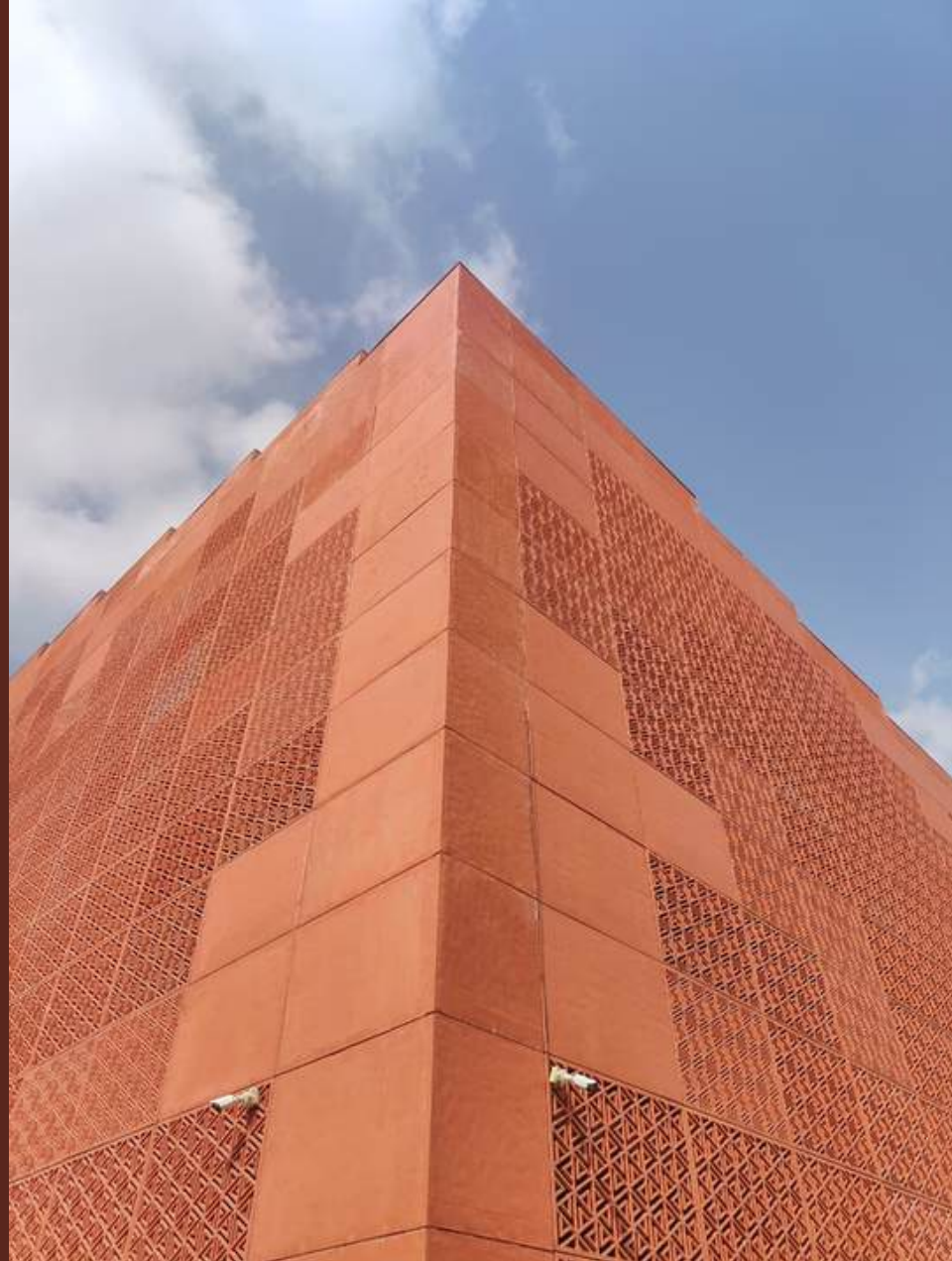
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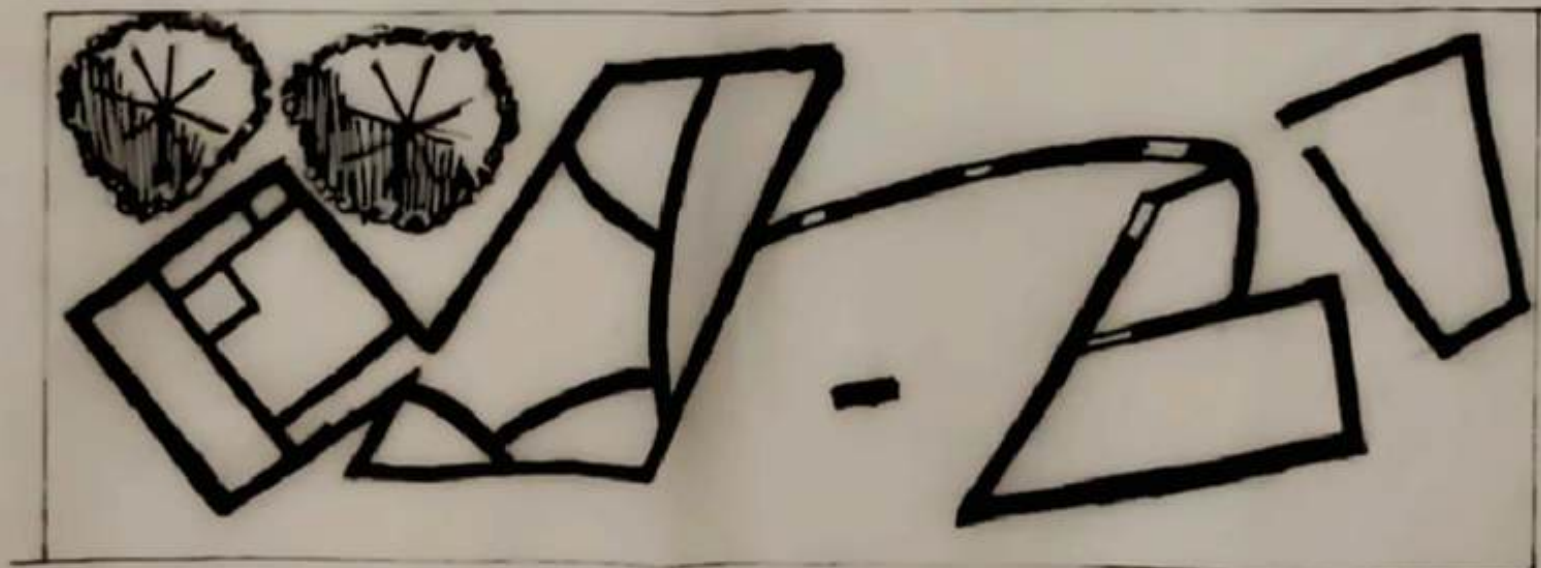
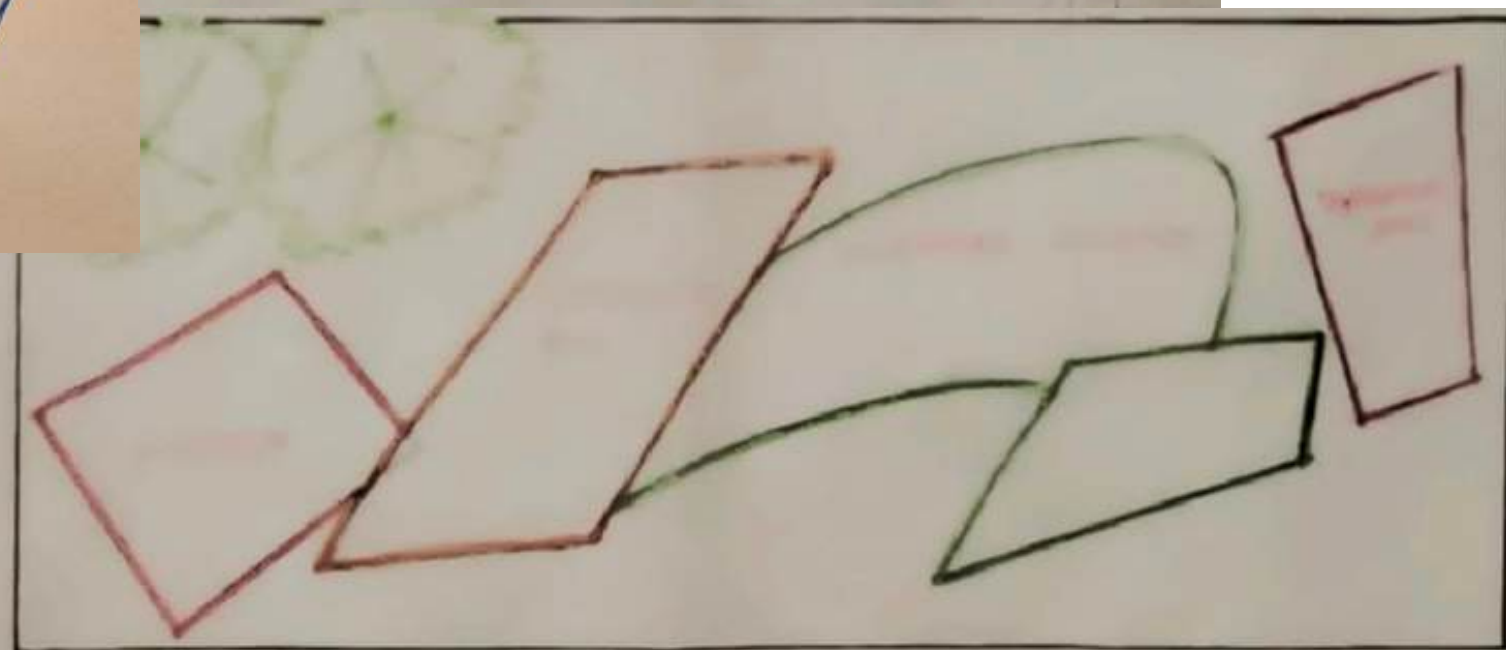
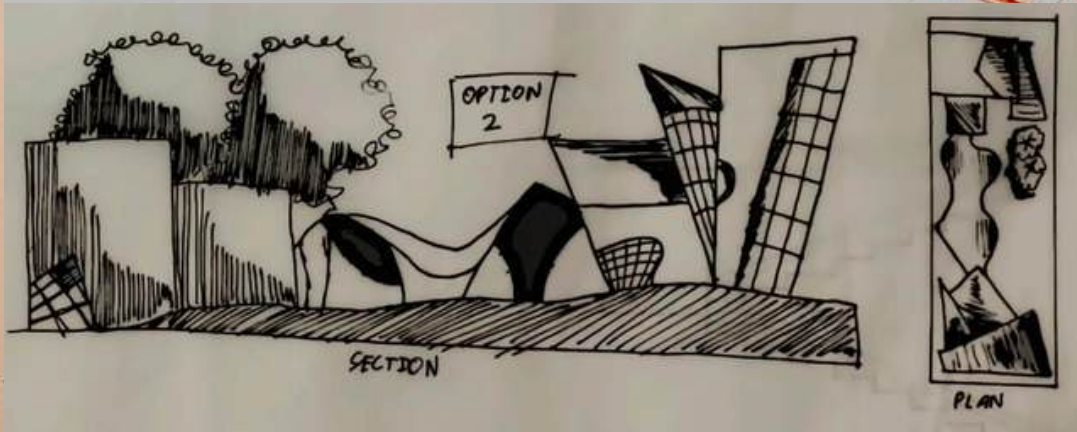
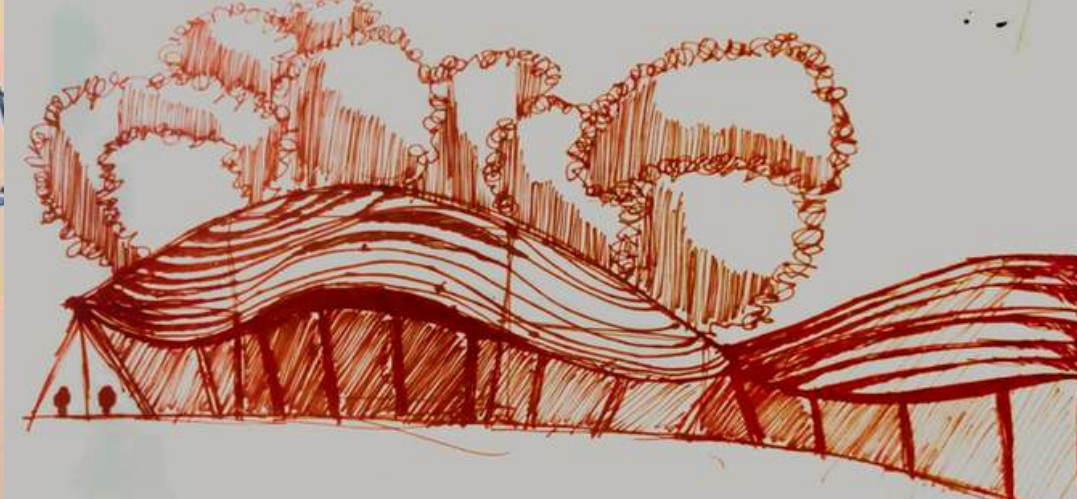
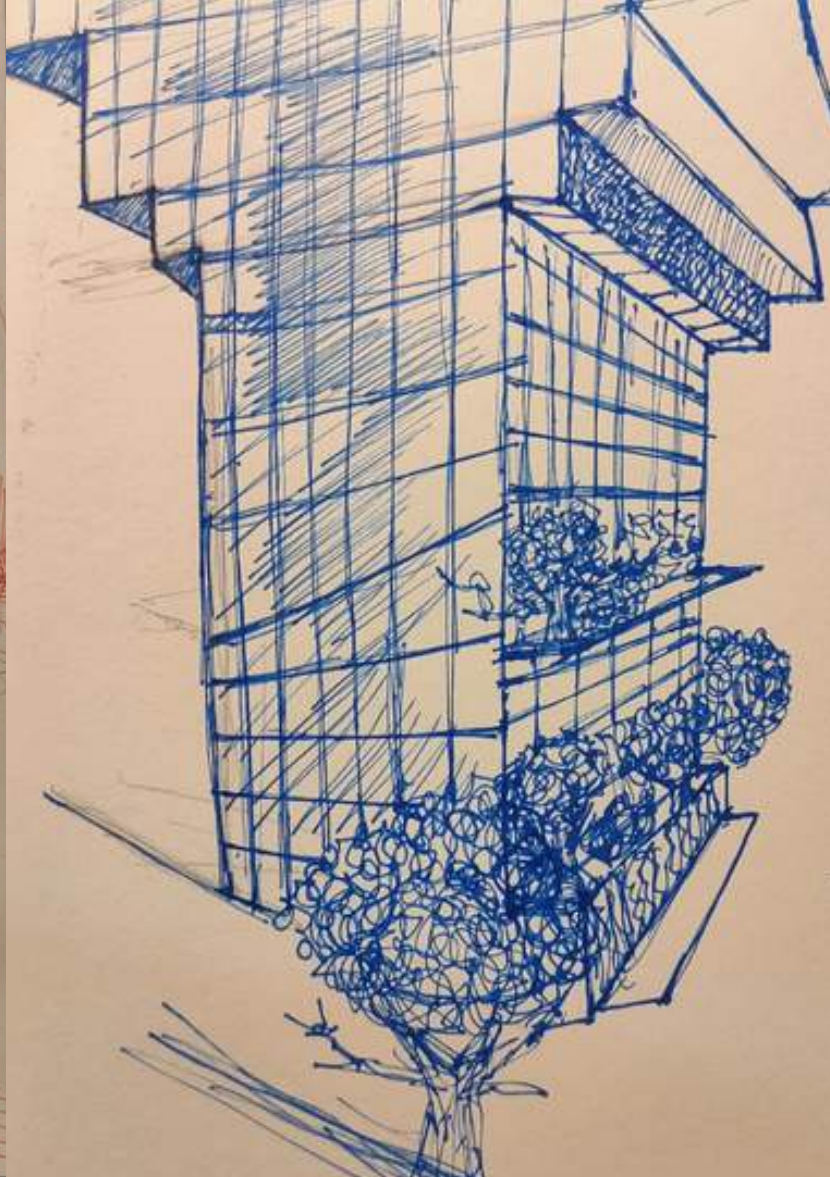
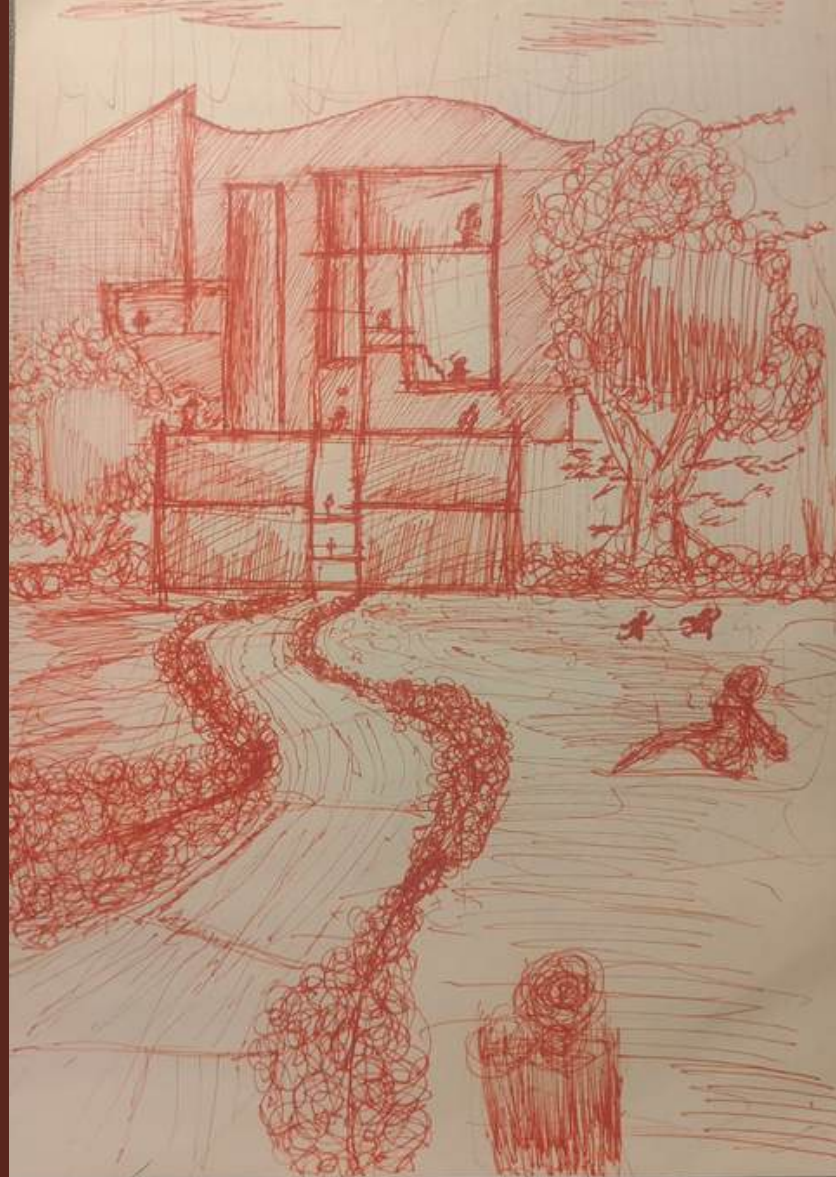


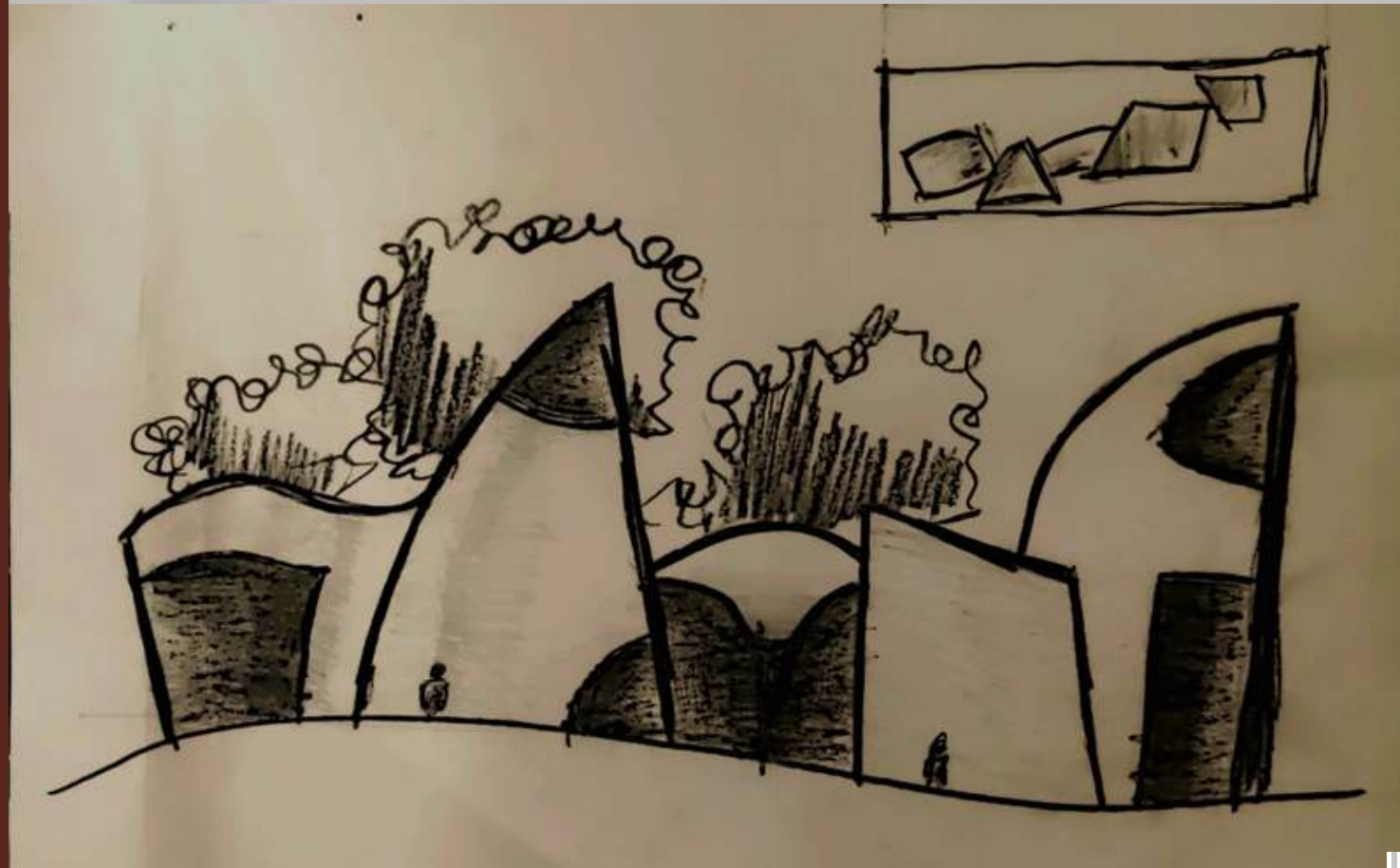
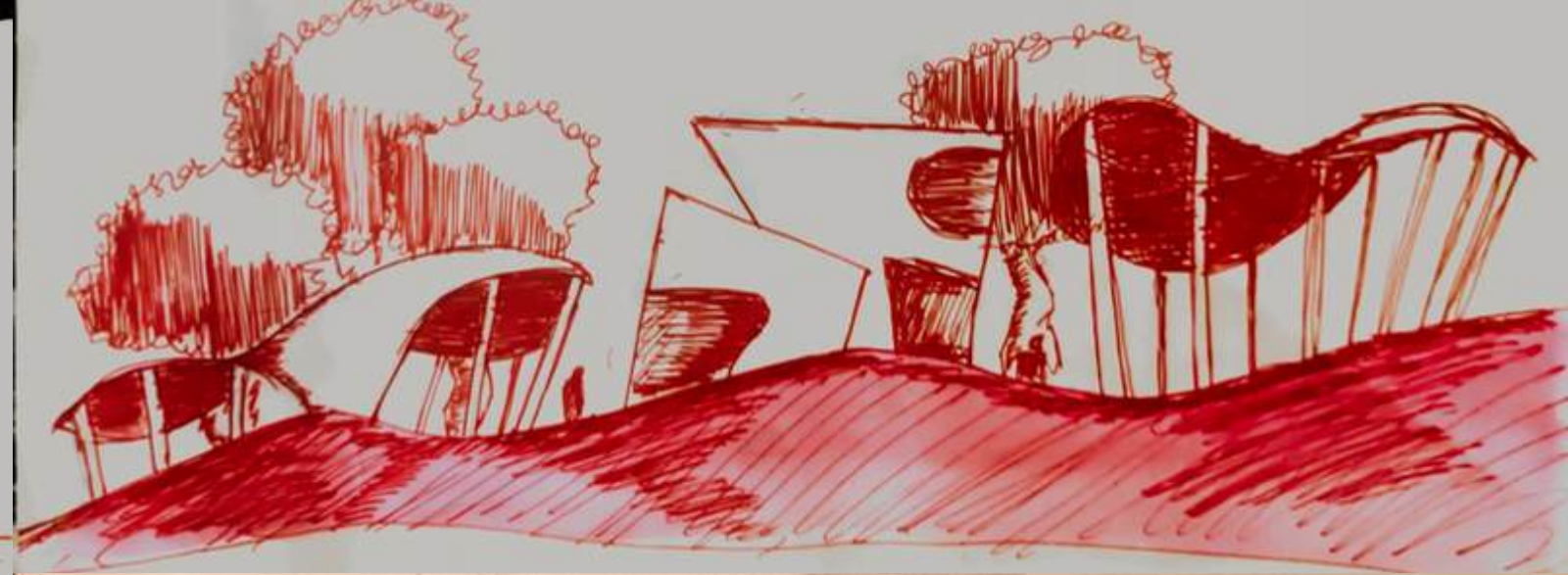
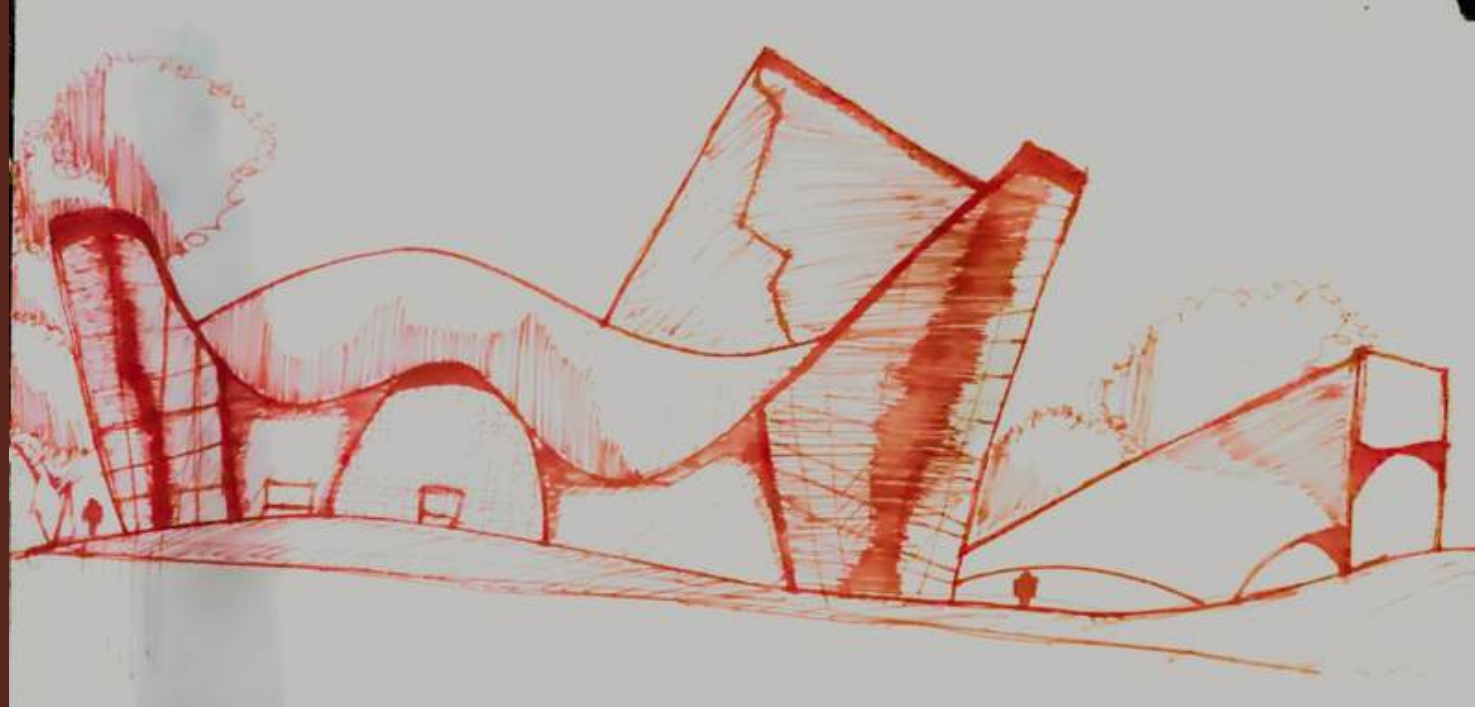


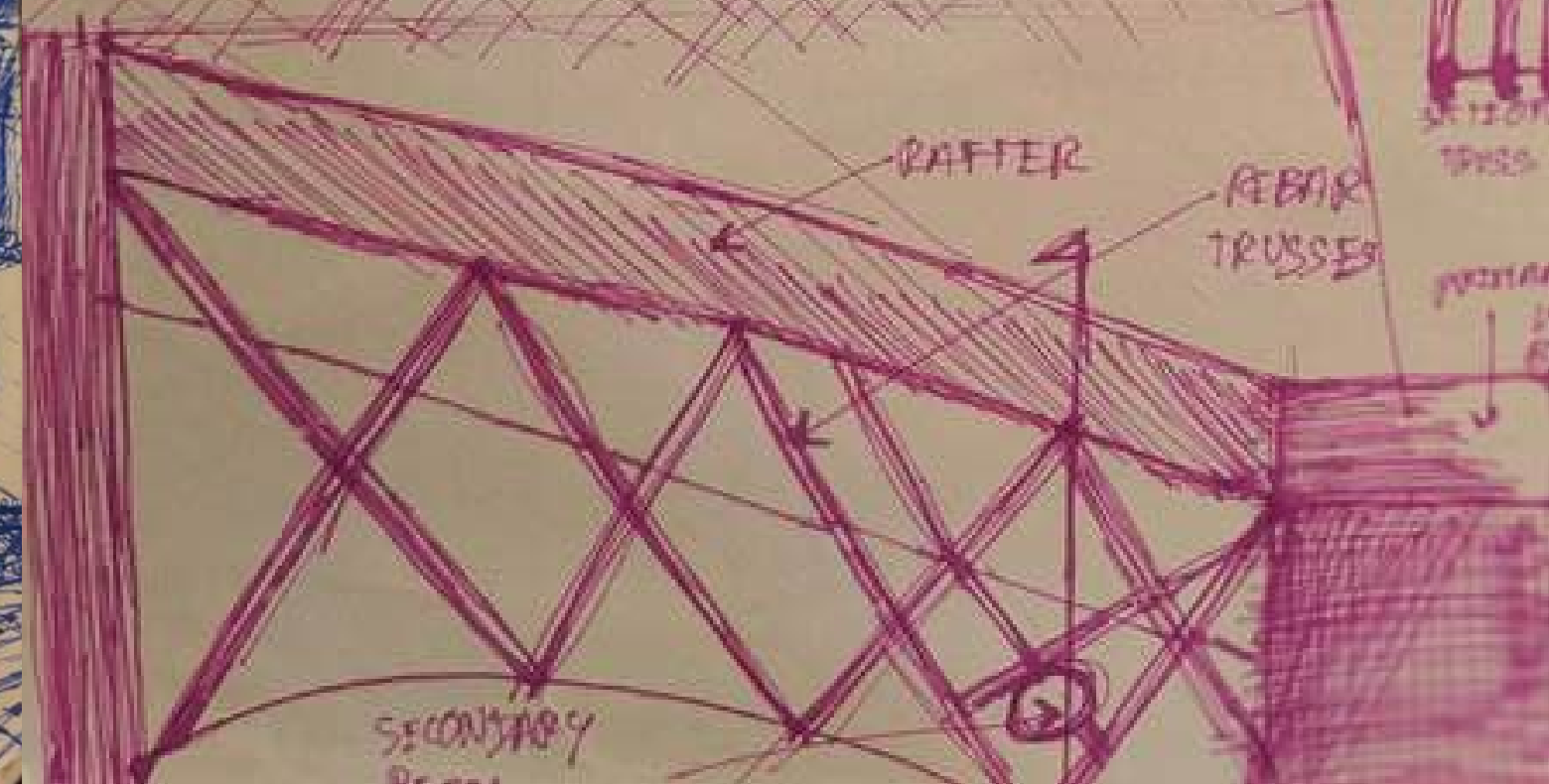
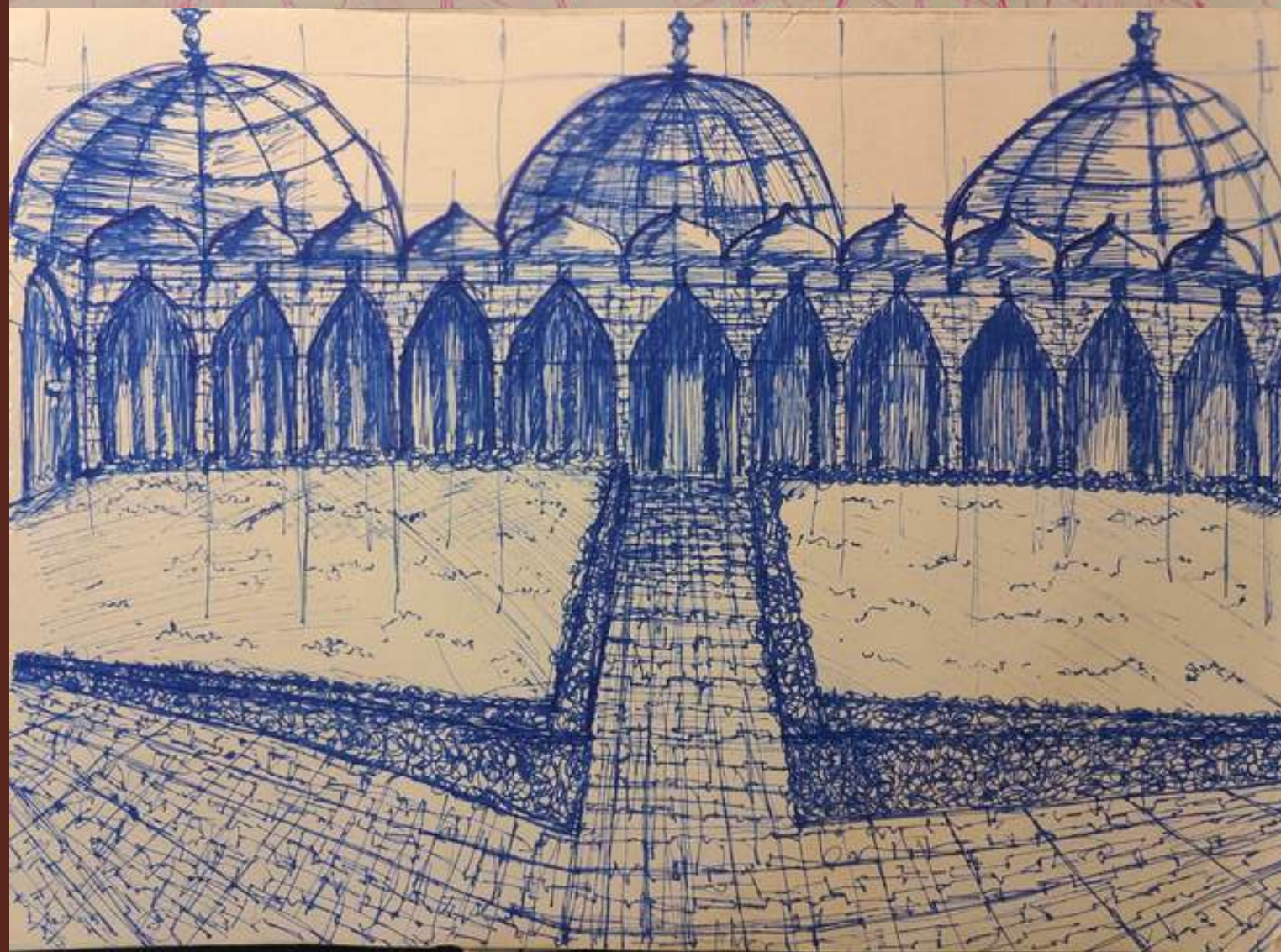
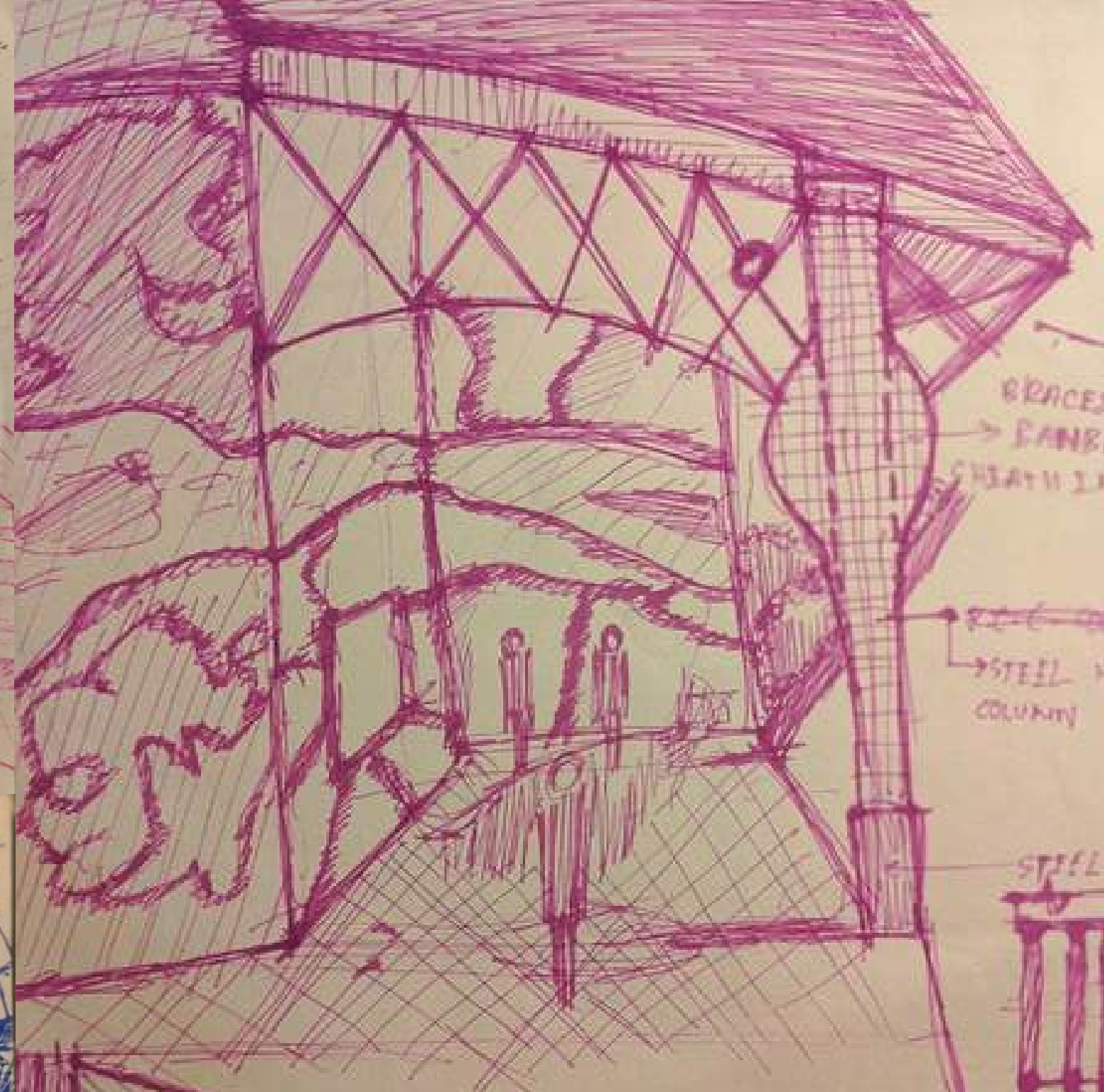
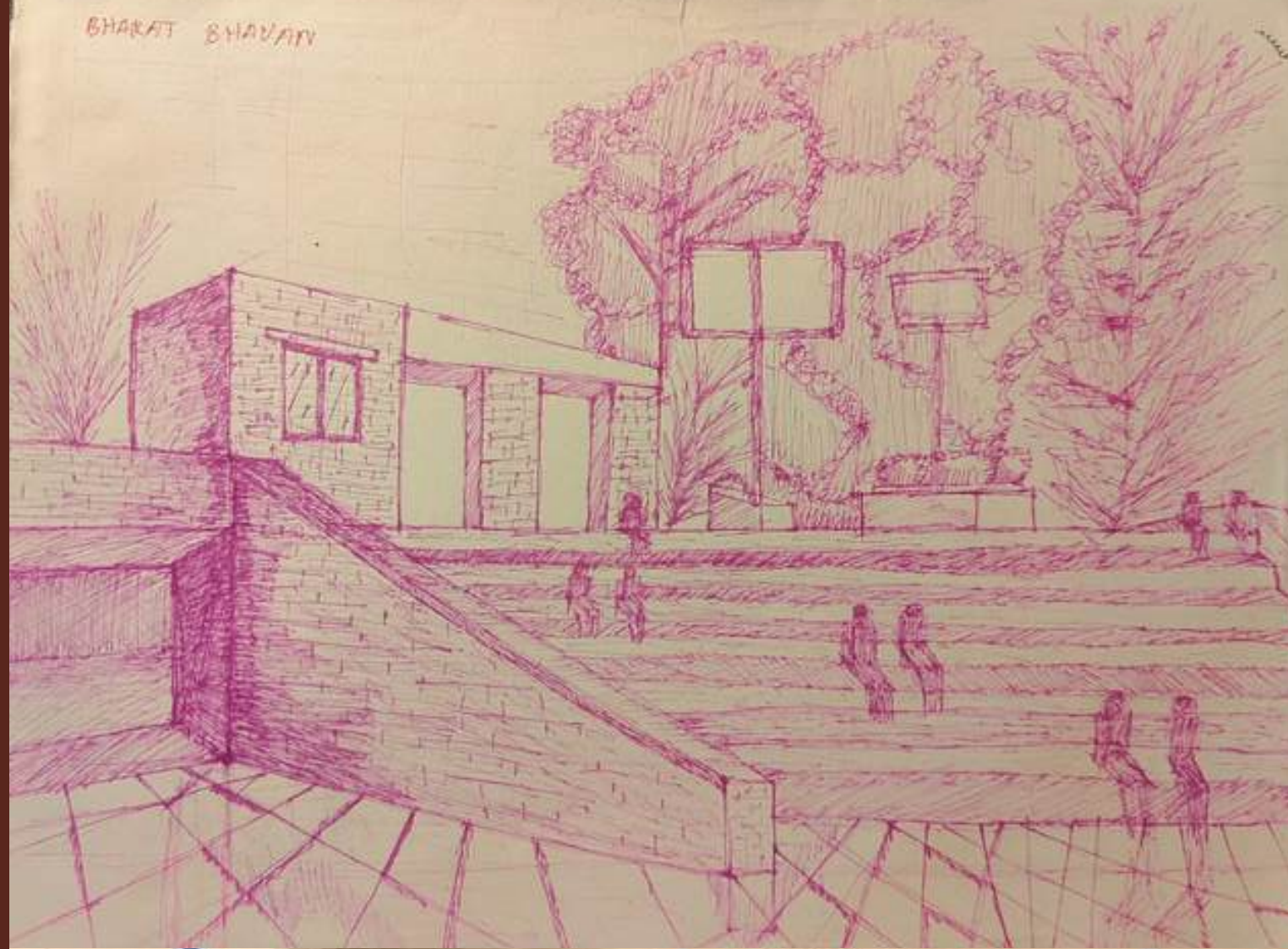


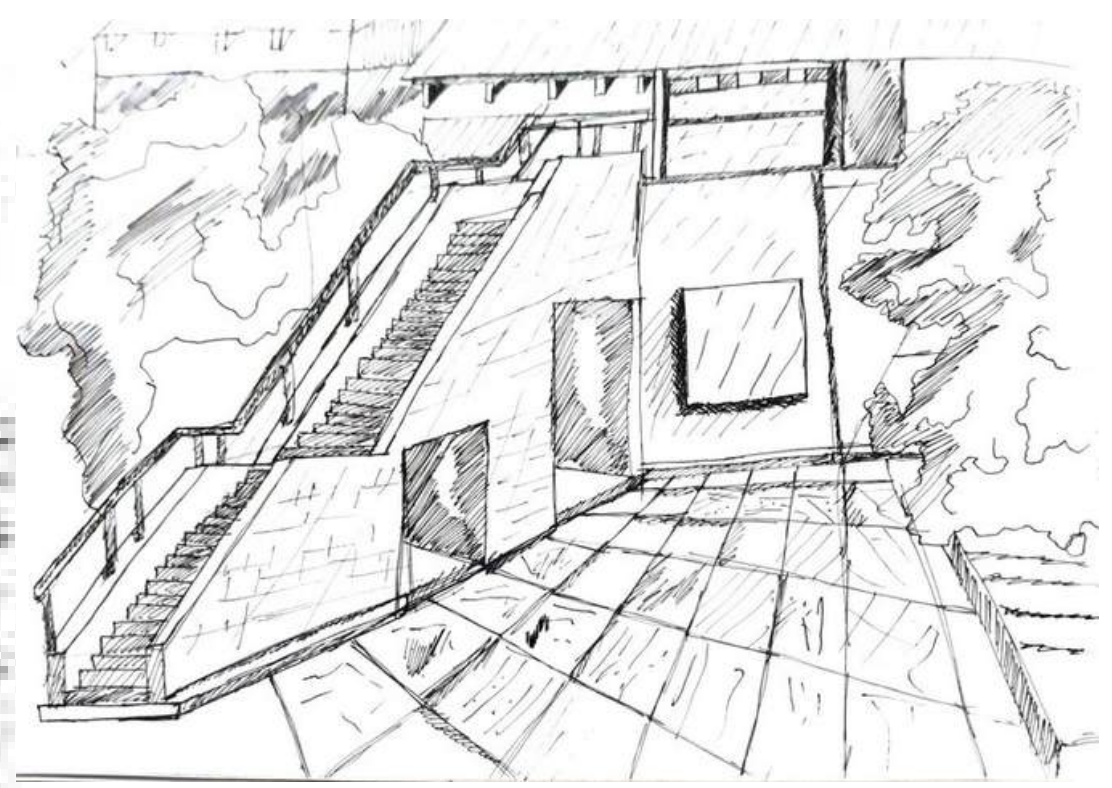
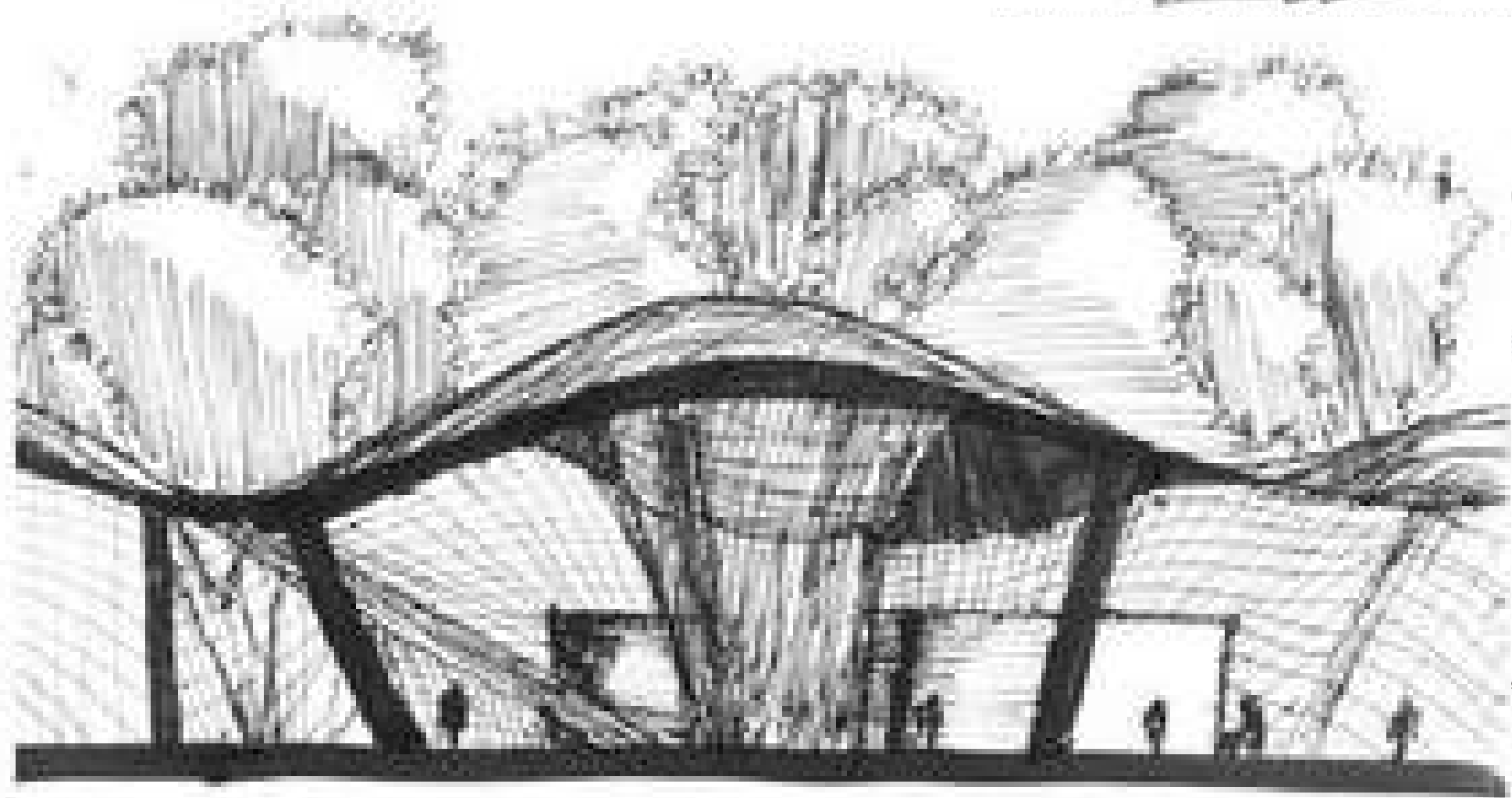
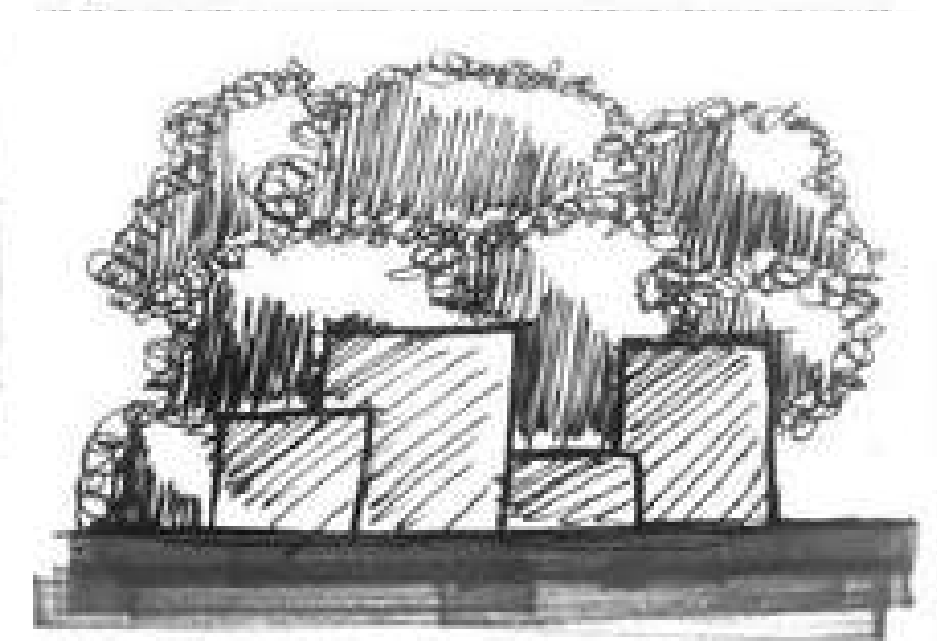
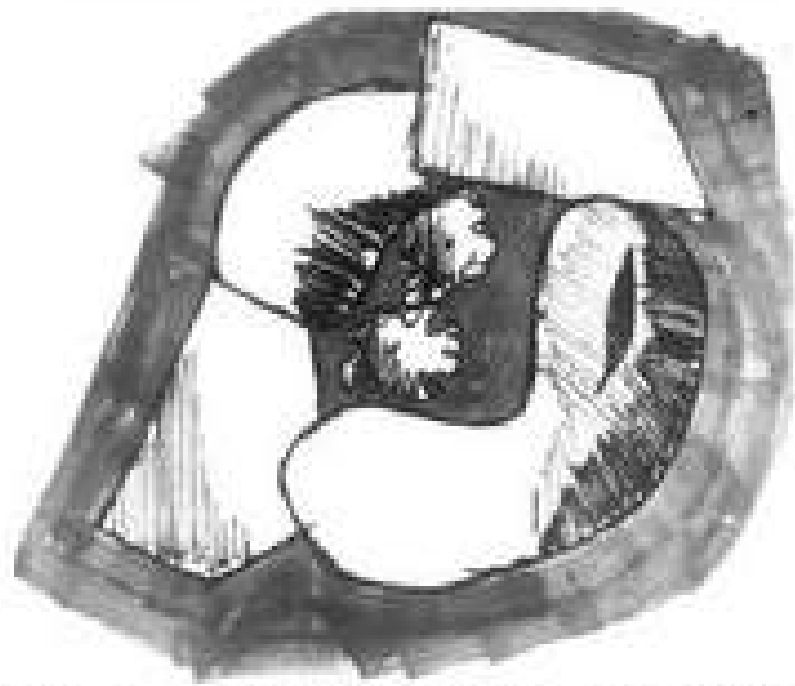
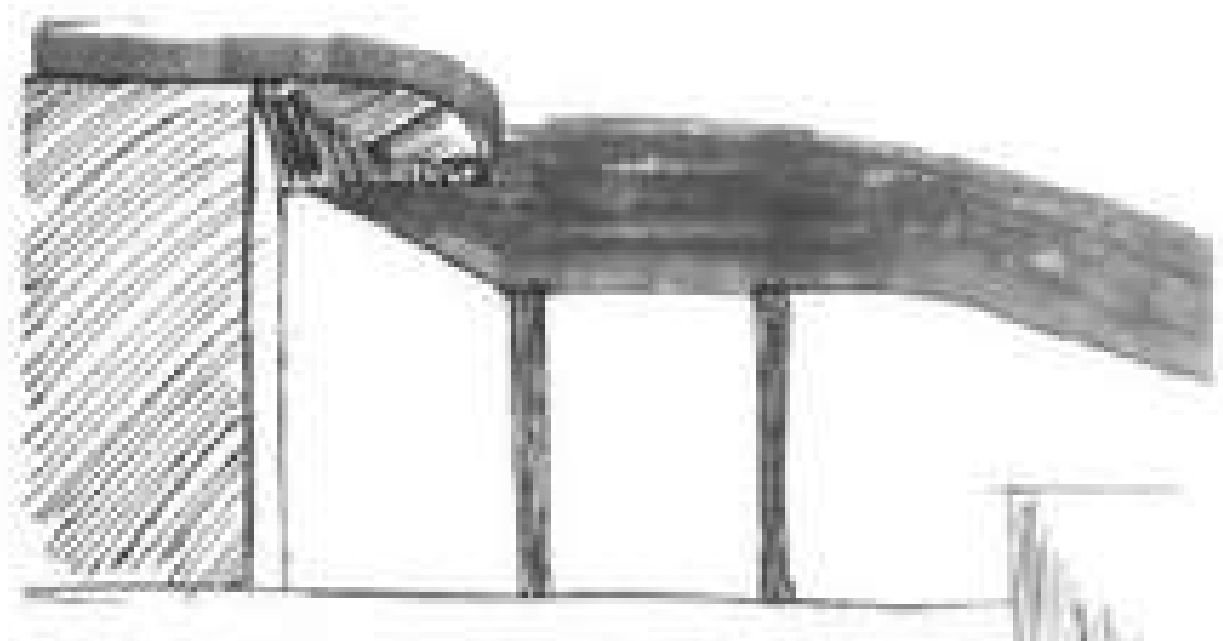
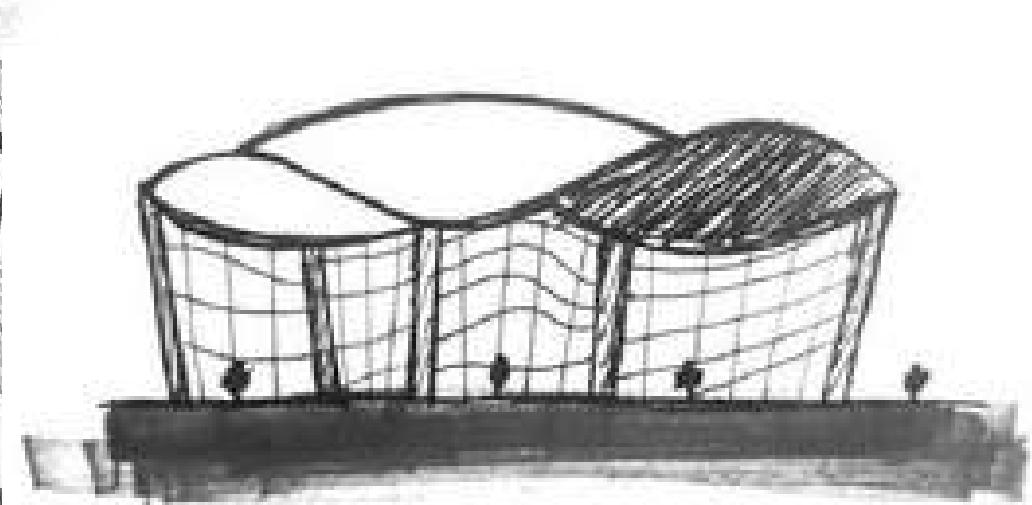
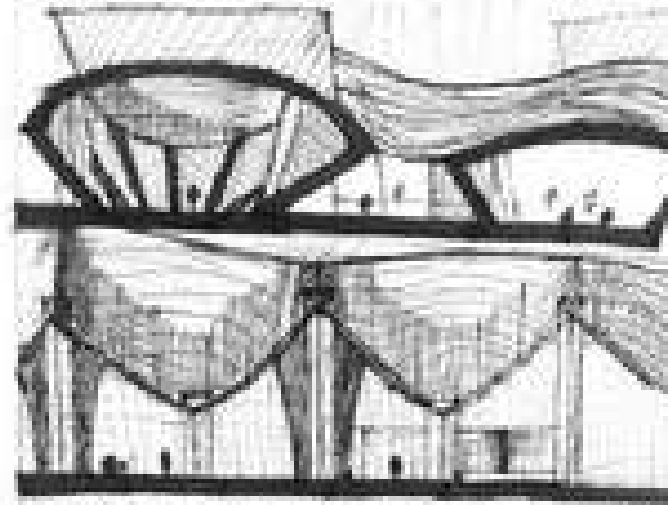
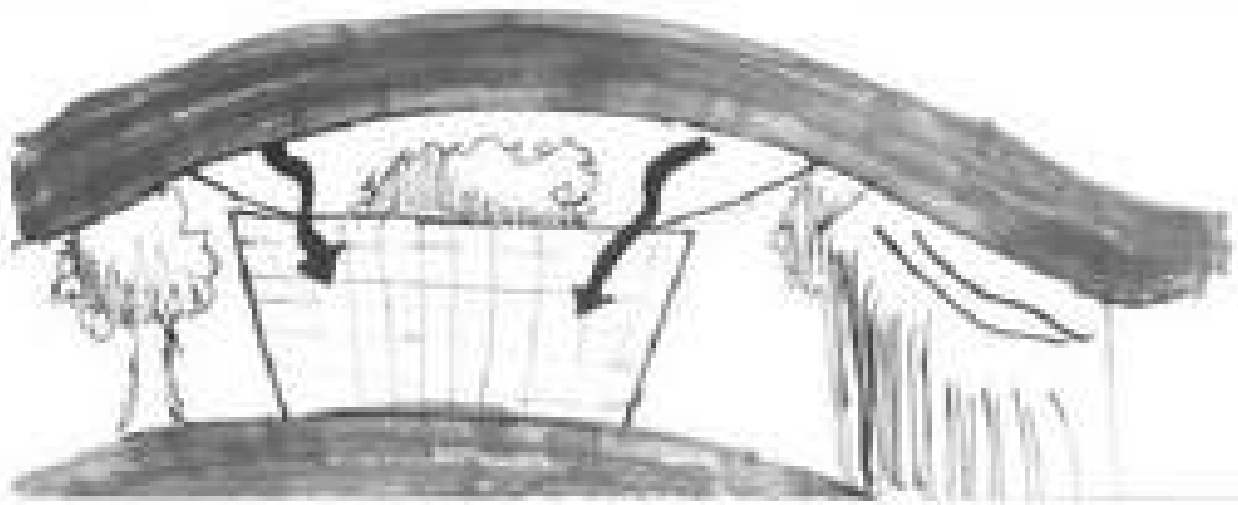


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Pedestrian path 
 Vehicle path 
 Vegetation 
 Landmark 
 Installation site 
 Waterbody 

MARINE DRIVE



Precious Elaish Nathan
AR23046

THEMATIC SPACES IN INDIAN ARCHITECTURE

PAVILLIONS



COURTYARDS



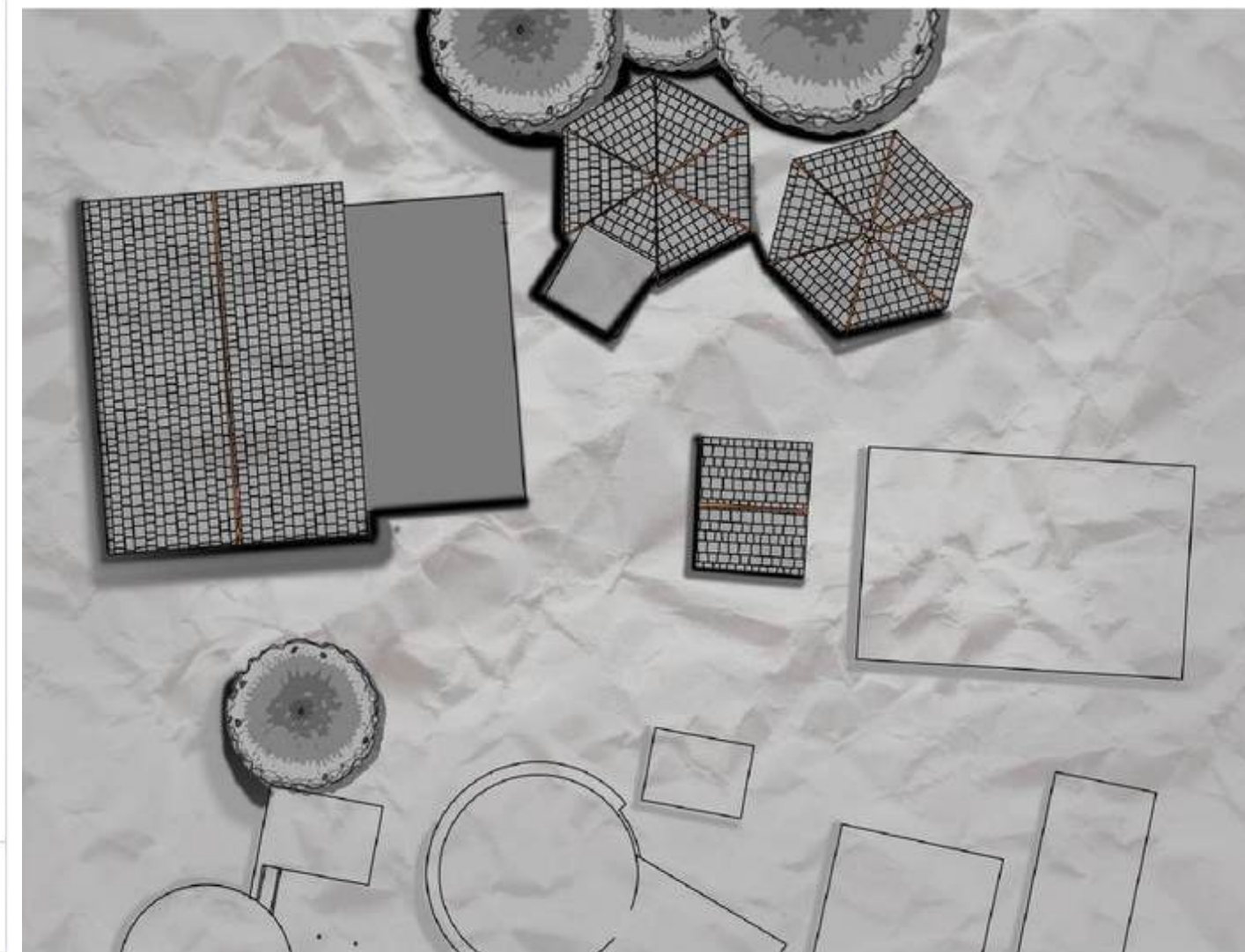
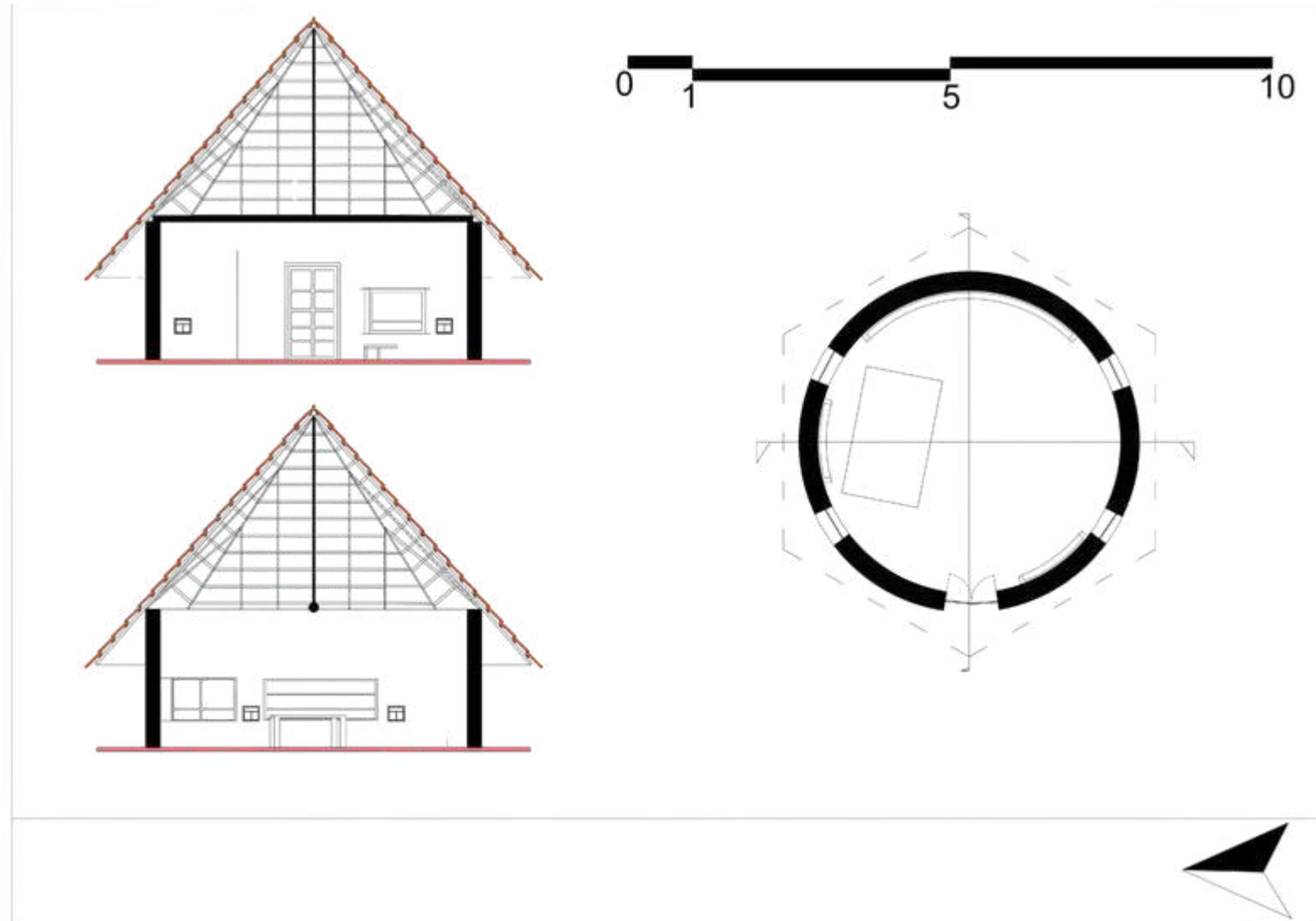
INFORMATION REFERENCE: "THEMATIC SPACES IN INDIAN ARCHITECTURE" BY KULBUSHAN JAIN

THE INDIAN EDUCATION SOCIETY'S GALLERY OF ARCHITECTURE

THEMATIC SPACES IN INDIAN ARCHITECTURE

STUDENT NAME: PRECIOUS ELAISH NATHAN

WORKSHOPS



EARTHQUAKE RESILIENT HOUSE TYPOLOGY
SOLUTIONS AND ANALYSIS OF BUILDINGS IN
SIKKIM

Precious Elaish Nathan¹, Om Maharana², Sahil Tambe³

¹ Precious Elaish, 2nd year B.Arch, student at VES College of Architecture

² Om, 3rd year B.Arch, student at VES College of Architecture

³ Sahil, 3rd year B.Arch, student at VES College of Architecture

Abstract

The 2011 Sikkim earthquake was a powerful shock-wave that badly damaged buildings and other structures. This study looks at how well different types of buildings in Sikkim can withstand earthquakes and give more ways to make them stronger for future earthquakes. The study found that poor building quality, weak earthquake design, and the specific terrain of the area made buildings more vulnerable. By looking at how existing buildings survived in the 2011 earthquake, it can be identified that areas where building design and construction did not get along, can be improved. The study comes up with using stronger building materials and techniques, like reinforced concrete, and strengthening older buildings. It also focuses the need to understand the specific earthquake risks in different parts of Sikkim and create stricter building construction and design necessities to ensure future buildings are safe and can withstand earthquakes.

Keywords: Earthquakes, Seismicity, Fault-lines, Shockwaves, House Typology, Tectonic Plates, Shing Khim, Ikra, Lepcha, RCC framed, Random Rubble, Tensile, Organic, Bamboo, Wood

1. Introduction

Earthquakes are one of the most devastating natural disasters that has cause immense havoc on the face of the Earth. An earthquake is a violent and abrupt shaking of the ground, caused by movement between tectonic plates along a fault line in the earth's crust. Earthquakes can result in the ground shaking, soil liquefaction, landslides, fissures, avalanches, fires and tsunamis (Ali, 2023). These groundbreaking after events due to these earthquakes are also witnessed and documented in India. Regions such as Gujarat and North-East India are earthquake prone areas (Santanu Baruah, 2018). Sikkim is one such region, which as suffered from severe damages due these constant earthquakes. The 2011 Sikkim earthquake, a magnitude of 6.9, caused significant damage to infrastructure and buildings in the region. This study investigates the seismic vulnerability of various building typologies in Sikkim and proposes design solutions to enhance their resilience against future earthquakes (Ghosh, 2001). The research analyses the factors contributing to building damage, including poor construction practices, inadequate seismic design standards, and site-specific geological conditions. By examining the seismic performance of existing structures and the impact of the 2011 earthquake, the study identifies areas for improvement in building typology (Dutta, Mukhopadhyay, Saha, & Nayak, 2015).

Given below is the table showcasing a few major earthquakes that have occurred in India:

No.	Region	Date & Time	Magnitude	Depth	Casualties	Cause	Reason of failure	Remedial action
1	Gujarat (Bhuj, Ahmedabad, Rajkot)	2/01/2001	7.7	40 km	2000	shallow strike slip fault within the Indian Plate	reinforcement and columns were not adequately even for gravity loads	IV
2	Sikkim	9/18/2011	6.9	10 km	111	Structural failure due to poor construction and landslide	inferior quality of raw materials and shoddy workmanship	IV
3	Kashmir (Srinagar, Baramulla, Anantnag)	5/5/2013	5.7	18 km	8	lies in the convergence zone of the Indian and Eurasian tectonic plates		V
4	Imphal, Manipur	1/10/2018	6.7	27 km	13		poor compliance with seismic codes, inferior quality of raw materials and shoddy workmanship	V
5	West Bengal	9/12/2018	5.3	10 km	1	located in a seismically active region, primarily due to the Indo-Burma plate		IV
6	Assam	6/28/2012	8	34 km	2	collision between the Indian Plate and the Eurasian Plate	highland and uncontrolled high urban population combined with poor quality and ill-maintained infrastructure	V

Table 1. Major Earthquakes in India

The map below shows earthquake mapped in different parts of India:

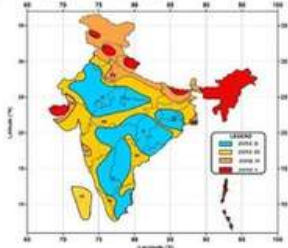


Figure 2. Zones with higher numbers indicate a higher potential for earthquakes. (Singh, 2021)

2.4 SCENARIO OF EARTHQUAKES IN SIKKIM: An earthquake of magnitude 6.8 occurred on 18th of September, 2011 at 18:11 hrs IST in Sikkim - Nepal Border region. This region has experienced moderate seismicity in the past, with 18 earthquakes of magnitude 5 or greater being experienced over the previous 35 years within 100 km of the epicenter of the September 18th event. The largest of these was magnitude 6.1 earthquake of 19th November, 1980 that had its epicenter around 75 km to the southeast of the 18th September event **Invalid source specified..** The preliminary hypo-central parameters of the 18th September, 2011 earthquake, as estimated by the Seismic Monitoring Network of India Meteorological Department (IMD) are as given below:

DATE AND TIME OF OCCURRENCE	MAGNITUDE	FOCAL DEPTH	EPICENTRE LATITUDE AND LONGITUDE	REGION
18/09/2011 at 18:11 hrs (IST)	6.8	10 KM	27° 42' N & 88° 12' E	Sikkim-Nepal Border Region

Table 2. Parameters of Earthquake occurrence at Sikkim-Nepal border (Rautela, 2012)

The region of Sikkim comes under in Zone IV of the Seismic Zone Map of India (IS1893, 2002). Sikkim and surrounding areas are situated in the active Alpine-Himalayan seismic belt, a region historically marked by four

The proposed design strategies include the use of earthquake-resistant construction techniques, such as reinforced concrete frames and shear walls, and the implementation of seismic retrofitting techniques for existing structures. Additionally, the study emphasizes the importance of site-specific seismic hazard assessments and the development of appropriate building codes and standards to ensure the safety and resilience of buildings in Sikkim.

2. Literature Review

2.1 EARTHQUAKES: Earthquake, it's a sudden and natural and powerful natural event. It can have major consequences, affecting human life, infrastructures, and also, it's national economies. The intensity of an earthquake's impact is directly linked to its magnitude. Most earthquakes, being of small to moderate magnitude, have minimal impact on built structures. These quakes often originate deep beneath the Earth's surface, at depths of around 800 kilometers, and their effects diminish with increasing distance from the epicenter. (Oluwafemi, Oluwatan, Oikunle, & Abolarin, 2018). The location from where the earthquake starts below the earth's surface is called Hypocenter, and the location which is directly above it is known as Epicenter. The most common ones are the tectonic earthquakes. Even though the earth's surface looks much stable, there's lot of Activity going beneath. The earth is basically made of four layer's (generally three), these are : a solid crust, a nearly solid , a liquid outer core and a solid inner Core A, a special class of **tectonic earthquake** is sometimes recognised as volcanic earthquake. However, these are confined to areas of active volcanoes. Earthquakes produced by stress changes in solid rock due to the injection or withdrawal of magma (molten rock) are called volcano earthquakes. These earthquakes can cause land to subside and can produce large ground cracks. These earthquakes can occur as rock is moving to fill in spaces where magma is no longer present. Volcano-tectonic earthquakes don't indicate that the volcano will be erupting but can occur at any time. In the areas of intense mining **activity**, sometimes the roofs of underground mines collapse causing minor tremors. These are called **collapse earthquakes**. Ground shaking may also occur due to the explosion of chemical or nuclear devices. Such tremors are called **explosion earthquakes**. The earthquakes that occur in the areas of large reservoirs are referred to as reservoir **induced earthquakes**. Invalid source specified.

2.2 SCENARIO OF EARTHQUAKES AROUND THE WORLD: Earthquakes are so devastating that it has left cities all around the world in shambles. Some examples of earthquakes that have been so devastating that it took everything for the governments of these nations to completely restore the affected cities are as follows:

- May 22, 1960 Chile Earthquake:** The most horribly impacting and devastating earthquake in the world was recorded in Chile on the 22nd of May, 1960. The earthquake had a magnitude value on the Richter scale to be 9.5. (Oluwafemi, Oluwatan, Oikunle, & Abolarin, 2018)
- The 1964 Alaska earthquake:** The earthquake of 1964 that occurred in Alaska is a very rare event in the history of world's earthquakes. (Oluwafemi, Oluwatan, Oikunle, & Abolarin, 2018).
- The Tangshan Earthquake:** On the 28th of July, 1976, a deadly earthquake referred to as the Tangshan earthquake, hit China in the province of Hebei. The event occurred early in the morning at about 3:42 am and it was measured to be 7.8 magnitude. The event was rated the deadliest earthquake. (Oluwafemi, Oluwatan, Oikunle, & Abolarin, 2018).

Earthquakes have caused widespread destruction to cities prone over the years. Between 1998-2017, earthquakes caused nearly 750 000 deaths globally, more than half of all deaths related to natural disasters. More than 125 million people were affected by earthquakes during this time period, meaning they were injured, made homeless, displaced or evacuated during the emergency phase of the disaster. (Ali, 2023).

significant earthquakes of magnitude 8.0 or greater. These seismic events are mainly triggered by the accumulation and sudden release of shock resulting from the northward tectonic motion of the Indian Plate against the Eurasian plate (Rautela, 2012). The National Geophysical Research Institute (NGRI) and the Indian Institute of Technology Kharagpur have been looking into the seismicity of Sikkim in the past recent times using local seismic networks, which have now begun to be evident again. (Rajendran, Rajendran, Natrajan, & Andrews, 2020). The shocks are generally released by activity along Himalayan faultlines and thrusts of regional areas of which Main Boundary Thrust (MBT) and Main Central Thrust (MCT) are important. Other important geological features in Sikkim include Tista lineament, Kunchenjunga lineament, Purnea - Everest lineament, Arun lineament and Dhubri fault in the southeast. Because of these seismic activities, Sikkim has faced devastating after-events after the earthquake of 2011. The earthquake caused a large number of landslides and triggered significant damage to buildings and infrastructure. Above that, there were three aftershocks 5.7, 5.1 and 4.6 on the Richter scale that were experienced after 75 minutes of the earthquake. The landslides, rock falls and mud slides were mainly responsible for the loss of life and damage to property. The total loss has been estimated at around US \$500 million. The state government opened 85 relief camps and carried out several rescue operations (Siakia, et al., 2013).



Figure 3. Faultlines of Sikkim (Tiawari, Tiawari, Jaishi, & Singh)

The map given below depicts the affected villages of Sikkim due to the earthquake:

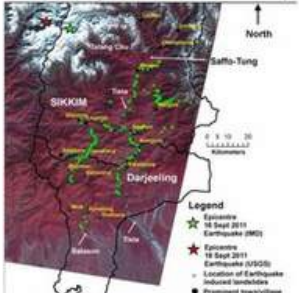


Figure 4. Impact of the 2011 Sikkim Earthquake: Landslide Distribution (Siakia, et al., 2013)

Given below is a mapping of earthquakes occurring around the world:

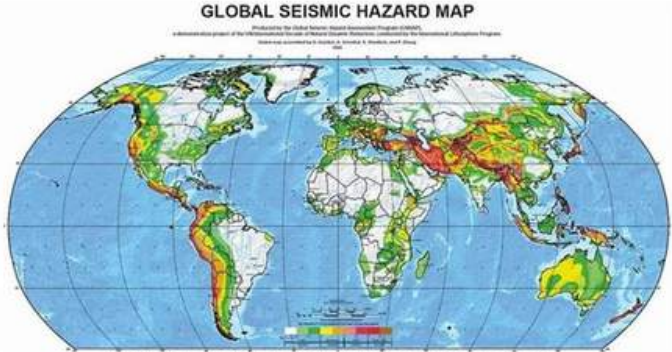


Figure 1. Global Seismic Hazard Map: Identifying Earthquake Risk Zones (Alden, 2024)

Even India has shown instances of susceptibility to destruction caused by earthquakes that occur in certain seismic zones.

2.3 SCENARIO OF EARTHQUAKES IN INDIA: Over the years, India has faced a huge amounts of loss due to frequent earthquakes across many states, few regions being **Bhuj in Gujarat, The Seven Sister States of North East India, Ladakh, Koyananagar in Maharashtra, and Gangtok in Sikkim**.

- The Bhuj earthquake of January 26, 2001** took place at 8:46 am IST and had a Richter scale magnitude reading of 7.7 (United States Geological Survey). Out of a population of 1,60,000, casualties have ranged from 1,00,000 to 1,50,000. (Ghosh, Observations From the Bhuj Earthquake of January 26, 2001, 2001).
- The North Eastern Region (NER) of India**, located in the Eastern Himalayas, is one of the most seismically active regions in the world (Das et al.,2012) and is placed in seismic zone V (highest-risk zone; except Sikkim) as per seismic zonation map of India (IS-1893, 2016). The seven sister states, especially Assam, still undergo frequent earthquakes of relatively low magnitudes ranging from 3-4 on the Richter scale. (Agrawal, Gupta, Dixit, & Kumar, 2022).
- An earthquake of magnitude M 5.5 occurred at 15:48:53 IST located in the **Zaskar, District Kargil, Ladakh**. The area is seismically very active associated with collision of tectonics where Indian plate collided with the Eurasian Plate. Following the main shock, four aftershocks ranging from M 3.8 to 4.8 were also reported. (Government of India, 2023).
- The Koyna region** is a highly active seismic zone. The region has very severe earthquakes continue to occur there four decades after the initial spurt in activity **Invalid source specified..** The earthquake occurred in this region in 1967 was the largest known earthquake in western India at the time with a magnitude of 6.3 on the Richter scale (Yadav, Bansal, & Pandey, 2016).
- The 18 September 2011, magnitude *Mw* 6.9 earthquake close to the **Nepal-Sikkim border** caused significant damage due to ground shaking and caused several landslides.

Many houses, mostly modern buildings (RCC framed structures), were completely destroyed due to the earthquake, but, the traditional houses of Sikkim actually survived with very minimal to no damages. The different types of structure constructed in Sikkim before the earthquake are:

- Shing-Khim Houses
- Ikra Houses (Assamese style)
- Random Rubble masonry house
- RCC Framed structures
- Lepcha House

These houses have unique structural integrity, out of which Shing-Khim, Ikra, Random Rubble masonry, and Lepcha houses have seen to have had better resistance to earthquakes than RCC structures **Invalid source specified..**

3. MATERIALS AND METHODS

Sikkim, located in northeastern India, is surrounded by the towering Himalayan mountains. It shares borders with Tibet, Bhutan, Nepal, and West Bengal, placing it in a seismically active region classified as Zone V. The region's seismic activity is primarily caused by the ongoing collision between the Indian Plate and the Eurasian Plate. One of the most devastating earthquakes to hit the region was the 2011 Sikkim Earthquake, which registered a magnitude of 6.9 on the Richter scale. The tremors were felt across neighbouring countries, including Nepal, Bhutan, and Bangladesh. The earthquake caused extensive damage to infrastructure, such as roads, bridges, and buildings, while landslides triggered by the quake worsened the situation by blocking roads and delaying rescue efforts. The disaster resulted in around 111 casualties. The seismic activity in this area is primarily attributed to thrust faults caused by the compressional forces of the plate collision (Rai, Mondal, Parool, & Pradhan, 2012). Building failures were a major contributor to the disaster's impact in Sikkim. The region features three primary types of construction practices: RC frame structures with infill walls, Shing-Khim houses, and Ikra houses (Rai, Mondal, Parool, & Pradhan, 2012). RC frame buildings with masonry infill are common in urban areas and typically do not exceed three stories. Shing-Khim houses are traditional single-story wooden structures made from planks connected with steel bolts and nails. These houses are supported on wooden posts or, in some cases, RC columns or masonry walls, which also serve as thermal insulation. Ikra houses are one- or two-story timber-framed structures with panels made of woven bamboo splints reinforced with cement or mud plaster. Both Shing-Khim and Ikra houses are often equipped with pitched roofs made of asbestos sheets or corrugated galvanized iron (CGI) sheets, designed to withstand the region's heavy rainfall, which averages around 3000 mm annually. These construction styles, though traditional and adapted to local materials, were especially vulnerable during the earthquake, with significant structural failures observed, particularly in the northern districts. With the help of this study we will be recommending design solution to reduce the risk of failure of these structures (Rai, Mondal, Parool, & Pradhan, 2012). It is observed that, the traditional houses of Sikkim, namely, Shing-Khim houses, Ikra houses, and Lepcha houses have retained their structural integrity after the massive shockwaves caused by the earthquake, and also the 3 after-shocks which occurred after 75 minutes when the earthquake ended.

Even Random rubble stone masonry houses were moderately able to retain structural integrity, but not as much as the traditional houses. RCC framed structures failed miserably in retaining their structural integrity when in comparison to the traditional houses, which suggests that traditional methods of construction which are developed over the years are preferred more than modern RCC framed buildings. Traditional houses of Sikkim are constructed using organic materials such as wood and bamboo. Since wood and bamboo are very tensile materials, it is best suited for building construction in earthquake prone areas

Following is the detailed review of each house typology which actually resisted the shockwave of the earthquake of 2011:

1). **Shing-Khim house:** It is one of the most common type tradition house typology found in Sikkim. It has a **stone foundation**, which exceeds the ground level 1-2 meters above. **Wooden planks** are set on the foundations to add more support. **Walls** are constructed using wood and **columns** are made of bamboo, to provide more tensile strength to the entire structure of the house. **Beams** are made of bamboo and structurally integrated into columns.



Figure 5. A Shing Khim house of Sikkim



Figure 6. Stone foundations used in Shing Khim



Figure 6. Bamboo beam-column joinery



Figure 11. Piling foundations (Piling Foundation Services)



Figure 12. R.C.C. Columns
(Piling Foundation Services)



Figure 13. R.C.C. Beams
(Piling Foundation Services)



Figure 14. R.C.C. Slabs (Piling Foundation Services)

2). **Ikra house:** Ikra houses are another type of common houses found in Sikkim that survived the earthquake of 2011. The typology of the house is actually inspired by the Bhutia houses of Assam, and it didn't fail due to the similar seismic situation of Assam.

The **foundation** consists of stone rubble masonry foundation very similar to Shing Khim house typology, **wooden wall framework**, brick masonry **walls**, and bamboo **beams and columns**. **Thatched roofs** are common in Ikra houses as well.



Figure 7. A standard Ikra house of Sikkim

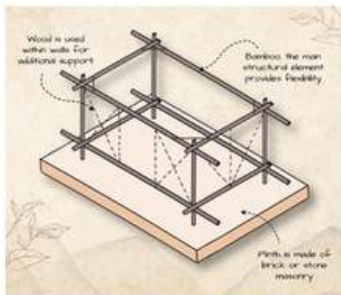


Figure 8. Wooden framework used in walls
(Chougule S.)

4). **Random Rubble Stone Masonry house:** Random rubble construction is one of the most prominent type of construction that could be seen in Sikkim. The stone masonry is known for its durability and longevity. These stone walls can withstand earthquakes to some extent, not as efficiently as Ikra construction or Shing-Khim construction (Schildkamp & Araki).

Its foundation is made of typically using stone and construction, plinth often raised to protect from moisture and pests. Walls are made of locally available stones of different sizes laid in courses in leveled to ensure stability bonded using lime, mud or mortar. For roofs, Thatched roofs are made using locally available materials like grass and bamboos. (Schildkamp & Araki).



Figure 15. Foundations of random rubble
stone masonry (Schildkamp & Araki)



Figure 17. Random rubble stone
masonry (Schildkamp & Araki)



Figure 16. Thatch roofs of random rubble houses
(Schildkamp & Araki)

5). **Lepcha houses:** Lepcha house which is also known as Dokemoo Lee is a traditional house for the Lepcha people. It's an indigenous tribe which are inhabitation the eastern Himalayan region The house is constructed using wood, bamboo, mud and thatch, The north-south side is larger compared to east-west side adding to its distinctive design (Chettri)

The Kaden long foundation, made of molded stones, elevates the wooden pillar structure, kaodampu, to protect against wildlife and accommodate livestock, with the number of pillars varying from eight to eighteen.

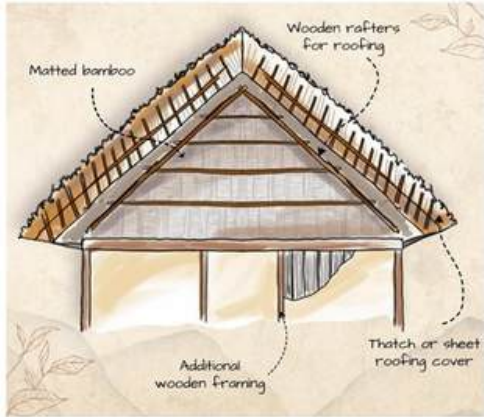


Figure 9. Thatch roof system in Ikra houses
(Chougule S.)

3). **R.C.C. Houses:** In contemporary urban settings, it is common to see buildings constructed with a reinforced concrete structural frame and filled-in spaces between the frame members with masonry materials like bricks or blocks (Rai, Mondal, Parool, & Pradhan, 2012).

R.C.C. pile foundations, R.C.C. wall frames, brick walls, and R.C.C. beams, columns, and cantilever slabs sums up the specifications of a typical R.C.C. framed building, constructed before the Sikkim earthquake. These buildings lacked structural integrity because of less design expertise in Sikkim. Only 35% of building in Sikkim actually had expert supervision during its construction (Vijaya Narayan, Goswami, & C.V.R., 2012). Even materials were not of standard quality, because of which, the buildings couldn't withstand the shockwaves of the earthquake (Rai, Mondal, Parool, & Pradhan, 2012).



Figure 10. R.C.C. Buildings in Sikkim (Vijaya Narayan, Goswami, & C.V.R., 2012)

Initially, walls were constructed using grass, mud, bamboo, or wooden panels. Later, cow dung mixed with red soil and bamboo weaving techniques were adopted to create thicker, more durable wall. Initially, bamboo frames were covered with long grass to form the roof, particularly suitable for areas with moderate temperatures and lower altitudes (Chettri).



Figure 17. Typical Lepcha house of Sikkim (Chettri)



Figure 18. Walls of Lepcha House
(Chettri)



Figure 21. Foundations of Lepcha House
(Chettri)



Figure 19. Wooden Framework in Lepcha house (Chettri)

4 ANALYSIS

The following table shows data about different structural elements of the different house typologies mapped in the review:

HOUSE TYPOLOGY	Structural					STATUS GOOD
	FOUNDATION	WALLS	ROOF	SLABS	BOARDS	
BRICK/STONE HOUSE	WOODEN POST SET ON STONE MASONRY PLATFOM	WOOD WALL AND BAMBOO COLUMNS	THATCHED ROOF	NA	IT IS A GOOD TYPE OF CONSTRUCTION STYLE FOR SIKKIM	
SHING-KHIM	BRICK OR STONE FOUNDATION	BAMBOO COLUMN WITH MASONRY WALL	CORRUGATED METAL SHEETS SUPPORTED BY BAMBOO FRAMEWORK	NA	IT IS A GOOD TYPE OF CONSTRUCTION STYLE FOR SIKKIM	
RANDOM RUBBLE MASONRY	RANDOM RUBBLE FOUNDATIONS	RANDOM RUBBLE PLASTERED BRICK MASONRY	SLATE ROOFS	NA	IT IS MODERATELY GOOD, DEPENDS ON WORKMANSHIP	
RCC FRAMED BUILDING	RCC FOUNDATIONS	CONCRETE	PLASTERED BRICK MASONRY	CONCRETE	FLAT CONCRETE FILLER SLAB	IT IS NOT A GOOD TYPE OF CONSTRUCTION
	KADEN LONG STONE FOUNDATION	KADEN LONG PLASTERED	WOOD WALL AND BAMBOO COLUMNS	NA	THATCHED ROOF	IT IS A GOOD TYPE OF CONSTRUCTION STYLE FOR SIKKIM

Table 3. Table comparing different house typologies and their structural elements

The 2011 Sikkim earthquake underscored the critical importance of earthquake-resistant building design and construction. While the region experienced significant damage, certain house typologies demonstrated remarkable resilience.

4.1 Key Observations:

- Traditional Houses Outperformed Modern Structures:
Shing-Khim and Ikra houses: These traditional wooden structures, characterized by their flexible and lightweight construction, exhibited exceptional resilience. Their ability to absorb seismic energy and distribute forces effectively contributed to their survival.
Random Rubble Masonry Houses: While not as resilient as traditional wooden houses, these structures, constructed with locally sourced stone and mortar, demonstrated moderate resistance to seismic forces.
- Modern RCC Framed Structures:

Vulnerability to Seismic Forces: These structures, despite being constructed with modern materials and techniques, were susceptible to damage due to factors such as poor construction quality, inadequate seismic design, and the specific geological conditions of the region.
Need for Improved Design and Construction Practices: To enhance the seismic performance of such structures, it is crucial to adhere to stringent building codes, employ quality materials, and engage experienced engineers and contractors.
Factors Contributing to the Resilience of Traditional Houses:
Flexible and Lightweight Construction: The use of wood and bamboo, which are flexible materials, allows these structures to absorb seismic energy and deform elastically, reducing the risk of collapse.
Primitive Knowledge and Experience: Traditional building techniques, developed over centuries, have incorporated knowledge of local seismic conditions and materials, resulting in structures that are well-suited to the region's seismic environment.
Site-Specific Design: Traditional houses are often designed to harmonize with the local topography and climate, further enhancing their resilience.

5 FINDINGS

The table given below, shows the observations and findings of each house typology in Sikkim:

HOUSE TYPE	STRENGTH	WEAKNESS	SUITABLE SOIL
Lepcha Houses	Adaptability to Terrain, Natural Materials, Flexibility, Low Maintenance	Fire Risk, Limited Load-Bearing Capacity, Susceptibility to Pests	Loamy soil
Shing-Khim Houses	Durability, Earthquake Resistance, Insulation, Good Ventilation	Construction Complexity, Material Availability, Fire Risk	clayey soil
Lepcha Houses	Lightweight Construction, Flexibility, Adaptability, Low Cost, Environmental Friendliness, Cultural Significance	Durability, Fire Risk, Limited Load-Bearing Capacity, Susceptibility to Pests, Requires Skilled Craftsmanship	Loamy soil
Random Rubble Masonry Houses	Durability, Stability	Weight, Cracking, Irregularity	Silty soil
RCC Framed Structures	High Strength and Stiffness, Ductility, Fire Resistance	More expensive compared to other traditional materials	Loamy soil

Table 4. Table depicting findings of each house typology

It can be certain after discussion that:
Lepcha Houses are well-adapted to the terrain and use natural materials, but are also vulnerable to pest infestations. They are well suited for hilly regions.
Shing-Khim Houses offer good insulation and earthquake resistance but are complex to construct and require specific materials. Can be used in various environments but may require foundation adjustments based on soil conditions.
RCC Houses are versatile and can be used in diverse environments, but require extremely high level of design expertise, workmanship, and supervision, especially in earthquake prone regions such as Sikkim.
Random Rubble Masonry Houses, though tough and durable, they pose to have heavy structural integrity. In general, they are moderately good for surviving earthquakes, due to their structural bulk.

Ikra Houses were originally staple to Assam, which comes under seismic zone 5. These houses were developed for generations in Assam, where the builders kept seismicity in mind. But these houses can only survive in few soil types.

6 CONCLUSIONS

From the above information, regarding Ikra houses, Lepcha houses, Shing-Khim houses, Random Rubble Stone Masonry and RCC framed structure, we can conclude that Ikra, Shing Khim, Random Rubble Stone Masonry and Lepcha Houses can survive earthquake attacks more efficiently than modern building methods. Considering different land typologies, north Sikkim consists of mainly mountains and the soil is mostly loamy. Because of these reasons, constructions of Lepcha and Shing- Khim houses are preferable.

In the west side of Sikkim sandy loam soil is very common as well as some other types of soil that contains acidic properties. That is why Lepcha and random rubble houses are preferable.

In south Sikkim, silty soil is common. That is why Ikra houses are preferable.
In east side of Sikkim, silty clay soil is common. that is why Ikra and shing khim houses are preferable. Since the east side of Sikkim is the farthest away from the Indian- Eurasian fault, RCC buildings can also be constructed provided that maximum supervision and workmanship is ensured.

Sikkim, located in northeastern India, is surrounded by Himalayan mountains and shares borders with Tibet, Bhutan, Nepal, and West Bengal., Sikkim Comes under seismically active region classified as Zone IV. The region's seismic activity is caused by the ongoing collision between the Indian and Eurasian Plates. due to this earthquakes are a major issue in Sikkim. One of the most destructive Earthquake was 2011 earthquake a magnitude 6.9 event, had devastating consequences for the region, causing widespread damage to infrastructure and leading to over 100 casualties. The collapse of numerous buildings, due to poor construction quality and inadequate seismic design, significantly increase the impact of the earthquake. The region's construction practices primarily involve RC frame structures with infill walls, Shing-Khim houses, and Ikra houses
The figure given below shows the map of Sikkim and the various house typologies that can be constructed in various regions across Sikkim:



Figure 20. House typologies in various regions of Sikkim

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QUANTUM GEOGRAPHIC INFORMATION SYSTEM

INTRODUCTION TO GIS:

GIS is a computer system that can capture, store, analyze, and display geographic information. It allows us to visualize, question, analyze, and interpret data to understand relationships, patterns, and trends. Imagine you have a map of your city. Now, imagine you could add layers of information on top of that map: where the schools are, where the hospitals are, where the parks are, and so on. This is essentially what a Geographic Information System (GIS) does. Examples of GIS software include ArcGIS, QGIS, and Google Earth.

IMPORTANCE OF GIS SOFTWARES:

GIS software is used in the analysis of certain regions where traveling to a particular area is not feasible multiple times. Predictive modeling by combining geographic data with statistical models. We can predict future trends, such as population growth or natural disaster risks. GIS software also automates many tasks, saving time and reducing errors. The accuracy of GIS data is crucial for decision-making. GIS software is used in various fields, including urban planning, environmental management, and disaster response.

TYPES OF GIS DATA:

Two types of GIS data are vector data and raster data. Vector data represents geographic features as points, lines, and polygons. Each feature has a specific location and can be associated with attributes like name, population, or land use. Raster data represents geographic features as a grid of cells, each with a specific value. The value can represent elevation, temperature, land cover, or other attributes.

COORDINATE REFERENCE SYSTEMS AND PROJECTIONS:

A coordinate reference system (CRS) is a framework that defines the position of geographic features on the Earth's surface. It's like a grid system that allows us to locate specific points on a map. Using the correct CRS ensures accurate measurements of distances, areas, and directions. Different CRS are used for different purposes. Some projections preserve area (e.g., Albers Equal Area), while others distort it. Some projections preserve shape (e.g., Mercator), while others distort it. Some projections preserve direction (e.g., UTM), while others distort it.

ATTRIBUTE TABLE AND FIELD CALCULATIONS:

ID	type	areas
1	commercial	5431
2	commercial	3777
3	commercial	2009
4	commercial	60289
5	commercial	17328

QUICK OSM:

OSM plugins are additional tools and functionalities that can be added to QGIS. They are developed by the QGIS community and offer a wide range of specialized tools for various GIS tasks. Plugins can be installed through the Plugin Manager within QGIS, allowing users to customize their QGIS experience and tailor it to specific needs. Examples include QuickOSM and Processing SAGA.

EXPORTING LAYERS AS DIFFERENT FORMATS:

To export a layer in QGIS to a different format, follow these steps: 1. Right-click on the layer in the Layers panel. 2. Select 'Export' and then 'Save Features As...'. 3. In the 'Save Features As' dialog box, choose the desired format. 4. Select any additional options that can store multiple layers in a single file.

EXPORTING MAPS:

Once you've finished your map design and analysis in QGIS, the final step is to export the map to a format that can be used in other applications. The 'Export' menu in QGIS provides several options for saving your map. You can export the map as a PDF, a high-resolution image, or a web map. The 'Export' menu also allows you to save the map as a QGIS project file, which can be used to open the map in QGIS at a later date.

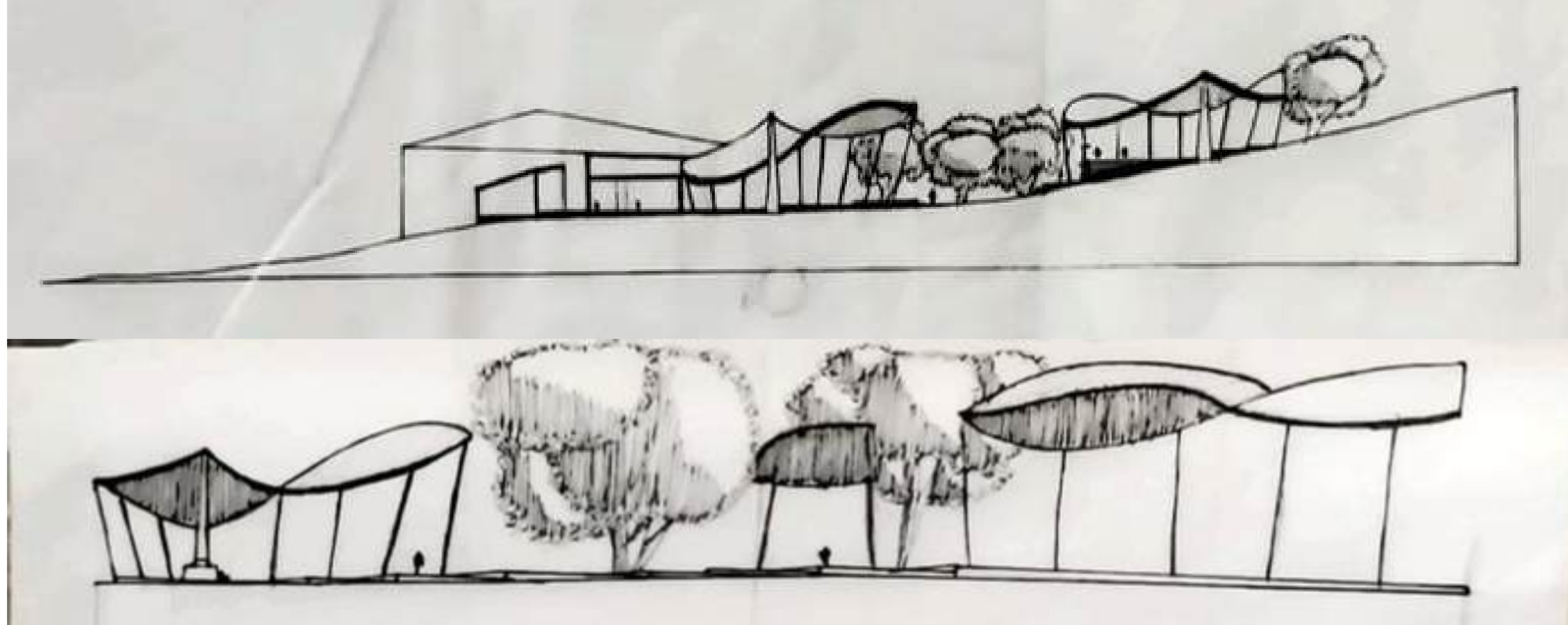
FINAL MAP LAYOUT

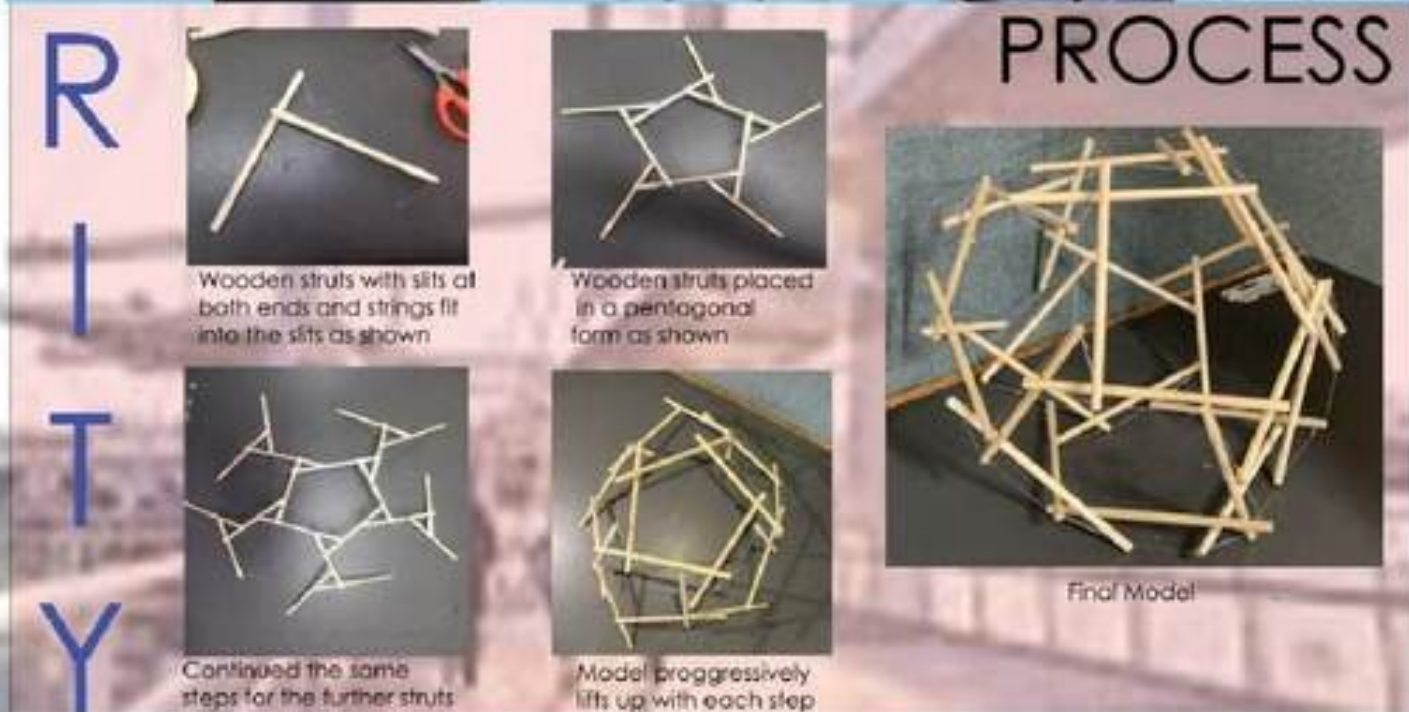
AY - 2024-25 | ARD WORKSHOP | QGIS (QUANTUM GEOGRAPHIC INFORMATION SYSTEM)

PRECIOUS ELASH NATHAN

AY - 2024-25 | ARD WORKSHOP | QGIS (QUANTUM GEOGRAPHIC INFORMATION SYSTEM)

PRECIOUS ELASH NATHAN





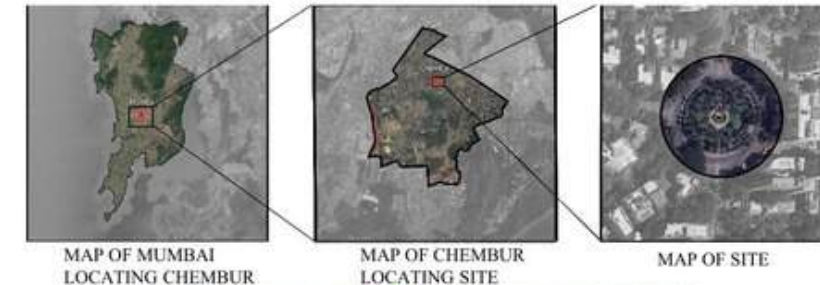
VIVEKANAND EDUCATION SOCIETY'S COLLEGE OF ARCHITECTURE

Barrier-free Public Spaces and Accessible Pathway Design

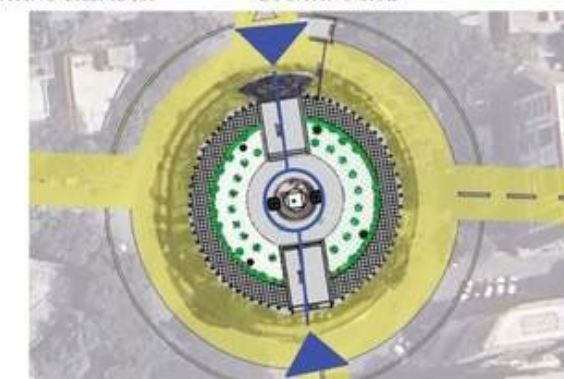
The site is located in Chembur, Mumbai. Ambedkar Garden is extremely visited by people paying respects to Dr. Babasaheb Ambedkar and is opened twice a year. Even though the site is extremely famous, it is still prone to some issues creating hurdles for persons with disabilities to access the site with ease.



YASHWANTRAO CHAVAN CENTRE
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- AMBEDKAR GARDEN, CHEMBUR SITE DETAILS:**
- OPEN ONLY FOR 2 DAYS IN A YEAR, 22 APRIL AND 6 DECEMBER.
 - OPEN FROM 8 AM TO 10 PM ON THESE DAYS.
 - 200-250 PEOPLE VISIT ON THESE DAYS.
 - PEOPLE VISIT FOR CELEBRATION OF WORKS DONE BY DR. B.R. AMBEDKAR AND DO POOJA IN HONOUR OF DR. AMBEDKAR.
 - ONLY MAINTENANCE STAFF ARE ALLOWED TO ENTER PARK THROUGHOUT THE YEAR.



AMBEDKAR GARDEN, CHEMBUR



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ACCESSIBILITY ASSESSMENT	RATING SCALE	VISUAL REPRESENTATION
HAZARDOUS, INACCESSIBLE AND UNSATISFACTORY	1	
INACCESSIBLE AND UNSATISFACTORY	2	
ACCESSIBLE AND UNSATISFACTORY	3	
ACCESSIBLE AND SATISFACTORY	4	
ACCESSIBLE AS BEST PRACTICE	5	

SITE ENTRANCE does the entrance have legible with clear signs are there TGSIs provided on walkway	Is map provided with braille/tactile features? Is the counter for security gate low height?
ACCESS ROUTES Is the minimum width of walkway (one-way traffic) 1200? Are there TGSIs provided on walkway?	Is the minimum width of walkway (two-way traffic) 1800? Are the level changes with contrasting color strips or distinct material change?
MAIN ENTRANCE Is the main entrance of the building accessible? Do the steps have a handrail?	Are there any steps at the entrance? Can the entrance door be operated independently?
DOOR Are there doors automated or assisted at the entrance?	Is the landing surface non slippery? Can the doors be operated without much effort?
STAIRCASE/STEPS Is the minimum width of the stairs 1500mm?	Are the height of the risers maximum of 150 mm & treads a minimum of 300 mm?



YASHWANTRAO CHAVAN CENTRE

ACCESSIBILITY AUDITS

SITE ENTRANCE	LEGIBLE CLEAR SIGNAGES SIGNAGES ARE DESCIPHERABLE	TGSI NO TGSIs PROVIDED	BRILLE/TACTILE FEATURES NO BRILLE/TACTILE FEATURES	LOW HEIGHT SECURITY GATE NO LOW HEIGHT GATE
ACCESS ROUTES	MIN 1200MM WIDTH WALKWAY (ONE WAY TRAFFIC) WALKWAY WIDTH MORE THAN IS 1200 MM	TGSI NO TGSIs PROVIDED	1800MM WIDTH WALKWAY (TWO WAY TRAFFIC) ACCESSIBLE WALKWAY	LEVEL CHANGE DEMARKATION LEVEL CHANGES ARE NOT DEMARKATED



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MAIN ENTRANCE	ACCESSIBLE STRUCTURE MAIN ENTRANCE IS NOT EASILY ACCESSIBLE	STEPS TO ENTRANCE THERE IS JUST ONE LEVEL DIFFERENCE AT ENTRANCE	HANDRAIL ON STEPS NO HANDRAILS	SLIPPERY LANDING SURFACE LANDING SURFACE IS SLIPPERY
KERBS/ RAMPS	RAMP AT ENTRANCE RAMP IS PRESENT	WALKWAY 1200MM WIDTH ACCESSIBLE WALKWAY	OBSTRUCTIONS OR CLEAR RAMP STAGNANT WATER AT ENTRANCE RAMPS	BRILLE ON HANDRAILS NO BRILLE ON HANDRAILS
DOORS	ASSISTED DOOR NO PROPER ASSISTANCE AT ENTRY	GLAZED DOORS NO GLAZED DOORS		



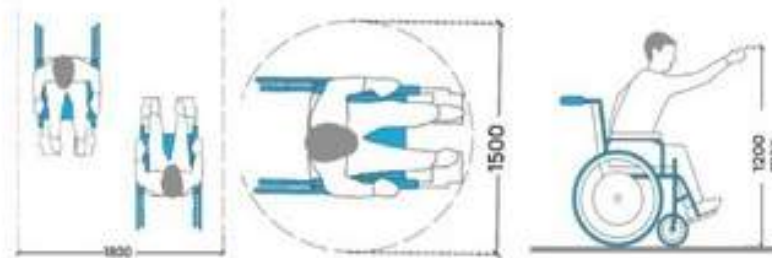
YASHWANTRAO CHAVAN CENTRE
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STAIRS/ STEPS	MINIMUM WIDTH 1500MM ACCESSIBLE PATHWAYS	MIN 150MM RISER AND 300MM TREAD NOT SATISFACTORY TREADS AND RISERS	ALTERNATIVE RAMP PROVISION NO ALTERNATIVE PROVISIONS	
SIGNAGE/ PARKING	SIGNAGES SIGNAGES NOT EVIDENT	ACCESSIBLE PARKING FACILITIES NO PROPER PARKING		
HANDRAILS	EASY GRIP TO HANDRAILS NO GRIP TO HANDRAILS	HANDRAILS GROUTED IN GROUND HANDRAILS NOT GROUTED	CONTRASTING COLOUR HANDRAIL NO CONTRASTING COLOURS	

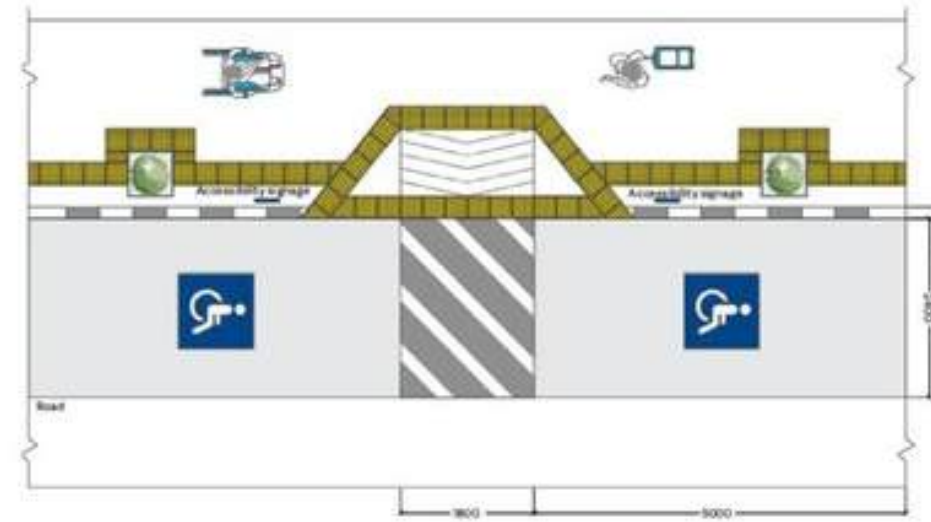


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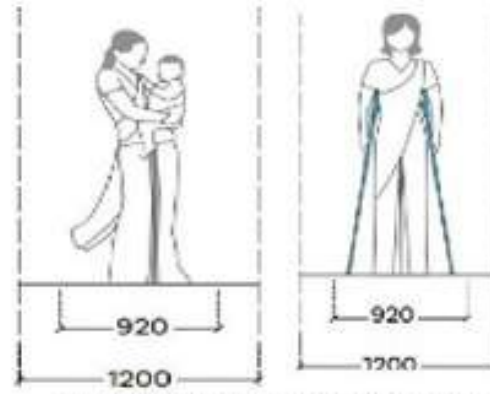
STANDARDS



ANTHROPOMETRY FOR WHEELCHAIR USERS



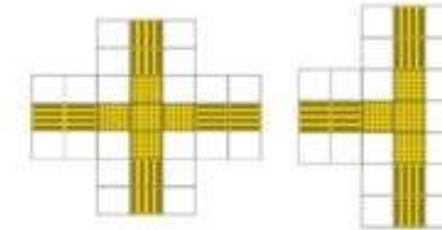
ACCESSIBILITY SIGNAGE FOR WHEELCHAIR USERS



ANTHROPOMETRY STUDY OF USERS



BRAILLE APPLICATIONS IN SIGNAGES



TSGI APPLICATIONS



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DESIGN PROPOSALS



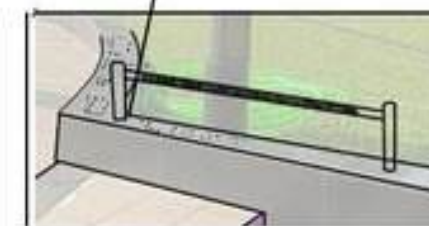
INCORPORATION OF ZEBRA CROSSING



INSTALLING SURFACE INDICATORS AT THE START OF RAMP



ADDED SIGNAGES



IMPLEMENTING BRAILLE IMPRINTS ON HANDRAILS



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A REVIEW ON ENERGY MANAGEMENT FOR CONTEMPORARY
AIRPORT DEVELOPMENT: SOLAR ENERGY UTILIZATION

A Research Report
Submitted in fulfilment of the requirements
of
Architectural Research Writing Elective
for the academic year 2025 - 26

By
Varas H. Patel and Precious Elaish J. Nathan
(Roll Number: **AR23046, AR23046**)

Conducted and Guided by::
Ar. Aparna Rao, Dr. Nitin Rane and Ar. Akanksha Katare
faculty – VESCOA
(Guides)



VIVEKANAND EDUCATION SOCIETY'S
COLLEGE OF ARCHITECTURE

2025-2026

A Review On Energy Management For
Contemporary Airport Development: Solar
Energy Utilization

Abstract: Airports of the 21st century are energy intensive centres, their energy demand can sometimes be comparable to small cities. Airports require large amounts of energy for terminal operations, airfield systems and ground transport activities. This high demand results in exponential rise in operating costs, carbon emissions and embodied energy. This paper examines the energy management approaches that help to transition towards more efficient and sustainable airport development, with emphasis on solar energy utilization. The study reviews infrastructural upgrades required to integrate renewable energy systems into airport environments. The review highlights the importance of resilient microgrids that enable airports to generate, store and distribute their own renewable energy aiding operational reliability. These measures support the emerging concept of airports as "energy hubs". Overall, the findings show that integration of solar energy can aid to transform airport infrastructure into a more sustainable, self-reliant and future-ready energy model.

Keywords: Airport infrastructure, Energy transition, Renewable Integration, Solar Energy, Decarbonization, Solar PV Panels.

1. Aim

To review integration of solar energy to enhance sustainability and operational efficiency in contemporary airport development.

2. Objective

- Examining the different types of energy efficient strategies used for sustainable airport design.
- Identifying the uses and benefits of solar energy capturing devices for sustainable airport design.
- Recording comparisons, advantages, and disadvantages achieved by airports which were able to integrate sustainable strategies for efficiency.

3. Introduction

Airports are highly complex infrastructures that are operating round the clock, supporting large volumes of passenger movements, aircraft operations, cargo handling and a wide range of commercial activities. The global aviation sector faces significant difficulties to match its projected growth with international climate goals (Zein, et al., 2025). Decreasing carbon emissions in flight operations and ground infrastructure is an important global task. Although the main focus has been on airborne emissions, reducing the immediate climate impacts from airport to ground operations is key. This area is often more controllable and allows for quicker action. The move toward sustainable energy management at airports is a necessary response to global targets and operational needs.

Carbon emissions, also called as Greenhouse Gas Emissions (GHG emissions), are released gasses that act like a layer of glass in a greenhouse (Ibeas, 2024). The greenhouse effect is essentially Earth's natural insulation. It is a process that is vital for life, enabling our planet to retain heat that arrives as



Certificate

This is to certify that the project report entitled "**A REVIEW ON ENERGY MANAGEMENT FOR CONTEMPORARY AIRPORT DEVELOPMENT: SOLAR ENERGY UTILIZATION**", submitted to the Vivekanand Education Society's College of Architecture, in fulfillment for the Architectural Research Writing Elective for the degree of Bachelor of Architecture, is a record of bona fide work carried out by:

Name of the Student: **PATEL VARAS HITESH, NATHAN PRECIOUS ELAISH JOHNSON**

Roll No: **AR23054, AR23046**

Academic Year: **2025-2026**

Under my supervision and guidance.

All references received from various sources have been duly acknowledged.

**Dr. Nitin Rane, Ar. Aparna Rao and
Ar. Akanksha Katare**
Professor
Research Guides

Dr. Prof. Anand Achari
Principal – VESCOA

Place: Mumbai

Date: _____

solar radiation, which includes both visible and ultraviolet light. Think of the atmosphere acting like the glass walls of a garden greenhouse: it allows the Sun's energy to pass through and warm the interior, but then it traps much of that heat from escaping, keeping the Earth warm enough to support liquid water and maintain a habitable temperature (Ibeas, 2024).

The global aviation industry is currently the cause for approximately 3% of worldwide Carbon emissions, a number that is bound to increase significantly unless immediate mitigation strategies are initiated (Yue, et al., 2021). Consequently, the sector is under intense regulatory and commercial pressure to achieve target of net-zero emissions. Airports that function as vast energy hubs that are comparable in consumption to small or medium-sized cities, require crucial intervention in this decarbonization effort (Yue, et al., 2021). The increase of air traffic and the multifunctional nature of modern airport spaces, which now include terminals, hotels, cargo centres, and different transport nodes, drive a continuous increase in electricity demand and, in correspondence to, an expansion of their carbon footprint. This urgency requires a transition from traditional energy strategies to localized, renewable energy generation (Nam, 2019).

The paper discusses harnessing solar energy and its integration into airport infrastructure through photovoltaic systems. It outlines the sequence of solar energy utilisation, mechanisms through which photovoltaic devices harness, convert, store and distribute energy across airport operations. Evaluation of different applications of the stored energy and how it can be converted into other useful forms. Additionally, it compares solar energy usage with conventional energy demand, examining the role of solar power in helping to reduce carbon emissions produced by airports. These discussions highlight the role and importance of solar energy in creating more sustainable and efficient airport environments.

4. Airports as Multi-Layered Urban Systems

The functioning of airports is as complex as multi-layered urban systems, operating far beyond the conventional definition of transport hubs. Airports are dense infrastructural ecosystems containing 'surveillance grids, airfield lighting networks, and uninterrupted 24/7 operational activities, all of which contribute to massive electricity consumption. Airports are classified under the "service building sector," a category that can account for 40% of nation's building related energy demand in developed economies, highlighting the intense energy profile associated with aviation infrastructure (Ortega & Manana, 2016). The continuous annual growth in passenger movement and aircraft operations in India has resulted in rapidly escalating terminal electricity loads, with systems such as cooling, lighting, communications, and passenger amenities contributing to long-term demand increases (Choudhary , et al., 2020). Airports are also regional economic anchors, stimulating tourism, job creation, trade flows, and urban expansion which are factors that reinforce the need for resilient, energy-efficient development strategies for contemporary airports (Melgar, et al., 2024).

Airports are particularly favourable for large-scale solar photovoltaic (PV) energy adoption due to their several inherent characteristics. Vast tracts of unused land that remain unshaded and structurally unobstructed such as buffer zones, approach areas, airside peripheries, and building rooftops that cannot host tall structures due to aviation safety regulations are ideal locations for solar farms, as highlighted in multinational case studies of airport PV deployment (Ortega & Manana, 2016) (Choudhary , et al., 2020) (Teofilu, et al., 2021). Load profile of airports is highly predictable and largely daytime-dominated, this aligns almost perfectly with the generation curve of solar PV systems. Studies from Indian, Australian, and Pakistani airports show that the majority of aviation energy

Declaration

"I hereby declare that this written submission entitled "**A REVIEW ON ENERGY MANAGEMENT FOR CONTEMPORARY AIRPORT DEVELOPMENT: SOLAR ENERGY UTILIZATION**" by **ADURKAR MANASI VINIT** represents my ideas in my own words and has not been taken from the work of others (as from books, articles, essays, dissertations, other media and online); and where others' ideas or words have been included, I have adequately cited and referenced the original sources. Direct quotations from books, journal articles, internet sources, other texts, or any other source whatsoever are acknowledged, and the source cited are identified in the dissertation references.

No material other than that cited and listed has been used. I have read and know the meaning of plagiarism and I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the consequences should I engage in plagiarism, collusion or copying. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact source in my submission. This work, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

Name of the Student: **PATEL VARAS HITESH, NATHAN PRECIOUS ELAISH**

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Place: Mumbai

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demand occurs during daylight when terminal operations, cooling loads, and airside movements are at their peak, enabling airports to consume solar energy directly without excessive storage requirements (Choudhary , et al., 2020) (Muhammad, et al., 2024) (Teofilu, et al., 2021). Typically in airports, strong distribution networks are already in place to handle large loads, these include robust transformers, ring main networks, and redundant feeds which simplify grid integration of solar power and supports higher renewable penetration (Melgar, et al., 2024) (Teofilu, et al., 2021).

There is an availability of large roof expanses at airports such as terminal roofs, hangar roofs, cargo sheds, and maintenance buildings that often exceed the usable roof area of standard commercial complexes. Australian federal airports, for instance, demonstrated the potential to generate nearly 466.7 GWh annually using rooftop systems alone, showing how aviation infrastructure naturally lends itself to distributed solar generation without additional land-use conflicts (Teofilu, et al., 2021). Compliance with aggressive sustainability targets, carbon-neutrality programs, and international aviation guidelines, motivates airport authorities to adopt solar energy as a strategic tool to reduce emissions and operational expenditures. Solar energy integration aligns with emerging models of sustainable airport design, as seen in Singapore's Changi Airport, where large-span roofs, open landscaped interiors, and bioclimatic design principles have already established architectural precedents for renewable-ready infrastructure (Seng, 2015). Another key aspect is the demand of extremely high reliability in energy supply for various airport infrastructure systems such as navigation aids, communication equipment, and runway lighting. It is observed from case studies from Pakistan and Nepal which reveal that solar PV contributes significantly in improving operational resilience in regions which are prone to grid instability, thus ensuring uninterrupted functioning of airport operations during power outages and reinforcing energy security (Muhammad, et al., 2024) (Yadav, et al., 2024). Collectively, these factors such as large land and roof availability, ideal solar exposure, predictable daytime loads, robust electrical infrastructure, sustainability-driven governance, and the need for operational resilience result in airports being exceptionally favourable environments for solar PV integration.

5. Working Principles of Photovoltaic (PV) Devices for harnessing solar energy

5.1. The Photovoltaic Effect

The functioning of a PV cell typically relies on a silicon semiconductor that responds to incoming sunlight through photovoltaic effect.

Absorption of photons occurs as the sunlight strikes the surface of semiconductor. Energy is imparted to electrons by photons, forming electron-hole pairs and these charge carriers begin moving across the PN junction. As they move, the metal contacts on the solar cell capture and guide their flow, forming an electric current within the circuit. Through this process, the photovoltaic device ultimately produces direct current (DC) electricity, which can then be converted for practical use.

5.2. Conversion flow: From Sunlight to Usable Electricity

At several stages of photovoltaic system, there are minor losses. Beginning at module level where performance reduction occurs due to factors such as high temperature and cell mismatch. Additional losses arise within the inverters due to inefficiencies in transforming DC power to AC. Resistance and transmission limitations in cables and connectors also add to these losses. Further environmental

conditions such as dust accumulation and shading decrease amount the amount of electricity produced by PV System.

6. PV Modules and Materials Used in Airport Solar Systems

6.1. Crystalline Silicon Modules

Crystalline silicon is one of the most widely used photovoltaic technologies in airport installations due to its durability, efficiency, and commercial availability. Monocrystalline silicon modules were adopted by Indian airport sustainability study because of their superior efficiency and performance under India's high solar irradiance conditions (Choudhary , et al., 2020). Although less efficient, polycrystalline modules remain suitable alternatives for airports aiming at cost-effectiveness without compromising on overall energy output.

6.2. Thin-Film Photovoltaic Modules

In certain climates that are characterized by high temperatures or diffused sunlight, strong performance is demonstrated by thin-film solar technologies. In Pakistan's 7E framework evaluation, thin-film modules equipped with single-axis tracking systems provided optimal results at Quetta Airport because of their favourable thermal performance and their ability to absorb solar energy across a broader spectral range, making them less sensitive to fluctuating radiation levels (Muhammad, et al., 2024). This indicates that thin-film PV can outperform crystalline options in specific environmental contexts.

6.3. Anti-Reflective and Low-Glare Glass

The risk of glint and glare from PV modules is a major safety concern in highly regulated aviation environments like airports. Reflections from solar panels can interfere with the line-of-sight of air-traffic controllers or impair pilot's visibility during approach and landing. It was observed in Egyptian PV feasibility study that utilising anti-reflective coatings and specialized low-glare glass can minimize reflective hazards while maintaining high electrical output (Mahmoud , et al., 2017). These material considerations aid to ensure compatibility of solar installations with aviation safety requirements.

6.4. Structural and Mounting Materials

PV arrays at airports require mounting structures that are capable of withstanding strong winds, thermal expansion and outdoor exposure. Lightweight but durable materials such as aluminium rails are commonly used in rooftop installations while galvanized steel supports are often employed in ground-mounted trackers and fixed-tilt arrays. The choice of structural material must meet stringent reliability standards to ensure longevity and reduce maintenance requirements because these operate in open airfield environments.

7. Components of Airport PV Systems

A complete airport PV system consists of the following modules:

7.1. PV Modules

Photovoltaic modules, also known as solar panels, are a capturing screen that captures solar energy to convert it into Direct Current electrical energy. A semiconductor material, usually silicon, is the main constituent of each individual solar cell. It is sensitive to light and generates electricity when hit by the rays of the sun because of a physical phenomenon called the photovoltaic effect.

7.2. Inverters

The direct current from PV modules is transformed into alternating current by inverters. High-quality inverters with strong maximum power point tracking (MPPT) capabilities ensure that the system's output remains optimized despite environmental variations.

7.3. Transformers and Distribution Equipment

Transformers adjust voltage levels to integrate solar electricity into the airport's distribution network. Switchgear and protection devices regulate electrical flows, prevent overloads, and ensure safe system operation.

7.4. Cabling, Grounding, and EMI-Shielding

The cabling and grounding systems of PV installations must be carefully designed in airport environments due to the potential for electromagnetic interference (EMI). Poorly shielded cables can interfere with sensitive aviation navigation equipment such as instrument landing systems (ILS), radar, and VOR stations. The Egyptian studies stress the necessity of EMI-safe installation practices to protect operational safety (Mahmoud , et al., 2017).

7.5. Monitoring and Supervisory Systems

Supervisory Control and Data Acquisition (SCADA) systems continuously monitor energy production, module temperature, inverter efficiency, and overall system performance. This enables real-time fault detection and data-driven maintenance.

7.6 Battery Storage Systems

Advanced airport microgrid designs incorporate battery energy storage to smooth output fluctuations, provide backup power, and enhance the penetration of renewable energy. The Copenhagen Airport MPC-based model demonstrates that integrated storage systems can help achieve renewable penetration rates exceeding eighty percent during favourable seasons (Goh, et al., 2024).

8. Airports as ideal sites for solar photovoltaics

It is observed that the unique combination of spatial, operational, and environmental characteristics offered by airports make them exceptionally well suited for solar energy adoption. Airports occupy large tracts of land, it provides large portion of space which includes exclusion zones, building

rooftops, parking areas, and buffer lands that are unused but receive unobstructed sunlight, providing an ideal condition to deploy ground-mounted or rooftop PV arrays. (Choudhary , et al., 2020) (Mahmoud , et al., 2017) (Muhammad, et al., 2024) (Yadav, et al., 2024) (Teofilo, et al., 2021)



Fig. 1. Spatial, Environmental, and Operational characteristics of Airport PV (Choudhary , et al., 2020) (Mahmoud , et al., 2017) (Muhammad, et al., 2024) (Yadav, et al., 2024) (Teofilo, et al., 2021)

High solar irradiance also benefits airport. Egypt, for example possesses airport solar potential exceeding three gigawatts due to its consistently high radiation levels (Mahmoud , et al., 2017). Similar favourable conditions are present in Australia, enabling more than 466.7 gigawatt-hours of potential annual rooftop solar production (Teofilo, et al., 2021).

Alignment between energy demand and solar output is another major advantage. There is a rise in airport energy consumption during daytime hours due to passenger processing, cooling loads, and operational activity. Since solar generation also peaks during the daytime, PV systems can satisfy a significant share of the airport's immediate electricity needs without requiring extensive storage solutions (Ortega & Manana, 2016).

Along with operational advantages, solar energy ensures energy security of airports located in regions with unstable power grids. Grid unreliability is identified as a serious operational risk by the Pakistani and Nepalese studies and PV systems aid in improving energy resilience and autonomy (Muhammad, et al., 2024) (Yadav, et al., 2024). Furthermore, airports adopting renewable energy contribute positively to their public image and national sustainability objectives, as illustrated by Changi Airport's integration of environmental design into its architectural identity.

9. Comparisons of international airport solar systems

Comparative evaluation of solar PV airports reveals significant variations in performance linked to irradiance, technology selection, and system design.

9.1. India

Feasibility studies for the Udaipur, Raipur, and Aurangabad airports revealed PV capacities between 2.08 and 3.20 megawatt-peak, delivering annual outputs between 4238 and 5012 megawatt-hours. These projects achieved performance ratios of approximately 77 to 79 % and reduced carbon dioxide emissions by 3168 to 4635 tons annually (Choudhary , et al., 2020).

9.2. Egypt

The proposed PV system at Hurghada International Airport exhibited an annual generation potential of 55,401 megawatt-hours and more than 27,700 tons of annual carbon savings. However, despite its strong technical feasibility, the project faces significant financial challenges due to high interest rates, inflation, and volatile currency conditions (Mahmoud , et al., 2017).

9.3. Pakistan

The multi-airport study identified Quetta Airport as the strongest candidate for PV deployment because of its high solar irradiance and favourable system performance. The system modelled for Quetta offered a reference yield of 2752 kWh per kilowatt installed, a CUF of 27.63 percent, an LCOE of 0.031 USD per kWh, and a projected reduction of 5730 tons of annual carbon dioxide emissions, making it one of the most economically advantageous studies (Muhammad, et al., 2024).

9.4. Australia

The assessment of twenty-one federal airports shows that they collectively possess the potential to generate 466.7 gigawatt-hours every year from rooftop solar alone, which would reduce carbon emissions by more than 151,600 tons annually (Teofilo, et al., 2021).

9.5. Nepal

The Biratnagar Airport hybrid solar–wind model demonstrates how small airports in developing countries can function primarily on renewable energy when systems are correctly sized and optimized using tools such as HOMER (Yadav, et al., 2024).

9.6. Copenhagen

The Copenhagen Airport represents one of the most advanced future models, where simulations indicate that integrating PV with wind and waste-to-energy sources under a model predictive control framework can achieve renewable penetration of up to eighty-one percent during summer months (Yue, et al., 2021).

AIRPORT AND LOCATION	PERFORMANCE METRICS	TECHNOLOGY AND SYSTEM DESIGN	AFFECTED FACTORS
India (Udaipur, Raipur, Aurangabad)	Energy Capacity: 2.08 to 3.20 MWp	Standard PV system (Implied)	Performance ratings (PR): ~77 to 79 % (Indicates good operational efficiency within Indian context).
Egypt (Hurghada International Airport)	High Annual Energy Generation Potential: 55,401 MWh/yr	Proposed large-scale PV system	Financial Challenges: Faces significant difficulties due to high interest rates, inflation, and volatile currency; despite strong technology.
Pakistan (Quetta Airport)	High Reference Yield: 2752 kWh/kW installed	Standard PV system (Implied)	Irradiance/Economic Advantage: Has the strongest potential due to high solar irradiance; Extremely low LCOE of 0.031 USD/kWh, making it one of the most economically advantageous studies.
Australia (Twenty-one Federal Airports)	High Collective Potential: 466.7 GW/yr overall	Only Roof Solar energy capture is possible. But there is scope for expansion.	High collective potential from existing infrastructure; Focus on leveraging large volumes of roof space.
Nepal (Biratnagar Airport)	Focus on primary function with renewable energy	Hybrid Solar–Wind Model: Optimized using tools like HOMER	Representative model for small airports in developing countries, emphasizing correct system sizing and optimization for renewable energy dependence.
Copenhagen (Copenhagen Airport)	High Renewable Penetration: Up to 81 % only during summers	Advanced Integration of PV combined with wind and reusable energy from energy sources; Managed by Model Predictive Control (MPC) framework.	Advanced future model; Focus on complex, multi-source system integration and smart control for maximizing renewable energy use.

Table 1. Comparison of airports (Choudhary , et al., 2020) (Muhammad, et al., 2024) (Teofilo, et al., 2021) (Yadav, et al., 2024) (Yue, et al., 2021)

10. Differences between solar-powered and non-solar airports

10.1. Non-solar airports

Airports that do not rely on solar energy remain heavily dependent on grid electricity and fossil fuels, which exposes them to rising energy costs, fluctuating fuel prices, and the operational risks associated with grid instability. This is particularly evident in the case of Pakistan and Nepal, where energy shortages and unreliable grids pose significant challenges to airport operation, safety, and passenger

comfort (Muhammad, et al., 2024), (Yadav, et al., 2024). Non-solar airports also contribute more heavily to carbon emissions and struggle to meet global sustainability targets.

10.2. Solar-powered airports

Solar-powered airports, on the other hand, gain substantial advantages by reducing grid dependency, lowering operating costs, and improving energy reliability. Indian airports with PV installations benefit from immediate reductions in electricity expenditure due to on-site generation [2]. Egyptian and Australian airports demonstrate large-scale carbon-reduction potential, with annual savings reaching tens of thousands of tons of CO₂ (Mahmoud , et al., 2017) , (Teofilo, et al., 2021). Airports adopting hybrid renewable-energy systems, such as those modelled for Copenhagen Airport, further enhance energy resilience and operational stability (Yue, et al., 2021). Finally, solar-powered airports improve their sustainability branding, contributing positively to national environmental commitments and public perception.

Changi Airport's sustainability-focused architectural strategies offer a vivid example of how renewable energy and environmental design can reinforce national identity (Seng, 2015).

FEATURES	NON-SOLAR AIRPORTS	SOLAR POWERED AIRPORTS
Energy Source Dependency	Heavy reliance on grid electricity and fossil fuels.	Reduced dependency on the grid, leveraging on-site PV generation and hybrid renewable systems.
Energy Costs & Exposure	Exposed to rising energy costs and fluctuating fuel prices.	Benefit from immediate reductions in electricity expenditure and greater cost stability.
Operational Reliability	Prone to operational risks associated with grid instability and energy shortages (e.g., Pakistan, Nepal).	Improved energy reliability and enhanced operational stability (especially with hybrid systems like the one modelled for Copenhagen).
Carbon Emissions	Contribute more heavily to carbon emissions.	Demonstrate large-scale carbon-reduction potential (e.g., tens of thousands of tons of CO ₂ savings in Egypt and Australia).
Sustainability & Branding	Struggle to meet global sustainability targets.	Improve sustainability branding, contributing positively to national environmental commitments and public perception (e.g., Changi Airport).
Impact on National Goals	Makes meeting national environmental commitments difficult.	Directly supports national environmental commitments and enhances reputation.

Table 2. Comparison between Non-Solar and Solar Airports (Muhammad, et al., 2024) (Yadav, et al., 2024) (Mahmoud , et al., 2017) (Teofilo, et al., 2021) (Yue, et al., 2021)

11. Challenges

11.1. Glint and Glare

One of the most significant concerns is glint and glare. Reflections from PV surfaces can affect pilot visibility or the operations of the air-traffic-control tower. The Egyptian studies underline the necessity of anti-reflective coatings, careful panel orientation, and glare analysis tools to address this risk effectively (Mahmoud , et al., 2017).

11.2. Electromagnetic interference

Electromagnetic interference (EMI) represents another significant challenge. Poorly designed PV electrical systems can disrupt crucial aviation navigation equipment, necessitating strict compliance with EMI safety guidelines and robust shielding of cables and inverters (Mahmoud , et al., 2017).

11.3. Wildlife

Airport solar installations also have the potential to influence wildlife behaviour, particularly by attracting birds, which increases the risk of bird strikes. Addressing this issue requires integrated wildlife management strategies and careful siting of PV arrays.

11.4. Finance and policy-related

Egypt's PV projects struggle with high financing costs, currency instability, and insufficient policy incentives, which hinder the implementation of even technically sound renewable-energy systems (Mahmoud , et al., 2017).

CHALLENGES	CAIRO INTERNATIONAL AIRPORT, EGYPT	HARTSFIELD-JACKSON ATLANTA AIRPORT, USA	COCHIN INTERNATIONAL AIRPORT, INDIA
Glint and Glare	High Concern. Requires anti-reflective coatings, careful panel orientation, and detailed glare analyst tools to protect pilot visibility and air-traffic-control tower operations.	High Concern. A primary safety roadblock. The FAA (Federal Aviation Administration) provides specific technical guidance and requires checklists/assessments to ensure no risk to operations.	Managed Concern. Generally addressed through prior glare assessment during the feasibility stage to ensure flight paths and the ATC tower are free from glare.
Electromagnetic Interference (EMI)	High Concern. Poorly designed PV electrical system pose a risk of disrupting crucial aviation navigation equipment like radar, ILS, MDR, and DME. Requires robust shielding of cables/inverters and strict con with safety guidelines.	High Concern. Interference with radar, communication, and navigation equipment is a ma. technical challenge. Mitigation includes maintaining safe separation distances (e.g., 150-500 feet) and using shielded components.	Managed Concern. The implementation of the mega-scale solar plant demonstrated the technical feasibility to avoid signal loss or false signals by following safety distance standards.
Wildlife	Concern. Solar installations can attract	Concern. Arrays can attract hazardous	Managed/Mitigated. The successful

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14. Research Gaps

Several research gaps remain unaddressed despite strong evidence supporting solar PV integration at airports. Firstly, while most studies focus on annual energy generation and technical feasibility only a limited amount of exertion assesses long term degradation, efficiency under extreme change in climate, and behaviour of life cycle to cost relationship trend for a PV system for airport scales. Secondly, analysing economics do not commonly implement real financing restrictions, fluctuation of interest rates, or uncertainty in policies that affect decision making in airports, which was previously demonstrated in Pakistan and Egypt. Thirdly, integration of PV in hybrid systems, storage, waste-to-energy systems, or wind lack the standardized frameworks for optimization which are rendered specifically for airport operational databases, especially for demands such as futuristic electric aircrafts and ground electric fleets. Fourth, environmental examinations usually focus decarbonization only, which caused overlooking broader impacts such as conflicts in land use, heat island effects, and considerations in biodiversity near runways. Finally, comparing global benchmarking throughout the airports is very limited, which restricts the development of universal design guidelines for energy ready efficient infrastructure of airports.

15. Results And Discussions

The findings gathered from global case studies of solar energy integration at airports reveal a broad consensus that from both technical and operational perspectives, airports are well suited for large scale photovoltaic systems integration. Airports demonstrate strong photovoltaic performance and significant potential for enhanced resilience, cost savings, and environmental improvement, referring to studies conducted in India, Egypt, Pakistan, Nepal, Australia, and Denmark. While performance values differ across climates, technologies, and economic contexts, the collective results consistently show that airports can harness solar energy to reduce dependence on grid electricity, lower carbon emissions, and progress toward more sustainable aviation infrastructure. This section synthesizes results from the literature and discusses emerging themes related to energy performance, economic feasibility, environmental impact, operational resilience, and futuristic energy models.

15.1. Global Findings

General trend observed across international studies indicates that airports with solar PV systems have exhibited strong annual yields, reliable performance ratios, and substantial environmental gains. Airports surveyed in India, including Udaipur, Raipur, and Aurangabad, achieved annual generation between 4238 and 5012 MWh, supported by performance ratios around 77% to 79%. These consistent values demonstrate well sized systems optimized for India's reliable irradiance profile (Choudhary , et al., 2020).

Hurghada airport in Egypt achieved highest technical performance among all reviewed cases, with projected annual generation exceeding 55,000 MWh. Egypt's geographical positioning within a global solar belt, providing consistently high solar irradiance throughout the year directly influences this output. However, an important take away from the Egyptian case reveals an important theme that is technical feasibility does not guarantee proper implementation if economic and policy frameworks remain unstable (Mahmoud , et al., 2017).

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	birds, increasing the risk of bird strikes. Requires integral wildlife management strategies and careful siting e arrays.	wildlife (e.g., birds and insects) by providing perches, shade, or by reflecting polarized light that mimics water. Requires periodic monitoring of bird activity and mitigation measures.	implementation suggests this was not a primary constraint. Cochin has an established system and has also implemented agro-photovoltaic activities (growing vegetables between panels), which is a unique approach to land use.
Financial and Policy-Related	High Barrier. Struggled with high financing costs, currency instability, and insufficient policy incentive which significantly hinder project implementation despite technical soundness.	Managed Barrier. Supported by strong government and corporate policy drivers (City of Atlanta's 100% Clean Energy goal, ACI Net-Zero commitment), making solar financially viable. Focus is on overcoming technical hurdles within a supportive policy framework	Initial barriers, including high photovoltaic system costs and uncertainty in power banking and settlement regulations with the state utility, were mitigated through supportive renewable energy directives issued by the State Regulatory Commission and the strong financial rationale created by elevated pre solar electricity tariffs.

Table 3. Challenges faced by Airports (Mahmoud , et al., 2017)

12. Futuristic models and approaches

The transition toward future, energy-autonomous airports requires sophisticated optimization and energy-management models.

12.1. Copenhagen Airport

Copenhagen Airport's integrated renewable-energy simulation demonstrates how PV, wind, and waste-to-energy systems can be co-ordinately operated using model predictive control to maximize renewable penetration and minimize operational costs (Yue, et al., 2021).

12.2. Dutch Airport

Dutch airport electrification studies using mixed-integer linear programming models show how airports can prepare for electrified aircraft, electric ground-handling vehicles, and expanded battery storage networks by optimizing the size and placement of renewable-energy infrastructure (Yue, et al., 2021).

12.3. Pakistan

Pakistan's 7E analytical framework provides a comprehensive approach to evaluating energy systems using criteria such as energy efficiency, exergy performance, environmental benefits, and economic viability. This holistic perspective helps airport planners identify the most sustainable energy options under different climatic and economic conditions (Muhammad, et al., 2024).

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A different layer of insight is observed in Pakistan's airports. From evaluation of Quetta Airport, it is revealed that thin-film PV technologies, combined with single-axis tracking, can outperform crystalline silicon modules in high-temperature, diffuse-light environments. With a CUF of 27.63 percent and an LCOE of only 0.031 USD/kWh, strong economic and technical performance is demonstrated by Quetta airport, showing that region-specific technology selection is crucial (Muhammad, et al., 2024).

Australia's federal airport rooftop analysis shows that aggregated airport networks provide considerable national renewable-energy capacity, with potential generation exceeding 466.7 GWh annually. This enormous figure demonstrates that airports can meaningfully contribute to national energy portfolios even without dedicated ground-mounted systems (Teofilo, et al., 2021).

Hybrid solar-wind model of Nepal highlights importance of multi-resource renewable systems in region with variable sunlight. Meanwhile Denmark's Copenhagen airport's advanced renewable energy simulation demonstrates the future direction of airport energy systems: intelligent, multi-source renewable microgrids with model predictive control achieving renewable penetration above eighty percent (Yadav, et al., 2024).

Collectively, it can be inferred from these findings that solar energy is technically viable and beneficial option for airports across diverse geographic and economic contexts.

AIRPORT REGION	FINDINGS SUMMARY	PV FEASIBILITY OUTCOME
India	Stable performance, moderate high irradiance, reliable power rating	Consistently successful operations; strong daytime alignment with energy loads
Egypt	Very high solar irradiance, highest generation potential among all the cases	Optimal feasibility, financially sensitive
Pakistan	Effective tracking, suited to hot, semi-arid climate	Climate reactive performance, subject to very low financial issues
Nepal	Variable climate, hybrid solar wind increases supply reliability	Technically viable under hybrid system
Australia	Large uninterrupted rooftop areas	High technical feasibility
Copenhagen	High renewable penetration through hybrid renewable energy predictive control	Model for future energy efficient airports

Table 4. Global Comparative Summary of PV findings in Airports (Choudhary , et al., 2020) (Mahmoud , et al., 2017)

15.2. Comparative Performance of Airport Solar PV Systems

15.2.1. Energy Generation Across Different Climates

Difference in climatic conditions strongly influenced energy generation outcomes across the studied airports. Desert climate of Egypt yielded the strongest technical performance due to its high global horizontal irradiance (GHI) and minimal season fluctuation (Mahmoud , et al., 2017). Australian airports demonstrated excellent generation stability because of consistent sunlight and low cloud cover (Teofilo, et al., 2021). India benefits from long annual sunshine duration. Even though it is not at the extreme end of solar irradiance, medium-sized systems can still deliver reliable results (Choudhary , et al., 2020). Pakistan's Quetta airport achieved exceptional CUF values due to the suitability of thin-film technology for high-temperature semi-arid climates (Muhammad, et al., 2024). Variable sun conditions of Nepal required hybridization with wind resources to ensure continuous

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13. Methodology

The findings of this review revolve around contemporary airports being technically and operationally ideal sites for large-scale solar photovoltaic implementation, providing a crucial route toward decarbonization and increased energy resilience. Airports function as energy filled, multi-layered urban systems with expansive, majorly unused, land and rooftop areas that are perfectly suited for PV panels implications (Choudhary , et al., 2020) (Mahmoud , et al., 2017) (Teofilo, et al., 2021). Furthermore, the airport's vast and predictable daytime electricity demand aligns adequately with the solar energy generated, minimizing the need for storing extensive energy (Ortega & Manana, 2016). International examples, from India and Australia, which have shown high generation yields (Choudhary , et al., 2020), and Pakistan and Nepal, which show improved energy security in unstable arrays, consistently confirm the operability and benefits (Muhammad, et al., 2024) (Yadav, et al., 2024). Solar energy greatly reduces reliance on grid electricity, lowers operational costs, and achieves substantial reduction of annual carbon emissions, thousands of tons of CO2 annually in Australia's case (Teofilo, et al., 2021). The most technical components, including crystalline silicon modules, specialized anti-reflective glass, and highly effective EMI-shielding, ensures compatibility with rigorous aviation safety standards. The future focus involves transitioning airports into self-reliant hubs of energy through advanced hybrid microgrids and Model Predictive Control systems (MPCS), overcoming challenges like glint and glare and financial barriers to establish a sustainable, future-ready energy model for worldwide aviation.

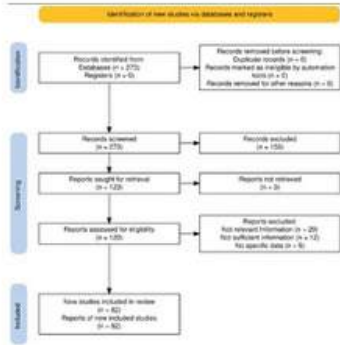


Fig.2. Prisma Diagram

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generation (Yadav, et al., 2024). It is confirmed by the comparative evidence that while solar resource availability is the strongest single determinant of PV performance, even moderate-irradiance regions attain stable outcomes with proper system optimization.

15.2.2. Performance Ratios and CUF Variations

Performance ratios and CUFs variations occurred depending on climate, technology selection and system design. Indian airports displayed stable performance ratio values, that reflected optimized array orientation and minimal shading (Choudhary , et al., 2020). Exceptionally high CUF was observed in Pakistan's Quetta Airport due to the combination of thin-film modules and single-axis tracking (Muhammad, et al., 2024). In contrast, Nepal's systems demonstrated moderate performance ratio values due to climatic variations but maintained high reliability when wind integration was included (Yadav, et al., 2024). Copenhagen Airport's high renewable penetration was not driven by climatic advantage but by advanced predictive-control architecture that increased the effective utilization of multiple renewable sources (Goh, et al., 2024). These results reveal that performance metrics are enhanced through a combination of optimal technology choice and intelligent control strategies.

15.2.3. Influence of Irradiance on Output

From the case studies it was observed that irradiance is the most influential variable influencing PV output. Egypt's exceptional irradiance produced the highest annual yield in the dataset (Mahmoud , et al., 2017), while Australia's stable irradiance patterns supported large-scale rooftop deployment with minimal performance degradation (Teofilo, et al., 2021). India demonstrated moderate but consistent irradiance, enabling predictable output patterns that align well with airport electricity demand (Choudhary , et al., 2020). The Pakistan and Nepal studies further showed that irradiance levels influence not only total output but also the optimal choice of module technology and system configuration (Muhammad, et al., 2024). (Yadav, et al., 2024). These results emphasize that irradiance acts as both a resource potential indicator and a guiding parameter for system design.

15.3. Technology-Specific Performance Differences

15.3.1. Crystalline Silicon vs Thin-Film Modules

Crystalline silicon is dominant technology across airport installations due to its quality of efficiency and robustness. However Indian airports preferentially use monocrystalline modules to maximize energy production from limited rooftop space (Choudhary , et al., 2020). Nevertheless, under high-heat, diffuse-light conditions typical of semi-arid climates thin-film modules can outperform crystalline silicon as illustrated by Pakistan's Quetta Airport (Muhammad, et al., 2024). Thin-film modules' lower temperature coefficient allows them to maintain efficiency in hotter environments, whereas crystalline silicon suffers from thermal losses. These contrasting outcomes highlight the necessity of climate-specific technology selection in airport PV projects.

15.3.2. Role of Tracking Systems

Tracking systems played a crucial role in enhancing the output of Pakistan's PV installations. The superiority of thin-film systems paired with single-axis trackers demonstrated increased daily energy yield and CUF (Muhammad, et al., 2024). In Egypt and Australia, fixed-tilt systems proved sufficient due to their stable irradiance patterns (Mahmoud , et al., 2017) (Teofilo, et al., 2021). It can be

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inferred that though tracking systems are most beneficial in regions with dynamic sun angles or intermittent irradiance, they are less critical where sunlight conditions are consistently optimal.

15.3.3. System Losses and Efficiency Factors

Factors such as thermal degradation, dust accumulation, inverter inefficiency, and cable resistive losses resulted in system losses across all airports. Egypt and Pakistan experienced higher dust deposition owing to desert environments, which necessitated more frequent maintenance cycles (Mahmoud , et al., 2017) (Muhammad, et al., 2024). In colder regions, inverter selection influenced system performance, while in hot climates thermal management remained a critical aspect. Overall, systems with well-designed maintenance strategies and high-quality components, maintained PR values above 75%, indicating strong engineering outcomes.

15.4. Economic and Financial Outcomes

15.4.1. Cost Competitiveness

Across reviewed regions, the economic performance of airport solar PV systems has revealed that solar power is one of the most cost-competitive energy options for aviation infrastructure. For instance, Quetta airport of Pakistan has achieved extremely low Levelized Cost of Electricity (LCOE) of 0.031 USD/kWh, attributed to strong natural irradiance, cost effective thin film modules and high CUF values enabled by single-axis tracking (Muhammad, et al., 2024). Rapidly declining module prices and favourable renewable energy policies are benefitting Indian airports and enabling mid-sized airports to adopt PV Systems with financially attractive payback periods (Choudhary , et al., 2020). In Australia, potential of just the rooftop-only PV power generation exceeds 466.7 GWh/year opens opportunity to leverage economies of scale and reduced Balance of System (BOS) expenses which aid in enhancing cost-effectivity of operations (Teofilo, et al., 2021). Hurghada airport in Egypt theoretically demonstrates strong economic potential due to high annual yields, but macroeconomic instability affects financial viability that surges borrowing costs and reduced investor confidence (Mahmoud , et al., 2017). It is inferred from the cross-regional comparisons that while solar PV is inherently cost-competitive due to minimal operational expenses and free solar resource availability, economic feasibility is significantly strengthened when paired with stable policy frameworks and accessible financing mechanisms. Airports thus benefit both operationally and financially from adopting solar energy.

15.4.2. Role of Policy and Financial Frameworks

National policy environments and financial frameworks strongly shape the success of solar PV integration within airports. India represents a model case in which supportive policies such as accelerated depreciation, tax benefits, and reduced GST for renewable components have accelerated the solar adoption despite moderate irradiance in some regions (Choudhary , et al., 2020). In contrast, Egypt’s exceptionally high solar resource cannot be harnessed to its full potential due to financial barriers such as high interest rates, limited funding access, and insufficient governmental incentives’ which have limited large scale airport PV installations (Mahmoud , et al., 2017). Inconsistent renewable-energy policies and limited long-term financing channels have been a constraint in Pakistan’s strong technical benefits, these factors impede national scale adoption despite high CUF and low LCOE demonstrated at Quetta Airport (Muhammad, et al., 2024). Stable policy ecosystem benefits Australian airports by enabling high rooftop adoption and long-term renewable expansion strategies (Teofilo, et al., 2021). Hybrid PV–wind model of Nepal highlights the need for international

renewable systems achieve high reliability and renewable penetration (Goh, et al., 2024). Australia’s distributed rooftop PV reduces dependence on transmission networks (Teofilo, et al., 2021). India benefits through reduced daytime grid load (Choudhary , et al., 2020). In this way PV aids in airport’s energy autonomy and infrastructure resilience.

15.7. Feasibility Across Different Regions

15.7.1. Airports in High-Irradiance Regions

In desert and sunbelt regions like Egypt and Australia the conditions offer strongest feasibility for airport solar PV by achieving very high yields with minimal seasonal variability (Teofilo, et al., 2021). Large available airport land and stable irradiance support utility-scale installations. These help to achieve short payback periods and strong system performance, making such regions ideal for large-scale PV device deployment.

15.7.2. Airports in Developing Economies

Practical barriers such as limited financing, high interest rates, and policy inconsistencies over power the technical potential in developing regions such as Pakistan and Nepal (Muhammad, et al., 2024). In case of Pakistan, the performance derived is excellent, nevertheless the financial and institutional landscape restricts widespread adoption of renewable energy systems (Muhammad, et al., 2024). Hybrid systems or external funding often become necessary for viability.

15.7.3. Airports in Transitional Economies

Through the example of India, it can be understood how declining module prices, stable policies, and favourable financial environments can help to achieve successful deployment of solar PV in transitional economies (Choudhary , et al., 2020). Consistent performance across multiple airports indicate that solar deployment is scalable and cost-effective. Strong national renewable policies complement technical suitability, making India a leading example among mid-income nations.

15.8. Hybrid And Advanced Renewable-Energy Models

15.8.1. Solar–Wind Hybrid Systems

In regions with fluctuating irradiance, hybrid solar–wind systems proved to enhance reliability. When PV output decreases during monsoon seasons, Nepal’s Biratnagar airport relies on hybridization to maintain supply (Yadav, et al., 2024). This reduces reliance on storage and ensures consistent availability for essential operations. Hybrid systems are therefore advantageous for airports facing marked seasonal variability.

15.8.2. Battery Storage Integration

Battery storage enhances renewable penetration by enabling load shifting, peak shaving, and backup support. Although many airports currently operate without large storage due to good daytime alignment, future electrification of ground vehicles will increase storage needs. Modelling shows storage is an essential requirement for high-penetration systems like Copenhagen’s predictive microgrid (Goh, et al., 2024). Storage supports long-term resilience and flexibility.

development funding to support the effort of renewable integration in lower income contexts (Yadav, et al., 2024). Collectively emphasis is laid on policy consistency, financial incentives, and institutional support are essential drivers of renewable integration in airports. Without these frameworks, strong technical feasibility alone cannot ensure implementation.

15.4.3. Regional LCOE Variation and Cost Drivers

Varying Levelized Cost of Electricity across different airports reflect the combined influence of climate, technology choice, financing conditions, and policy design. Extremely low LCOE at Pakistan’s Quetta Airport is a result of high irradiance, low installation costs, and the choice of thin-film modules optimized for local climatic conditions (Muhammad, et al., 2024). Declining module prices, maturing solar markets, and supportive government policies reduce investment risk and encourage competitive bidding which contributes to maintain stable LCOE in India (Teofilo, et al., 2021). In Australia, economies of scale and large installation surfaces significantly reduce BOS costs, resulting in competitive LCOE values across federal airports (Teofilo, et al., 2021). In case of Egypt, high LCOE is not a result of poor solar performance, rather it is due to national financial instability that rises the cost of capital and reduces investor confidence (Mahmoud , et al., 2017). Seasonal wind patterns cause variations in costs of Nepal’s hybrid system which although makes financial performance more complex but still viable when evaluated through long-term resilience benefits (Yadav, et al., 2024). Through these findings it is demonstrated that LCOE is not merely a technological measure but a broader reflection of national economic conditions.

REGION	ECONOMIC STRENGTHS	ECONOMIC LIMITATIONS	POLICY INFLUENCE ON PV ADOPTION
India	Declining module cost; consistent financial environment	Limited impetus at smaller airports	Strong national renewable-energy policies
Egypt	Very high solar resource	High borrowing costs and financial instability	Weak support schemes
Pakistan	Low LCOE wherever implemented	Policy inconsistency and expensive loans	Renewable adoption limited by financing
Australia	High economies of scale and strong energy market	High costs for large roofs	Clear long-term sustainability roadmap
Nepal	International funding accessibility	Limited domestic funding	Hybrid systems favoured under development projects

Table 5. Economic factors affecting airport PV (Choudhary , et al., 2020) (Mahmoud , et al., 2017) (Teofilo, et al., 2021) (Yadav, et al., 2024)

15.8.3. Intelligent Microgrids and MPC-Based Control

Predictive control incorporated by intelligent microgrids significantly improve renewable integration. PV, wind, and waste-to-energy sources are optimized by Copenhagen Airport’s MPC model to achieve renewable penetration of 49–81% depending on the season (Goh, et al., 2024). Such frameworks enable smart, adaptive airport energy ecosystems.

15.9. Cross Comparative Synthesis Of Findings

Egypt and Australia lead in output, this shows irradiance as the strongest driver of PV performance. Technological optimizations such as thin-film, tracking or hybridization enable strong performance in varying climates (Yadav, et al., 2024). Financing strength and policy stability greatly influence economic viability, explaining India and Australia’s success versus Egypt and Pakistan’s constraints. Overall, solar PV provides significant environmental and operational benefits across global airports.

THEME	GLOBAL INSIGHT	AIRPORTS THAT SHOW TREND
Climate Influence	High irradiance gives high PV output	Egypt, Australia
Technology Optimization	Thin-film tracking is more efficient in hot climates	Pakistan
Hybrid Requirement	Variable climates require hybrid renewable energy	Nepal, Copenhagen
Policy as Enabler	State policy results in rapid adoption	India, Australia
Financial Barrier	High interest results in low viability	Egypt, Pakistan
Sustainability Impact	PV results in major CO2 reduction	India, Egypt, Australia, Copenhagen

Table 6. Cross Comparative Combination of Global findings (Teofilo, et al., 2021) (Muhammad, et al., 2024) (Yadav, et al., 2024) (Goh, et al., 2024) (Choudhary , et al., 2020)

16. Conclusion

The paper demonstrates that solar photovoltaic (PV) systems are technically robust, economically viable, and environmentally transformative solution for contemporary airport energy management. Due to large unobstructed land areas, high daytime energy demand, and predictable load profiles airports consistently exhibit strong solar compatibility across the globe, ranging from Egypt’s sunbelt to India’s semi-arid zones and Pakistan’s hot, diffuse-radiation environments. Factors such as climate, module selection, and system configuration result in variations in performance outcome. However, renewable penetration in challenging environments can be significantly enhanced by employing thin-film modules, tracking systems, hybrid solar–wind models, and predictive microgrid controls. Financing conditions and policy support strongly influence economic feasibility of solar PV systems, thus airports in countries with stable regulatory frameworks show rapid adoption and long-term cost benefits. Solar airports also contribute substantial CO2 reductions, strengthening national decarbonization goals. Overall, solar PV integration positions airports as critical nodes in advancing sustainable aviation infrastructure and transitioning toward resilient, low-carbon energy ecosystems.

15.5. Environmental and Sustainability Impacts

15.5.1. Carbon Emission Reduction Potential

Substantial reductions in carbon emissions were observed across all airports. Indian airports had an annual reduction of carbon emissions by 3168-4365 tons (Choudhary , et al., 2020). Egypt’s Hurghada airport’s reduction in carbon emission exceeded 27,700 tons (Mahmoud , et al., 2017). The overall carbon savings of Australian federal airports was above 151,600 tons per year (Teofilo, et al., 2021). These figures highlight the role of PV systems in a meaningful progress towards decarbonization while being environmentally transformative and technically feasible.

15.5.2. Contribution to National Climate Commitments

By reducing grid dependence and operational emissions, airports significantly contribute towards supporting national renewable-energy commitments. India integrates solar airports into national energy transition goals (Choudhary , et al., 2020). Egypt and Australia strengthen national decarbonization pathways through airport PV deployment (Mahmoud , et al., 2017). Renewable integration in variable climates is illustrated by hybrid and predictive systems in Nepal and Copenhagen (Goh, et al., 2024). Therefore, airports play an important role in national climate agendas.

15.5.3. Sustainability Positioning and Public-Perception Benefits

Solar airports project modern, environmentally responsible infrastructure. Changi Airport exemplifies this through bioclimatic architecture and daylighting that enhance its green identity (Seng, 2015). India uses solar airports to demonstrate its commitment towards the environment (Choudhary , et al., 2020), Pakistan and Nepal benefit from improved operational confidence through increased reliability (Yadav, et al., 2024). Along with improving environmental performance, PV adoption enhances public perception while improving performance.

15.6. Operational Impacts On Airport Functioning

15.6.1. Load Matching with Solar Generation Profiles

Airports’ daytime dominated energy demand aligns naturally with solar generation peaks which aid in reducing reliance on external grids and improving self-sustainability. Heavy HVAC loads during day enables Indian airports to benefit from this synchrony (Choudhary , et al., 2020). This favourable aspect enhances operational stability maximizing PV performance without requiring large storage systems.

15.6.2. Enhancing Reliability in Grid-Unstable Regions

In regions facing energy issues such as grid instability, the incorporation of solar PV reinforces operational reliability which reduces dependence on diesel generator, as observed in Pakistan (Muhammad, et al., 2024), while Nepal’s hybrid setup of PV and wind, ensures reliability during instances of climate variability (Yadav, et al., 2024).

15.6.3. Energy Security and Infrastructure Stability

Solar PV enhances long-term energy security by reducing exposure to fuel-price volatility and grid disruptions. Predictive microgrid model adopted in Copenhagen illustrates how multi-source

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